Hartshorne, Chapter III: Cohomology

Solutions to the exercises

Conventions. Schemes are noetherian where the text assumes it; the base field is not assumed algebraically closed unless an exercise says so. Results of the text are cited by their numbers in Hartshorne ("Thm. 5.2", "Prop. 9.7", ...), and results of other chapters as "I, ..." or "II, ...". The numbering and statements of the exercises follow the edition scanned in this repository (in particular Section 1 has no exercises in this edition); each exercise is restated before its solution, and starred exercises are marked "Starred in the book" at the end of the statement.

Honest notes. Several printed statements need an extra hypothesis or correction, and the solutions say what is assumed or proved instead: char $\neq 2$ in Ex. 10.6 and 10.8; the printed values in Ex. 12.3 are false for the untwisted ideal sheaf; Ex. 10.7(b) omits a third case in its classification; Ex. 12.5 needs rank ≥ 2. Ex. 7.4 depends on a choice of normalization of the trace maps (stated as an assumption), and a few solutions use standard facts beyond the text (Matsumura's local flatness criteria, comparison theorems), which are named where used.

1. Derived functors

Section 1 of Chapter III (pp. 202–206 of the book) contains no exercises: the exercises of this chapter begin at the end of Section 2 (numbered 2.1–2.7) and are solved in the next section. The results of this section that are used below are cited by their numbers: Thm. 1.1A (existence of the right derived functors R^iF of a left exact functor F on a category with enough injectives, and the long exact sequences), Prop. 1.2A (acyclic resolutions compute R^iF), Thm. 1.3A (effaceable δ-functors are universal) and Cor. 1.4.

2. Cohomology of sheaves

Conventions. $\mathrm{Ab}(X)$ is the category of sheaves of abelian groups on X; it has enough injectives (Cor. 2.3). For an open set $U \subseteq X$ with inclusion i, and a sheaf $\mathcal{F}$, $\mathcal{F}_U = i_!(\mathcal{F}|_U)$ is the extension by zero (II, Ex. 1.19); for a closed set Y with inclusion j, $\mathcal{F}_Y = j_*(\mathcal{F}|_Y)$. For every sheaf $\mathcal{G}$ one has $\mathrm{Hom}(\mathbf{Z}_U, \mathcal{G}) = \mathcal{G}(U)$ (this was used in the proof of Lemma 2.4). We use freely: injective sheaves are flasque (Lemma 2.4); flasque sheaves are acyclic for $\Gamma(X, \cdot)$ (Prop. 2.5), hence cohomology can be computed from flasque resolutions (Prop. 1.2A); if $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is exact with $\mathcal{F}', \mathcal{F}$ flasque then so is $\mathcal{F}''$, and if $\mathcal{F}'$ is flasque then $\Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}'')$ is surjective (II, Ex. 1.16(b),(c)); restrictions of flasque sheaves to open subsets are flasque; a constant sheaf on an irreducible space is flasque (II, Ex. 1.16(a)).

Exercise 2.1

Problem.

(a) Let $X = \mathbf{A}^1_k$ be the affine line over an infinite field k. Let P, Q be distinct closed points of X, and let $U = X - \{P, Q\}$. Show that $H^1(X, \mathbf{Z}_U) \neq 0$.

(b) More generally, let $Y \subseteq X = \mathbf{A}^n_k$ be the union of $n+1$ hyperplanes in suitably general position, and let $U = X - Y$. Show that $H^n(X, \mathbf{Z}_U) \neq 0$. Thus the result of (2.7) is the best possible.

(Part (b) is starred in the book.)

Solution. (a) Let $Y = \{P, Q\}$ (closed, discrete). By II, Ex. 1.19 there is an exact sequence $0 \to \mathbf{Z}_U \to \mathbf{Z} \to \mathbf{Z}_Y \to 0$ (stalkwise: at $x \in U$ it reads $0 \to \mathbf{Z} \to \mathbf{Z} \to 0 \to 0$, at P it reads $0 \to 0 \to \mathbf{Z} \to \mathbf{Z} \to 0$, similarly at Q, and at other points everything is 0 in the last two places). Here $\mathbf{Z}_Y = \mathbf{Z}_P \oplus \mathbf{Z}_Q$ is a sum of two skyscraper sheaves, so $\Gamma(X, \mathbf{Z}_Y) = \mathbf{Z} \oplus \mathbf{Z}$, and $\Gamma(X, \mathbf{Z}) \to \Gamma(X, \mathbf{Z}_Y)$ is the diagonal map $\mathbf{Z} \to \mathbf{Z} \oplus \mathbf{Z}$. The space X is irreducible, so $\mathbf{Z}$ is flasque and $H^1(X, \mathbf{Z}) = 0$ (Prop. 2.5). The long exact sequence gives

$$\Gamma(X, \mathbf{Z}) \xrightarrow{\Delta} \Gamma(X, \mathbf{Z}_Y) \to H^1(X, \mathbf{Z}_U) \to H^1(X, \mathbf{Z}) = 0,$$

so $H^1(X, \mathbf{Z}_U) \cong (\mathbf{Z} \oplus \mathbf{Z})/\Delta(\mathbf{Z}) \cong \mathbf{Z} \neq 0$. (Only irreducibility of X was used; the hypothesis that k is infinite is not needed.)

(b) Reading of "suitably general position". Let $H_0, \ldots, H_n \subseteq X = \mathbf{A}^n_k$ be hyperplanes such that for every $I \subseteq \{0, \ldots, n\}$ with $1 \leq |I| \leq n$ the intersection $Y_I = \bigcap_{i \in I} H_i$ is nonempty (it is then an affine linear subspace, in particular irreducible), while $\bigcap_{i=0}^n H_i = \emptyset$. Such hyperplanes exist: take $H_i = \{x_i = 0\}$ for $1 \leq i \leq n$ and $H_0 = \{x_1 + \cdots + x_n = 1\}$. (Indeed for $0 \notin I$, Y_I is a coordinate subspace; for $0 \in I$, Y_I is defined by $x_i = 0$ ($i \in I - \{0\}$) and $\sum_{j \notin I} x_j = 1$, which is nonempty since $n - (|I| - 1) \geq 1$ coordinates remain; and all $n+1$ together would force $0 = 1$.) For $n = 1$ this is the situation of (a). Put $Y = \bigcup H_i$, $U = X - Y$. We assume $n \geq 2$; the case $n = 1$ is (a).

For a nonempty closed subvariety W that is irreducible, the sheaf $\mathbf{Z}_W$ has $\mathbf{Z}_W(V) = \mathbf{Z}$ if $V \cap W \neq \emptyset$ (a locally constant function on the irreducible space $V \cap W$ is constant) and 0 otherwise, with restriction maps identity or zero. Hence $\mathbf{Z}_W$ is flasque and $\Gamma(X, \mathbf{Z}_W) = \mathbf{Z}$. For $I \subseteq J$ there is the natural restriction map $\mathbf{Z}_{Y_I} \to \mathbf{Z}_{Y_J}$ (as $Y_J \subseteq Y_I$); on global sections it is the identity of $\mathbf{Z}$ when $Y_J \neq \emptyset$.

Define a complex of sheaves $C^p = \bigoplus_{|I| = p+1} \mathbf{Z}_{Y_I}$ for $0 \leq p \leq n-1$ (the term with $|I| = n+1$ is $\mathbf{Z}_\emptyset = 0$), with the usual alternating-sign (Cech-type) differential, and an augmentation $\mathbf{Z}_Y \to C^0 = \bigoplus_i \mathbf{Z}_{H_i}$ given by the restriction maps. We claim

$$0 \to \mathbf{Z}_Y \to C^0 \to C^1 \to \cdots \to C^{n-1} \to 0$$

is exact. Check on stalks at $x \in X$. If $x \notin Y$ all stalks are 0. If $x \in Y$, let $S = \{i : x \in H_i\}$; then $S \neq \emptyset$ and $|S| \leq n$. The stalk of $\mathbf{Z}_{Y_I}$ at x is $\mathbf{Z}$ if $I \subseteq S$ and 0 otherwise, so the stalk complex is $\mathbf{Z} \to \bigoplus_{i \in S} \mathbf{Z} \to \bigoplus_{i<j \in S} \mathbf{Z} \to \cdots$, the augmented simplicial cochain complex of the full simplex on the vertex set $S \neq \emptyset$ (all subsets $I \subseteq S$ have $|I| \leq n$, so no term is lost), which is exact (it

has a contracting homotopy: fix $s \in S$, and let h delete s from a set containing s, up to sign). So the sequence is exact.

All C^p are flasque, so this is a flasque resolution of $\mathbf{Z}_Y$, and $H^p(X, \mathbf{Z}_Y) = h^p(\Gamma(X, C^\bullet))$ (Prop. 1.2A and Prop. 2.5). Now $\Gamma(X, C^p) = K^p := \bigoplus_{|I|=p+1} \mathbf{Z}$, and the differentials are the simplicial coboundaries. Let $K^\bullet$ be the full complex K^p, $0 \le p \le n$, where $K^n = \mathbf{Z}$ corresponds to $I = \{0, \dots, n\}$, augmented by $\mathbf{Z} \to K^0$; it is the augmented cochain complex of the n-simplex, hence exact. Our complex $\Gamma(X, C^\bullet)$ is its truncation $K^0 \to \cdots \to K^{n-1}$. Therefore, for $n \ge 2$: $h^p = 0$ for $1 \le p \le n-2$ by exactness of $K^\bullet$, $h^0 = \ker(K^0 \to K^1) \cong \mathbf{Z}$, and

$$H^{n-1}(X, \mathbf{Z}_Y) = h^{n-1} = K^{n-1}/\operatorname{im}(K^{n-2} \to K^{n-1}) = K^{n-1}/\ker(K^{n-1} \to K^n) \cong K^n = \mathbf{Z},$$

using exactness of $K^\bullet$ at K^{n-1} and surjectivity of $K^{n-1} \to K^n$ (exactness at K^n).

Now use $0 \to \mathbf{Z}_U \to \mathbf{Z} \to \mathbf{Z}_Y \to 0$. Since X is irreducible, $H^i(X, \mathbf{Z}) = 0$ for $i > 0$, and $n - 1 \ge 1$, so the long exact sequence gives

$$0 = H^{n-1}(X, \mathbf{Z}) \to H^{n-1}(X, \mathbf{Z}_Y) \to H^n(X, \mathbf{Z}_U) \to H^n(X, \mathbf{Z}) = 0,$$

hence $H^n(X, \mathbf{Z}_U) \cong H^{n-1}(X, \mathbf{Z}_Y) \cong \mathbf{Z} \neq 0$. Since $\dim X = n$, Grothendieck's vanishing theorem (2.7) says $H^i = 0$ for $i > n$ and this shows that n cannot be lowered. □

Exercise 2.2

> **Problem.** Let $X = \mathbf{P}^1_k$ be the projective line over an algebraically closed field k. Show that the exact sequence $0 \to \mathcal{O} \to \mathcal{K} \to \mathcal{K}/\mathcal{O} \to 0$ of (II, Ex. 1.21d) is a flasque resolution of $\mathcal{O}$. Conclude from (II, Ex. 1.21e) that $H^i(X, \mathcal{O}) = 0$ for all $i > 0$.

Solution. Recall from II, Ex. 1.21(d),(e): $\mathcal{K}$ is the constant sheaf associated with the function field K of X, $\mathcal{O} \to \mathcal{K}$ is injective, $\mathcal{K}/\mathcal{O} \cong \bigoplus_P i_P(I_P)$ where P runs over the closed points, $I_P = K/\mathcal{O}_P$ and $i_P(I_P)$ is the skyscraper sheaf, and $0 \to \Gamma(X, \mathcal{O}) \to \Gamma(X, \mathcal{K}) \to \Gamma(X, \mathcal{K}/\mathcal{O}) \to 0$ is exact.

$\mathcal{K}$ is a constant sheaf on the irreducible space X, hence flasque (II, Ex. 1.16(a)).

$\mathcal{K}/\mathcal{O}$ is flasque. Let $\mathcal{S} = \bigoplus_P i_P(I_P)$ (sheaf direct sum). Consider the presheaf $\mathcal{S}'(V) = \bigoplus_{P \in V} I_P$ (direct sum of abelian groups over the closed points of V). It is a sheaf: locality is clear, and if $V = \bigcup V_\lambda$ and sections $s_\lambda \in \mathcal{S}'(V_\lambda)$ agree on overlaps, they define a family $s = (s_P)_{P \in V}$; if all V_λ are empty there is nothing to prove, and otherwise some $V_{\lambda_0} \neq \emptyset$ is cofinite in X (open nonempty subsets of $\mathbf{P}^1$ are complements of finite sets), so $V - V_{\lambda_0}$ is finite, and s has finite support because $s|_{V_{\lambda_0}} = s_{\lambda_0}$ does. Since $\mathcal{S}'$ is a sheaf which is the direct sum of the sheaves $i_P(I_P)$ (it satisfies the universal property: sheaf homomorphisms from $\mathcal{S}'$ correspond to compatible families on the summands), $\mathcal{S} = \mathcal{S}'$. Its restriction maps for $W \subseteq V$ are the projections $\bigoplus_{P \in V} I_P \to \bigoplus_{P \in W} I_P$, which are surjective. So $\mathcal{K}/\mathcal{O} \cong \mathcal{S}$ is flasque.

Thus $0 \to \mathcal{O} \to \mathcal{K} \to \mathcal{K}/\mathcal{O} \to 0$ is a flasque resolution of $\mathcal{O}$ (Prop. 2.5, Rem. 2.5.1 and Prop. 1.2A), and $H^i(X, \mathcal{O}) = h^i$ of the complex $0 \to \Gamma(X, \mathcal{K}) \to \Gamma(X, \mathcal{K}/\mathcal{O}) \to 0$ placed in degrees $0, 1$. In degrees $i \ge 2$ this complex is 0, so $H^i(X, \mathcal{O}) = 0$ for $i \ge 2$. In degree 1, $H^1(X, \mathcal{O}) = \operatorname{coker}(\Gamma(X, \mathcal{K}) \to \Gamma(X, \mathcal{K}/\mathcal{O})) = 0$ by the surjectivity in II, Ex. 1.21(e). □

Exercise 2.3

Problem. *Cohomology with Supports* (Grothendieck [7]). Let X be a topological space, let Y be a closed subset, and let $\mathcal{F}$ be a sheaf of abelian groups on X. Let $\Gamma_Y(X,\mathcal{F})$ denote the group of sections of $\mathcal{F}$ with support in Y (II, Ex. 1.20).

(a) Show that $\Gamma_Y(X,\cdot)$ is a left exact functor from $\mathrm{Ab}(X)$ to Ab. We denote the right derived functors by $H^i_Y(X,\cdot)$. They are the *cohomology groups of X with supports in Y*, and coefficients in a given sheaf.

(b) If $0\to\mathcal{F}'\to\mathcal{F}\to\mathcal{F}''\to 0$ is an exact sequence of sheaves, with $\mathcal{F}'$ flasque, show that $0\to\Gamma_Y(X,\mathcal{F}')\to\Gamma_Y(X,\mathcal{F})\to\Gamma_Y(X,\mathcal{F}'')\to 0$ is exact.

(c) Show that if $\mathcal{F}$ is flasque, then $H^i_Y(X,\mathcal{F})=0$ for all $i>0$.

(d) If $\mathcal{F}$ is flasque, show that the sequence $0\to\Gamma_Y(X,\mathcal{F})\to\Gamma(X,\mathcal{F})\to\Gamma(X-Y,\mathcal{F})\to 0$ is exact.

(e) Let $U=X-Y$. Show that for any $\mathcal{F}$, there is a long exact sequence of cohomology groups

$$0\to H^0_Y(X,\mathcal{F})\to H^0(X,\mathcal{F})\to H^0(U,\mathcal{F}|_U)\to H^1_Y(X,\mathcal{F})\to H^1(X,\mathcal{F})\to H^1(U,\mathcal{F}|_U)\to H^2_Y(X,\mathcal{F})\to\cdots.$$

(f) *Excision.* Let V be an open subset of X containing Y. Then there are natural functorial isomorphisms, for all i and $\mathcal{F}$, $H^i_Y(X,\mathcal{F})\cong H^i_Y(V,\mathcal{F}|_V)$.

Solution. Throughout $U=X-Y$, and for a section $s\in\mathcal{F}(W)$ its support is the closed subset of W of points where the germ is nonzero; thus $\Gamma_Y(X,\mathcal{F})=\{s\in\Gamma(X,\mathcal{F}):s|_U=0\}=\ker\big(\Gamma(X,\mathcal{F})\to\Gamma(U,\mathcal{F})\big)$ (a section vanishing on U has all germs at points of U zero, i.e. support in Y, and conversely, by the sheaf axiom).

(a) $\Gamma_Y(X,\cdot)$ is an additive functor, the kernel of the natural transformation $\Gamma(X,\cdot)\to\Gamma(U,\cdot|_U)$. Let $0\to\mathcal{F}'\xrightarrow{\alpha}\mathcal{F}\xrightarrow{\beta}\mathcal{F}''$ be exact. Since $\Gamma(X,\alpha)$ is injective, so is $\Gamma_Y(\alpha)$. Clearly $\Gamma_Y(\beta)\Gamma_Y(\alpha)=0$. If $s\in\Gamma_Y(X,\mathcal{F})$ has $\beta(s)=0$, then $s=\alpha(t)$ for a unique $t\in\Gamma(X,\mathcal{F}')$; and $\alpha(t|_U)=s|_U=0$ gives $t|_U=0$ by injectivity of $\Gamma(U,\alpha)$. So $t\in\Gamma_Y(X,\mathcal{F}')$. Hence Γ_Y is left exact, and its derived functors $H^i_Y(X,\cdot)$ exist by Cor. 2.3 and Thm. 1.1A.

(b) Exactness on the left and in the middle is (a). Let $s''\in\Gamma_Y(X,\mathcal{F}'')$. As $\mathcal{F}'$ is flasque, $\Gamma(X,\mathcal{F})\to\Gamma(X,\mathcal{F}'')$ is surjective, so $s''=\beta(s)$ for some $s\in\Gamma(X,\mathcal{F})$. Now $\beta(s|_U)=s''|_U=0$, so $s|_U=\alpha(t)$ with $t\in\Gamma(U,\mathcal{F}')$. Since $\mathcal{F}'$ is flasque, t extends to $\tilde{t}\in\Gamma(X,\mathcal{F}')$. Put $s_1=s-\alpha(\tilde{t})$. Then $\beta(s_1)=s''$ and $s_1|_U=s|_U-\alpha(t)=0$, i.e. $s_1\in\Gamma_Y(X,\mathcal{F})$ is a preimage of s''.

(c) Embed $\mathcal{F}\hookrightarrow\mathcal{I}$ with $\mathcal{I}$ injective (Cor. 2.3), and let $\mathcal{G}=\mathcal{I}/\mathcal{F}$. Then $\mathcal{I}$ is flasque (Lemma 2.4), hence $\mathcal{G}$ is flasque (II, Ex. 1.16(c)). By (b), $0\to\Gamma_Y(\mathcal{F})\to\Gamma_Y(\mathcal{I})\to\Gamma_Y(\mathcal{G})\to 0$ is exact. As $H^i_Y(X,\mathcal{I})=0$ for $i>0$ (Thm. 1.1A(e)), the long exact sequence gives $H^1_Y(X,\mathcal{F})=\mathrm{coker}(\Gamma_Y(\mathcal{I})\to\Gamma_Y(\mathcal{G}))=0$, and $H^i_Y(X,\mathcal{F})\cong H^{i-1}_Y(X,\mathcal{G})$ for $i\geq 2$. Since $\mathcal{G}$ is again flasque, induction on i gives $H^i_Y(X,\mathcal{F})=0$ for all $i>0$, exactly as in Prop. 2.5.

(d) Injectivity on the left is the definition of Γ_Y, exactness in the middle is the description $\Gamma_Y=\ker(\Gamma(X,\cdot)\to\Gamma(U,\cdot))$, and surjectivity on the right is the flasque property of $\mathcal{F}$.

(e) Let $\mathcal{F}\to\mathcal{I}^\bullet$ be an injective resolution in $\mathrm{Ab}(X)$. Then $H^i_Y(X,\mathcal{F})=h^i(\Gamma_Y(X,\mathcal{I}^\bullet))$ and $H^i(X,\mathcal{F})=h^i(\Gamma(X,\mathcal{I}^\bullet))$. Restriction to U is exact, so $\mathcal{F}|_U\to\mathcal{I}^\bullet|_U$ is a resolution, by sheaves $\mathcal{I}^j|_U$ which are flasque (restriction of flasque sheaves), hence acyclic for $\Gamma(U,\cdot)$ (Prop. 2.5). By Prop. 1.2A, $H^i(U,\mathcal{F}|_U)=h^i(\Gamma(U,\mathcal{I}^\bullet|_U))$. By (d), applied to each flasque $\mathcal{I}^j$, we have a short exact sequence of complexes

$$0\to\Gamma_Y(X,\mathcal{I}^\bullet)\to\Gamma(X,\mathcal{I}^\bullet)\to\Gamma(U,\mathcal{I}^\bullet|_U)\to 0,$$

whose long exact cohomology sequence is the required one. (It starts with 0 because h^0 of the left complex is $H^0_Y = \Gamma_Y$ and the complexes live in degrees ≥ 0; the maps $H^i(X,\mathcal{F}) \to H^i(U,\mathcal{F}|_U)$ are the restriction maps.)

(f) Restriction of sections gives a map $\rho : \Gamma_Y(X,\mathcal{G}) \to \Gamma_Y(V,\mathcal{G}|_V)$, natural in $\mathcal{G}$ (Y is also closed in V). It is an isomorphism for every sheaf $\mathcal{G}$: if $s \in \Gamma_Y(X,\mathcal{G})$ and $s|_V = 0$, then s vanishes on V and on $X - Y$ (by definition), and these two open sets cover X because $Y \subseteq V$, so $s = 0$; if $t \in \Gamma_Y(V,\mathcal{G}|_V)$, then t and $0 \in \Gamma(X-Y,\mathcal{G})$ agree on $V \cap (X-Y) = V - Y$, so they glue to $s \in \Gamma(X,\mathcal{G})$ with $s|_{X-Y} = 0$, i.e. $s \in \Gamma_Y(X,\mathcal{G})$, and $\rho(s) = t$.

Now take an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in $\mathrm{Ab}(X)$. Then $\mathcal{F}|_V \to \mathcal{I}^\bullet|_V$ is a resolution of $\mathcal{F}|_V$ by flasque sheaves on V, which are acyclic for $\Gamma_Y(V,\cdot)$ by (c) (applied to V and its closed subset Y). Prop. 1.2A is stated for an arbitrary left exact functor on an abelian category with enough injectives, so it applies to $\Gamma_Y(V,\cdot)$ on $\mathrm{Ab}(V)$ (acyclic meaning $H^i_Y(V,\cdot) = 0$ for $i > 0$) and gives $H^i_Y(V,\mathcal{F}|_V) = h^i(\Gamma_Y(V,\mathcal{I}^\bullet|_V))$. By ρ, the complexes $\Gamma_Y(X,\mathcal{I}^\bullet)$ and $\Gamma_Y(V,\mathcal{I}^\bullet|_V)$ are isomorphic, whence $H^i_Y(X,\mathcal{F}) \cong H^i_Y(V,\mathcal{F}|_V)$. The isomorphism is induced by the restriction maps, which are natural in $\mathcal{F}$ (use a morphism of injective resolutions over a given $\mathcal{F} \to \mathcal{F}'$), and independent of the resolution because any two resolutions are related by comparison maps. □

Exercise 2.4

> **Problem.** *Mayer–Vietoris Sequence.* Let Y_1, Y_2 be two closed subsets of X. Then there is a long exact sequence of cohomology with supports
>
> $$\cdots \to H^i_{Y_1\cap Y_2}(X,\mathcal{F}) \to H^i_{Y_1}(X,\mathcal{F}) \oplus H^i_{Y_2}(X,\mathcal{F}) \to H^i_{Y_1\cup Y_2}(X,\mathcal{F}) \to H^{i+1}_{Y_1\cap Y_2}(X,\mathcal{F}) \to \cdots .$$

Solution. Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution in $\mathrm{Ab}(X)$; all $\mathcal{I}^j$ are flasque (Lemma 2.4). For any closed Z, $H^i_Z(X,\mathcal{F}) = h^i(\Gamma_Z(X,\mathcal{I}^\bullet))$. Notice that for a section s of any sheaf, $\operatorname{supp} s \subseteq Y_1 \cap Y_2$ iff $\operatorname{supp} s \subseteq Y_1$ and $\subseteq Y_2$, so $\Gamma_{Y_1\cap Y_2} = \Gamma_{Y_1} \cap \Gamma_{Y_2}$ inside $\Gamma(X,\cdot)$; also $\Gamma_{Y_i} \subseteq \Gamma_{Y_1\cup Y_2}$. Claim: for every flasque sheaf $\mathcal{I}$,

$$0 \to \Gamma_{Y_1\cap Y_2}(X,\mathcal{I}) \xrightarrow{a\mapsto(a,a)} \Gamma_{Y_1}(X,\mathcal{I}) \oplus \Gamma_{Y_2}(X,\mathcal{I}) \xrightarrow{(b,c)\mapsto b-c} \Gamma_{Y_1\cup Y_2}(X,\mathcal{I}) \to 0$$

is exact. The first map is injective, the composite is 0, and if $b - c = 0$ then $b = c \in \Gamma_{Y_1} \cap \Gamma_{Y_2} = \Gamma_{Y_1\cap Y_2}$, proving exactness in the middle. Surjectivity: let $s \in \Gamma_{Y_1\cup Y_2}(X,\mathcal{I})$, and put $U_i = X - Y_i$. Then $s|_{U_1\cap U_2} = 0$ since $U_1 \cap U_2 = X - (Y_1 \cup Y_2)$. Hence there is a section b' of $\mathcal{I}$ over $U_1 \cup U_2$ with $b'|_{U_1} = 0$ and $b'|_{U_2} = s|_{U_2}$ (they agree on $U_1 \cap U_2$). As $\mathcal{I}$ is flasque, b' extends to $b \in \Gamma(X,\mathcal{I})$. Then $b|_{U_1} = 0$, i.e. $b \in \Gamma_{Y_1}$; and $c := b - s$ satisfies $c|_{U_2} = b'|_{U_2} - s|_{U_2} = 0$, i.e. $c \in \Gamma_{Y_2}$, and $s = b - c$.

Applying this to each $\mathcal{I}^j$ gives a short exact sequence of complexes, whose long exact cohomology sequence is the required one (the maps being induced by the inclusions $\Gamma_{Y_1\cap Y_2} \subseteq \Gamma_{Y_i} \subseteq \Gamma_{Y_1\cup Y_2}$). □

Exercise 2.5

> **Problem.** Let X be a Zariski space (II, Ex. 3.17). Let $P \in X$ be a closed point, and let X_P be the subset of X consisting of all points $Q \in X$ such that $P \in \overline{\{Q\}}$. We call X_P the *local*

space of X at P, and give it the induced topology. Let $j : X_P \to X$ be the inclusion, and for any sheaf $\mathcal{F}$ on X, let $\mathcal{F}_P = j^{-1}\mathcal{F}$. Show that for all i, $\mathcal{F}$, we have

$$H^i_P(X, \mathcal{F}) = H^i_P(X_P, \mathcal{F}_P).$$

Solution. Recall that a Zariski space is a noetherian topological space in which every nonempty irreducible closed subset has a unique generic point. Here H^i_P denotes cohomology with supports in the closed set $\{P\}$ (closed in X, hence in X_P). The subspace X_P is noetherian. The sheaf $\mathcal{F}_P = j^{-1}\mathcal{F}$ is the sheaf associated to the presheaf $W \mapsto \varinjlim_{V \supseteq W} \mathcal{F}(V)$ ($W \subseteq X_P$ open, V ranging over open subsets of X containing W). Note that if $V \subseteq X$ is open and contains a point of $\overline{\{Q\}}$, then $Q \in V$; hence every open neighborhood V of P contains X_P, and the open sets of X containing X_P are exactly the open neighborhoods of P.

Step 1: the presheaf $G(W) = \varinjlim_{V \supseteq W} \mathcal{F}(V)$ is a sheaf on X_P, so $\Gamma(W, \mathcal{F}_P) = G(W)$ for all open $W \subseteq X_P$. We know G is a presheaf with $G(\emptyset) = 0$. *Locality:* let $W = \bigcup W_\lambda$ and $s \in G(W)$ with $s|_{W_\lambda} = 0$ for all λ. Represent s by $s \in \mathcal{F}(V)$, $V \supseteq W$. Then $s|_{V_\lambda} = 0$ for some open V_λ with $W_\lambda \subseteq V_\lambda \subseteq V$, so s vanishes on $\bigcup V_\lambda \supseteq W$ and the germ is 0. *Gluing for two sets:* let $W_1, W_2 \subseteq X_P$ be open and $s_i \in \mathcal{F}(V_i)$ germs on W_i that agree on $W_1 \cap W_2$. Choose open $V_i^0 \subseteq X$ with $V_i^0 \cap X_P = W_i$ and replace V_i by $V_i \cap V_i^0$; so we may assume $V_i \cap X_P = W_i$. Put $V_{12} = V_1 \cap V_2$, so $V_{12} \cap X_P = W_1 \cap W_2$. By hypothesis there is an open $N \subseteq V_{12}$ with $N \supseteq W_1 \cap W_2$ and $s_1|_N = s_2|_N$. Let $Z = V_{12} - N$, closed in V_{12}; then $Z \cap X_P = \emptyset$. Let $\overline{Z}$ be the closure of Z in X and let $Z_1, \ldots, Z_m$ be its irreducible components with generic points ζ_k. Then $\overline{Z} \cap V_{12} = Z$. If $Z_k \cap V_{12} \neq \emptyset$, this is a nonempty open subset of Z_k, so it contains ζ_k; thus $\zeta_k \in Z$, so $\zeta_k \notin X_P$, i.e. $P \notin \overline{\{\zeta_k\}} = Z_k$. Let Z'' be the union of those Z_k with $Z_k \cap V_{12} \neq \emptyset$. Then Z'' is closed, $Z \subseteq Z''$, $P \notin Z''$, and $Z'' \cap X_P = \emptyset$ (if $Q \in X_P \cap Z_k$ then $P \in \overline{\{Q\}} \subseteq Z_k$). Put $U_i = V_i - Z''$. Then $U_i \supseteq V_i \cap X_P = W_i$, and $U_1 \cap U_2 \subseteq V_{12} - Z \subseteq N$, so $s_1|_{U_1}$ and $s_2|_{U_2}$ agree on $U_1 \cap U_2$ and glue to a section over $U_1 \cup U_2 \supseteq W_1 \cup W_2$, whose germ restricts to s_1, s_2. *Gluing in general:* let $W = \bigcup W_\lambda$ with germs t_λ agreeing on overlaps. W is quasi-compact (noetherian), so $W = W_1 \cup \cdots \cup W_r$ for finitely many of the W_λ. By induction on r using the two-set case and locality, the t_i ($i \leq r$) glue to $t \in G(W)$: glue $W_1, \ldots, W_{r-1}$ to t' on $W' = \bigcup_{i<r} W_i$, then t' and t_r agree on $W' \cap W_r = \bigcup_{i<r}(W_i \cap W_r)$ by locality, and apply the two-set case. For any λ, $t|_{W_\lambda}$ and t_λ agree on each $W_\lambda \cap W_i$, hence on W_λ by locality. So G is a sheaf.

Two consequences. (i) $\Gamma(X_P, \mathcal{F}_P) = \varinjlim_{V \ni P} \mathcal{F}(V)$ is the stalk $\mathcal{F}_P$ at P (so the notation is consistent). (ii) If $\mathcal{I}$ is a flasque sheaf on X then $\mathcal{I}_P := j^{-1}\mathcal{I}$ is flasque on X_P. Indeed let $W' \subseteq W$ be open in X_P and let $s \in \mathcal{I}(V'')$ represent a germ on W' ($V'' \supseteq W'$ open in X). Choose V open in X with $V \cap X_P = W$. Since $\mathcal{I}$ is flasque, s extends to $\hat{s} \in \mathcal{I}(V'' \cup V)$, and $V'' \cup V \supseteq W' \cup W = W$; the germ of $\hat{s}$ on W restricts to the germ of s on W'.

Step 2: cofinality. If $V' \subseteq X$ is open and $V' \supseteq X_P - \{P\}$, then there is an open $V \ni P$ with $V - \{P\} \subseteq V'$. Proof: let $C = X - V'$, closed, with $C \cap X_P \subseteq \{P\}$. Let $C_1, \ldots, C_m$ be its irreducible components with generic points γ_k. If $P \in C_k$, then $\gamma_k \in X_P \cap C \subseteq \{P\}$, so $\gamma_k = P$ and $C_k = \{P\}$. Put $V = X - \bigcup\{C_k : P \notin C_k\}$. This is an open neighborhood of P, and $V \cap C \subseteq \{P\}$, i.e. $V - \{P\} \subseteq V'$.

Step 3: $\Gamma_P(X, \mathcal{F}) \cong \Gamma_P(X_P, \mathcal{F}_P)$, naturally in $\mathcal{F}$. The map θ sends $s \in \Gamma_P(X, \mathcal{F})$ to the section of $\mathcal{F}_P$ over X_P given by the germ of s at P; its restriction to $X_P - \{P\}$ is the germ of $s|_{X-P} = 0$, so $\theta(s) \in \Gamma_P(X_P, \mathcal{F}_P)$. Injective: if $\theta(s) = 0$ then $s|_V = 0$ for an open $V \ni P$, and $s|_{X-\{P\}} = 0$; since $V \cup (X - \{P\}) = X$, $s = 0$. Surjective: let $\sigma \in \Gamma_P(X_P, \mathcal{F}_P)$, represented

by $s \in \mathcal{F}(V)$ for an open neighborhood V of P (consequence (i)). Its restriction to $X_P - \{P\}$ is zero in $G(X_P - \{P\}) = \varinjlim_{V' \supseteq X_P - P} \mathcal{F}(V')$, so $s|_{V'} = 0$ for some open $V' \supseteq X_P - \{P\}$ with $V' \subseteq V$. By Step 2 there is an open $V_1 \ni P$ with $V_1 - \{P\} \subseteq V'$; shrinking, $V_1 \subseteq V$. Then $s|_{V_1 - \{P\}} = 0$. So $s|_{V_1}$ and $0 \in \mathcal{F}(X - \{P\})$ agree on $V_1 \cap (X - \{P\})$ and glue to $\tilde{s} \in \Gamma_P(X, \mathcal{F})$ with $\theta(\tilde{s}) = \sigma$.

Step 4: conclusion. Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution in $\mathrm{Ab}(X)$. Then $H^i_P(X, \mathcal{F}) = h^i(\Gamma_P(X, \mathcal{I}^\bullet))$. The functor j^{-1} is exact (it does not change stalks), so $\mathcal{F}_P \to \mathcal{I}^\bullet_P$ is a resolution on X_P, by flasque sheaves by Step 1(ii). By Ex. 2.3(c) and Prop. 1.2A applied to $\Gamma_P(X_P, \cdot)$ (as in 2.3(f)), $H^i_P(X_P, \mathcal{F}_P) = h^i(\Gamma_P(X_P, \mathcal{I}^\bullet_P))$. The maps θ of Step 3 are isomorphisms commuting with the differentials (naturality), so they induce an isomorphism of complexes, hence $H^i_P(X, \mathcal{F}) \cong H^i_P(X_P, \mathcal{F}_P)$, natural in $\mathcal{F}$. □

Exercise 2.6

> **Problem.** Let X be a noetherian topological space, and let $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ be a direct system of injective sheaves of abelian groups on X. Then $\varinjlim \mathcal{I}_\alpha$ is also injective. [*Hints:* First show that a sheaf $\mathcal{I}$ is injective if and only if for every open set $U \subseteq X$, and for every subsheaf $\mathcal{R} \subseteq \mathbf{Z}_U$, and for every map $f : \mathcal{R} \to \mathcal{I}$, there exists an extension of f to a map $\mathbf{Z}_U \to \mathcal{I}$. Secondly, show that any such sheaf $\mathcal{R}$ is finitely generated, so any map $\mathcal{R} \to \varinjlim \mathcal{I}_\alpha$ factors through one of the $\mathcal{I}_\alpha$.]

Solution. Write $\mathcal{I} = \varinjlim \mathcal{I}_\alpha$. We use two standard facts: $\mathrm{Hom}(\mathbf{Z}_V, \mathcal{G}) = \mathcal{G}(V)$ for any open V and sheaf $\mathcal{G}$ (via $\varphi \mapsto \varphi(1)$, $1 \in \mathbf{Z}_V(V)$); and, since X is noetherian, $\mathcal{I}(V) = \varinjlim_\alpha \mathcal{I}_\alpha(V)$ for every open V (II, Ex. 1.11).

Step 1 (criterion). Let $\mathcal{I}$ be a sheaf satisfying the extension property of the hint. We show it is injective, i.e. every map $f : \mathcal{A} \to \mathcal{I}$ from a subsheaf $\mathcal{A} \subseteq \mathcal{B}$ extends to $\mathcal{B}$ (the converse direction of the equivalence is trivial, since $\mathbf{Z}_U$ is a sheaf and $\mathcal{R} \subseteq \mathbf{Z}_U$). Consider the set of pairs $(\mathcal{A}', f')$ with $\mathcal{A} \subseteq \mathcal{A}' \subseteq \mathcal{B}$ and $f'|_{\mathcal{A}} = f$, ordered by extension. A chain has an upper bound: the union of the subsheaves (the subsheaf of $\mathcal{B}$ generated by their sections, i.e. the sheaf associated to the union presheaf), with the map induced from the f' via the universal property of the associated sheaf. By Zorn's lemma there is a maximal $(\mathcal{A}', f')$. Suppose $\mathcal{A}' \neq \mathcal{B}$; there are an open U and $s \in \mathcal{B}(U) - \mathcal{A}'(U)$. Let $\varphi : \mathbf{Z}_U \to \mathcal{B}$ be the map with $\varphi(1) = s$, and let $\mathcal{R} = \varphi^{-1}(\mathcal{A}') = \ker(\mathbf{Z}_U \to \mathcal{B}/\mathcal{A}') \subseteq \mathbf{Z}_U$. By hypothesis $f' \circ \varphi|_{\mathcal{R}} : \mathcal{R} \to \mathcal{I}$ extends to $g : \mathbf{Z}_U \to \mathcal{I}$. Let $\mathcal{A}'' = \mathcal{A}' + \varphi(\mathbf{Z}_U)$, a subsheaf of $\mathcal{B}$ with $\mathcal{A}''(U) \ni s$, so $\mathcal{A}'' \supsetneq \mathcal{A}'$. The map $\psi : \mathcal{A}' \oplus \mathbf{Z}_U \to \mathcal{A}''$, $(a, t) \mapsto a + \varphi(t)$, is an epimorphism with kernel $L = \{(-\varphi(t), t) : t \in \mathcal{R}\}$, and $\theta(a, t) = f'(a) + g(t)$ vanishes on L because $g|_{\mathcal{R}} = f'\varphi|_{\mathcal{R}}$. So θ factors through $\mathcal{A}''$, giving an extension of f' to $\mathcal{A}''$: contradicting maximality. Hence $\mathcal{A}' = \mathcal{B}$.

Step 2 (finite generation). Say that a sheaf $\mathcal{R}$ is finitely generated if there are finitely many opens $O_1, \ldots, O_N$ and sections $\sigma_k \in \mathcal{R}(O_k)$ such that the induced map $\bigoplus_k \mathbf{Z}_{O_k} \to \mathcal{R}$ is an epimorphism. *Claim:* every subsheaf $\mathcal{R}$ of $\mathcal{P} = \bigoplus_{k=1}^r \mathbf{Z}_{V_k}$ (V_k open) is finitely generated.

Case $r = 1$. Let $\mathcal{R} \subseteq \mathbf{Z}_V$. For $y \in V$ write $\mathcal{R}_y = d(y)\mathbf{Z} \subseteq \mathbf{Z}$ with $d(y) \geq 0$; for $y \notin V$, $\mathcal{R}_y = 0$. For an integer $m \geq 1$ let $D_m = \{y \in V : m \in \mathcal{R}_y\} = \{y \in V : d(y) \mid m\}$, where m denotes the constant section of $\mathbf{Z}_V$ over V. D_m is open: if $m_y \in \mathcal{R}_y$, there is an open $V_0 \ni y$ and $r \in \mathcal{R}(V_0)$ with $r_y = m_y$, and after shrinking $V_0 \subseteq V$, $r = m|_{V_0}$, so $m_z \in \mathcal{R}_z$ for all $z \in V_0$. Moreover $m|_{D_m} \in \mathcal{R}(D_m)$, since a section of $\mathbf{Z}_V$ all of whose germs lie in $\mathcal{R}$ lies in

$\mathcal{R}$. The open set $V_+ = \{y \in V : d(y) > 0\}$ equals $\bigcup_{m\geq 1} D_m$ (as $y \in D_{d(y)}$ if $d(y) > 0$). Since X is noetherian, among the finite unions of the D_m there is a maximal one, and it must equal V_+: $V_+ = D_{m_1} \cup \cdots \cup D_{m_s}$. For $y \in D_{m_i}$ we have $d(y) \mid m_i$, so d takes only finitely many values $e_1, \ldots, e_t > 0$ on V_+. Let $\mathcal{R}' \subseteq \mathcal{R}$ be the image of $\bigoplus_l \mathbf{Z}_{D_{e_l}} \to \mathcal{R}$ given by the sections $e_l|_{D_{e_l}} \in \mathcal{R}(D_{e_l})$. For $y \in V_+$, $d(y) = e_l$ for some l and $y \in D_{e_l}$, so $\mathcal{R}'_y \ni e_l$ and $\mathcal{R}'_y = \mathcal{R}_y$; for other y, $\mathcal{R}_y = 0$. By stalks, $\mathcal{R}' = \mathcal{R}$.

Induction on r. Let $r \geq 2$, $\mathcal{A} = \mathbf{Z}_{V_r}$, $\mathcal{B} = \bigoplus_{k<r} \mathbf{Z}_{V_k}$, $\pi : \mathcal{P} = \mathcal{A} \oplus \mathcal{B} \to \mathcal{B}$, and $\mathcal{R} \subseteq \mathcal{P}$. By the case $r = 1$, $\mathcal{R} \cap \mathcal{A}$ is generated by finitely many sections $a_l \in \mathcal{R}(O_l)$, and by the induction hypothesis the subsheaf $\pi(\mathcal{R}) \subseteq \mathcal{B}$ is generated by finitely many $b_j \in \pi(\mathcal{R})(W_j)$. As $\mathcal{R} \to \pi(\mathcal{R})$ is an epimorphism, each W_j is covered by opens on which b_j lifts to $\mathcal{R}$; W_j is quasi-compact, so finitely many such opens W_{ji} and lifts $c_{ji} \in \mathcal{R}(W_{ji})$ suffice. Let $\mathcal{R}'$ be the subsheaf generated by the a_l and the c_{ji}. For $y \in X$ and $x \in \mathcal{R}_y$, $\pi(x) = \sum n_j (b_j)_y$ with $n_j \in \mathbf{Z}$ and $(b_j)_y = \pi((c_{ji})_y)$ for some i (depending on j); then $x - \sum n_j (c_{ji})_y \in \ker \pi \cap \mathcal{R}_y = (\mathcal{R} \cap \mathcal{A})_y \subseteq \mathcal{R}'_y$. So $\mathcal{R}' = \mathcal{R}$, proving the claim.

Step 3 (factoring through some $\mathcal{I}_\alpha$). Let $\mathcal{R}$ be a subsheaf of $\mathbf{Z}_U$ and $f : \mathcal{R} \to \mathcal{I} = \varinjlim \mathcal{I}_\alpha$. By Step 2 there is an epimorphism $\varepsilon : \mathcal{P} = \bigoplus_{k=1}^r \mathbf{Z}_{V_k} \to \mathcal{R}$, and its kernel $\mathcal{K} \subseteq \mathcal{P}$ is again finitely generated by Step 2: there are opens $O_1, \ldots, O_q$ and $\kappa_j \in \mathcal{K}(O_j) \subseteq \mathcal{P}(O_j)$ with $\bigoplus \mathbf{Z}_{O_j} \to \mathcal{K}$ onto. The map $g = f\varepsilon : \mathcal{P} \to \mathcal{I}$ is given by sections $g_k \in \mathcal{I}(V_k) = \varinjlim_\alpha \mathcal{I}_\alpha(V_k)$; choose α_0 and $t_k \in \mathcal{I}_{\alpha_0}(V_k)$ mapping to g_k (take α_0 above the finitely many indices needed), and let $\psi : \mathcal{P} \to \mathcal{I}_{\alpha_0}$ be the corresponding map. Then $\lambda_{\alpha_0}\psi = g$, where $\lambda_\alpha : \mathcal{I}_\alpha \to \mathcal{I}$ is the canonical map. Now $\lambda_{\alpha_0}(\psi(\kappa_j)) = g(\kappa_j) = 0$ in $\mathcal{I}(O_j) = \varinjlim \mathcal{I}_\alpha(O_j)$, so there is $\beta_j \geq \alpha_0$ such that $\psi(\kappa_j) \mapsto 0$ in $\mathcal{I}_{\beta_j}(O_j)$. Take $\beta \geq$ all β_j and set $\psi_\beta : \mathcal{P} \to \mathcal{I}_{\alpha_0} \to \mathcal{I}_\beta$. Then $\psi_\beta(\kappa_j) = 0$ for all j, so ψ_β kills the subsheaf $\mathcal{K}$ they generate, hence factors as $\psi_\beta = f_\beta \varepsilon$ with $f_\beta : \mathcal{R} \to \mathcal{I}_\beta$ (ε is an epimorphism with kernel $\mathcal{K}$). And $\lambda_\beta f_\beta \varepsilon = \lambda_\beta \psi_\beta = \lambda_{\alpha_0}\psi = g = f\varepsilon$, so $\lambda_\beta f_\beta = f$ because ε is an epimorphism.

Step 4. Let U be open, $\mathcal{R} \subseteq \mathbf{Z}_U$ and $f : \mathcal{R} \to \mathcal{I}$. By Step 3, $f = \lambda_\beta f_\beta$ with $f_\beta : \mathcal{R} \to \mathcal{I}_\beta$. As $\mathcal{I}_\beta$ is injective, f_β extends to $F_\beta : \mathbf{Z}_U \to \mathcal{I}_\beta$, and then $\lambda_\beta F_\beta$ extends f. By Step 1, $\mathcal{I} = \varinjlim \mathcal{I}_\alpha$ is injective. □

Exercise 2.7

Problem. Let S^1 be the circle (with its usual topology), and let $\mathbf{Z}$ be the constant sheaf $\mathbf{Z}$.

(a) Show that $H^1(S^1, \mathbf{Z}) \cong \mathbf{Z}$, using our definition of cohomology.

(b) Now let $\mathcal{R}$ be the sheaf of germs of continuous real-valued functions on S^1. Show that $H^1(S^1, \mathcal{R}) = 0$.

Solution. We prove (b) first, then use it for (a).

(b) Let $\mathcal{G}$ be the sheaf of discontinuous sections of $\mathcal{R}$: $\mathcal{G}(W) = \prod_{x\in W} \mathcal{R}_x$ (II, Ex. 1.16(e)). It is flasque, and $\mathcal{R} \to \mathcal{G}$ ($s \mapsto$ its family of germs) is injective. Let $\mathcal{Q} = \mathcal{G}/\mathcal{R}$. The long exact sequence for $0 \to \mathcal{R} \to \mathcal{G} \to \mathcal{Q} \to 0$ and $H^1(S^1, \mathcal{G}) = 0$ (Prop. 2.5) give $H^1(S^1, \mathcal{R}) \cong \operatorname{coker}(\Gamma(S^1, \mathcal{G}) \to \Gamma(S^1, \mathcal{Q}))$. So we must show that every $q \in \Gamma(S^1, \mathcal{Q})$ lifts to $\Gamma(S^1, \mathcal{G})$.

Locally q lifts: there is an open cover of S^1 and $g_i \in \mathcal{G}(U_i)$ mapping to $q|_{U_i}$; since S^1 is compact we may take the cover finite, $U_1, \ldots, U_m$. Then $c_{ij} := g_i - g_j \in \mathcal{G}(U_i \cap U_j)$ maps to 0 in $\mathcal{Q}$, hence lies in $\mathcal{R}(U_i \cap U_j)$, i.e. is a continuous function on $U_i \cap U_j$, and $c_{ij} + c_{jk} = c_{ik}$, $c_{ij} = -c_{ji}$.

Choose a continuous partition of unity $\varphi_1, \dots, \varphi_m$ subordinate to the cover, i.e. $\varphi_k : S^1 \to [0,1]$ continuous, $\operatorname{supp}\varphi_k \subseteq U_k$ (closed in S^1), $\sum \varphi_k = 1$ (it exists since S^1 is compact Hausdorff, hence normal). For each k, the function $\varphi_k c_{ik}$ on $U_i \cap U_k$ extends by 0 to a continuous function on U_i (it vanishes on $U_i - \operatorname{supp}\varphi_k$, and $U_i = (U_i \cap U_k) \cup (U_i - \operatorname{supp}\varphi_k)$ is an open cover of U_i on which the definitions agree). Put $b_i = \sum_k \varphi_k c_{ik} \in \mathcal{R}(U_i)$. Then on $U_i \cap U_j$

$$b_i - b_j = \sum_k \varphi_k(c_{ik} - c_{jk}) = \sum_k \varphi_k(c_{ik} + c_{kj}) = \sum_k \varphi_k c_{ij} = c_{ij}.$$

Hence $g_i - b_i = g_j - b_j$ on $U_i \cap U_j$, so the sections $g_i - b_i \in \mathcal{G}(U_i)$ glue to $g \in \Gamma(S^1, \mathcal{G})$. Its image in $\mathcal{Q}$ restricts on U_i to the class of $g_i - b_i$, which equals that of g_i because $b_i \in \mathcal{R}(U_i)$; so g maps to q. Hence $H^1(S^1, \mathcal{R}) = 0$.

(a) Write $S^1 = \mathbf{R}/\mathbf{Z}$ and let $\mathcal{S}$ be the sheaf of continuous functions to the group $\mathbf{R}/\mathbf{Z}$ (a sheaf of abelian groups). Consider $0 \to \mathbf{Z} \xrightarrow{\iota} \mathcal{R} \xrightarrow{p} \mathcal{S} \to 0$, where ι regards a locally constant integer-valued function as a continuous real function, and p composes with $\mathbf{R} \to \mathbf{R}/\mathbf{Z}$. Exactness: $\ker p$ consists of the continuous functions with values in $\mathbf{Z} \subset \mathbf{R}$, which are the locally constant integer-valued functions, i.e. the constant sheaf $\mathbf{Z}$; and p is surjective on stalks, because $\mathbf{R} \to \mathbf{R}/\mathbf{Z}$ is a covering map, so any continuous map from a small enough (connected, simply connected) neighborhood of a point to $\mathbf{R}/\mathbf{Z}$ lifts to $\mathbf{R}$. The long exact sequence and (b) give

$$\Gamma(S^1, \mathcal{R}) \xrightarrow{p} \Gamma(S^1, \mathcal{S}) \to H^1(S^1, \mathbf{Z}) \to H^1(S^1, \mathcal{R}) = 0,$$

so $H^1(S^1, \mathbf{Z}) \cong C(S^1, \mathbf{R}/\mathbf{Z})/p\big(C(S^1, \mathbf{R})\big)$, where $C(\cdot,\cdot)$ denotes continuous maps.

We compute this quotient via the degree. Let $\pi : \mathbf{R} \to S^1 = \mathbf{R}/\mathbf{Z}$ be the quotient map. For $h \in C(S^1, \mathbf{R}/\mathbf{Z})$, the map $h\pi : \mathbf{R} \to \mathbf{R}/\mathbf{Z}$ lifts (path lifting for the covering, $\mathbf{R}$ simply connected) to a continuous $H : \mathbf{R} \to \mathbf{R}$ with $H \bmod \mathbf{Z} = h\pi$, unique up to an additive integer constant. The function $t \mapsto H(t+1) - H(t)$ is continuous with values in $\mathbf{Z}$ (as $h\pi(t+1) = h\pi(t)$), hence a constant $\deg h \in \mathbf{Z}$, independent of the choice of H. If H_1, H_2 lift h_1, h_2 then $H_1 + H_2$ lifts $h_1 + h_2$, so $\deg : C(S^1, \mathbf{R}/\mathbf{Z}) \to \mathbf{Z}$ is a homomorphism. It is surjective: $h_n(t) = nt \bmod \mathbf{Z}$ (well defined on $\mathbf{R}/\mathbf{Z}$) has lift $H(t) = nt$ and $\deg h_n = n$. Its kernel is $p(C(S^1, \mathbf{R}))$: if $\deg h = 0$ then $H(t+1) = H(t)$, so H descends to a continuous $\tilde{H} : S^1 \to \mathbf{R}$ with $p(\tilde{H}) = h$; conversely $p(f)$ has lift $f\pi$ of degree 0. Therefore $C(S^1, \mathbf{R}/\mathbf{Z})/p(C(S^1, \mathbf{R})) \cong \mathbf{Z}$, i.e. $H^1(S^1, \mathbf{Z}) \cong \mathbf{Z}$. □

3. Cohomology of a noetherian affine scheme

Conventions and results used. For an ideal $\mathfrak{a}$ of a ring A and an A-module M, $\Gamma_{\mathfrak{a}}(M) = \{m \in M \mid \mathfrak{a}^n m = 0 \text{ for some } n > 0\}$ (II, Ex. 5.6). Open subschemes and closed subschemes of noetherian schemes are noetherian. If $i : Y \to X$ is a closed immersion, then $H^p(X, i_*\mathcal{G}) = H^p(Y, \mathcal{G})$ (Lemma 2.10), and a quasi-coherent $\mathcal{O}_X$-module annihilated by the ideal sheaf of Y is of the form $i_*\mathcal{G}$ with $\mathcal{G}$ quasi-coherent on Y (II, Ex. 5.10). A closed subscheme of an affine scheme is affine. Throughout, "Thm. 3.5" means that quasi-coherent sheaves on a noetherian affine scheme have no higher cohomology, and Thm. 3.7 is Serre's criterion: a noetherian scheme X is affine if and only if $H^1(X, \mathcal{I}) = 0$ for every coherent sheaf of ideals $\mathcal{I}$.

Exercise 3.1

Problem. Let X be a noetherian scheme. Show that X is affine if and only if X_{red} is affine (II, Ex. 2.3). [Hint: Use (3.7), and for every coherent sheaf $\mathcal{F}$ on X, consider the filtration $\mathcal{F} \supseteq \mathcal{N} \cdot \mathcal{F} \supseteq \mathcal{N}^2 \cdot \mathcal{F} \supseteq \ldots$, where $\mathcal{N}$ is the sheaf of nilpotent elements on X.]

Solution. ($\Rightarrow$) X_{red} is a closed subscheme of X, hence affine if X is.

($\Leftarrow$) Let $i : X_{\mathrm{red}} \to X$ be the closed immersion, $\mathcal{N}$ the nilradical sheaf (the ideal sheaf of X_{red}); it is coherent because X is noetherian. On each affine open $\operatorname{Spec} A_j$ of a finite affine cover the nilradical of the noetherian ring A_j is nilpotent, say $\mathcal{N}^{n_j} = 0$ there; so $\mathcal{N}^n = 0$ for $n = \max n_j$.

Let $\mathcal{F}$ be a coherent sheaf on X. We claim $H^p(X, \mathcal{F}) = 0$ for all $p > 0$. Consider the filtration $\mathcal{F} \supseteq \mathcal{N}\mathcal{F} \supseteq \cdots \supseteq \mathcal{N}^n\mathcal{F} = 0$. Its quotients $\mathcal{Q}_k = \mathcal{N}^k\mathcal{F}/\mathcal{N}^{k+1}\mathcal{F}$ are coherent and annihilated by $\mathcal{N}$, hence $\mathcal{Q}_k = i_*\mathcal{G}_k$ with $\mathcal{G}_k$ quasi-coherent on X_{red}. Since X_{red} is a noetherian affine scheme, Lemma 2.10 and Thm. 3.5 give $H^p(X, \mathcal{Q}_k) = H^p(X_{\mathrm{red}}, \mathcal{G}_k) = 0$ for $p > 0$. Now show by descending induction on k that $H^p(X, \mathcal{N}^k\mathcal{F}) = 0$ for $p > 0$: it is clear for $k = n$; and the exact sequence $0 \to \mathcal{N}^{k+1}\mathcal{F} \to \mathcal{N}^k\mathcal{F} \to \mathcal{Q}_k \to 0$ and the long exact sequence give it for k from $k + 1$. For $k = 0$ we get the claim.

In particular $H^1(X, \mathcal{I}) = 0$ for all coherent ideal sheaves $\mathcal{I}$, so X is affine by Thm. 3.7. □

Exercise 3.2

Problem. Let X be a reduced noetherian scheme. Show that X is affine if and only if each irreducible component of X is affine.

Solution. Irreducible components are given their reduced induced structure (they are then integral closed subschemes).

($\Rightarrow$) Closed subschemes of affine schemes are affine.

($\Leftarrow$) Induction on the number n of irreducible components $X_1, \ldots, X_n$ of X (all affine). If $n = 1$, $X = X_1$. Let $n \geq 2$, $Y = X_1$ and $Z = X_2 \cup \cdots \cup X_n$ with the reduced structure. Then Z is a reduced noetherian scheme whose irreducible components are $X_2, \ldots, X_n$, so Z is affine by induction; Y is affine by hypothesis. Let $\mathcal{I}_Y, \mathcal{I}_Z \subseteq \mathcal{O}_X$ be their ideal sheaves.

Claim: $\mathcal{I}_Y \cap \mathcal{I}_Z = 0$. On an affine open $\operatorname{Spec} A$ of X (A a reduced noetherian ring), $\mathcal{I}_Y, \mathcal{I}_Z$ correspond to the radical ideals $\mathfrak{a} = \bigcap_{\mathfrak{p} \in Y} \mathfrak{p}$, $\mathfrak{b} = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}$, so $\mathfrak{a} \cap \mathfrak{b} = \bigcap_{\mathfrak{p} \in Y \cup Z} \mathfrak{p} = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p} = \mathfrak{N}(A) = 0$.

Now let $\mathcal{J} \subseteq \mathcal{O}_X$ be any coherent ideal sheaf. We show $H^1(X, \mathcal{J}) = 0$; then X is affine by Thm. 3.7. Consider $0 \to \mathcal{J} \cap \mathcal{I}_Y \to \mathcal{J} \to \mathcal{J}/(\mathcal{J} \cap \mathcal{I}_Y) \to 0$.

- $\mathcal{J}/(\mathcal{J} \cap \mathcal{I}_Y) \cong (\mathcal{J} + \mathcal{I}_Y)/\mathcal{I}_Y$ is a coherent sheaf killed by $\mathcal{I}_Y$, i.e. it is i_{Y*} of a quasi-coherent sheaf on Y. As Y is affine, its H^1 vanishes by Lemma 2.10 and Thm. 3.5.
- $\mathcal{J} \cap \mathcal{I}_Y \subseteq \mathcal{I}_Y$ is coherent and killed by $\mathcal{I}_Z$, since $\mathcal{I}_Z \cdot (\mathcal{J} \cap \mathcal{I}_Y) \subseteq \mathcal{I}_Z\mathcal{I}_Y \subseteq \mathcal{I}_Z \cap \mathcal{I}_Y = 0$. So it is i_{Z*} of a quasi-coherent sheaf on the affine scheme Z, and its H^1 vanishes in the same way.

The long exact sequence now gives $H^1(X, \mathcal{J}) = 0$. □

Exercise 3.3

Problem. Let A be a noetherian ring, and let $\mathfrak{a}$ be an ideal of A.

(a) Show that $\Gamma_{\mathfrak{a}}(\cdot)$ (II, Ex. 5.6) is a left-exact functor from the category of A-modules to itself. We denote its right derived functors, calculated in $\mathfrak{Mod}(A)$, by $H^i_{\mathfrak{a}}(\cdot)$.

(b) Now let $X = \operatorname{Spec} A$, $Y = V(\mathfrak{a})$. Show that for any A-module M, $H^i_{\mathfrak{a}}(M) = H^i_Y(X, \widetilde{M})$, where $H^i_Y(X, \cdot)$ denotes cohomology with supports in Y (Ex. 2.3).

(c) For any i, show that $\Gamma_{\mathfrak{a}}(H^i_{\mathfrak{a}}(M)) = H^i_{\mathfrak{a}}(M)$.

Solution. (a) $\Gamma_{\mathfrak{a}}$ is an additive functor (a homomorphism sends $\mathfrak{a}$-power torsion elements to such). Let $0 \to M' \xrightarrow{u} M \xrightarrow{v} M''$ be exact. If $m' \in \Gamma_{\mathfrak{a}}(M')$ with $\mathfrak{a}^n m' = 0$ then $\mathfrak{a}^n u(m') = 0$; u is injective, so $\Gamma_{\mathfrak{a}}(u)$ is injective. If $m \in \Gamma_{\mathfrak{a}}(M)$ and $v(m) = 0$, then $m = u(m')$ and $u(\mathfrak{a}^n m') = \mathfrak{a}^n m = 0$, so $\mathfrak{a}^n m' = 0$, i.e. $m' \in \Gamma_{\mathfrak{a}}(M')$. Hence $0 \to \Gamma_{\mathfrak{a}}(M') \to \Gamma_{\mathfrak{a}}(M) \to \Gamma_{\mathfrak{a}}(M'')$ is exact. (Also $\Gamma_{\mathfrak{a}}(\cdot)$ is A-linear.) The category of A-modules has enough injectives, so the right derived functors $H^i_{\mathfrak{a}}$ exist.

(b) First, for any A-module M we have $\Gamma_Y(X, \widetilde{M}) = \Gamma_{\mathfrak{a}}(M)$ as submodules of $M = \Gamma(X, \widetilde{M})$. Indeed, for $m \in M$ the support of the section m is $\{\mathfrak{p} \mid m/1 \neq 0 \text{ in } M_{\mathfrak{p}}\} = \{\mathfrak{p} \mid \operatorname{Ann}(m) \subseteq \mathfrak{p}\} = V(\operatorname{Ann} m)$. So $\operatorname{Supp}(m) \subseteq V(\mathfrak{a})$ iff $\mathfrak{a} \subseteq r(\operatorname{Ann} m)$ iff $\mathfrak{a}^n \subseteq \operatorname{Ann}(m)$ for some n ($\mathfrak{a}$ is finitely generated), i.e. iff $m \in \Gamma_{\mathfrak{a}}(M)$.

Let $0 \to M \to I^\bullet$ be an injective resolution in $\mathfrak{Mod}(A)$. The functor $\widetilde{\ }$ is exact, so $0 \to \widetilde{M} \to \widetilde{I}^\bullet$ is a resolution, and each $\widetilde{I}^p$ is flasque by Prop. 3.4. By Ex. 2.3(c), $H^i_Y(X, \widetilde{I}^p) = 0$ for $i > 0$, so flasque sheaves are acyclic for $\Gamma_Y(X, \cdot)$ and this resolution computes $H^i_Y(X, \widetilde{M})$ (Prop. 1.2A). Thus

$$H^i_Y(X, \widetilde{M}) = h^i(\Gamma_Y(X, \widetilde{I}^\bullet)) = h^i(\Gamma_{\mathfrak{a}}(I^\bullet)) = H^i_{\mathfrak{a}}(M).$$

(c) Let $M \to I^\bullet$ be an injective resolution. Then $H^i_{\mathfrak{a}}(M) = \ker(\Gamma_{\mathfrak{a}}(I^i) \to \Gamma_{\mathfrak{a}}(I^{i+1}))/\operatorname{im}$, a subquotient of $\Gamma_{\mathfrak{a}}(I^i)$. Every element z of the kernel satisfies $\mathfrak{a}^n z = 0$ for some n, hence so does its class $[z]$. So every element of $H^i_{\mathfrak{a}}(M)$ is killed by a power of $\mathfrak{a}$, i.e. $\Gamma_{\mathfrak{a}}(H^i_{\mathfrak{a}}(M)) = H^i_{\mathfrak{a}}(M)$. □

Exercise 3.4

Problem. *Cohomological interpretation of depth.* If A is a ring, $\mathfrak{a}$ an ideal, and M an A-module, then $\operatorname{depth}_{\mathfrak{a}} M$ is the maximum length of an M-regular sequence $x_1, \ldots, x_r$, with all $x_i \in \mathfrak{a}$. This generalizes the notion of depth introduced in (II, Section 8).

(a) Assume that A is noetherian. Show that if $\operatorname{depth}_{\mathfrak{a}} M \geq 1$, then $\Gamma_{\mathfrak{a}}(M) = 0$, and the converse is true if M is finitely generated. [Hint: When M is finitely generated, both conditions are equivalent to saying that $\mathfrak{a}$ is not contained in any associated prime of M.]

(b) Show inductively, for M finitely generated, that for any $n \geq 0$, the following conditions are equivalent: (i) $\operatorname{depth}_{\mathfrak{a}} M \geq n$; (ii) $H^i_{\mathfrak{a}}(M) = 0$ for all $i < n$.

For more details, and related results, see Grothendieck [7].

Solution. *Convention.* Here $x_1, \dots, x_r \in \mathfrak{a}$ is called M-regular if x_i is a nonzerodivisor on $M/(x_1, \dots, x_{i-1})M$ for each i. The extra requirement $M/(x_1, \dots, x_r)M \neq 0$ that is often added is *not* imposed (and it must not be, in order that (b) be true: for $\mathfrak{a} = A$ one has $\Gamma_\mathfrak{a} = 0$, so $H^i_\mathfrak{a}(M) = 0$ for all i, while A contains regular sequences $1, 1, 1, \dots$ of any length). Depth may be ∞; "$\operatorname{depth}_\mathfrak{a} M \geq n$" means that an M-regular sequence of length n in $\mathfrak{a}$ exists.

(a) Suppose $x \in \mathfrak{a}$ is M-regular and $m \in \Gamma_\mathfrak{a}(M)$, $\mathfrak{a}^n m = 0$. Then $x^n m = 0$, so $m = 0$ as x is a nonzerodivisor on M. Hence $\Gamma_\mathfrak{a}(M) = 0$.

Conversely let M be finitely generated and $\Gamma_\mathfrak{a}(M) = 0$. We prove: $\Gamma_\mathfrak{a}(M) = 0$ iff $\mathfrak{a} \not\subseteq \mathfrak{p}$ for every $\mathfrak{p} \in \operatorname{Ass} M$. If $\mathfrak{a} \subseteq \mathfrak{p} = \operatorname{Ann}(m)$ with $0 \neq m$, then $\mathfrak{a}m = 0$, so $0 \neq m \in \Gamma_\mathfrak{a}(M)$. Conversely if $0 \neq m \in \Gamma_\mathfrak{a}(M)$, the nonzero module $Am \subseteq \Gamma_\mathfrak{a}(M)$ has an associated prime $\mathfrak{p} = \operatorname{Ann}(m')$, $m' \in Am$; as $\mathfrak{a}^n m' = 0$ for some n, $\mathfrak{a}^n \subseteq \mathfrak{p}$ hence $\mathfrak{a} \subseteq \mathfrak{p}$, and $\mathfrak{p} \in \operatorname{Ass}(M)$. Now $\operatorname{Ass} M$ is finite and the set of zerodivisors of M is $\bigcup_{\mathfrak{p} \in \operatorname{Ass} M} \mathfrak{p}$ (Matsumura, Thm. 6.1). If $\mathfrak{a} \not\subseteq \mathfrak{p}$ for all $\mathfrak{p} \in \operatorname{Ass} M$, prime avoidance gives $x \in \mathfrak{a}$ not in any associated prime, i.e. x is M-regular, so $\operatorname{depth}_\mathfrak{a} M \geq 1$.

(b) We prove by induction on n: $P(n)$: *for every finitely generated M,* $\operatorname{depth}_\mathfrak{a} M \geq n \iff H^i_\mathfrak{a}(M) = 0$ *for $i < n$.* $P(0)$ is trivial and $P(1)$ is (a) ($H^0_\mathfrak{a} = \Gamma_\mathfrak{a}$). Assume $P(n)$ for some $n \geq 1$; we prove $P(n+1)$ for any finitely generated M. For an M-regular $x \in \mathfrak{a}$ the sequence $0 \to M \xrightarrow{x} M \to M/xM \to 0$ gives, by the long exact sequence of the δ-functor $H^\bullet_\mathfrak{a}$,

$$\cdots \to H^{i-1}_\mathfrak{a}(M/xM) \to H^i_\mathfrak{a}(M) \xrightarrow{x} H^i_\mathfrak{a}(M) \to H^i_\mathfrak{a}(M/xM) \to H^{i+1}_\mathfrak{a}(M) \to \dots \qquad (*)$$

(the map induced by multiplication by x on M is multiplication by x on $H^i_\mathfrak{a}(M)$, because $\Gamma_\mathfrak{a}$ is A-linear; lift $x\cdot$ to the injective resolution as $x\cdot$).

($\Leftarrow$) Suppose $H^i_\mathfrak{a}(M) = 0$ for $i < n+1$. In particular $\Gamma_\mathfrak{a}(M) = 0$, so by (a) there is an M-regular $x \in \mathfrak{a}$. For $i < n$ we have $H^i_\mathfrak{a}(M) = 0$ and $H^{i+1}_\mathfrak{a}(M) = 0$, so by $(*)$ $H^i_\mathfrak{a}(M/xM) = 0$. By $P(n)$ applied to the finitely generated module M/xM, $\operatorname{depth}_\mathfrak{a} M/xM \geq n$, so $\operatorname{depth}_\mathfrak{a} M \geq n+1$.

($\Rightarrow$) Suppose $x_1, \dots, x_{n+1} \in \mathfrak{a}$ is M-regular. Put $x = x_1$, $M' = M/xM$; then $x_2, \dots, x_{n+1}$ is M'-regular, so $\operatorname{depth}_\mathfrak{a} M' \geq n$ and by $P(n)$ $H^i_\mathfrak{a}(M') = 0$ for $i < n$. By $(*)$, for $i \leq n$ the map $x : H^i_\mathfrak{a}(M) \to H^i_\mathfrak{a}(M)$ is injective (its kernel is the image of $H^{i-1}_\mathfrak{a}(M') = 0$). But every element of $H^i_\mathfrak{a}(M)$ is killed by a power of $\mathfrak{a}$, hence by a power of x (3.3(c)); so an injective x forces $H^i_\mathfrak{a}(M) = 0$. Thus $H^i_\mathfrak{a}(M) = 0$ for $i \leq n$, i.e. for $i < n+1$. □

Exercise 3.5

> **Problem.** Let X be a noetherian scheme, and let P be a closed point of X. Show that the following conditions are equivalent: (i) $\operatorname{depth} \mathcal{O}_P \geq 2$; (ii) if U is any open neighborhood of P, then every section of $\mathcal{O}_X$ over $U - P$ extends uniquely to a section of $\mathcal{O}_X$ over U. This generalizes (I, Ex. 3.20), in view of (II, 8.22A).

Solution. Here $\operatorname{depth} \mathcal{O}_P$ is the depth of the noetherian local ring $R = \mathcal{O}_{X,P}$ with respect to its maximal ideal $\mathfrak{m}_P$, i.e. $\operatorname{depth}_{\mathfrak{m}_P} R$ in the sense of 3.4.

Step 1: reduction to affine neighbourhoods. For an open $U \ni P$ let $\rho_U : \mathcal{O}_X(U) \to \mathcal{O}_X(U - P)$ be the restriction ($U - P$ is open as P is closed). Claim: (ii) holds iff ρ_U is bijective for every *affine* open neighbourhood U of P. One direction is trivial. Conversely, let $U \ni P$ be open and

consider the morphism of sheaves $\varphi : \mathcal{O}_U \to j_*(\mathcal{O}_{U-P})$ on U ($j : U - P \to U$). Let $\mathcal{B}$ be the collection of opens $W \subseteq U$ that either do not contain P or are affine neighbourhoods of P. This is a basis of the topology of U: a neighbourhood of $x \neq P$ can be shrunk inside $U - P$, and every neighbourhood of P contains an affine neighbourhood. For $W \in \mathcal{B}$ not containing P, $\varphi(W)$ is the identity; for W affine containing P, $\varphi(W) = \rho_W$ is bijective by assumption. A morphism of sheaves which is bijective on a basis is an isomorphism of sheaves, hence bijective on all open sets, in particular ρ_U is bijective.

Step 2: the affine case. Let $U = \operatorname{Spec} A$ be an affine open neighbourhood of P and $\mathfrak{m} \subset A$ the maximal ideal of P (the point P is closed in U). Ex. 2.3(e) for $Y = \{P\}$ gives an exact sequence

$$0 \to H^0_P(U, \mathcal{O}_U) \to A \xrightarrow{\rho_U} \mathcal{O}(U - P) \to H^1_P(U, \mathcal{O}_U) \to H^1(U, \mathcal{O}_U) = 0,$$

the last group vanishing by Thm. 3.5. By 3.3(b), $H^i_P(U, \mathcal{O}_U) = H^i_{\mathfrak{m}}(A)$. Hence ρ_U is bijective iff $H^0_{\mathfrak{m}}(A) = H^1_{\mathfrak{m}}(A) = 0$.

Step 3: localization. Let $R = A_{\mathfrak{m}} = \mathcal{O}_P$. Claim: $H^i_{\mathfrak{m}}(A) \cong H^i_{\mathfrak{m}R}(R)$. For a multiplicative set $S \subset A$, we have $S^{-1}\Gamma_{\mathfrak{a}}(M) = \Gamma_{S^{-1}\mathfrak{a}}(S^{-1}M)$ ($\subseteq$ is clear; if z/s is killed by $(S^{-1}\mathfrak{a})^n$, then with $\mathfrak{a}^n = (g_1, \ldots, g_k)$ there are $t_l \in S$ with $t_l g_l z = 0$, and $t = \prod t_l$ gives $\mathfrak{a}^n(tz) = 0$ and $z/s = tz/ts$). Moreover localization preserves injectives over the noetherian ring A: if I is injective and $\mathfrak{b}' = S^{-1}\mathfrak{b}$ ($\mathfrak{b}$ a finitely generated ideal) is an ideal of $S^{-1}A$, then since $\mathfrak{b}$ is finitely presented, every $\varphi : S^{-1}\mathfrak{b} \to S^{-1}I$ has the form ψ/s with $\psi \in \operatorname{Hom}_A(\mathfrak{b}, I)$; ψ extends to $\tilde{\psi} : A \to I$ and $\tilde{\psi}/s$ extends φ; Baer's criterion applies. Thus S^{-1} of an injective resolution of M is an injective resolution of $S^{-1}M$ over $S^{-1}A$, and since S^{-1} is exact, $S^{-1}H^i_{\mathfrak{a}}(M) = H^i_{S^{-1}\mathfrak{a}}(S^{-1}M)$. Take $S = A - \mathfrak{m}$. By 3.3(c) $H^i_{\mathfrak{m}}(A)$ is $\mathfrak{m}$-power torsion, and then every $s \notin \mathfrak{m}$ acts bijectively on it: if z is killed by $\mathfrak{m}^k$ then s is a unit modulo $\mathfrak{m}^k$, $gs \equiv 1 \pmod{\mathfrak{m}^k}$, so $z = gsz$; this shows $sz = 0 \Rightarrow z = 0$ and $z = s(gz)$. Therefore $H^i_{\mathfrak{m}}(A) = H^i_{\mathfrak{m}}(A)_{\mathfrak{m}} = H^i_{\mathfrak{m}R}(R)$.

Conclusion. By Steps 2 and 3, for an affine neighbourhood U: ρ_U is bijective iff $H^0_{\mathfrak{m}R}(R) = H^1_{\mathfrak{m}R}(R) = 0$. This condition depends only on R. By 3.4(b) with $n = 2$ (applied to the noetherian local ring R, ideal $\mathfrak{m}R$, $M = R$) it is equivalent to $\operatorname{depth}_{\mathfrak{m}R} R \geq 2$. (For the maximal ideal of a noetherian local ring, $R/(x_1, \ldots, x_r)R \neq 0$ automatically by Nakayama, so the convention of 3.4 agrees with the usual definition of the depth of a local ring, II, Section 8.) Together with Step 1, (ii) holds iff $\operatorname{depth} \mathcal{O}_P \geq 2$. (Existence of an affine neighbourhood U is clear.) □

Exercise 3.6

Problem. Let X be a noetherian scheme.

(a) Show that the sheaf $\mathcal{G}$ constructed in the proof of (3.6) is an injective object in the category $\mathfrak{Qco}(X)$ of quasi-coherent sheaves on X. Thus $\mathfrak{Qco}(X)$ has enough injectives.

(b) Show that any injective object of $\mathfrak{Qco}(X)$ is flasque. [Hints: The method of proof of (2.4) will not work, because $\mathcal{O}_U$ is not quasi-coherent on X in general. Instead, use (II, Ex. 5.15) to show that if $\mathcal{I} \in \mathfrak{Qco}(X)$ is injective, and if $U \subseteq X$ is an open subset, then $\mathcal{I}|_U$ is an injective object of $\mathfrak{Qco}(U)$. Then cover X with open affines...]

(c) Conclude that one can compute cohomology as the derived functors of $\Gamma(X, \cdot)$, considered as a functor from $\mathfrak{Qco}(X)$ to $\mathfrak{Ab}$.

(Part (b) is starred in the book.)

Solution. Recall the construction in 3.6: a finite affine open cover $X = \bigcup U_i$, $U_i = \operatorname{Spec} A_i$, $f_i : U_i \to X$ the inclusions, $\mathcal{F}|_{U_i} = \widetilde{M_i}$, $M_i \hookrightarrow I_i$ an injective A_i-module, and $\mathcal{G} = \bigoplus_i f_{i*}(\widetilde{I_i})$, with the monomorphism $\mathcal{F} \to \mathcal{G}$. The sheaf $\mathcal{G}$ is quasi-coherent (II, 5.8) and flasque (3.4, II Ex. 1.16(d)).

(a) $\mathfrak{Qco}(X)$ is a full subcategory of $\mathfrak{Mod}(X)$, closed under kernels and cokernels (II, 5.7); so a monomorphism in it is an injective sheaf map. Let $\mathcal{A} \hookrightarrow \mathcal{B}$ be a monomorphism in $\mathfrak{Qco}(X)$; we must show $\operatorname{Hom}(\mathcal{B}, \mathcal{G}) \to \operatorname{Hom}(\mathcal{A}, \mathcal{G})$ is surjective. Since the sum is finite, it suffices to treat each $f_{i*}\widetilde{I_i}$. By the adjunction between f_i^* (restriction to U_i) and f_{i*}, and by the equivalence $\mathfrak{Qco}(U_i) \simeq \mathfrak{Mod}(A_i)$ (II, 5.5),

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{A}, f_{i*}\widetilde{I_i}) \cong \operatorname{Hom}_{\mathcal{O}_{U_i}}(\mathcal{A}|_{U_i}, \widetilde{I_i}) \cong \operatorname{Hom}_{A_i}(\Gamma(U_i, \mathcal{A}), I_i),$$

naturally in $\mathcal{A}$. The map $\Gamma(U_i, \mathcal{A}) \to \Gamma(U_i, \mathcal{B})$ is injective (Γ is left exact), so, I_i being injective, $\operatorname{Hom}_{A_i}(\Gamma(U_i, \mathcal{B}), I_i) \to \operatorname{Hom}_{A_i}(\Gamma(U_i, \mathcal{A}), I_i)$ is surjective. Hence $\mathcal{G}$ is injective in $\mathfrak{Qco}(X)$. Since every quasi-coherent $\mathcal{F}$ embeds in such a $\mathcal{G}$ (3.6), $\mathfrak{Qco}(X)$ has enough injectives.

(b) (We do not need the hinted route: (a) gives a short proof.) Let $\mathcal{J}$ be an injective object of $\mathfrak{Qco}(X)$. By (3.6) there is a monomorphism $u : \mathcal{J} \to \mathcal{G}$ in $\mathfrak{Qco}(X)$ with $\mathcal{G}$ as above, which is flasque. Since $\mathcal{J}$ is injective in $\mathfrak{Qco}(X)$, the identity of $\mathcal{J}$ extends along u: there is $r : \mathcal{G} \to \mathcal{J}$ with $ru = \mathrm{id}_{\mathcal{J}}$. Now let $V \subseteq X$ be open and $s \in \mathcal{J}(V)$. Then $u(s) \in \mathcal{G}(V)$ lifts to $t \in \mathcal{G}(X)$ since $\mathcal{G}$ is flasque, and $r(t) \in \mathcal{J}(X)$ satisfies $r(t)|_V = r(t|_V) = r(u(s)) = s$. So $\mathcal{J}(X) \to \mathcal{J}(V)$ is surjective for all V: $\mathcal{J}$ is flasque (a direct summand of a flasque sheaf is flasque).

(c) Let $\mathcal{F}$ be quasi-coherent and $0 \to \mathcal{F} \to \mathcal{J}^\bullet$ an injective resolution in $\mathfrak{Qco}(X)$ (exists by (a)). Since $\mathfrak{Qco}(X)$ is closed under kernels and cokernels in $\mathfrak{Mod}(X)$ (II, 5.7), the sequence is exact as a sequence of sheaves, and by (b) each $\mathcal{J}^p$ is flasque. By Prop. 2.5 (flasque resolutions compute cohomology), $H^p(X, \mathcal{F}) = h^p(\Gamma(X, \mathcal{J}^\bullet))$, which is by definition the p-th right derived functor of $\Gamma(X, \cdot) : \mathfrak{Qco}(X) \to \mathfrak{Ab}$ evaluated at $\mathcal{F}$. (The identification is natural in $\mathcal{F}$, and compatible with connecting morphisms for short exact sequences of quasi-coherent sheaves.) □

Exercise 3.7

Problem. Let A be a noetherian ring, let $X = \operatorname{Spec} A$, let $\mathfrak{a} \subseteq A$ be an ideal, and let $U \subseteq X$ be the open set $X - V(\mathfrak{a})$.

(a) For any A-module M, establish the following formula of Deligne:

$$\Gamma(U, \widetilde{M}) \cong \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M).$$

(b) Apply this in the case of an injective A-module I, to give another proof of (3.4).

Solution. Choose generators $\mathfrak{a} = (f_1, \ldots, f_r)$ (A is noetherian). Then $U = \bigcup_i D(f_i)$ and $D(f) \subseteq U$ for every $f \in \mathfrak{a}$, and $\{D(f)\}_{f \in \mathfrak{a}}$ covers U.

(a) *The map.* For $\varphi \in \operatorname{Hom}_A(\mathfrak{a}^n, M)$ and $f \in \mathfrak{a}$, put $s_f = \varphi(f^n)/f^n \in M_f = \widetilde{M}(D(f))$. For $f, g \in \mathfrak{a}$, in M_{fg} we have $s_f = s_g$, because $g^n\varphi(f^n) = \varphi(f^n g^n) = f^n\varphi(g^n)$. So the s_f glue to a section $\sigma(\varphi) \in \Gamma(U, \widetilde{M})$ with $\sigma(\varphi)|_{D(f)} = \varphi(f^n)/f^n$ for all $f \in \mathfrak{a}$. If $\varphi' = \varphi|_{\mathfrak{a}^{n+1}}$ then

$\varphi'(f^{n+1})/f^{n+1} = f\varphi(f^n)/f^{n+1} = \varphi(f^n)/f^n$, so $\sigma(\varphi') = \sigma(\varphi)$. Hence σ induces an A-linear map $\sigma : \varinjlim_n \mathrm{Hom}_A(\mathfrak{a}^n, M) \to \Gamma(U, \widetilde{M})$.

Injectivity. Let $\sigma(\varphi) = 0$, $\varphi \in \mathrm{Hom}(\mathfrak{a}^n, M)$. Then $\varphi(f_i^n)/f_i^n = 0$ in M_{f_i}, so there is k (uniform in i) with $f_i^k \varphi(f_i^n) = 0$, i.e. $\varphi(f_i^{n+k}) = 0$ for all i. Put $N = r(n+k)$. The ideal $\mathfrak{a}^N$ is generated by the monomials $f^\alpha = \prod f_i^{\alpha_i}$ with $|\alpha| = N$; each has some $\alpha_i \geq n+k$, so $f^\alpha = g f_i^{n+k}$ with $g \in A$ and $\varphi(f^\alpha) = g\varphi(f_i^{n+k}) = 0$. So $\varphi|_{\mathfrak{a}^N} = 0$: the class of φ in the limit is 0.

Surjectivity. Let $s \in \Gamma(U, \widetilde{M})$. Then $s|_{D(f_i)} = m_i/f_i^n$ for some $m_i \in M$ and a common $n \geq 1$ (we may always increase n). On $D(f_i f_j)$, $m_i/f_i^n = m_j/f_j^n$, so there is k with $(f_i f_j)^k (f_j^n m_i - f_i^n m_j) = 0$ for all i, j. Replacing n by $n+k$ and m_i by $f_i^k m_i$ (which does not change m_i/f_i^n) we may assume

$$f_j^n m_i = f_i^n m_j \ \text{ in } M \quad \text{for all } i, j. \tag{1}$$

Let $F = A^r$ with basis $e_1, \ldots, e_r$, let $\pi : F \to \mathfrak{a}^{(n)} := (f_1^n, \ldots, f_r^n)$, $e_i \mapsto f_i^n$, with kernel R (finitely generated, A noetherian), and let $\psi : F \to M$, $e_i \mapsto m_i$. If $\rho = \sum c_i e_i \in R$, then $z = \psi(\rho) = \sum c_i m_i$ satisfies, in M_{f_j}, using (1) ($m_i/1 = f_i^n m_j / f_j^n$):

$$z/1 = \Big(\sum c_i f_i^n \Big) \frac{m_j}{f_j^n} = 0.$$

Thus every element of $\psi(R)$ is killed by a power of each f_j, hence by a power of $\mathfrak{a}$; since $\psi(R)$ is finitely generated there is T with $\mathfrak{a}^T \psi(R) = 0$. By Krull's theorem (3.1A) applied to $R \subseteq F$ there is $T' \geq T$ with $R \cap \mathfrak{a}^{T'} F \subseteq \mathfrak{a}^T R$, so $\psi(R \cap \mathfrak{a}^{T'} F) = 0$. Hence ψ restricted to $\mathfrak{a}^{T'} F$ factors through $\mathfrak{a}^{T'} F / (R \cap \mathfrak{a}^{T'} F) \cong \pi(\mathfrak{a}^{T'} F) = \mathfrak{a}^{T'} \mathfrak{a}^{(n)}$: we get $\varphi_0 : \mathfrak{a}^{T'} \mathfrak{a}^{(n)} \to M$ with $\varphi_0(\sum c_i f_i^n) = \sum c_i m_i$ for $c_i \in \mathfrak{a}^{T'}$.

Now $\mathfrak{a}^{N_0} \subseteq \mathfrak{a}^{(n)}$ for $N_0 = r(n-1)+1$ (pigeonhole on monomials), hence $\mathfrak{a}^{T'+N_0} \subseteq \mathfrak{a}^{T'} \mathfrak{a}^{(n)}$. Let $N = T' + N_0$ and $\varphi = \varphi_0|_{\mathfrak{a}^N} \in \mathrm{Hom}(\mathfrak{a}^N, M)$. For j we have $N \geq T' + n$ (as $N_0 \geq n$) and $f_j^N = f_j^{N-n} \cdot f_j^n$ with $f_j^{N-n} \in \mathfrak{a}^{T'}$, so $\varphi(f_j^N) = f_j^{N-n} m_j$ and

$$\sigma(\varphi)|_{D(f_j)} = \varphi(f_j^N)/f_j^N = m_j/f_j^n = s|_{D(f_j)}.$$

The $D(f_j)$ cover U, so $\sigma(\varphi) = s$.

(b) Let I be an injective A-module, $\mathfrak{a}$ any ideal. Since I is injective, restriction $I = \mathrm{Hom}_A(A, I) \to \mathrm{Hom}_A(\mathfrak{a}^n, I)$ is surjective for each n: every $\varphi \in \mathrm{Hom}(\mathfrak{a}^n, I)$ is $x \mapsto xm$ for some $m \in I$. For such φ, $\sigma(\varphi)|_{D(f)} = f^n m/f^n = m|_{D(f)}$, so $\sigma(\varphi) = m|_U$, the restriction to U of the global section $m \in I = \Gamma(X, \widetilde{I})$. By (a), σ is surjective, so $\Gamma(X, \widetilde{I}) \to \Gamma(U, \widetilde{I})$ is surjective. Every open subset U of X is of the form $X - V(\mathfrak{a})$ for some ideal $\mathfrak{a}$. Thus $\widetilde{I}(X) \to \widetilde{I}(U)$ is surjective for every open U; if $V \subseteq U$ are open and $s \in \widetilde{I}(V)$, lift s to $t \in \widetilde{I}(X)$, and then $t|_U$ extends s. So $\widetilde{I}$ is flasque, which is Prop. 3.4. (This proof uses Krull's theorem 3.1A, as did Lemma 3.2, but not Lemmas 3.2 and 3.3 or noetherian induction.) □

Exercise 3.8

Problem. Without the noetherian hypothesis, (3.3) and (3.4) are false. Let $A = k[x_0, x_1, x_2, \ldots]$ with the relations $x_0^n x_n = 0$ for $n = 1, 2, \ldots$. Let I be an injective A-module containing A. Show that $I \to I_{x_0}$ is not surjective.

Solution. Let $\theta : I \to I_{x_0}$, $z \mapsto z/1$, and regard $A \subseteq I$. Consider the element $1/x_0 \in I_{x_0}$ (the class of $1 \in A \subseteq I$). Suppose $\theta(z) = 1/x_0$ for some $z \in I$. Then $z/1 = 1/x_0$ in I_{x_0} means that for some $m \geq 0$, $x_0^m(x_0 z - 1) = 0$ in I, i.e.

$$x_0^{m+1} z = x_0^m \cdot 1 \quad \text{in } I.$$

Multiply by $x_{m+1} \in A$ and use $x_0^{m+1} x_{m+1} = 0$ in A:

$$x_{m+1} x_0^m \cdot 1 = x_{m+1} x_0^{m+1} z = 0 \quad \text{in } I.$$

Since $A \subseteq I$, this says $x_0^m x_{m+1} = 0$ in A. But $A = k[x_0, x_1, \dots]/\mathfrak{b}$ with $\mathfrak{b} = (x_0^n x_n \mid n \geq 1)$ a monomial ideal, and the monomial $x_0^m x_{m+1}$ is divisible by no generator $x_0^n x_n$ (it involves x_{m+1}, so we would need $n = m+1$, but then x_0 would have to occur to the power $\geq m+1$). Hence $x_0^m x_{m+1} \notin \mathfrak{b}$, i.e. it is nonzero in A, a contradiction. Therefore $1/x_0 \notin \theta(I)$ and $I \to I_{x_0}$ is not surjective.

(Remark: the argument works for any A-module I containing A; the failure is caused by the strictly increasing chain $\operatorname{Ann}(x_0) \subsetneq \operatorname{Ann}(x_0^2) \subsetneq \cdots$, $x_n \in \operatorname{Ann}(x_0^n) \setminus \operatorname{Ann}(x_0^{n-1})$, which violates the stabilization used in Lemma 3.3. Consequently the proof of Prop. 3.4 does not go through; in fact for this I the surjectivity $\widetilde{I}(X) \to \widetilde{I}(D(x_0)) = I_{x_0}$ fails, so $\widetilde{I}$ is not flasque.) □

4. Cech cohomology

Conventions and results used. For a topological space X, an open covering $\mathfrak{U} = (U_i)_{i \in I}$ with I well-ordered, and a sheaf $\mathcal{F}$ of abelian groups, $C^\bullet(\mathfrak{U}, \mathcal{F})$ is the Cech complex and $\check{H}^p(\mathfrak{U}, \mathcal{F})$ its cohomology. By Remark 4.0.1 we may regard a cochain $\alpha \in C^p(\mathfrak{U}, \mathcal{F})$ as a family $\alpha_{i_0 \dots i_p} \in \mathcal{F}(U_{i_0 \dots i_p})$ indexed by *all* $(p+1)$-tuples, alternating (zero if two indices coincide, changing sign under a transposition), and the formula for d is then valid for all $(p+2)$-tuples. We use: Lemma 4.1 ($\check{H}^0(\mathfrak{U}, \mathcal{F}) = \Gamma(X, \mathcal{F})$), Lemma 4.2 (the sheafified complex $\mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F})$ is a resolution of $\mathcal{F}$), Prop. 4.3 (flasque sheaves have $\check{H}^p = 0$, $p > 0$), the natural maps $\check{H}^p(\mathfrak{U}, \mathcal{F}) \to H^p(X, \mathcal{F})$ of Lemma 4.4, and Thm. 4.5. In a separated scheme the intersection of two open affines is affine (II, Ex. 4.3). If X is noetherian and separated, $\mathfrak{U}$ an open affine cover and $\mathcal{F}$ quasi-coherent, then $\check{H}^p(\mathfrak{U}, \mathcal{F}) = H^p(X, \mathcal{F})$ (Thm. 4.5).

Exercise 4.1

> **Problem.** Let $f : X \to Y$ be an affine morphism of noetherian separated schemes (II, Ex. 5.17). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X, there are natural isomorphisms for all $i \geq 0$,
>
> $$H^i(X, \mathcal{F}) \cong H^i(Y, f_* \mathcal{F}).$$
>
> [Hint: Use (II, 5.8).]

Solution. Let $\mathfrak{V} = (V_j)_{j \in J}$ be a finite open affine cover of Y and $U_j = f^{-1}(V_j)$; since f is affine, each U_j is affine, so $\mathfrak{U} = (U_j)$ is an open affine cover of X. Moreover $U_{j_0 \dots j_p} = f^{-1}(V_{j_0 \dots j_p})$. The sheaf $f_* \mathcal{F}$ is quasi-coherent on Y (II, 5.8(c); X is noetherian), and

$$\mathcal{F}(U_{j_0 \dots j_p}) = \mathcal{F}(f^{-1} V_{j_0 \dots j_p}) = (f_* \mathcal{F})(V_{j_0 \dots j_p}).$$

Hence (with the same ordering of J) $C^\bullet(\mathfrak{U},\mathcal{F}) = C^\bullet(\mathfrak{V}, f_*\mathcal{F})$ as complexes. Applying Thm. 4.5 to X and to Y,

$$H^i(X,\mathcal{F}) \cong \check{H}^i(\mathfrak{U},\mathcal{F}) = \check{H}^i(\mathfrak{V}, f_*\mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

(Naturality in $\mathcal{F}$ is clear since the Cech isomorphisms are natural in the sheaf.) □

Exercise 4.2

Problem. Prove Chevalley's theorem: Let $f : X \to Y$ be a finite surjective morphism of noetherian separated schemes, with X affine. Then Y is affine.

(a) Let $f : X \to Y$ be a finite surjective morphism of integral noetherian schemes. Show that there is a coherent sheaf $\mathcal{M}$ on X, and a morphism of sheaves $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{M}$ for some $r > 0$, such that α is an isomorphism at the generic point of Y.

(b) For any coherent sheaf $\mathcal{F}$ on Y, show that there is a coherent sheaf $\mathcal{G}$ on X, and a morphism $\beta : f_*\mathcal{G} \to \mathcal{F}^r$ which is an isomorphism at the generic point of Y. [Hint: Apply $\mathcal{H}om(\cdot,\mathcal{F})$ to α and use (II, Ex. 5.17e).]

(c) Now prove Chevalley's theorem. First use (Ex. 3.1) and (Ex. 3.2) to reduce to the case X and Y integral. Then use (3.7), (Ex. 4.1), consider $\ker\beta$ and $\operatorname{coker}\beta$, and use noetherian induction on Y.

Solution. (a) Let η and ξ be the generic points of Y and X, $K(Y) = \mathcal{O}_{Y,\eta}$, $K(X) = \mathcal{O}_{X,\xi}$. Let $V = \operatorname{Spec} A \subseteq Y$ be an affine open; $f^{-1}V = \operatorname{Spec} B$ with B a domain which is a finite A-module, A a domain. Since f is surjective, $A \to B$ is injective: if $a \in \ker$, then a lies in every prime of A (every prime of A is the contraction of a prime of B), so $a \in \mathfrak{N}A = 0$. Hence $B \otimes_A K(Y) = B_{(A-0)}$ is a domain finite over the field $K(Y)$, hence a field, and therefore equal to $\operatorname{Frac} B = K(X)$. So $K(X)$ is a finite extension of $K(Y)$; let $r = [K(X) : K(Y)] \geq 1$, and choose a basis $s_1, \ldots, s_r$ of $K(X)$ over $K(Y)$. For a quasi-coherent $\mathcal{M}$ on X, with $N = \Gamma(f^{-1}V, \mathcal{M})$ (a B-module), $(f_*\mathcal{M})_\eta = N \otimes_A K(Y) = N \otimes_B (B \otimes_A K(Y)) = N \otimes_B K(X) = \mathcal{M}_\xi$.

Let $\mathcal{K}$ be the constant sheaf $K(X)$ on X; it is quasi-coherent (on an affine open $\operatorname{Spec} B$ it is $\widetilde{K(X)}$ with $K(X) = \operatorname{Frac} B$, a B-module). Let $\mathcal{M} \subseteq \mathcal{K}$ be the image of $\mathcal{O}_X^r \to \mathcal{K}$, $e_i \mapsto s_i$. It is quasi-coherent and finitely generated, hence coherent, and the s_i are global sections of $\mathcal{M}$. Define $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{M}$ by $e_i \mapsto s_i \in \Gamma(X,\mathcal{M}) = \Gamma(Y, f_*\mathcal{M})$. At the generic point, $\alpha_\eta : K(Y)^r \to (f_*\mathcal{M})_\eta = \mathcal{M}_\xi = \sum K(X)s_i = K(X)$ sends $e_i \mapsto s_i$, an isomorphism because $s_1, \ldots, s_r$ is a basis.

(b) The sheaf $f_*\mathcal{M}$ is a coherent $\mathcal{O}_Y$-module (finite morphism), and a module over the sheaf of algebras $\mathcal{A} = f_*\mathcal{O}_X$. Hence $\mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{M},\mathcal{F})$ is a coherent $\mathcal{O}_Y$-module with a natural $\mathcal{A}$-module structure $((a\cdot\varphi)(m) = \varphi(am))$. By II, Ex. 5.17(e) (equivalence between quasi-coherent $\mathcal{O}_X$-modules and quasi-coherent $\mathcal{A}$-modules, via f_*; f is affine, II Ex. 5.17(b)) there is a quasi-coherent $\mathcal{O}_X$-module $\mathcal{G}$ with $f_*\mathcal{G} \cong \mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{M},\mathcal{F})$ as $\mathcal{A}$-modules; $\mathcal{G}$ is coherent since $f_*\mathcal{G}$ is coherent over $\mathcal{O}_Y$ (locally $\mathcal{G} = \widetilde{N}$ with N a B-module finitely generated over A, hence over B). Applying $\mathcal{H}om_{\mathcal{O}_Y}(\cdot,\mathcal{F})$ to α we get

$$\beta : \ f_*\mathcal{G} \cong \mathcal{H}om(f_*\mathcal{M},\mathcal{F}) \xrightarrow{\alpha^*} \mathcal{H}om(\mathcal{O}_Y^r,\mathcal{F}) = \mathcal{F}^r.$$

Since $f_*\mathcal{M}$ is coherent, the stalk of $\mathcal{H}om(f_*\mathcal{M},\mathcal{F})$ at η is $\operatorname{Hom}_{K(Y)}((f_*\mathcal{M})_\eta, \mathcal{F}_\eta)$, and $\beta_\eta = \operatorname{Hom}(\alpha_\eta, \mathcal{F}_\eta)$ is an isomorphism because α_η is.

(c) Let $f : X \to Y$ be finite and surjective, X affine, X, Y noetherian separated. Suppose Y is not affine; we derive a contradiction.

Reduction to a minimal counterexample. Among the closed subsets $Z \subseteq Y$ such that Z (with its reduced structure) is not affine – a nonempty family, since Y belongs to it – choose a minimal one, Z_0 (the space Y is noetherian). Let $Y_0 = Z_0$ with the reduced structure and $X_0 = (f^{-1}Z_0)_{\mathrm{red}}$, a closed subscheme of the affine X, hence affine. Since f is surjective, $f(f^{-1}Z_0) = Z_0$. The reduced scheme X_0 maps into Y_0 (a morphism from a reduced scheme to Y with set-theoretic image in the closed subset Z_0 factors through Y_0), and $X_0 \to Y_0$ is finite (a closed subscheme of $X \times_Y Y_0$, which is finite over Y_0) and surjective. So replacing f by $X_0 \to Y_0$ we may assume: Y is reduced, noetherian, separated, not affine, and every proper closed subset of Y, with reduced structure, is affine; X is affine; f finite surjective.

Y is integral. If Y had several irreducible components, each would be a proper closed subset, hence affine, and then Y would be affine by Ex. 3.2: contradiction. So Y is irreducible, hence integral.

X may be taken integral. $f(X) = Y$ is the finite union of the closed irreducible sets $f(X_j)$ (X_j the irreducible components of X; f is closed as it is finite), so $Y = f(X_j)$ for some j (Y is irreducible). Replace X by X_j with the reduced structure: it is integral, affine (closed in X) and $X_j \to Y$ is finite and surjective.

The cohomological argument. Now $f : X \to Y$ is a finite surjective morphism of integral noetherian separated schemes, X affine, Y not affine, all proper closed subsets of Y (reduced structure) affine. Let $\mathcal{F}$ be any coherent sheaf on Y. By (b) there are $r > 0$, a coherent sheaf $\mathcal{G}$ on X and $\beta : f_*\mathcal{G} \to \mathcal{F}^r$ which is an isomorphism at the generic point η. Let $\mathcal{K} = \ker \beta$, $\mathcal{B} = \operatorname{im} \beta$, $\mathcal{C} = \operatorname{coker} \beta$; these are coherent, and

$$0 \to \mathcal{K} \to f_*\mathcal{G} \to \mathcal{B} \to 0, \qquad 0 \to \mathcal{B} \to \mathcal{F}^r \to \mathcal{C} \to 0$$

are exact. As $\mathcal{K}_\eta = 0 = \mathcal{C}_\eta$, the supports of $\mathcal{K}$ and $\mathcal{C}$ are closed subsets of Y not containing η, so are proper closed subsets.

Claim: $H^p(Y, \mathcal{L}) = 0$ for $p > 0$ if $\mathcal{L}$ is coherent with $\operatorname{Supp} \mathcal{L} \neq Y$. Let $\mathcal{J} = \operatorname{Ann} \mathcal{L}$, a coherent ideal sheaf with $V(\mathcal{J}) = \operatorname{Supp} \mathcal{L}$, and Z' the closed subscheme defined by $\mathcal{J}$. Its reduced scheme Z'_{red} is the proper closed subset $\operatorname{Supp} \mathcal{L}$ with reduced structure, hence affine by minimality. By Ex. 3.1, the noetherian scheme Z' is affine. Then $\mathcal{L} = i_*\mathcal{L}'$ with $\mathcal{L}'$ coherent on Z' (II, Ex. 5.10), and $H^p(Y, \mathcal{L}) = H^p(Z', \mathcal{L}') = 0$ for $p > 0$ by Lemma 2.10 and Thm. 3.5.

Now f is finite, hence affine, so $H^p(Y, f_*\mathcal{G}) = H^p(X, \mathcal{G}) = 0$ for $p > 0$ by Ex. 4.1 and Thm. 3.5 (X is affine). Thus: from the first sequence, $H^1(Y, \mathcal{B}) \hookrightarrow H^2(Y, \mathcal{K}) = 0$ (as $H^1(f_*\mathcal{G}) = 0$), so $H^1(Y, \mathcal{B}) = 0$; from the second, $H^1(Y, \mathcal{B}) \to H^1(Y, \mathcal{F})^r \to H^1(Y, \mathcal{C}) = 0$, so $H^1(Y, \mathcal{F})^r = 0$ and $H^1(Y, \mathcal{F}) = 0$. This holds for every coherent $\mathcal{F}$, in particular for every coherent ideal sheaf; by Serre's criterion (3.7), Y is affine. This contradicts the assumption. Hence Y is affine. □

Exercise 4.3

Problem. Let $X = \mathbf{A}^2_k = \operatorname{Spec} k[x, y]$, and let $U = X - \{(0, 0)\}$. Using a suitable cover of U by open affine subsets, show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k-vector space spanned by $\{x^i y^j \mid i, j < 0\}$. In particular, it is infinite-dimensional. (Using (3.5), this provides another proof that U is not affine — cf. (I, Ex. 3.6).)

Solution. Put $A = k[x, y]$, $U_1 = D(x)$, $U_2 = D(y)$, with $U_1 \cup U_2 = X - V(x, y) = U$. These are affine, $U_1 \cap U_2 = D(xy)$ is affine, and U is noetherian and separated. By Thm. 4.5, $H^i(U, \mathcal{O}_U) = \check{H}^i(\mathfrak{U}, \mathcal{O}_U)$ for $\mathfrak{U} = (U_1, U_2)$. The Cech complex has only two nonzero terms:

$$0 \to A_x \oplus A_y \xrightarrow{d} A_{xy} \to 0, \qquad d(a, b) = b - a.$$

$A_{xy} = k[x, y, x^{-1}, y^{-1}]$ has k-basis the monomials $x^i y^j$, $(i, j) \in \mathbf{Z}^2$; A_x is spanned by those with $j \geq 0$, A_y by those with $i \geq 0$. Hence $\operatorname{im} d$ is spanned by the monomials with $i \geq 0$ or $j \geq 0$ and

$$H^1(U, \mathcal{O}_U) = \operatorname{coker} d \cong \bigoplus_{i<0,\, j<0} k \cdot x^i y^j,$$

an infinite-dimensional vector space ($H^0 = \ker d = A$, and $H^p = 0$ for $p \geq 2$). If U were affine, $H^1(U, \mathcal{O}_U) = 0$ by Thm. 3.5; hence U is not affine. □

Exercise 4.4

Problem. On an arbitrary topological space X with an arbitrary abelian sheaf $\mathcal{F}$, Cech cohomology may not give the same result as the derived functor cohomology. But here we show that for H^1, there is an isomorphism if one takes the limit over all coverings.

(a) Let $\mathfrak{U} = (U_i)_{i \in I}$ be an open covering of the topological space X. A *refinement* of $\mathfrak{U}$ is a covering $\mathfrak{V} = (V_j)_{j \in J}$, together with a map $\lambda : J \to I$ of the index sets, such that for each $j \in J$, $V_j \subseteq U_{\lambda(j)}$. If $\mathfrak{V}$ is a refinement of $\mathfrak{U}$, show that there is a natural induced map on Cech cohomology, for any abelian sheaf $\mathcal{F}$, and for each i,

$$\lambda^i : \check{H}^i(\mathfrak{U}, \mathcal{F}) \to \check{H}^i(\mathfrak{V}, \mathcal{F}).$$

The coverings of X form a partially ordered set under refinement, so we can consider the Cech cohomology in the limit

$$\varinjlim_{\mathfrak{U}} \check{H}^i(\mathfrak{U}, \mathcal{F}).$$

(b) For any abelian sheaf $\mathcal{F}$ on X, show that the natural maps (4.4) for each covering

$$\check{H}^i(\mathfrak{U}, \mathcal{F}) \to H^i(X, \mathcal{F})$$

are compatible with the refinement maps above.

(c) Now prove the following theorem. Let X be a topological space, $\mathcal{F}$ a sheaf of abelian groups. Then the natural map

$$\varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U}, \mathcal{F}) \to H^1(X, \mathcal{F})$$

is an isomorphism. [Hint: Embed $\mathcal{F}$ in a flasque sheaf $\mathcal{G}$, and let $\mathcal{R} = \mathcal{G}/\mathcal{F}$, so that we have an exact sequence $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{R} \to 0$. Define a complex $D^\bullet(\mathfrak{U})$ by $0 \to C^\bullet(\mathfrak{U}, \mathcal{F}) \to C^\bullet(\mathfrak{U}, \mathcal{G}) \to D^\bullet(\mathfrak{U}) \to 0$. Then use the exact cohomology sequence of this sequence of complexes, and the natural map of complexes $D^\bullet(\mathfrak{U}) \to C^\bullet(\mathfrak{U}, \mathcal{R})$, and see what happens under refinement.]

Solution. (a) *The map.* Let $\mathfrak{V} = (V_j)$ refine $\mathfrak{U} = (U_i)$ via λ. Use alternating cochains (Remark 4.0.1) and define $\lambda^{\#} : C^p(\mathfrak{U}, \mathcal{F}) \to C^p(\mathfrak{V}, \mathcal{F})$ by

$$(\lambda^{\#}\alpha)_{j_0 \dots j_p} = \alpha_{\lambda(j_0)\dots\lambda(j_p)}\big|_{V_{j_0\dots j_p}}$$

(note $V_{j_0\dots j_p} \subseteq U_{\lambda(j_0)\dots\lambda(j_p)}$). This is again alternating (if two j's coincide so do two $\lambda(j)$'s, and a transposition of the j's transposes the $\lambda(j)$'s). It commutes with d:

$$(d\lambda^{\#}\alpha)_{j_0\dots j_{p+1}} = \sum_k (-1)^k \alpha_{\lambda(j_0)\dots\widehat{\lambda(j_k)}\dots\lambda(j_{p+1})}\Big| = (\lambda^{\#} d\alpha)_{j_0\dots j_{p+1}},$$

using that the formula for d holds for arbitrary tuples. So $\lambda^{\#}$ is a morphism of complexes, natural in $\mathcal{F}$, and induces $\lambda^i : \check{H}^i(\mathfrak{U}, \mathcal{F}) \to \check{H}^i(\mathfrak{V}, \mathcal{F})$.

Independence of λ. Let $\mu : J \to I$ be another map with $V_j \subseteq U_{\mu(j)}$. Then $\lambda^i = \mu^i$. Indeed define $h : C^p(\mathfrak{U}, \mathcal{F}) \to C^{p-1}(\mathfrak{V}, \mathcal{F})$ by

$$(h\alpha)_{j_0\dots j_{p-1}} = \sum_{k=0}^{p-1} (-1)^k \alpha_{\lambda(j_0)\dots\lambda(j_k)\mu(j_k)\dots\mu(j_{p-1})}\big|_{V_{j_0\dots j_{p-1}}}$$

(meaningful, since $V_{j_0\dots j_{p-1}}$ lies in $U_{\lambda j_0} \cap \dots \cap U_{\lambda j_k} \cap U_{\mu j_k} \cap \dots \cap U_{\mu j_{p-1}}$). We define $h\alpha$ by this formula for $j_0 < \dots < j_{p-1}$ and extend it to all tuples by alternation (the formula itself need not be alternating). To verify the identity below it suffices to take $j_0 < \dots < j_p$, since both sides are alternating cochains; then every sub-tuple of $(j_0, \dots, j_p)$ is increasing, so $h\alpha$ is only evaluated on increasing tuples, while $d\alpha$ may be evaluated on the non-increasing tuples w_k, which is legitimate by Remark 4.0.1. We claim $dh + hd = \mu^{\#} - \lambda^{\#}$. Fix $j_0, \dots, j_p$ and put $w_k = (\lambda j_0, \dots, \lambda j_k, \mu j_k, \dots, \mu j_p)$, $0 \le k \le p$ (a $(p+2)$-tuple). Then

$$(hd\alpha)_{j_0\dots j_p} = \sum_{k=0}^{p} (-1)^k (d\alpha)_{w_k} = \sum_{k=0}^{p} \sum_{l=0}^{p+1} (-1)^{k+l} \alpha_{w_k \setminus l},$$

where $w_k \setminus l$ omits the entry in position l; and $(dh\alpha)_{j_0\dots j_p} = \sum_{l=0}^{p} (-1)^l (h\alpha)_{j_0\dots\widehat{j_l}\dots j_p} = \sum_{l=0}^{p} \sum_{k=0}^{p-1} (-1)^{l+k} \alpha_{t(l,k)}$, where $t(l,k)$ is the tuple built from the list $j_0, \dots, \widehat{j_l}, \dots, j_p$ by applying λ to its first $k+1$ entries and μ to its entries number $k, \dots, p-1$. Sort the terms of $hd\alpha$ by the position l relative to k (in w_k, the positions $0, \dots, k$ hold $\lambda j_0, \dots, \lambda j_k$ and positions $k+1, \dots, p+1$ hold $\mu j_k, \dots, \mu j_p$).

- $l = k+1$ (omit μj_k) and $l = k$ (omit λj_k): $w_k \setminus (k+1) = (\lambda j_0, \dots, \lambda j_k, \mu j_{k+1}, \dots, \mu j_p) = w_{k+1} \setminus (k+1)$, with signs -1 and $+1$ respectively, so for $0 \le k \le p-1$ the term $(k, k+1)$ cancels the term $(k+1, k+1)$. What is left of these two kinds are $(k,l) = (0,0)$, giving $+\alpha_{\mu j_0 \dots \mu j_p}$, and $(k,l) = (p, p+1)$, giving $-\alpha_{\lambda j_0 \dots \lambda j_p}$.

- $l < k$ (omit λj_l): $w_k \setminus l = (\lambda j_0, .., \widehat{\lambda j_l}, .., \lambda j_k, \mu j_k, \dots, \mu j_p) = t(l, k-1)$ (here $l \le k-1$, so the $(k-1)$-st entry of the shortened list is j_k), and the sign $(-1)^{k+l}$ is opposite to $(-1)^{l+k-1}$. So these cancel exactly the terms of $dh\alpha$ with inner index $k' \ge l$.

- $l \ge k+2$ (omit μj_{l-1}): $w_k \setminus l = (\lambda j_0, \dots, \lambda j_k, \mu j_k, .., \widehat{\mu j_{l-1}}, .., \mu j_p) = t(l-1, k)$ (inner index $k < l-1$), with sign $(-1)^{k+l}$ opposite to $(-1)^{(l-1)+k}$. So these cancel exactly the terms of $dh\alpha$ with inner index $k' <$ omitted index.

Every term of $dh\alpha$ is matched exactly once (a pair (l',k') has either $k' \geq l'$ or $k' < l'$). The only terms left are those displayed in the first item. Hence $dh + hd = \mu^{\#} - \lambda^{\#}$, so $\lambda^i = \mu^i$ on cohomology. Consequently the maps λ^i depend only on the refinement relation, they compose ($(\lambda\mu')^{\#} = \mu'^{\#}\lambda^{\#}$), and since any two coverings have a common refinement ($(U_i \cap U'_{i'})_{(i,i')}$), the groups $\check{H}^i(\mathfrak{U},\mathcal{F})$ form a direct system over the directed set of coverings, whose limit $\varinjlim_{\mathfrak{U}} \check{H}^i(\mathfrak{U},\mathcal{F})$ is what is considered. (To avoid set-theoretic problems one may restrict to coverings indexed by subsets of the power set of X.)

(b) Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in $\mathfrak{Ab}(X)$. The natural map of (4.4) is obtained as follows: by Lemma 4.2, $\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F})$ is a resolution of $\mathcal{F}$, so there is a morphism of complexes $\theta_{\mathfrak{U}} : \mathcal{C}^\bullet(\mathfrak{U},\mathcal{F}) \to \mathcal{I}^\bullet$ over $\mathrm{id}_{\mathcal{F}}$, unique up to homotopy; one applies $\Gamma(X,\cdot)$ and takes h^p, using $\Gamma(X,\mathcal{C}^p(\mathfrak{U},\mathcal{F})) = C^p(\mathfrak{U},\mathcal{F})$. Let $\mathfrak{V}$ refine $\mathfrak{U}$ via λ. The formula for $\lambda^{\#}$ defines a morphism of sheaf complexes $\lambda^{\#} : \mathcal{C}^\bullet(\mathfrak{U},\mathcal{F}) \to \mathcal{C}^\bullet(\mathfrak{V},\mathcal{F})$ (on an open W: $(\lambda^{\#}\alpha)_{j_0\dots j_p} = \alpha_{\lambda j_0 \dots \lambda j_p}|_{W \cap V_{j_0\cdots j_p}}$, the same computation as in (a)), which commutes with the augmentations $\varepsilon : \mathcal{F} \to \mathcal{C}^0$, since $\varepsilon(s)_j = s|_{V_j}$ and $(\lambda^{\#}\varepsilon s)_j = s|_{U_{\lambda j}}|_{V_j} = s|_{V_j}$. Therefore $\theta_{\mathfrak{V}} \circ \lambda^{\#}$ and $\theta_{\mathfrak{U}}$ are two morphisms of complexes from the resolution $\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F})$ of $\mathcal{F}$ to the injective resolution $\mathcal{I}^\bullet$ lying over $\mathrm{id}_{\mathcal{F}}$; hence they are homotopic (Hilton–Stammbach IV.4.4). Applying $\Gamma(X,\cdot)$ (and $\Gamma(X,\lambda^{\#}) = \lambda^{\#}$ on Cech cochains) we get that on cohomology the composite $\check{H}^i(\mathfrak{U},\mathcal{F}) \xrightarrow{\lambda^i} \check{H}^i(\mathfrak{V},\mathcal{F}) \to H^i(X,\mathcal{F})$ equals the natural map for $\mathfrak{U}$. This is the compatibility. In particular there is a well-defined map $\varinjlim_{\mathfrak{U}} \check{H}^i(\mathfrak{U},\mathcal{F}) \to H^i(X,\mathcal{F})$.

(c) Embed $\mathcal{F} \hookrightarrow \mathcal{G}$ with $\mathcal{G}$ flasque (e.g. injective; Lemma 2.4) and put $\mathcal{R} = \mathcal{G}/\mathcal{F}$. Fix a covering $\mathfrak{U}$. The maps $C^p(\mathfrak{U},\mathcal{F}) \to C^p(\mathfrak{U},\mathcal{G})$ are injective (products of the injections $\mathcal{F}(W) \hookrightarrow \mathcal{G}(W)$), so with $D^p = D^p(\mathfrak{U}) = C^p(\mathfrak{U},\mathcal{G})/C^p(\mathfrak{U},\mathcal{F}) = \prod \mathcal{G}(U_{i_0\dots i_p})/\mathcal{F}(U_{i_0\dots i_p})$ there is a short exact sequence of complexes

$$0 \to C^\bullet(\mathfrak{U},\mathcal{F}) \to C^\bullet(\mathfrak{U},\mathcal{G}) \to D^\bullet(\mathfrak{U}) \to 0.$$

Since $\mathcal{G}(W)/\mathcal{F}(W) \hookrightarrow \mathcal{R}(W)$ (left exactness of sections), $D^p \hookrightarrow C^p(\mathfrak{U},\mathcal{R})$, and this is a morphism (injective) of complexes. Because $D^1 \to C^1(\mathfrak{U},\mathcal{R})$ is injective,

$$H^0(D^\bullet(\mathfrak{U})) = \ker(D^0 \to D^1) = D^0 \cap \ker(d : C^0(\mathfrak{U},\mathcal{R}) \to C^1(\mathfrak{U},\mathcal{R})) = \{t \in \Gamma(X,\mathcal{R}) \mid t|_{U_i} \in \mathrm{im}(\mathcal{G}(U_i) \to \mathcal{R}(U_i))\ \forall i$$

using Lemma 4.1 ($\ker d = \Gamma(X,\mathcal{R})$). So $j : H^0(D^\bullet(\mathfrak{U})) \hookrightarrow \Gamma(X,\mathcal{R})$ is injective with image $\Gamma_{\mathfrak{U}}$. By Lemma 4.1, $\check{H}^0(\mathfrak{U},\mathcal{G}) = \Gamma(X,\mathcal{G})$ and, as $\mathcal{G}$ is flasque, $\check{H}^1(\mathfrak{U},\mathcal{G}) = 0$ (Prop. 4.3). The long exact sequence of the sequence of complexes therefore reads

$$\Gamma(X,\mathcal{G}) \to H^0(D^\bullet(\mathfrak{U})) \xrightarrow{\delta_{\check{C}}} \check{H}^1(\mathfrak{U},\mathcal{F}) \to 0,$$

exact, where the first map followed by j is the map $\Gamma(X,\mathcal{G}) \to \Gamma(X,\mathcal{R})$. On the other side, the sequence $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{R} \to 0$ and $H^1(X,\mathcal{G}) = 0$ (Prop. 2.5) give an exact sequence

$$\Gamma(X,\mathcal{G}) \to \Gamma(X,\mathcal{R}) \xrightarrow{\delta} H^1(X,\mathcal{F}) \to 0.$$

Claim: $\theta_{\mathfrak{U}} \circ \delta_{\check{C}} = \delta \circ j$, where $\theta_{\mathfrak{U}} : \check{H}^1(\mathfrak{U},\mathcal{F}) \to H^1(X,\mathcal{F})$ is the natural map. Granting it, we have a commutative diagram with exact rows

$$\begin{array}{ccccccc}
\Gamma(X,\mathcal{G}) & \to & H^0(D^\bullet(\mathfrak{U})) & \xrightarrow{\delta_{\check{C}}} & \check{H}^1(\mathfrak{U},\mathcal{F}) & \to & 0 \\
\| & & \downarrow j & & \downarrow \theta_{\mathfrak{U}} & & \\
\Gamma(X,\mathcal{G}) & \to & \Gamma(X,\mathcal{R}) & \xrightarrow{\delta} & H^1(X,\mathcal{F}) & \to & 0
\end{array}$$

Proof of the claim. Take injective resolutions $\mathcal{I}_F^\bullet$ of $\mathcal{F}$ and $\mathcal{I}_R^\bullet$ of $\mathcal{R}$; by the horseshoe lemma there is an injective resolution $\mathcal{I}_G^\bullet = \mathcal{I}_F^\bullet \oplus \mathcal{I}_R^\bullet$ of $\mathcal{G}$ with $0 \to \mathcal{I}_F^p \to \mathcal{I}_G^p \to \mathcal{I}_R^p \to 0$ split exact and compatible with $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{R} \to 0$. After applying Γ one still has short exact sequences, and the connecting homomorphism $H^0(\Gamma(\mathcal{I}_R^\bullet)) = \Gamma(X, \mathcal{R}) \to H^1(\Gamma(\mathcal{I}_F^\bullet)) = H^1(X, \mathcal{F})$ is δ. Choose $\theta_F : \mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F}) \to \mathcal{I}_F^\bullet$ as in 4.4 and let $\iota : \mathcal{I}_F^\bullet \to \mathcal{I}_G^\bullet$. Choose any morphism $\theta_G : \mathcal{C}^\bullet(\mathfrak{U}, \mathcal{G}) \to \mathcal{I}_G^\bullet$ over $\mathrm{id}_\mathcal{G}$. The two morphisms $\theta_G|_{\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F})}$ and $\iota\theta_F$ from the resolution $\mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F})$ of $\mathcal{F}$ to the injective resolution $\mathcal{I}_G^\bullet$ of $\mathcal{G}$ lie over $\mathcal{F} \to \mathcal{G}$, so they are homotopic, by some $s^p : \mathcal{C}^p(\mathfrak{U}, \mathcal{F}) \to \mathcal{I}_G^{p-1}$ with $\theta_G|_{\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F})} - \iota\theta_F = ds + sd$. As $\mathcal{I}_G^{p-1}$ is injective and $\mathcal{C}^p(\mathfrak{U}, \mathcal{F}) \to \mathcal{C}^p(\mathfrak{U}, \mathcal{G})$ is a monomorphism, s extends to $\tilde{s}^p : \mathcal{C}^p(\mathfrak{U}, \mathcal{G}) \to \mathcal{I}_G^{p-1}$; replacing θ_G by $\theta_G - d\tilde{s} - \tilde{s}d$ (still a morphism of complexes, and still over $\mathrm{id}_\mathcal{G}$: $\tilde{s}^0 = 0$, and $d \circ \varepsilon = 0$ for the augmentation $\varepsilon : \mathcal{G} \to \mathcal{C}^0(\mathfrak{U}, \mathcal{G})$) we may assume $\theta_G|_{\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F})} = \iota\theta_F$. Applying Γ gives a morphism of short exact sequences of complexes

$$\begin{array}{ccccccccc} 0 & \to & C^\bullet(\mathfrak{U}, \mathcal{F}) & \to & C^\bullet(\mathfrak{U}, \mathcal{G}) & \to & D^\bullet(\mathfrak{U}) & \to & 0 \\ & & \downarrow \theta_F & & \downarrow \theta_G & & \downarrow \bar{\theta} & & \\ 0 & \to & \Gamma(\mathcal{I}_F^\bullet) & \to & \Gamma(\mathcal{I}_G^\bullet) & \to & \Gamma(\mathcal{I}_R^\bullet) & \to & 0 \end{array}$$

where $\bar{\theta}$ is induced on quotients. By naturality of connecting homomorphisms, $\delta \circ H^0(\bar{\theta}) = H^1(\theta_F) \circ \delta_{\check{C}} = \theta_\mathfrak{U} \circ \delta_{\check{C}}$. It remains to see $H^0(\bar{\theta}) = j$ (identifying $H^0(\Gamma(\mathcal{I}_R^\bullet)) = \Gamma(X, \mathcal{R})$ via the augmentation $\mathcal{R} \to \mathcal{I}_R^0$). Let $z \in H^0(D^\bullet)$ be represented by $(g_i) \in \prod \mathcal{G}(U_i)$, with $t_i \in \mathcal{R}(U_i)$ the image of g_i and $t \in \Gamma(X, \mathcal{R})$ the section glued from the t_i, so $j(z) = t$. Let $g = (g_i) \in C^0(\mathfrak{U}, \mathcal{G})$ and $t' = \theta_G^0(g) \in \Gamma(X, \mathcal{I}_G^0)$; then $H^0(\bar{\theta})(z)$ is the image of t' in $\Gamma(\mathcal{I}_R^0)$. On U_i: $g|_{U_i} = (g_k|_{U_{ik}})_k$ differs from $\varepsilon(g_i) = (g_i|_{U_{ik}})_k$ by $((g_k - g_i)|_{U_{ik}})_k \in \mathcal{C}^0(\mathfrak{U}, \mathcal{F})(U_i)$ (because $g_k - g_i \mapsto t_k - t_i = 0$ on U_{ik}). Since θ_G^0 maps $\mathcal{C}^0(\mathfrak{U}, \mathcal{F})$ into $\mathcal{I}_F^0$ and $\theta_G^0\varepsilon = \eta_G$ (the augmentation of $\mathcal{I}_G^\bullet$), $t'|_{U_i} \equiv \eta_G(g_i)$ (mod $\mathcal{I}_F^0(U_i)$), so the image of t' in $\mathcal{I}_R^0(U_i)$ is $\eta_R(t_i) = \eta_R(t)|_{U_i}$. Hence the image of t' in $\Gamma(\mathcal{I}_R^0)$ is $\eta_R(t)$, i.e. $H^0(\bar{\theta})(z) = t = j(z)$. This proves the claim.

Conclusion. For every covering $\mathfrak{U}$, $\theta_\mathfrak{U}$ is *injective*: if $c = \delta_{\check{C}}(z)$ and $\theta_\mathfrak{U}(c) = \delta(j(z)) = 0$, then $j(z)$ is the image of some $g \in \Gamma(X, \mathcal{G})$; the image z' of g in $H^0(D^\bullet(\mathfrak{U}))$ has $j(z') = j(z)$, hence $z' = z$ (j injective), and $c = \delta_{\check{C}}(z') = 0$ by exactness of the top row. As a direct limit of injective maps (over a directed set, compatibly by (b)) is injective, $\varinjlim_\mathfrak{U} \check{H}^1(\mathfrak{U}, \mathcal{F}) \to H^1(X, \mathcal{F})$ is injective. It is surjective: let $c' \in H^1(X, \mathcal{F})$; by the bottom row $c' = \delta(t)$ for some $t \in \Gamma(X, \mathcal{R})$. Since $\mathcal{G} \to \mathcal{R}$ is a surjection of sheaves, there is an open covering $\mathfrak{U}_t$ on which t lifts locally: $t|_{U_i} \in \mathrm{im}(\mathcal{G}(U_i) \to \mathcal{R}(U_i))$. Then $t \in \Gamma_{\mathfrak{U}_t} = j(H^0(D^\bullet(\mathfrak{U}_t)))$, say $t = j(z)$, and $c' = \delta(j(z)) = \theta_{\mathfrak{U}_t}(\delta_{\check{C}} z)$ lies in the image of $\theta_{\mathfrak{U}_t}$. Hence the limit map is an isomorphism. □

Exercise 4.5

> **Problem.** For any ringed space $(X, \mathcal{O}_X)$, let $\mathrm{Pic}\, X$ be the group of isomorphism classes of invertible sheaves on X (II, Section 6). Show that $\mathrm{Pic}\, X \cong H^1(X, \mathcal{O}_X^*)$, where $\mathcal{O}_X^*$ denotes the sheaf whose sections over an open set U are the units in the ring $\Gamma(U, \mathcal{O}_X)$, with multiplication as the group operation. [Hint: For any invertible sheaf $\mathcal{L}$ on X, cover X by open sets U_i on which $\mathcal{L}$ is free, and fix isomorphisms $\varphi_i : \mathcal{O}_{U_i} \xrightarrow{\sim} \mathcal{L}|_{U_i}$. Then on $U_i \cap U_j$, we get an isomorphism $\varphi_i^{-1} \circ \varphi_j$ of $\mathcal{O}_{U_i \cap U_j}$ with itself. These isomorphisms give an element of $\check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$. Now use (Ex. 4.4).]

Solution. Write the group law of $\mathcal{O}_X^*$ multiplicatively; its Cech complex $C^\bullet(\mathfrak{U}, \mathcal{O}_X^*)$ then has $(d\alpha)_{ijk} = \alpha_{jk}\alpha_{ik}^{-1}\alpha_{ij}$ for $p = 1$ and $(d\beta)_{ij} = \beta_j\beta_i^{-1}$ for $p = 0$. Note that an automorphism of the $\mathcal{O}_W$-module $\mathcal{O}_W$ is multiplication by a unit of $\Gamma(W, \mathcal{O}_X)$, i.e. $\mathrm{Aut}(\mathcal{O}_W) = \Gamma(W, \mathcal{O}_X^*)$.

The map. Let $\mathcal{L}$ be invertible, $\mathfrak{U} = (U_i)$ a covering with isomorphisms $\varphi_i : \mathcal{O}_{U_i} \to \mathcal{L}|_{U_i}$ (a "trivialization"). Define $g_{ij} = \varphi_i^{-1}\varphi_j \in \Gamma(U_{ij}, \mathcal{O}_X^*)$. Then $g_{ij}g_{jk} = \varphi_i^{-1}\varphi_j\varphi_j^{-1}\varphi_k = g_{ik}$ on U_{ijk}, so, putting $\alpha_{ij} = g_{ij}$ for $i < j$, we have $(d\alpha)_{ijk} = g_{jk}g_{ik}^{-1}g_{ij} = 1$: α is a cocycle (and $g_{ji} = g_{ij}^{-1}$, $g_{ii} = 1$, consistent with Remark 4.0.1). If we change the trivializations to $\varphi_i' = \varphi_i u_i$, $u_i \in \Gamma(U_i, \mathcal{O}_X^*)$, then $g_{ij}' = u_i^{-1}g_{ij}u_j = g_{ij} \cdot (du)_{ij}$, so the class $[\alpha] \in \check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$ is independent of the choice of the φ_i for fixed $\mathfrak{U}$. If $\mathfrak{V}$ refines $\mathfrak{U}$ via λ, the trivializations $\varphi_{\lambda(j)}|_{V_j}$ of $\mathcal{L}$ on $\mathfrak{V}$ give the cocycle $\lambda^{\#}\alpha$; any trivializing covering has a common refinement with any other (take $(U_i \cap U_{i'}')$, trivialized by restriction), and the class in $\check{H}^1(\mathfrak{V}, \mathcal{O}_X^*)$ is independent of trivialization. Thus $\mathcal{L}$ determines a well-defined element $c(\mathcal{L})$ of $\varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$, which equals $H^1(X, \mathcal{O}_X^*)$ by Ex. 4.4(c) ($\mathcal{O}_X^*$ is a sheaf of abelian groups on the topological space X). If $\psi : \mathcal{L} \to \mathcal{L}'$ is an isomorphism, then $\psi\varphi_i$ are trivializations of $\mathcal{L}'$ with the same g_{ij}; so c is defined on $\operatorname{Pic} X$.

Homomorphism. If $\mathcal{L}, \mathcal{L}'$ are trivialized by φ_i, φ_i' on the same covering (use a common refinement), then $\varphi_i \otimes \varphi_i' : \mathcal{O}_{U_i} = \mathcal{O} \otimes \mathcal{O} \to (\mathcal{L} \otimes \mathcal{L}')|_{U_i}$ has transition functions $g_{ij}g_{ij}'$. So $c(\mathcal{L} \otimes \mathcal{L}') = c(\mathcal{L})c(\mathcal{L}')$.

Injective. Suppose $c(\mathcal{L}) = 0$. Then for some covering $\mathfrak{U}$ with trivializations φ_i the cocycle $\alpha = (g_{ij})$ is a coboundary: there are $u_i \in \Gamma(U_i, \mathcal{O}_X^*)$ with $g_{ij} = u_j u_i^{-1}$ (after replacing the covering by a refinement, using the definition of the direct limit). Put $\varphi_i' = \varphi_i \circ (u_i^{-1})$. Then $\varphi_i'^{-1}\varphi_j' = u_i g_{ij} u_j^{-1} = 1$, i.e. $\varphi_i' = \varphi_j'$ on U_{ij}. By the sheaf property of $\mathcal{H}om(\mathcal{O}_X, \mathcal{L})$ the φ_i' glue to a global isomorphism $\mathcal{O}_X \to \mathcal{L}$. So $\mathcal{L} \cong \mathcal{O}_X$.

Surjective. Let $\xi \in H^1(X, \mathcal{O}_X^*)$. By Ex. 4.4(c) it comes from some $\check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$, i.e. is represented by a cocycle $\alpha = (\alpha_{ij})_{i<j}$, $\alpha_{ij} \in \Gamma(U_{ij}, \mathcal{O}_X^*)$, with $\alpha_{jk}\alpha_{ik}^{-1}\alpha_{ij} = 1$ for $i < j < k$. Extend alternatingly ($\alpha_{ii} = 1$, $\alpha_{ji} = \alpha_{ij}^{-1}$); then $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$ for all i, j, k (Remark 4.0.1). Glue the sheaves $\mathcal{O}_{U_i}$ along the isomorphisms $\psi_{ij} : \mathcal{O}_{U_j}|_{U_{ij}} \to \mathcal{O}_{U_i}|_{U_{ij}}$, "multiplication by α_{ij}": $\psi_{ii} = \mathrm{id}$ and $\psi_{ik} = \psi_{ij}\psi_{jk}$ on U_{ijk}. By the gluing of sheaves (II, Ex. 1.22) there is a sheaf of $\mathcal{O}_X$-modules $\mathcal{L}$ with isomorphisms $\varphi_i : \mathcal{O}_{U_i} \to \mathcal{L}|_{U_i}$ such that $\varphi_i^{-1}\varphi_j = \psi_{ij}$. Then $\mathcal{L}$ is invertible with $c(\mathcal{L}) = [\alpha] = \xi$.

Thus $c : \operatorname{Pic} X \to H^1(X, \mathcal{O}_X^*)$ is an isomorphism of groups. □

Exercise 4.6

> **Problem.** Let $(X, \mathcal{O}_X)$ be a ringed space, let $\mathcal{I}$ be a sheaf of ideals with $\mathcal{I}^2 = 0$, and let X_0 be the ringed space $(X, \mathcal{O}_X/\mathcal{I})$. Show that there is an exact sequence of sheaves of abelian groups on X,
>
> $$0 \to \mathcal{I} \to \mathcal{O}_X^* \to \mathcal{O}_{X_0}^* \to 0,$$
>
> where $\mathcal{O}_X^*$ (respectively $\mathcal{O}_{X_0}^*$) denotes the sheaf of (multiplicative) groups of units in the sheaf of rings $\mathcal{O}_X$ (respectively $\mathcal{O}_{X_0}$); the map $\mathcal{I} \to \mathcal{O}_X^*$ is defined by $a \mapsto 1 + a$, and $\mathcal{I}$ has its usual (additive) group structure. Conclude there is an exact sequence of abelian groups
>
> $$\cdots \to H^1(X, \mathcal{I}) \to \operatorname{Pic} X \to \operatorname{Pic} X_0 \to H^2(X, \mathcal{I}) \to \ldots.$$

Solution. *The map $\mathcal{I} \to \mathcal{O}_X^*$.* For a local section a of $\mathcal{I}$, $(1 + a)(1 - a) = 1 - a^2 = 1$, so $1 + a$ is a unit. And $(1 + a)(1 + b) = 1 + a + b + ab = 1 + a + b$ since $ab \in \mathcal{I}^2 = 0$. So $a \mapsto 1 + a$ is a homomorphism of sheaves of groups $\mathcal{I} \to \mathcal{O}_X^*$, clearly injective.

The map $\mathcal{O}_X^* \to \mathcal{O}_{X_0}^*$ is induced by the quotient map $\pi : \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}$, a ring homomorphism, and it is a group homomorphism. Its kernel consists of the units u with $\pi(u) = 1$, i.e. $u - 1 \in \mathcal{I}$, i.e. $u = 1 + a$ with $a \in \mathcal{I}$; this is the image of $\mathcal{I}$. So the sequence is exact at $\mathcal{I}$ and at $\mathcal{O}_X^*$.

Surjectivity of $\mathcal{O}_X^* \to \mathcal{O}_{X_0}^*$ (as sheaves): it suffices to check on stalks. The stalk of $\mathcal{O}_{X_0}^*$ at x is the group of units of $\mathcal{O}_{X,x}/\mathcal{I}_x$ (a germ of a section is a unit iff it is invertible in the stalk, since an inverse of the germ is defined near x and is an inverse on a neighbourhood). Let $\bar{u}$ be such a unit with inverse $\bar{v}$, and lift to $u, v \in \mathcal{O}_{X,x}$. Then $uv = 1 + a$ with $a \in \mathcal{I}_x$, and $1 + a$ is a unit with inverse $1 - a$. So $u \cdot v(1 - a) = 1$, and u is a unit of $\mathcal{O}_{X,x}$ mapping to $\bar{u}$.

The cohomology sequence. The long exact cohomology sequence of this short exact sequence of sheaves of abelian groups on X reads

$$\cdots \to H^1(X, \mathcal{I}) \to H^1(X, \mathcal{O}_X^*) \to H^1(X, \mathcal{O}_{X_0}^*) \to H^2(X, \mathcal{I}) \to \dots$$

(X_0 has the same underlying space.) By Ex. 4.5 applied to the ringed spaces X and X_0, $H^1(X, \mathcal{O}_X^*) \cong \operatorname{Pic} X$ and $H^1(X, \mathcal{O}_{X_0}^*) \cong \operatorname{Pic} X_0$, which gives the desired sequence (it can be continued to the left by H^0-terms). □

Exercise 4.7

Problem. Let X be a subscheme of $\mathbf{P}_k^2$ defined by a single homogeneous equation $f(x_0, x_1, x_2) = 0$ of degree d. (Do not assume f is irreducible.) Assume that the point $(1, 0, 0)$ is not on X. Then show that X can be covered by the two open affine subsets $U = X \cap \{x_1 \neq 0\}$ and $V = X \cap \{x_2 \neq 0\}$. Now calculate the Cech complex

$$\Gamma(U, \mathcal{O}_X) \oplus \Gamma(V, \mathcal{O}_X) \to \Gamma(U \cap V, \mathcal{O}_X)$$

explicitly, and thus show that

$$\dim_k H^0(X, \mathcal{O}_X) = 1, \qquad \dim_k H^1(X, \mathcal{O}_X) = \tfrac{1}{2}(d-1)(d-2).$$

Solution. Here k is a field (arbitrary), $S = k[x_0, x_1, x_2]$, $T = S/(f)$, $X = \operatorname{Proj} T$, and $d \geq 1$.

Covering. The closed subset $\{x_1 = x_2 = 0\}$ of $\mathbf{P}_k^2$ is the single rational point $(1, 0, 0)$. It is not on X, so $X \cap \{x_1 = x_2 = 0\} = \emptyset$ and $X = U \cup V$, where $U = D_+(x_1) \cap X = \operatorname{Spec}(T_{x_1})_0$ and $V = D_+(x_2) \cap X = \operatorname{Spec}(T_{x_2})_0$ are affine; $U \cap V = \operatorname{Spec}(T_{x_1 x_2})_0$. Also X is noetherian and separated (closed in $\mathbf{P}^2$), so Thm. 4.5 gives $H^i(X, \mathcal{O}_X) = \check{H}^i(\mathfrak{U}, \mathcal{O}_X)$ for $\mathfrak{U} = (U, V)$.

Structure of T. Since $f(1, 0, 0) \neq 0$, the coefficient of x_0^d in f is a nonzero constant; scaling, f is monic of degree d in x_0 over $k[x_1, x_2]$. Hence $T = k[x_1, x_2][x_0]/(f)$ is a free $k[x_1, x_2]$-module with basis $1, x_0, \dots, x_0^{d-1}$, and these elements are homogeneous. Localizing, $T_{x_1 x_2}$ is free over $k[x_1^{\pm 1}, x_2^{\pm 1}]$, T_{x_1} free over $k[x_1^{\pm 1}, x_2]$, T_{x_2} free over $k[x_1, x_2^{\pm 1}]$, all with the same basis x_0^m ($0 \leq m \leq d-1$); the maps $T_{x_1} \to T_{x_1 x_2} \leftarrow T_{x_2}$ are injective, the inclusions of these free modules coefficientwise. An element of degree 0 is $\sum_m x_0^m r_m$ with r_m homogeneous of degree $-m$ in the Laurent ring. A k-basis of the degree-0 parts is therefore:

- $\Gamma(U, \mathcal{O}_X) = (T_{x_1})_0$: the elements $x_0^m x_1^p x_2^q$ with $0 \leq m \leq d-1$, $p \in \mathbf{Z}$, $q \geq 0$, $p + q = -m$;
- $\Gamma(V, \mathcal{O}_X) = (T_{x_2})_0$: the elements $x_0^m x_1^p x_2^q$ with $p \geq 0$, $q \in \mathbf{Z}$, $p + q = -m$;

- $\Gamma(U \cap V, \mathcal{O}_X) = (T_{x_1 x_2})_0$: the elements $x_0^m x_1^p x_2^q$ with $p, q \in \mathbf{Z}$, $p + q = -m$.

The Cech differential is $(a, b) \mapsto b - a$, so its image is spanned by the basis elements with $q \geq 0$ or $p \geq 0$, and its kernel (the elements $a = b$) by the basis elements common to U and V, i.e. $p, q \geq 0$ and $p + q = -m \leq 0$, which forces $m = p = q = 0$.

Conclusion. $H^0(X, \mathcal{O}_X) = \ker d = k \cdot 1$, of dimension 1. And $H^1(X, \mathcal{O}_X) = \operatorname{coker} d$ has basis $x_0^m x_1^{-s} x_2^{-t}$ with $0 \leq m \leq d-1$, $s, t \geq 1$, $s + t = m$. For given m there are $m - 1$ pairs (s, t) if $m \geq 2$, and none if $m \leq 1$. So

$$\dim_k H^1(X, \mathcal{O}_X) = \sum_{m=2}^{d-1} (m-1) = \frac{(d-2)(d-1)}{2},$$

(which is 0 for $d = 1, 2$). □

Exercise 4.8

Problem. *Cohomological Dimension* (Hartshorne [3]). Let X be a noetherian separated scheme. We define the *cohomological dimension* of X, denoted $\operatorname{cd}(X)$, to be the least integer n such that $H^i(X, \mathcal{F}) = 0$ for all quasi-coherent sheaves $\mathcal{F}$ and all $i > n$. Thus for example, Serre's theorem (3.7) says that $\operatorname{cd}(X) = 0$ if and only if X is affine. Grothendieck's theorem (2.7) implies that $\operatorname{cd}(X) \leq \dim X$.

(a) In the definition of $\operatorname{cd}(X)$, show that it is sufficient to consider only coherent sheaves $\mathcal{F}$. Use (II, Ex. 5.15) and (2.9).

(b) If X is quasi-projective over a field k, then it is even sufficient to consider only locally free coherent sheaves $\mathcal{F}$ on X. Use (II, 5.18).

(c) Suppose X has a covering by $r + 1$ open affine subsets. Use Cech cohomology to show that $\operatorname{cd}(X) \leq r$.

(d) If X is a quasi-projective scheme of dimension r over a field k, then X can be covered by $r + 1$ open affine subsets. Conclude (independently of (2.7)) that $\operatorname{cd}(X) \leq \dim X$. (Starred in the book.)

(e) Let Y be a set-theoretic complete intersection (I, Ex. 2.17) of codimension r in $X = \mathbf{P}_k^n$. Show that $\operatorname{cd}(X - Y) \leq r - 1$.

Solution. (a) Suppose $n \geq 0$ is such that $H^i(X, \mathcal{F}) = 0$ for all coherent $\mathcal{F}$ and all $i > n$; we show the same for quasi-coherent $\mathcal{F}$. By II, Ex. 5.15(e), $\mathcal{F}$ is the union of its coherent subsheaves $\mathcal{F}_\alpha$; this family is directed under inclusion (the sum of two coherent subsheaves is a coherent subsheaf), so $\mathcal{F} = \varinjlim \mathcal{F}_\alpha$. As X is a noetherian topological space, cohomology commutes with direct limits (2.9), so $H^i(X, \mathcal{F}) = \varinjlim H^i(X, \mathcal{F}_\alpha) = 0$ for $i > n$. Hence $\operatorname{cd}(X)$ is the least n with vanishing for coherent sheaves.

(b) Let X be quasi-projective over k, with $\mathcal{O}_X(1)$ very ample; $N = \dim X < \infty$. Suppose $H^i(X, \mathcal{E}) = 0$ for all locally free coherent $\mathcal{E}$ and all $i > n$. We prove by descending induction on i that $H^j(X, \mathcal{F}) = 0$ for all coherent $\mathcal{F}$ and all $j \geq i$, for each $i \geq n + 1$. For $i = N + 1$ this is Grothendieck vanishing (2.7). Suppose it holds for $i + 1$, where $n + 1 \leq i \leq N$. Given coherent $\mathcal{F}$, by II, 5.18 there is a surjection $\mathcal{E} \to \mathcal{F}$ with $\mathcal{E} = \bigoplus \mathcal{O}_X(-m_l)$ (finite sum), locally free; let $\mathcal{G}$ be its kernel (coherent). The exact sequence $H^i(X, \mathcal{E}) \to H^i(X, \mathcal{F}) \to H^{i+1}(X, \mathcal{G})$ has outer

terms 0 (the first since $i > n$, the second by induction), so $H^i(X, \mathcal{F}) = 0$. For $i = n + 1$ we get the claim; with (a), $\operatorname{cd}(X) \leq n$.

(c) Let $X = U_0 \cup \cdots \cup U_r$ with U_l open affine. Since X is noetherian and separated, by Thm. 4.5 for any quasi-coherent $\mathcal{F}$, $H^p(X, \mathcal{F}) = \check{H}^p(\mathfrak{U}, \mathcal{F})$ with $\mathfrak{U} = (U_0, \ldots, U_r)$. The Cech complex $C^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0 < \cdots < i_p} \mathcal{F}(U_{i_0 \ldots i_p})$ is zero for $p > r$ (there are only $r+1$ indices). So $H^p(X, \mathcal{F}) = 0$ for $p > r$, i.e. $\operatorname{cd}(X) \leq r$.

(d) Let X be an open subscheme of a closed subscheme P of $\mathbf{P}^n_k$, of dimension r, $X \neq \emptyset$ (this is what quasi-projective means); let $\bar{X}$ be the closure of X in $\mathbf{P}^n_k$ (a closed subset; the construction below only uses underlying sets until the last paragraph), so X is open in $\bar{X}$, $\dim \bar{X} = r$ (each irreducible component of $\bar{X}$ meets the dense open subset X in a dense open subset of equal dimension), and $Z = \bar{X} - X$ is closed. Let $S = k[x_0, \ldots, x_n]$ and let $J \subseteq S$ be the ideal generated by all homogeneous f of positive degree with $Z \subseteq V_+(f)$. Then $V_+(J) = Z$: "$\supseteq$" is clear, and if $Z = V_+(\mathfrak{a})$, $\mathfrak{a}$ homogeneous, then $\mathfrak{a} \cap S_+ \subseteq J$ and $V_+(\mathfrak{a} \cap S_+) = Z$. If $W \subseteq \mathbf{P}^n$ is an irreducible closed subset with homogeneous prime $\mathfrak{p}_W$ and $W \not\subseteq Z$, then $J \not\subseteq \mathfrak{p}_W$ (else $W = V_+(\mathfrak{p}_W) \subseteq V_+(J) = Z$).

Graded prime avoidance. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$ be homogeneous primes with $J \not\subseteq \mathfrak{p}_j$ for all j. Then there is a homogeneous $y \in J$ (of positive degree, as $J \subseteq S_+$) with $y \notin \mathfrak{p}_j$ for all j. Proof: discarding primes contained in others we may assume there are no inclusions among the $\mathfrak{p}_j$. For $s = 1$ pick a homogeneous generator of J outside $\mathfrak{p}_1$. For $s \geq 2$ and each j the homogeneous ideal $J \prod_{i \neq j} \mathfrak{p}_i$ is not contained in $\mathfrak{p}_j$ (as $\mathfrak{p}_j$ is prime, $J \not\subseteq \mathfrak{p}_j$ and $\mathfrak{p}_i \not\subseteq \mathfrak{p}_j$), so it contains a homogeneous element $y_j \notin \mathfrak{p}_j$; thus $y_j \in J$, $y_j \in \mathfrak{p}_i$ $(i \neq j)$, $\deg y_j = e_j > 0$. With e a common multiple of the e_j, $y = \sum_j y_j^{e/e_j} \in J$ is homogeneous and $y \equiv y_i^{e/e_i} \not\equiv 0 \pmod{\mathfrak{p}_i}$ for each i.

Construction. We choose homogeneous $f_0, \ldots, f_r \in J$ of positive degree inductively. Put $Y_0 = \bar{X}$ and $Y_{i+1} = Y_i \cap V_+(f_i)$. Given Y_i, let $W_1, \ldots, W_s$ be the irreducible components of Y_i not contained in Z, and choose by graded prime avoidance a homogeneous $f_i \in J$ of positive degree not in any $\mathfrak{p}_{W_l}$ (if $s = 0$, take any $f_i \in J$ of positive degree). Let d_i be the maximal dimension of an irreducible component of Y_i not contained in Z ($d_i = -\infty$ if none); $d_0 = r$. We claim $d_{i+1} \leq d_i - 1$. Let V' be an irreducible component of Y_{i+1} not contained in Z. It lies in some component W of Y_i, and $W \not\subseteq Z$; V' is an irreducible component of $W \cap V_+(f_i)$ (it is maximal irreducible in the bigger set $Y_{i+1} \supseteq W \cap V_+(f_i) \supseteq V'$). Since $f_i \notin \mathfrak{p}_W$, f_i does not vanish identically on the integral scheme W, and by Krull's Hauptidealsatz (and the dimension formula for finitely generated k-algebras, applied on an affine chart $D_+(x_l) \cap W$ meeting V') $\dim V' = \dim W - 1 \leq d_i - 1$. So $d_{r+1} \leq r - (r+1) < 0$: every irreducible component of $Y_{r+1} = \bar{X} \cap V_+(f_0, \ldots, f_r)$ lies in Z, i.e. $Y_{r+1} \subseteq Z$.

Conclusion. Put $U_i = X \cap D_+(f_i)$, an open subscheme of X. As f_i vanishes on Z, $\bar{X} \cap D_+(f_i) \subseteq \bar{X} - Z = X$, so as a set $U_i = \bar{X} \cap D_+(f_i)$. Now $P \cap D_+(f_i) = \operatorname{Spec} C$ is a closed subscheme of the affine scheme $D_+(f_i)$, and U_i is a subset of it which is open (it is $X \cap D_+(f_i)$) and closed (it is $\bar{X} \cap D_+(f_i)$). A clopen subset of $\operatorname{Spec} C$ is $D(e)$ for an idempotent e, so $U_i = \operatorname{Spec} C_e$ is affine. Thus U_i is an open affine subscheme of X. Finally $X \cap V_+(f_0, \ldots, f_r) \subseteq Y_{r+1} \cap X \subseteq Z \cap X = \emptyset$, so $X = \bigcup_{i=0}^r U_i$. By (c), $\operatorname{cd}(X) \leq r = \dim X$ (this uses only Cech cohomology, Thm. 4.5, not (2.7)).

(e) "Y is a set-theoretic complete intersection of codimension r in $\mathbf{P}^n_k$" means $Y = H_1 \cap \cdots \cap H_r$ with $H_l = V_+(f_l)$ hypersurfaces, f_l homogeneous of positive degree. Then $X - Y = \mathbf{P}^n_k - \bigcap H_l =$

$\bigcup_{l=1}^{r} D_+(f_l)$, a union of r open affine subsets. $X - Y$ is noetherian and separated. By (c) (with $r-1$ in place of r), $\operatorname{cd}(X-Y) \le r-1$. □

Exercise 4.9

> **Problem.** Let $X = \operatorname{Spec} k[x_1, x_2, x_3, x_4]$ be affine four-space over a field k. Let Y_1 be the plane $x_1 = x_2 = 0$ and let Y_2 be the plane $x_3 = x_4 = 0$. Show that $Y = Y_1 \cup Y_2$ is not a set-theoretic complete intersection in X. Therefore the projective closure $\bar{Y}$ in $\mathbf{P}^4_k$ is also not a set-theoretic complete intersection. [Hints: Use an affine analogue of (Ex. 4.8e). Then show that $H^2(X - Y, \mathcal{O}_X) \neq 0$, by using (Ex. 2.3) and (Ex. 2.4). If $P = Y_1 \cap Y_2$, imitate (Ex. 4.3) to show $H^3(X - P, \mathcal{O}_X) \neq 0$.]

Solution. Put $A = k[x_1, \dots, x_4]$. All cohomology below is of $\mathcal{O}_X$ (or its restriction).

Step 1: affine analogue of 4.8(e). Suppose $Y = V(h_1, h_2)$ for two polynomials $h_1, h_2 \in A$ (this is what "set-theoretic complete intersection" means here, Y having codimension 2; fewer equations are covered by allowing some $h_l = 0$). Then $X - Y = D(h_1) \cup D(h_2)$ is a union of two open affines, so by 4.8(c) (the scheme $X - Y$ is noetherian and separated) $H^2(X - Y, \mathcal{O}_X) = 0$. So it suffices to show $H^2(X - Y, \mathcal{O}_X) \neq 0$.

Step 2: cohomology with supports. By Ex. 2.3(e), for a closed subset $W \subseteq X$ we have an exact sequence $H^{i-1}(X, \mathcal{O}) \to H^{i-1}(X - W, \mathcal{O}) \to H^i_W(X, \mathcal{O}) \to H^i(X, \mathcal{O})$. As X is affine, $H^i(X, \mathcal{O}) = 0$ for $i > 0$ (Thm. 3.5), so

$$H^{i-1}(X - W, \mathcal{O}) \cong H^i_W(X, \mathcal{O}) \qquad (i \ge 2). \tag{1}$$

(a) $W = Y_1$: $X - Y_1 = D(x_1) \cup D(x_2)$, covered by two open affines; its Cech complex (for this cover, Thm. 4.5) has no terms in degrees ≥ 2, so $H^q(X - Y_1, \mathcal{O}) = 0$ for $q \ge 2$, and by (1) $H^i_{Y_1}(X, \mathcal{O}) = 0$ for $i \ge 3$. Likewise $H^i_{Y_2}(X, \mathcal{O}) = 0$ for $i \ge 3$ (cover $X - Y_2 = D(x_3) \cup D(x_4)$).

(b) $W = P = Y_1 \cap Y_2 = V(x_1, \dots, x_4)$, the origin. The Mayer–Vietoris sequence with supports (Ex. 2.4), for the closed sets Y_1, Y_2, contains

$$H^3_{Y_1}(X, \mathcal{O}) \oplus H^3_{Y_2}(X, \mathcal{O}) \to H^3_Y(X, \mathcal{O}) \to H^4_P(X, \mathcal{O}) \to H^4_{Y_1}(X, \mathcal{O}) \oplus H^4_{Y_2}(X, \mathcal{O}).$$

By (a) both outer groups are 0, so $H^3_Y(X, \mathcal{O}) \cong H^4_P(X, \mathcal{O})$. Using (1) three times,

$$H^2(X - Y, \mathcal{O}) \cong H^3_Y(X, \mathcal{O}) \cong H^4_P(X, \mathcal{O}) \cong H^3(X - P, \mathcal{O}).$$

Step 3: $H^3(X - P, \mathcal{O}) \neq 0$. The open sets $U_l = D(x_l)$, $l = 1, \dots, 4$, are affine, cover $X - P = X - V(x_1, \dots, x_4)$, and all their intersections are affine; $X - P$ is noetherian separated, so $H^3(X - P, \mathcal{O}) = \check{H}^3(\mathfrak{U}, \mathcal{O})$ (Thm. 4.5). The Cech complex has $C^3 = A_{x_1x_2x_3x_4}$ (one term) and $C^2 = \bigoplus_{l=1}^4 A_{x_1 \cdots \widehat{x_l} \cdots x_4}$, and, writing $\alpha_{\hat{l}}$ for the component omitting the index l, $d(\alpha) = \alpha_{\hat{1}} - \alpha_{\hat{2}} + \alpha_{\hat{3}} - \alpha_{\hat{4}}$ (restricted to U_{1234}). $A_{x_1x_2x_3x_4}$ has k-basis the Laurent monomials $x^a = x_1^{a_1} \cdots x_4^{a_4}$, $a \in \mathbf{Z}^4$, and the summand $A_{x_1 \cdots \widehat{x_l} \cdots x_4}$ is spanned by those with $a_l \ge 0$. Hence the image of d is spanned by the monomials with some $a_l \ge 0$, and

$$H^3(X - P, \mathcal{O}) = \operatorname{coker} d \cong \bigoplus_{a_1, \dots, a_4 < 0} k\, x^a \neq 0$$

(for example $(x_1x_2x_3x_4)^{-1}$ gives a nonzero class). Hence $H^2(X-Y,\mathcal{O}) \neq 0$ by Step 2, and by Step 1, Y is not a set-theoretic complete intersection in X.

The projective closure. Identify $X = D_+(x_0) \subseteq \mathbf{P}^4_k$. The closure $\bar{Y}$ of Y satisfies $\bar{Y} \cap X = Y$ (as Y is closed in X); $\bar{Y}$ has dimension 2, so codimension 2 in $\mathbf{P}^4_k$. If $\bar{Y} = V_+(g_1, g_2)$ for homogeneous g_1, g_2, then $Y = \bar{Y} \cap D_+(x_0) = V(g_1(1, x_1, \ldots, x_4), g_2(1, x_1, \ldots, x_4))$ would be cut out in X by two polynomials (possibly zero or constant ones), which is impossible by what we proved. So $\bar{Y}$ is not a set-theoretic complete intersection in $\mathbf{P}^4_k$. □

Exercise 4.10

> **Problem.** Let X be a nonsingular variety over an algebraically closed field k, and let $\mathcal{F}$ be a coherent sheaf on X. Show that there is a one-to-one correspondence between the set of infinitesimal extensions of X by $\mathcal{F}$ (II, Ex. 8.7), up to isomorphism, and the group $H^1(X, \mathcal{F} \otimes \mathcal{T})$, where $\mathcal{T}$ is the tangent sheaf of X (II, Section 8). [Hint: Use (II, Ex. 8.6) and (4.5).] (Starred in the book.)

Solution. *Definitions and conventions.* An infinitesimal extension of X by $\mathcal{F}$ (II, Ex. 8.7) is a scheme X' over k together with a sheaf of ideals $\mathcal{J} \subseteq \mathcal{O}_{X'}$ with $\mathcal{J}^2 = 0$, an identification of k-schemes $(X', \mathcal{O}_{X'}/\mathcal{J}) = (X, \mathcal{O}_X)$ (so X' and X have the same underlying space and $\mathcal{O}_{X'} \to \mathcal{O}_X$ is the quotient map), and an isomorphism of $\mathcal{O}_X$-modules $\mathcal{J} \cong \mathcal{F}$ ($\mathcal{J}$ is an $\mathcal{O}_{X'}/\mathcal{J}$-module because $\mathcal{J}^2 = 0$). I use the (standard) notion of isomorphism: X', X'' are isomorphic extensions if there is an isomorphism of k-schemes $X' \to X''$ (the identity on the space) which is compatible with the identifications $\mathcal{O}_{X'}/\mathcal{J}' = \mathcal{O}_X = \mathcal{O}_{X''}/\mathcal{J}''$ and with $\mathcal{J}' \cong \mathcal{F} \cong \mathcal{J}''$. The trivial extension is $\mathcal{O}_X \oplus \mathcal{F}$ with $(a, f)(a', f') = (aa', af' + a'f)$.

Since X is a nonsingular variety, $\Omega = \Omega_{X/k}$ is locally free of finite rank (II, 8.15), so $\mathcal{T} = \mathcal{H}om(\Omega, \mathcal{O}_X)$ is locally free and $\mathcal{F} \otimes \mathcal{T} \cong \mathcal{H}om(\Omega, \mathcal{F}) =: \mathcal{H}$ (canonical map, iso for Ω locally free of finite rank). Also $\mathcal{H}(W) = \mathrm{Der}_k(\mathcal{O}_W, \mathcal{F}|_W)$ for open W (universal property of Ω, II, Section 8, applied on affine opens: $\mathrm{Hom}_B(\Omega_{B/k}, M) = \mathrm{Der}_k(B, M)$). $\mathcal{H}$ is coherent. Also X is separated.

Step 1: local triviality. Let $(X', \mathcal{J})$ be an infinitesimal extension, and $V = \operatorname{Spec} A \subseteq X$ an affine open. Then $V' = (V, \mathcal{O}_{X'}|_V)$ is an open subscheme of X'. It is noetherian (locally $\operatorname{Spec} B'$ with $B'/I = A_\alpha$ noetherian and $I = \mathcal{J}(\cdot)$ a finitely generated module over A_α, so B' is noetherian) and $V'_{\mathrm{red}} = V_{\mathrm{red}}$ is affine; hence V' is affine by Ex. 3.1. Let $B' = \Gamma(V', \mathcal{O}_{X'})$, $I = \Gamma(V, \mathcal{J})$; $\mathcal{J}$ is quasi-coherent on V' (ideal sheaf of the closed subscheme $V \subseteq V'$), so $0 \to I \to B' \to A \to 0$ is exact, $I^2 = 0$, and B' is a k-algebra. Since A is the coordinate ring of a nonsingular affine variety, the infinitesimal lifting property (II, Ex. 8.6) applied to $\mathrm{id} : A \to A$ gives a k-algebra homomorphism $s : A \to B'$ with $\pi s = \mathrm{id}$ ($\pi : B' \to A$). So every affine open V has a k-algebra section s; it extends to a morphism of sheaves of k-algebras $\sigma : \mathcal{O}_V \to \mathcal{O}_{X'}|_V$ with $\pi\sigma = \mathrm{id}$ (for $a \in A$, $s(a)$ restricts to a unit on $D(a)$ since its image in $\mathcal{O}_X(D(a))$ is a unit and $\mathcal{J}$ is nilpotent, so s extends to $A_a \to \mathcal{O}_{X'}(D(a))$ by the universal property of localization). Call such σ a *splitting* over V.

Step 2: the class. Let σ, σ' be two splittings over an open W. Then $\sigma' - \sigma : \mathcal{O}_W \to \mathcal{J}|_W$ (values in $\mathcal{J}$ as both lift the identity), and it is a k-derivation: for local sections a, b,

$$\sigma'(ab) - \sigma(ab) = \sigma'(a)\big(\sigma'(b) - \sigma(b)\big) + \big(\sigma'(a) - \sigma(a)\big)\sigma(b) = a\,(\sigma' - \sigma)(b) + b\,(\sigma' - \sigma)(a),$$

because $\mathcal{J}$ is an $\mathcal{O}_{X'}/\mathcal{J} = \mathcal{O}_X$-module (so $\sigma'(a)\cdot j = a\,j$ for $j \in \mathcal{J}$). Thus $\sigma' - \sigma \in \Gamma(W, \mathcal{H})$, $\mathcal{H} \cong \mathcal{H}om(\Omega, \mathcal{J}) = \mathcal{H}om(\Omega, \mathcal{F})$ via $\mathcal{J} \cong \mathcal{F}$. Let $\mathfrak{U} = (U_i)$ be a cover of X by open affines, with splittings σ_i over U_i (Step 1), and put $\theta_{ij} = \sigma_j - \sigma_i \in \Gamma(U_{ij}, \mathcal{H})$. Then $\theta_{ij} + \theta_{jk} = \theta_{ik}$, i.e. $\theta = (\theta_{ij})_{i<j}$ is a Cech 1-cocycle ($(d\theta)_{ijk} = \theta_{jk} - \theta_{ik} + \theta_{ij} = 0$). Changing σ_i to $\sigma_i + \eta_i$, $\eta_i \in \Gamma(U_i, \mathcal{H})$ (every splitting over U_i is of this form), changes θ to $\theta + d\eta$ (as $(d\eta)_{ij} = \eta_j - \eta_i$). So we get a class $c_{\mathfrak{U}}(X') \in \check{H}^1(\mathfrak{U}, \mathcal{H})$. Since X is noetherian and separated and $\mathcal{H}$ is quasi-coherent, $\check{H}^1(\mathfrak{U}, \mathcal{H}) \cong H^1(X, \mathcal{H})$ (Thm. 4.5). If $\mathfrak{V}$ is another open affine cover, then $\mathfrak{W} = (U_i \cap V_j)$ is an open affine cover refining both; restricting splittings gives the restricted cocycles, i.e. $c_{\mathfrak{U}}$ maps to $c_{\mathfrak{W}}$ under the refinement map (and likewise for $\mathfrak{V}$), and by Ex. 4.4(b) the images in $H^1(X, \mathcal{H})$ coincide. Hence we get a well-defined class $\Psi(X') \in H^1(X, \mathcal{H})$. If $\phi : X' \to X''$ is an isomorphism of extensions, then $\phi^\flat \circ \sigma_i''$ (with $\phi^\flat : \mathcal{O}_{X''} \to \mathcal{O}_{X'}$) are splittings of X' whose differences are the same θ_{ij} (as $\phi^\flat$ is the identity on $\mathcal{J} \cong \mathcal{F}$), so $\Psi(X') = \Psi(X'')$. We get a well-defined map $\Psi : \{\text{extensions}\}/\cong \to H^1(X, \mathcal{F} \otimes \mathcal{T})$.

Step 3: Ψ is injective. Let X', X'' be extensions with $\Psi(X') = \Psi(X'')$. Choose an affine cover $\mathfrak{U}$ and splittings σ_i of X', τ_i of X'' over U_i (Step 1); with $\theta'_{ij} = \sigma_j - \sigma_i$, $\theta''_{ij} = \tau_j - \tau_i$, the classes of θ', θ'' coincide in $\check{H}^1(\mathfrak{U}, \mathcal{H}) = H^1(X, \mathcal{H})$ (Thm. 4.5 for injectivity), so $\theta' - \theta'' = d\eta$ and replacing τ_i by $\tau_i + \eta_i$ ($\eta = (\eta_i)$) we may assume $\theta'_{ij} = \theta''_{ij}$. Over U_i, $\mathcal{O}_{X'}|_{U_i} = \sigma_i(\mathcal{O}_{U_i}) \oplus \mathcal{J}'$ (every $u = \sigma_i(\pi u) + (u - \sigma_i \pi u)$), with product $(\sigma_i(a) + x)(\sigma_i(b) + y) = \sigma_i(ab) + ay + bx$, and similarly for X'' with τ_i. So $\Phi_i : \sigma_i(a) + x \mapsto \tau_i(a) + x$ (using $\mathcal{J}' \cong \mathcal{F} \cong \mathcal{J}''$) is an isomorphism of sheaves of k-algebras $\mathcal{O}_{X'}|_{U_i} \to \mathcal{O}_{X''}|_{U_i}$, the identity on $\mathcal{J}$ and on the quotients $\mathcal{O}_X$. On U_{ij}: $\Phi_i(\sigma_j(a) + x) = \Phi_i(\sigma_i(a) + \theta_{ij}(a) + x) = \tau_i(a) + \theta_{ij}(a) + x = \tau_j(a) + x = \Phi_j(\sigma_j(a) + x)$. So the Φ_i glue to an isomorphism of sheaves of rings $\mathcal{O}_{X'} \to \mathcal{O}_{X''}$, i.e. (same underlying space) an isomorphism of k-schemes $X'' \to X'$ (and its inverse $X' \to X''$) compatible with all the identifications: $X' \cong X''$ as extensions.

Step 4: Ψ is surjective. Let $\xi \in H^1(X, \mathcal{H})$. Take any open affine cover $\mathfrak{U}$; by Thm. 4.5 ξ is represented by a cocycle $\theta = (\theta_{ij})_{i<j}$, $\theta_{ij} \in \Gamma(U_{ij}, \mathcal{H}) = \mathrm{Der}_k(\mathcal{O}_{U_{ij}}, \mathcal{F})$; extend alternatingly ($\theta_{ii} = 0$, $\theta_{ji} = -\theta_{ij}$), so that $\theta_{ij} + \theta_{jk} = \theta_{ik}$ for all i, j, k. On U_i let $\mathcal{A}_i = \mathcal{O}_{U_i} \oplus \mathcal{F}|_{U_i}$ be the trivial extension; $(U_i, \mathcal{A}_i)$ is a scheme: over an affine open $V = \operatorname{Spec} A' \subseteq U_i$ it is $\operatorname{Spec}(A' \oplus \mathcal{F}(V))$, because the localization of $A' \oplus M$ at $(a, 0)$ is $A'_a \oplus M_a$, and $D((a, m)) = D((a, 0))$ since $(0, m)$ is nilpotent. On U_{ij} define $g_{ij} : \mathcal{A}_j|_{U_{ij}} \to \mathcal{A}_i|_{U_{ij}}$, $g_{ij}(a, f) = (a, f + \theta_{ij}(a))$. It is a ring isomorphism: $g_{ij}(a, f)\, g_{ij}(b, h) = (ab, a(h + \theta_{ij}b) + b(f + \theta_{ij}a)) = (ab, ah + bf + \theta_{ij}(ab)) = g_{ij}\big((a, f)(b, h)\big)$, using the Leibniz rule; it is k-linear, kills nothing, and has inverse g_{ji}. It is the identity on $0 \oplus \mathcal{F}$ and on the quotient $\mathcal{O}_X$. Moreover $g_{ij} g_{jk} = g_{ik}$ by the cocycle condition and $g_{ii} = \mathrm{id}$. Gluing (II, Ex. 1.22, for sheaves of rings) we get a sheaf of rings $\mathcal{O}_{X'}$ on X, locally isomorphic to the $\mathcal{A}_i$, so $X' = (X, \mathcal{O}_{X'})$ is a scheme; it is a k-scheme ($k \to \mathcal{O}_X \to \mathcal{A}_i$, $c \mapsto (c, 0)$, is preserved by g_{ij} since derivations over k kill k). The ideal $\mathcal{J} = 0 \oplus \mathcal{F}$ glues to a sheaf of ideals with $\mathcal{J}^2 = 0$, $\mathcal{O}_{X'}/\mathcal{J} = \mathcal{O}_X$ and $\mathcal{J} \cong \mathcal{F}$ as $\mathcal{O}_X$-modules. So X' is an infinitesimal extension of X by $\mathcal{F}$. The maps $\sigma_i : a \mapsto (a, 0) \in \mathcal{A}_i$ are splittings over U_i, and on U_{ij}, $\sigma_j(a) = g_{ij}(a, 0) = (a, \theta_{ij}(a))$, so $\sigma_j - \sigma_i = \theta_{ij}$. Hence $\Psi(X') = \xi$.

Thus Ψ is a bijection between isomorphism classes of infinitesimal extensions of X by $\mathcal{F}$ and $H^1(X, \mathcal{F} \otimes \mathcal{T})$ (the trivial extension corresponding to 0). □

Exercise 4.11

Problem. This exercise shows that Cech cohomology will agree with the usual cohomology

whenever the sheaf has no cohomology on any of the open sets. More precisely, let X be a topological space, $\mathcal{F}$ a sheaf of abelian groups, and $\mathfrak{U} = (U_i)$ an open cover. Assume for any finite intersection $V = U_{i_0} \cap \cdots \cap U_{i_p}$ of open sets of the covering, and for any $k > 0$, that $H^k(V, \mathcal{F}|_V) = 0$. Then prove that for all $p \geq 0$, the natural maps

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \to H^p(X, \mathcal{F})$$

of (4.4) are isomorphisms. Show also that one can recover (4.5) as a corollary of this more general result.

Solution. Choose an injective resolution $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ in $\mathfrak{Ab}(X)$. Injective sheaves are flasque (Lemma 2.4) and flasque sheaves restrict to flasque sheaves, so for every open V, $\mathcal{I}^\bullet|_V$ is a flasque resolution of $\mathcal{F}|_V$ and computes $H^q(V, \mathcal{F}|_V)$ (Prop. 2.5). Form the double complex of abelian groups

$$K^{p,q} = C^p(\mathfrak{U}, \mathcal{I}^q) = \prod_{i_0 < \cdots < i_p} \mathcal{I}^q(U_{i_0 \ldots i_p}),$$

with horizontal differential the Cech d (p-direction) and vertical differential induced by $\mathcal{I}^q \to \mathcal{I}^{q+1}$ (q-direction); they commute. Let $\operatorname{Tot} K$ be its total complex (first quadrant, so finite sums in each degree). There are two augmentations:

- $\alpha : C^\bullet(\mathfrak{U}, \mathcal{F}) \to \operatorname{Tot} K$, induced by $\mathcal{F} \to \mathcal{I}^0$, landing in $K^{p,0}$;
- $\beta : \Gamma(X, \mathcal{I}^\bullet) \to \operatorname{Tot} K$, induced by restriction $\Gamma(X, \mathcal{I}^q) \to C^0(\mathfrak{U}, \mathcal{I}^q) = K^{0,q}$.

Both are quasi-isomorphisms. For fixed p, the augmented row $0 \to C^p(\mathfrak{U}, \mathcal{F}) \to K^{p,0} \to K^{p,1} \to \cdots$ is the product over $i_0 < \cdots < i_p$ of the complexes $0 \to \mathcal{F}(V) \to \mathcal{I}^0(V) \to \mathcal{I}^1(V) \to \cdots$ with $V = U_{i_0 \ldots i_p}$, whose cohomology is $H^q(V, \mathcal{F}|_V) = 0$ for $q > 0$ by the hypothesis, and $\mathcal{F}(V)$ in degree 0; products of exact sequences are exact, so all augmented rows are exact. For fixed q, the augmented column $0 \to \Gamma(X, \mathcal{I}^q) \to C^0(\mathfrak{U}, \mathcal{I}^q) \to C^1(\mathfrak{U}, \mathcal{I}^q) \to \cdots$ is exact by Lemma 4.1 and Prop. 4.3 ($\mathcal{I}^q$ is flasque). The standard lemma on first-quadrant double complexes with exact augmented rows (resp. columns) (Weibel, Lemma 2.7.3, "acyclic assembly") says that the augmentation of the rows (resp. columns) to the total complex is a quasi-isomorphism. So $\alpha_* : \check{H}^p(\mathfrak{U}, \mathcal{F}) \to h^p(\operatorname{Tot} K)$ and $\beta_* : H^p(X, \mathcal{F}) = h^p(\Gamma(X, \mathcal{I}^\bullet)) \to h^p(\operatorname{Tot} K)$ are isomorphisms.

Identification with the natural map of (4.4). The natural map $\theta : \check{H}^p(\mathfrak{U}, \mathcal{F}) \to H^p(X, \mathcal{F})$ is $h^p(\Gamma(\phi))$ where $\phi : \mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F}) \to \mathcal{I}^\bullet$ is a morphism of complexes over $\mathrm{id}_\mathcal{F}$. We show $\beta_* \theta = \alpha_*$, whence $\theta = \beta_*^{-1} \alpha_*$ is an isomorphism. Consider the sheafified double complex $\mathcal{K}^{p,q} = \mathcal{C}^p(\mathfrak{U}, \mathcal{I}^q) = \prod_{i_0 < \cdots < i_p} f_*(\mathcal{I}^q|_{U_{i_0 \ldots i_p}})$ (with f the inclusion of $U_{i_0 \ldots i_p}$), so $\Gamma(X, \mathcal{K}^{p,q}) = K^{p,q}$. Each $\mathcal{K}^{p,q}$ is an injective sheaf (f_* of an injective sheaf under an open immersion is injective since it is right adjoint to the exact restriction functor; products of injectives are injective), hence so is each (finite sum) term of $\operatorname{Tot} \mathcal{K}$. For fixed q, $\mathcal{C}^\bullet(\mathfrak{U}, \mathcal{I}^q)$ is a resolution of $\mathcal{I}^q$ (Lemma 4.2), so $\varepsilon : \mathcal{I}^\bullet \to \operatorname{Tot} \mathcal{K}$, induced by $\mathcal{I}^q \to \mathcal{C}^0(\mathfrak{U}, \mathcal{I}^q)$, is a quasi-isomorphism; therefore $\operatorname{Tot} \mathcal{K}$ is an injective resolution of $\mathcal{F}$ (via $\mathcal{F} \to \mathcal{I}^0 \to \mathcal{K}^{0,0}$). The two morphisms of complexes $\tilde{\alpha} : \mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F}) \to \operatorname{Tot} \mathcal{K}$ (induced by $\mathcal{F} \to \mathcal{I}^0$) and $\varepsilon\phi$ both lie over $\mathrm{id}_\mathcal{F}$ (that is, they are compatible with the augmentations from $\mathcal{F}$), the source is a resolution of $\mathcal{F}$ (Lemma 4.2) and the target an injective resolution; so they are homotopic. Applying $\Gamma(X, \cdot)$, $\alpha = \tilde{\alpha}(X)$ and $\beta \circ \Gamma(\phi) = (\varepsilon\phi)(X)$ are homotopic, i.e. $\alpha_* = \beta_* \theta$.

Hence $\theta : \check{H}^p(\mathfrak{U}, \mathcal{F}) \to H^p(X, \mathcal{F})$ is an isomorphism for all $p \geq 0$.

Recovering (4.5). Let X be a noetherian separated scheme, $\mathfrak{U}$ an open affine cover and $\mathcal{F}$ quasi-coherent. Every finite intersection $V = U_{i_0 \dots i_p}$ is affine (II, Ex. 4.3) and noetherian, and $\mathcal{F}|_V$ is quasi-coherent, so $H^k(V, \mathcal{F}|_V) = 0$ for $k > 0$ by Thm. 3.5. The hypothesis of the exercise holds, hence $\check{H}^p(\mathfrak{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$ for all p, which is Thm. 4.5 (and without using the embedding in flasque quasi-coherent sheaves). (The same argument applies to any topological space and covering by open sets whose finite intersections are acyclic for $\mathcal{F}$; e.g. for Example 4.0.4, the circle covered by two semicircles, whose intersection and the semicircles have no higher cohomology for the constant sheaf.) □

5. Cohomology of projective space

Throughout, "Thm. 5.1", "Thm. 5.2", "Prop. 5.3" refer to the results of Chapter III, Section 5 of the text: Thm. 5.1 computes $H^i(\mathbf{P}^r_A, \mathcal{O}(n))$, Thm. 5.2 (Serre) says that for $\mathcal{F}$ coherent on a projective scheme over a noetherian ring A the groups $H^i(X, \mathcal{F})$ are finitely generated A-modules and $H^i(X, \mathcal{F}(n)) = 0$ for $i > 0$, $n \gg 0$, and Prop. 5.3 says that, for X proper over $\operatorname{Spec} A$, an invertible sheaf $\mathcal{L}$ is ample iff for every coherent $\mathcal{F}$ one has $H^i(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0(\mathcal{F})$. If $i\colon Y \to X$ is a closed immersion and $\mathcal{G}$ is a sheaf of abelian groups on Y then $H^i(Y, \mathcal{G}) = H^i(X, i_*\mathcal{G})$ (Lemma 2.10), and we use this tacitly. If $X \subseteq \mathbf{P}^r_k$ is a closed subscheme then $H^i(X, \mathcal{F}) = 0$ for $i > r$ (Ex. 4.8(c), or Thm. 4.5 with the cover by the $r+1$ standard affines).

Exercise 5.1

Problem. Let X be a projective scheme over a field k, and let $\mathcal{F}$ be a coherent sheaf on X. We define the *Euler characteristic* of $\mathcal{F}$ by

$$\chi(\mathcal{F}) = \sum_{i \geq 0} (-1)^i \dim_k H^i(X, \mathcal{F}).$$

If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is a short exact sequence of coherent sheaves on X, show that $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$.

Solution. By Thm. 5.2(a) every $H^i(X, \mathcal{G})$, $\mathcal{G}$ coherent, is a finitely generated k-module, i.e. a finite dimensional vector space, and $H^i(X, \mathcal{G}) = 0$ for $i > r$ if $X \subseteq \mathbf{P}^r_k$. So χ is well defined for coherent sheaves ($\mathcal{F}', \mathcal{F}''$ are coherent by hypothesis). The long exact cohomology sequence

$$0 \to H^0(\mathcal{F}') \to H^0(\mathcal{F}) \to H^0(\mathcal{F}'') \to H^1(\mathcal{F}') \to \cdots \to H^r(\mathcal{F}') \to H^r(\mathcal{F}) \to H^r(\mathcal{F}'') \to 0$$

is a finite exact sequence of finite dimensional k-vector spaces (it ends with 0 because $H^{r+1}(\mathcal{F}') = 0$). For an exact sequence $0 \to V_0 \to V_1 \to \cdots \to V_m \to 0$ of finite dimensional vector spaces one has $\sum (-1)^j \dim V_j = 0$ (rank–nullity, applied to each map). Grouping the terms in threes, this reads $\sum_i (-1)^i \big(\dim H^i(\mathcal{F}') - \dim H^i(\mathcal{F}) + \dim H^i(\mathcal{F}'')\big) = 0$, i.e. $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$. □

Exercise 5.2

Problem.

(a) Let X be a projective scheme over a field k, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X over k, and let $\mathcal{F}$ be a coherent sheaf on X. Show that there is a polynomial $P(z) \in \mathbf{Q}[z]$, such that $\chi(\mathcal{F}(n)) = P(n)$ for all $n \in \mathbf{Z}$. We call P the *Hilbert polynomial* of $\mathcal{F}$ with respect to the sheaf $\mathcal{O}_X(1)$. [*Hints*: Use induction on $\dim \operatorname{Supp} \mathcal{F}$, general properties of numerical polynomials (I, 7.3), and suitable exact sequences $0 \to \mathcal{R} \to \mathcal{F}(-1) \to \mathcal{F} \to \mathcal{Q} \to 0$.]

(b) Now let $X = \mathbf{P}^r_k$, and let $M = \Gamma_*(\mathcal{F})$, considered as a graded $S = k[x_0, \dots, x_r]$-module. Use (5.2) to show that the Hilbert polynomial of $\mathcal{F}$ just defined is the same as the Hilbert polynomial of M defined in (I, Section 7).

Solution. (a) Fix a closed immersion $i\colon X \to \mathbf{P}^r_k$ with $i^*\mathcal{O}(1) \cong \mathcal{O}_X(1)$ (II, 5.16.1).

Step 0: reduction to k infinite. Let $K \supseteq k$ be an infinite field extension (e.g. an algebraic closure), $X_K = X \times_k K$, $\mathcal{F}_K$ the pull-back of $\mathcal{F}$. Then X_K is a projective scheme over K (closed in $\mathbf{P}^r_K$), $\mathcal{O}_{X_K}(1)$ is very ample, $\mathcal{F}_K$ is coherent, and $(\mathcal{F}(n))_K = \mathcal{F}_K(n)$. I claim $H^i(X_K, \mathcal{F}_K) \cong H^i(X, \mathcal{F}) \otimes_k K$. Indeed let $U_j = X \cap D_+(x_j)$; these are affine, finitely many, and all finite intersections are affine (X is separated). If $U = \operatorname{Spec} B$ and $\mathcal{F}|_U = \tilde{M}$ then $U_K = \operatorname{Spec}(B \otimes_k K)$ and $\Gamma(U_K, \mathcal{F}_K) = M \otimes_k K$. Hence the Cech complex of $\mathcal{F}_K$ for the cover $(U_{j,K})$ is the Cech complex of $\mathcal{F}$ for (U_j) tensored with K. As K is flat over k, cohomology commutes with $\otimes_k K$, and Thm. 4.5 (for X and for X_K, both noetherian separated) gives the claim. Consequently $\chi_K(\mathcal{F}_K(n)) = \chi_k(\mathcal{F}(n))$ for all n. So it suffices to prove the assertion over K for all coherent sheaves on X_K, i.e. we may and do assume *k is infinite*.

Step 1: a lemma on polynomials. Let $f\colon \mathbf{Z} \to \mathbf{Q}$ be a function such that $f(n) - f(n-1) = g(n)$ for all $n \in \mathbf{Z}$, where $g \in \mathbf{Q}[z]$. Then f is a polynomial function. Indeed, the polynomials $\binom{z+m}{m+1}$, $m \geq 0$, satisfy $\binom{z+m}{m+1} - \binom{z+m-1}{m+1} = \binom{z+m-1}{m}$, and the polynomials $\binom{z+m-1}{m}$ (of degree m, $m \geq 0$) form a $\mathbf{Q}$-basis of $\mathbf{Q}[z]$, so there is $h \in \mathbf{Q}[z]$ with $h(z) - h(z-1) = g(z)$. Then $f - h$ is constant on $\mathbf{Z}$, so $f = h + c$.

Step 2: induction. We prove: for every coherent $\mathcal{F}$ on X the function $n \mapsto \chi(\mathcal{F}(n))$ is a polynomial, by induction on $d = \dim \operatorname{Supp} \mathcal{F} \in \{-1, 0, 1, \dots\}$ ($d = -1$ meaning $\mathcal{F} = 0$, where $P = 0$). Let $d \geq 0$. We use the following standard facts about the finite set $\operatorname{Ass} \mathcal{F}$ of associated points of a coherent sheaf on a noetherian scheme ($p \in \operatorname{Ass} \mathcal{F}$ iff $\mathfrak{m}_p \in \operatorname{Ass}(\mathcal{F}_p)$): the generic point of each irreducible component of $\operatorname{Supp} \mathcal{F}$ lies in $\operatorname{Ass} \mathcal{F}$; a nonzero coherent subsheaf $\mathcal{R} \subseteq \mathcal{F}$ has $\emptyset \neq \operatorname{Ass} \mathcal{R} \subseteq \operatorname{Ass} \mathcal{F}$.

For $\ell \in V = \Gamma(\mathbf{P}^r_k, \mathcal{O}(1)) = \bigoplus k x_i$ and $p \in X$, let $\mathfrak{q}_p \subset S$ be the homogeneous prime of $i(p)$; the section $i^*\ell$ of $\mathcal{O}_X(1)$ does not vanish at p (i.e. generates the stalk $\mathcal{O}_X(1)_p$) iff $\ell \notin \mathfrak{q}_p$. Since $\mathfrak{q}_p$ does not contain all x_i, $V \cap \mathfrak{q}_p$ is a proper subspace of V. As k is infinite, V is not a finite union of proper subspaces, so there is ℓ with $x := i^*\ell$ nonvanishing at every point of the finite set $\operatorname{Ass} \mathcal{F}$. Multiplication by x gives $x\colon \mathcal{F}(-1) \to \mathcal{F}$ and an exact sequence

$$0 \to \mathcal{R} \to \mathcal{F}(-1) \to \mathcal{F} \to \mathcal{Q} \to 0$$

with $\mathcal{R}, \mathcal{Q}$ coherent. At $p \in \operatorname{Ass} \mathcal{F}$ the map x_p is an isomorphism, so $\mathcal{R}_p = 0$; if $\mathcal{R} \neq 0$ it would have an associated point in $\operatorname{Ass} \mathcal{F}(-1) = \operatorname{Ass} \mathcal{F}$, a contradiction. So $\mathcal{R} = 0$. Moreover $\operatorname{Supp} \mathcal{Q} \subseteq \operatorname{Supp} \mathcal{F} \cap Z(x)$, where $Z(x)$ is the closed set where x vanishes (x_p is an isomorphism off $Z(x)$). Let T be an irreducible component of $\operatorname{Supp} \mathcal{F}$ with generic point $\eta \in \operatorname{Ass} \mathcal{F}$; then $x(\eta) \neq 0$, so $T \cap Z(x)$ is a proper closed subset of T, hence has dimension $< \dim T \leq d$ (a

chain of irreducible closed subsets in it can be prolonged by T). Thus $\dim \operatorname{Supp} \mathcal{Q} < d$ and by induction $\chi(\mathcal{Q}(n)) = P_{\mathcal{Q}}(n)$ for a polynomial $P_{\mathcal{Q}}$.

Twisting $0 \to \mathcal{F}(-1) \to \mathcal{F} \to \mathcal{Q} \to 0$ by n and using Ex. 5.1 we get $\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = P_{\mathcal{Q}}(n)$ for all $n \in \mathbf{Z}$, and Step 1 shows $n \mapsto \chi(\mathcal{F}(n))$ is polynomial. Its polynomial is unique since it is prescribed on infinitely many integers. □

(b) For $n \gg 0$, Thm. 5.2(b) gives $H^i(\mathbf{P}^r_k, \mathcal{F}(n)) = 0$ for $i > 0$, so $\chi(\mathcal{F}(n)) = \dim_k H^0(\mathcal{F}(n)) = \dim_k M_n$. Thus the Hilbert polynomial $P_{\mathcal{F}}$ of (a) satisfies $P_{\mathcal{F}}(n) = \dim_k M_n$ for all $n \gg 0$, which is exactly the property defining the Hilbert polynomial of M (I, Section 7: the unique polynomial with $\varphi_M(n) = P_M(n)$ for $n \gg 0$). A remark on finite generation: $\Gamma_*(\mathcal{F})$ need not be finitely generated (for a skyscraper sheaf at a rational point, $\Gamma_*(\mathcal{F})_n = k$ for all $n \in \mathbf{Z}$), but (II, 5.15) gives a finitely generated graded N with $\tilde{N} \cong \mathcal{F}$, and then $N_n \to \Gamma(\mathcal{F}(n))$ is an isomorphism for $n \gg 0$ (II, Ex. 5.9); so N has the same Hilbert polynomial $P_{\mathcal{F}}$, and this is the finitely generated module to which (I, 7.5) literally applies. □

Exercise 5.3

Problem. *Arithmetic Genus.* Let X be a projective scheme of dimension r over a field k. We define the *arithmetic genus* p_a of X by $p_a(X) = (-1)^r(\chi(\mathcal{O}_X) - 1)$. Note that it depends only on X, not on any projective embedding.

(a) If X is integral, and k algebraically closed, show that $H^0(X, \mathcal{O}_X) \cong k$, so that $p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k H^{r-i}(X, \mathcal{O}_X)$. In particular, if X is a curve, we have $p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$. [*Hint*: Use (I, 3.4).]

(b) If X is a closed subvariety of $\mathbf{P}^r_k$, show that this $p_a(X)$ coincides with the one defined in (I, Ex. 7.2), which apparently depended on the projective embedding.

(c) If X is a nonsingular projective curve over an algebraically closed field k, show that $p_a(X)$ is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree $d \geq 3$ is not rational. (This gives another proof of (II, 8.20.3) where we used the geometric genus.)

Solution. (a) $\Gamma(X, \mathcal{O}_X)$ is an integral domain (X integral) and a finite dimensional k-algebra (Thm. 5.2(a), or Remark 5.2.1). A domain finite over k is a field finite over k, hence equals k since k is algebraically closed. So $h^0 := \dim H^0(X, \mathcal{O}_X) = 1$. Since $\dim X = r$, $H^i(X, \mathcal{O}_X) = 0$ for $i > r$ (Grothendieck, Thm. 2.7). Hence

$$\chi(\mathcal{O}_X) - 1 = \sum_{i=1}^{r} (-1)^i \dim H^i(X, \mathcal{O}_X), \qquad p_a(X) = \sum_{i=1}^{r} (-1)^{r+i} \dim H^i.$$

Putting $i = r - j$ ($0 \leq j \leq r-1$), $(-1)^{r+i} = (-1)^{2r-j} = (-1)^j$, which is the stated formula. For $r = 1$ it reads $p_a = \dim H^1(X, \mathcal{O}_X)$.

(b) Let $X \subseteq \mathbf{P}^r_k$ be a variety of dimension s with homogeneous ideal $I \subset S = k[x_0, \ldots, x_r]$. In (I, Ex. 7.2) $p_a(X) = (-1)^s(P_{S/I}(0) - 1)$, where $P_{S/I}$ is the Hilbert polynomial of the graded module S/I (I, Section 7). Now $X = \operatorname{Proj} S/I$ and $i_*\mathcal{O}_X = (S/I)^\sim$ (II, Section 5), so by Ex. 5.2(b) (with the finitely generated module $N = S/I$, $\tilde{N} = i_*\mathcal{O}_X$) $P_{S/I}(n) = \chi(X, \mathcal{O}_X(n))$ for all n. Evaluating at $n = 0$ gives $P_{S/I}(0) = \chi(\mathcal{O}_X)$, so the two definitions agree.

(c) If X, X' are birational nonsingular projective curves over the algebraically closed field k, then $X \cong X'$ (I, Cor. 6.12; II, 6.7). Since $p_a(X) = (-1)^1(\chi(\mathcal{O}_X) - 1)$ is defined from the scheme X alone, $p_a(X) = p_a(X')$, i.e. p_a is a birational invariant. (Here, as in Ch. I–II, a curve is integral.)

Now let $X \subset \mathbf{P}^2_k$ be a nonsingular plane curve of degree $d \geq 3$, $X = \operatorname{Proj} k[x_0, x_1, x_2]/(f)$, $\deg f = d$ (it is connected by Ex. 5.5(b) below, hence irreducible since it is nonsingular: its local rings are domains, so distinct components cannot meet; so it is an integral curve). Since f is a nonzerodivisor in S, we have the exact sequence $0 \to \mathcal{O}_{\mathbf{P}^2}(-d) \to \mathcal{O}_{\mathbf{P}^2} \to \mathcal{O}_X \to 0$. By Thm. 5.1, H^0 and H^1 of $\mathcal{O}(-d)$ vanish, $H^2(\mathcal{O}(-d)) \cong H^0(\mathcal{O}(d-3))^\vee$ has dimension $\binom{d-1}{2} = \frac{(d-1)(d-2)}{2}$, and $\chi(\mathcal{O}_{\mathbf{P}^2}) = 1$. Hence $\chi(\mathcal{O}_X) = 1 - \frac{(d-1)(d-2)}{2}$ and $p_a(X) = \frac{(d-1)(d-2)}{2} \geq 1$ for $d \geq 3$. If X were rational, it would be birational, hence isomorphic, to $\mathbf{P}^1$, which has $p_a = \dim H^1(\mathbf{P}^1, \mathcal{O}) = 0$ by Thm. 5.1. This contradiction shows X is not rational. □

Exercise 5.4

Problem. Recall from (II, Ex. 6.10) the definition of the Grothendieck group $K(X)$ of a noetherian scheme X.

(a) Let X be a projective scheme over a field k, and let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X. Show that there is a (unique) additive homomorphism $P\colon K(X) \to \mathbf{Q}[z]$ such that for each coherent sheaf $\mathcal{F}$ on X, $P(\gamma(\mathcal{F}))$ is the Hilbert polynomial of $\mathcal{F}$ (Ex. 5.2).

(b) Now let $X = \mathbf{P}^r_k$. For each $i = 0, 1, \ldots, r$, let L_i be a linear space of dimension i in X. Then show that (1) $K(X)$ is the free abelian group generated by $\{\gamma(\mathcal{O}_{L_i}) \mid i = 0, \ldots, r\}$, and (2) the map $P\colon K(X) \to \mathbf{Q}[z]$ is injective. [*Hint*: Show (1) ⇒ (2). Then prove (1) and (2) simultaneously by induction on r, using (II, Ex. 6.10c).]

Solution. Recall: $K(X)$ is the free abelian group on the coherent sheaves modulo the subgroup generated by $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$ for all exact $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$; $\gamma(\mathcal{F})$ is the class of $\mathcal{F}$.

(a) Let $P_\mathcal{F}$ be the Hilbert polynomial (Ex. 5.2(a)). Isomorphic sheaves have the same $P_\mathcal{F}$. If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is exact, so is its twist by n for every n, and Ex. 5.1 gives $\chi(\mathcal{F}(n)) = \chi(\mathcal{F}'(n)) + \chi(\mathcal{F}''(n))$ for all $n \in \mathbf{Z}$; two polynomials agreeing at all integers are equal, so $P_\mathcal{F} = P_{\mathcal{F}'} + P_{\mathcal{F}''}$. Hence $\mathcal{F} \mapsto P_\mathcal{F}$ extends to a homomorphism from the free abelian group on coherent sheaves to $\mathbf{Q}[z]$ killing all relations, thus descending to $P\colon K(X) \to \mathbf{Q}[z]$. Uniqueness: $K(X)$ is generated by the $\gamma(\mathcal{F})$ and $P(\gamma(\mathcal{F}))$ is prescribed.

(b) Let $L \subseteq X = \mathbf{P}^r_k$ be a linear subspace of dimension i, $L \cong \mathbf{P}^i_k$ with $\mathcal{O}_X(1)|_L = \mathcal{O}_L(1)$. By Lemma 2.10 and Thm. 5.1, for $n \geq 0$ and $i \geq 1$: $H^j(L, \mathcal{O}(n)) = 0$ for $j > 0$ (for $0 < j < i$ by 5.1(b), for $j = i$ since $H^i(\mathcal{O}(n)) \cong H^0(\mathcal{O}(-n-i-1))^\vee = 0$), and $\dim H^0(\mathcal{O}_L(n)) = \binom{n+i}{i}$; for $i = 0$ this is trivial. Hence

$$P(\gamma(\mathcal{O}_L)) = \binom{z+i}{i},$$

a polynomial of degree exactly i, *independent of the choice of L.*

(1) ⇒ (2). If $K(X)$ is free on the $\gamma(\mathcal{O}_{L_i})$, then P maps this basis to the polynomials $\binom{z+i}{i}$, $0 \leq i \leq r$, which have distinct degrees and so are linearly independent over $\mathbf{Q}$; hence P is injective on $K(X)$.

Generation, by induction on r. Claim: $K(\mathbf{P}^r_k)$ is generated by $\gamma(\mathcal{O}_{L_i})$, $0 \le i \le r$, for some choice of linear spaces L_i ($L_r = X$). For $r = 0$, $X = \operatorname{Spec} k$, a coherent sheaf is a finite dimensional vector space V, which has a filtration with $\dim V$ quotients k, so $\gamma(V) = \dim V \cdot \gamma(\mathcal{O}_X)$ and $L_0 = X$.

Let $r \ge 1$ and assume the claim for $\mathbf{P}^{r-1}$. Let $H = \{x_r = 0\} \cong \mathbf{P}^{r-1}$, $U = X \setminus H \cong \mathbf{A}^r_k$. By (II, Ex. 6.10c) the sequence $K(H) \to K(X) \to K(U) \to 0$ is exact (first map: extension by zero, second: restriction). I claim $K(U) = \mathbf{Z} \cdot \gamma(\mathcal{O}_U)$: a coherent sheaf on $U = \operatorname{Spec} A$, $A = k[t_1, \dots, t_r]$, is $\tilde{M}$ with M finitely generated, and by Hilbert's syzygy theorem M has a finite free resolution $0 \to F_m \to \cdots \to F_0 \to M \to 0$, so $\gamma(\tilde{M}) = \sum_j (-1)^j \operatorname{rank}(F_j)\gamma(\mathcal{O}_U)$ (the functor $\tilde{}$ is exact). Now let $\mathcal{F}$ be coherent on X; then $\gamma(\mathcal{F}|_U) = m\gamma(\mathcal{O}_U)$ for some $m \in \mathbf{Z}$, so $\gamma(\mathcal{F}) - m\gamma(\mathcal{O}_X)$ restricts to 0 in $K(U)$, hence comes from $K(H)$. By the induction hypothesis $K(H)$ is generated by $\gamma(\mathcal{O}_{L_i})$, $0 \le i \le r-1$, for linear subspaces $L_i \subseteq H$ of dimension i (which are linear subspaces of X); their images in $K(X)$ are the classes of the same sheaves viewed on X. So $K(X)$ is generated by $\gamma(\mathcal{O}_X)$ and $\gamma(\mathcal{O}_{L_i})$, $0 \le i \le r-1$, proving the claim.

Freeness. For these L_i the homomorphism $\mathbf{Z}^{r+1} \to K(X)$, $e_i \mapsto \gamma(\mathcal{O}_{L_i})$, is surjective, and its composite with P is injective (images of e_i are $\binom{z+i}{i}$, linearly independent over $\mathbf{Q}$). Hence $\mathbf{Z}^{r+1} \to K(X)$ is an isomorphism, and P is injective. Finally, for an *arbitrary* choice of linear subspaces L'_i of dimension i, $P(\gamma(\mathcal{O}_{L'_i})) = \binom{z+i}{i} = P(\gamma(\mathcal{O}_{L_i}))$, so by injectivity of P, $\gamma(\mathcal{O}_{L'_i}) = \gamma(\mathcal{O}_{L_i})$. This proves (1) and (2) for every choice of the L_i. □

Exercise 5.5

Problem. Let k be a field, let $X = \mathbf{P}^r_k$, and let Y be a closed subscheme of dimension $q \ge 1$, which is a complete intersection (II, Ex. 8.4). Then:

(a) for all $n \in \mathbf{Z}$, the natural map $H^0(X, \mathcal{O}_X(n)) \to H^0(Y, \mathcal{O}_Y(n))$ is surjective. (This gives a generalization and another proof of (II, Ex. 8.4c), where we assumed Y was normal.)

(b) Y is connected;

(c) $H^i(Y, \mathcal{O}_Y(n)) = 0$ for $0 < i < q$ and all $n \in \mathbf{Z}$;

(d) $p_a(Y) = \dim_k H^q(Y, \mathcal{O}_Y)$.

[*Hint*: Use exact sequences and induction on the codimension, starting from the case $Y = X$ which is (5.1).]

Solution. Write $S = k[x_0, \dots, x_r]$ and $c = r - q$ for the codimension. By definition (II, Ex. 8.4), the homogeneous ideal I of Y is generated by c elements; since I is homogeneous we may take them homogeneous, $f_1, \dots, f_c$ of degrees d_j (graded Nakayama), and $d_j \ge 1$ because $Y \ne \emptyset$ ($q \ge 1$). As S is a finitely generated domain over k of dimension $r+1$, $\operatorname{ht} J + \dim S/J = r+1$ for every proper ideal J (apply $\operatorname{ht} P + \dim S/P = r+1$ to the minimal primes of J), and $\dim S/(J + (g)) \ge \dim S/J - 1$ for every g. Here $\dim S/I = q+1$ (as $\dim Y = q$), so $\operatorname{ht} I = r - q = c$. Then $\operatorname{ht}(f_1, \dots, f_j) = j$ for all j: Krull's height theorem gives $\le j$, and $q+1 = \dim S/I \ge \dim S/(f_1, \dots, f_j) - (c-j)$ gives $\operatorname{ht}(f_1, \dots, f_j) = r+1-\dim S/(f_1, \dots, f_j) \ge r+1-(q+1)-(c-j) = j$. Since S is Cohen–Macaulay, $f_1, \dots, f_c$ is an $S_{\mathfrak{m}}$-regular sequence ($\mathfrak{m} = (x_0, \dots, x_r)$; Matsumura, Thm. 17.4; this is the unmixedness theorem of the hint to Ex. 8.4(a)), and then it is S-regular, because for a homogeneous element f and a graded module

N the kernel of f on N is a graded submodule, which vanishes iff it vanishes after localizing at $\mathfrak{m}$.

Let $Y_j = \operatorname{Proj} S/(f_1, \ldots, f_j)$, $Y_0 = X$, $Y_c = Y$, with $\mathcal{O}_{Y_j} = (S/(f_1, \ldots, f_j))^{\sim}$. Regularity gives exact sequences of graded modules $0 \to (S/(f_1, \ldots, f_{j-1}))(-d_j) \xrightarrow{f_j} S/(f_1, \ldots, f_{j-1}) \to S/(f_1, \ldots, f_j) \to 0$, hence exact sequences of sheaves on X

$$0 \to \mathcal{O}_{Y_{j-1}}(m - d_j) \to \mathcal{O}_{Y_{j-1}}(m) \to \mathcal{O}_{Y_j}(m) \to 0 \qquad (m \in \mathbf{Z}). \tag{$*$}$$

Claim (A_j), $0 \le j \le c$: $H^i(Y_j, \mathcal{O}_{Y_j}(m)) = 0$ for $0 < i < r - j$ and all $m \in \mathbf{Z}$, and $H^0(X, \mathcal{O}_X(m)) \to H^0(Y_j, \mathcal{O}_{Y_j}(m))$ is surjective for all m. For $j = 0$ this is Thm. 5.1(a),(b). Assume A_{j-1} and $j \le c$; then $s := r - j + 1 \ge q + 1 \ge 2$. The long exact sequence of $(*)$ gives

$$H^i(Y_{j-1}, \mathcal{O}(m)) \to H^i(Y_j, \mathcal{O}(m)) \to H^{i+1}(Y_{j-1}, \mathcal{O}(m - d_j)).$$

For $0 < i < r - j = s - 1$ both outer terms vanish by A_{j-1} (as $0 < i$ and $i + 1 < s$), so $H^i(Y_j, \mathcal{O}(m)) = 0$. For $i = 0$ the right-hand term is $H^1(Y_{j-1}, \mathcal{O}(m - d_j))$, which vanishes since $0 < 1 < s$; hence $H^0(Y_{j-1}, \mathcal{O}(m)) \to H^0(Y_j, \mathcal{O}(m))$ is surjective, and composing with the surjection from A_{j-1} gives the surjectivity for Y_j. This proves A_j for all $j \le c$.

Taking $j = c$ ($r - c = q$) gives (a) and (c).

(b) By (a) with $n = 0$, $\Gamma(Y, \mathcal{O}_Y)$ is a quotient of $\Gamma(X, \mathcal{O}_X) = k$; it is nonzero as $Y \neq \emptyset$. So $\Gamma(Y, \mathcal{O}_Y) = k$. If $Y = Y' \sqcup Y''$ with Y', Y'' nonempty open and closed, then $\Gamma(Y, \mathcal{O}_Y) = \Gamma(Y', \mathcal{O}) \times \Gamma(Y'', \mathcal{O})$ has dimension ≥ 2, a contradiction. So Y is connected.

(d) Y has dimension $q \ge 1$, so $H^i(Y, \mathcal{O}_Y) = 0$ for $i > q$ (Thm. 2.7), $H^i = 0$ for $0 < i < q$ by (c), and $\dim H^0 = 1$ by (b). Hence $\chi(\mathcal{O}_Y) = 1 + (-1)^q \dim H^q(Y, \mathcal{O}_Y)$ and $p_a(Y) = (-1)^q(\chi(\mathcal{O}_Y) - 1) = \dim_k H^q(Y, \mathcal{O}_Y)$. □

Exercise 5.6

Problem. *Curves on a Nonsingular Quadric Surface.* Let Q be the nonsingular quadric surface $xy = zw$ in $X = \mathbf{P}^3_k$ over a field k. We will consider locally principal subschemes Y of Q. These correspond to Cartier divisors on Q by (II, 6.17.1). On the other hand, we know that $\operatorname{Pic} Q \cong \mathbf{Z} \oplus \mathbf{Z}$, so we can talk about the type (a, b) of Y (II, 6.16) and (II, 6.6.1). Let us denote the invertible sheaf $\mathcal{L}(Y)$ by $\mathcal{O}_Q(a, b)$. Thus for any $n \in \mathbf{Z}$, $\mathcal{O}_Q(n) = \mathcal{O}_Q(n, n)$.

(a) Use the special cases $(q, 0)$ and $(0, q)$, with $q > 0$, when Y is a disjoint union of q lines $\mathbf{P}^1$ in Q, to show: (1) if $|a - b| \le 1$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$; (2) if $a, b < 0$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$; (3) if $a \le -2$, then $H^1(Q, \mathcal{O}_Q(a, 0)) \neq 0$.

(b) Now use these results to show: (1) if Y is a locally principal closed subscheme of type (a, b), with $a, b > 0$, then Y is connected; (2) now assume k is algebraically closed. Then for any $a, b > 0$, there exists an irreducible nonsingular curve Y of type (a, b). Use (II, 7.6.2) and (II, 8.18); (3) an irreducible nonsingular curve Y of type (a, b), $a, b > 0$ on Q is projectively normal (II, Ex. 5.14) if and only if $|a - b| \le 1$. In particular, this gives lots of examples of nonsingular, but not projectively normal curves in $\mathbf{P}^3$. The simplest is the one of type $(1, 3)$, which is just the rational quartic curve (I, Ex. 3.18).

(c) If Y is a locally principal subscheme of type (a, b) in Q, show that $p_a(Y) = ab - a - b + 1$. [*Hint*: Calculate Hilbert polynomials of suitable sheaves, and again use the special case $(q, 0)$ which is a disjoint union of q copies of $\mathbf{P}^1$. See (V, 1.5.2) for another method.]

Solution. Conventions. The Segre embedding $\mathbf{P}^1 \times_k \mathbf{P}^1 \to \mathbf{P}^3$, $((s:t),(s':t')) \mapsto (ss' : tt' : st' : ts')$ (coordinates x, y, z, w) identifies $\mathbf{P}^1 \times \mathbf{P}^1$ with Q over any field. We write p_1, p_2 for the projections and $\mathcal{O}_Q(a,b) = p_1^*\mathcal{O}(a) \otimes p_2^*\mathcal{O}(b)$; then $\mathcal{O}_Q(1) = \mathcal{O}_Q(1,1)$, $\operatorname{Pic} Q = \mathbf{Z} \oplus \mathbf{Z}$ via (a,b) (II, 6.6.1), and a locally principal Y has type (a,b) iff its ideal sheaf is $\mathcal{L}(Y)^{-1} \cong \mathcal{O}_Q(-a,-b)$, i.e. there is an exact sequence $0 \to \mathcal{O}_Q(-a,-b) \to \mathcal{O}_Q \to \mathcal{O}_Y \to 0$. The automorphism of Q swapping the factors interchanges (a,b) and (b,a); so every statement below is symmetric in the two factors (the labelling of the rulings is immaterial). We use two facts on Q, both from Ex. 5.5 (applied to the hypersurface $Q \subset \mathbf{P}^3$, a complete intersection of dimension 2): (i) $H^1(Q, \mathcal{O}_Q(n)) = 0$ for all n; (ii) $H^0(\mathbf{P}^3, \mathcal{O}(n)) \to H^0(Q, \mathcal{O}_Q(n))$ is surjective for all n and $H^0(Q, \mathcal{O}_Q) = k$.

The curves of type $(q,0)$, $(0,q)$. Let $p_1, \dots, p_q \in \mathbf{P}^1(k)$ be distinct rational points, $Z \subset \mathbf{P}^1$ the reduced subscheme $\{p_1, \dots, p_q\}$; its ideal sheaf is $\mathcal{O}_{\mathbf{P}^1}(-q)$ (the zero scheme of a product of q distinct linear forms). Put $Y = p_1^{-1}(Z) = Z \times \mathbf{P}^1 = \bigsqcup_{i=1}^q L_i$, $L_i = \{p_i\} \times \mathbf{P}^1$, a disjoint union of q lines of Q. As p_1 is flat, $\mathcal{I}_Y = p_1^*\mathcal{O}(-q) = \mathcal{O}_Q(-q,0)$, so Y has type $(q,0)$. Also $\mathcal{O}_Q(n,n)|_{L_i} = \mathcal{O}_{\mathbf{P}^1}(n)$ ($p_1^*\mathcal{O}(n)$ is trivial on L_i). So twisting $0 \to \mathcal{O}_Q(-q,0) \to \mathcal{O}_Q \to \mathcal{O}_Y \to 0$ by $\mathcal{O}_Q(n)$ gives

$$0 \to \mathcal{O}_Q(n-q,n) \to \mathcal{O}_Q(n) \to \bigoplus_{i=1}^{q} \mathcal{O}_{\mathbf{P}^1}(n) \to 0.$$

Using (i) the long exact sequence yields

$$H^0(Q, \mathcal{O}_Q(n)) \xrightarrow{\rho} \bigoplus_{i=1}^{q} H^0(\mathbf{P}^1, \mathcal{O}(n)) \to H^1(Q, \mathcal{O}_Q(n-q,n)) \to H^1(Q, \mathcal{O}_Q(n)) = 0. \qquad (**)$$

Hence $H^1(Q, \mathcal{O}_Q(n-q,n)) \cong \operatorname{coker} \rho$, a quotient of $\bigoplus_i H^0(\mathbf{P}^1, \mathcal{O}(n))$. Similarly, with p_2 in place of p_1 we get $H^1(Q, \mathcal{O}_Q(n,n-q)) \cong \operatorname{coker} \rho'$ with the analogous ρ'.

Since cohomology commutes with the flat base change $k \to \bar{k}$ (proof as in Ex. 5.2, Step 0, and $\mathcal{O}_Q(a,b)$ is compatible with base change), the dimensions of $H^1(Q, \mathcal{O}_Q(a,b))$ do not change; so for $q \geq 2$ below we may assume k has at least q rational points (e.g. k algebraically closed).

(a)(1). If $a = b$, use (i). Let $a - b = -1$, i.e. $(a,b) = (n-1, n)$, $n = b$. Take $q = 1$: $Y = L$ is a line in $\mathbf{P}^3$, and ρ is the composite $H^0(Q, \mathcal{O}_Q(n)) \to H^0(L, \mathcal{O}_L(n))$. The restriction $H^0(\mathbf{P}^3, \mathcal{O}(n)) \to H^0(L, \mathcal{O}_L(n))$ is surjective (L is a linear subspace, $S \to S/(\ell_1, \ell_2) \cong k[s,t]$ is onto, and $H^0(L, \mathcal{O}(n)) = k[s,t]_n$ for $n \geq 0$ by Thm. 5.1(a), zero for $n < 0$), and it factors through ρ. So ρ is surjective and $H^1(Q, \mathcal{O}_Q(n-1, n)) = 0$ by $(**)$. The case $a - b = 1$ is symmetric (use ρ').

(a)(2). Let $a, b < 0$. If $a \leq b$, put $n = b < 0$, $q = b - a \geq 0$ ($q = 0$ is (i)). In $(**)$, $H^0(\mathbf{P}^1, \mathcal{O}(n)) = 0$ since $n < 0$, so $H^1(Q, \mathcal{O}_Q(a,b)) = H^1(Q, \mathcal{O}_Q(n-q,n)) = 0$. If $a \geq b$, put $n = a < 0$, $q = a - b$ and use ρ' in the same way.

(a)(3). Let $a \leq -2$, $q = -a \geq 2$, $n = 0$. Then $H^0(\mathbf{P}^1, \mathcal{O})^{\oplus q} = k^q$, and the image of ρ: $H^0(Q, \mathcal{O}_Q) = k \to k^q$ has dimension ≤ 1. By $(**)$, $\dim H^1(Q, \mathcal{O}_Q(-q,0)) = \dim \operatorname{coker} \rho \geq q - 1 \geq 1$. So $H^1(Q, \mathcal{O}_Q(a,0)) \neq 0$; by symmetry also $H^1(Q, \mathcal{O}_Q(0,b)) \neq 0$ for $b \leq -2$.

(b)(1). Let Y be locally principal of type (a,b), $a, b > 0$. From $0 \to \mathcal{O}_Q(-a,-b) \to \mathcal{O}_Q \to \mathcal{O}_Y \to 0$ and $H^1(Q, \mathcal{O}_Q(-a,-b)) = 0$ (by (a)(2)), $H^0(Q, \mathcal{O}_Q) = k \to H^0(Y, \mathcal{O}_Y)$ is surjective. $Y \neq \emptyset$: otherwise $\mathcal{O}_Q(-a,-b) \cong \mathcal{O}_Q$, but restricted to the line $\mathbf{P}^1 \times \{pt\}$ this is $\mathcal{O}_{\mathbf{P}^1}(-a) \ncong \mathcal{O}$.

So $\Gamma(Y, \mathcal{O}_Y) = k$, and as in Ex. 5.5(b), Y is connected (a decomposition $Y = Y' \sqcup Y''$ would give $\dim \Gamma(\mathcal{O}_Y) \geq 2$).

(b)(2). Let k be algebraically closed, $a, b > 0$. The sheaf $\mathcal{O}_Q(a, b)$ is very ample: it is the pull-back of $\mathcal{O}(1)$ under the closed immersion $\mathbf{P}^1 \times \mathbf{P}^1 \to \mathbf{P}^a \times \mathbf{P}^b \to \mathbf{P}^N$, the product of the Veronese embeddings (II, Ex. 5.13; $v_a^* \mathcal{O}(1) = \mathcal{O}(a)$) followed by the Segre embedding (II, Ex. 2.14, 5.11); cf. (II, 7.6.2). So Q is embedded in $\mathbf{P}^N$ with $\mathcal{O}_Q(1)_{\mathbf{P}^N} = \mathcal{O}_Q(a, b)$. By Bertini (II, 8.18) there is a hyperplane $H \subset \mathbf{P}^N$, not containing Q, such that $Y = H \cap Q$ is regular at every point. Y is the zero scheme of a nonzero section s of $\mathcal{O}_Q(a, b)$ on the integral scheme Q, so it is a locally principal subscheme with ideal sheaf $\cong \mathcal{O}_Q(-a, -b)$, i.e. of type (a, b) (and of pure dimension 1). By (b)(1), Y is connected. A nonsingular connected scheme is irreducible (the local rings are regular, hence domains, so every point lies on a unique component; the finitely many components are then disjoint closed sets, so connectedness leaves one). So Y is an irreducible nonsingular (integral) curve of type (a, b).

(b)(3). Let k be algebraically closed and $Y \subset Q \subset \mathbf{P}^3$ be an irreducible nonsingular curve of type (a, b), $a, b > 0$. Y is normal and connected (b)(1), so by (II, Ex. 5.14(d)) Y is projectively normal iff $H^0(\mathbf{P}^3, \mathcal{O}(n)) \to H^0(Y, \mathcal{O}_Y(n))$ is surjective for every $n \geq 0$. Twist $0 \to \mathcal{O}_Q(-a, -b) \to \mathcal{O}_Q \to \mathcal{O}_Y \to 0$ by $\mathcal{O}_Q(n)$:

$$0 \to \mathcal{O}_Q(n - a, n - b) \to \mathcal{O}_Q(n) \to \mathcal{O}_Y(n) \to 0.$$

By (ii) the map $H^0(\mathbf{P}^3, \mathcal{O}(n)) \to H^0(Q, \mathcal{O}_Q(n))$ is onto, so our map is onto iff $\sigma\colon H^0(Q, \mathcal{O}_Q(n)) \to H^0(Y, \mathcal{O}_Y(n))$ is onto, iff the connecting map $\partial\colon H^0(\mathcal{O}_Y(n)) \to H^1(Q, \mathcal{O}_Q(n - a, n - b))$ is zero. But $\operatorname{im} \partial = \ker(H^1(\mathcal{O}_Q(n-a, n-b)) \to H^1(\mathcal{O}_Q(n)))$ and $H^1(\mathcal{O}_Q(n)) = 0$, so ∂ is onto. Therefore

$$Y \text{ is projectively normal } \iff H^1(Q, \mathcal{O}_Q(n - a, n - b)) = 0 \text{ for all } n \geq 0.$$

If $|a - b| \leq 1$ then $|(n - a) - (n - b)| \leq 1$ and the right side holds by (a)(1). If $|a - b| \geq 2$: when $a \geq b + 2$, take $n = b > 0$: $H^1(\mathcal{O}_Q(b - a, 0)) \neq 0$ by (a)(3); when $b \geq a + 2$, take $n = a$: $H^1(\mathcal{O}_Q(0, a - b)) \neq 0$ by (a)(3) (symmetric form). So Y is not projectively normal. This proves (3). For type $(1, 3)$: $|1 - 3| = 2$, so the curve (which exists and is nonsingular by (2), has degree $(1, 3) \cdot (1, 1) = 4$ and $p_a = (1 - 1)(3 - 1) = 0$, i.e. is the rational quartic) is nonsingular but not projectively normal.

(c) Let $Y \neq \emptyset$ be locally principal of type (a, b) (so $a, b \geq 0$, not both 0, and Y is an effective Cartier divisor on the integral surface Q, of dimension 1). Let $\chi(m, n) = \chi(Q, \mathcal{O}_Q(m, n))$. First $\chi(0, 0) = 1$: from $0 \to \mathcal{O}_{\mathbf{P}^3}(-2) \to \mathcal{O}_{\mathbf{P}^3} \to \mathcal{O}_Q \to 0$ and Ex. 5.1, $\chi(\mathcal{O}_Q) = \chi(\mathcal{O}_{\mathbf{P}^3}) - \chi(\mathcal{O}_{\mathbf{P}^3}(-2)) = 1 - 0$, since all cohomology of $\mathcal{O}_{\mathbf{P}^3}(-2)$ vanishes by Thm. 5.1 ($H^3(\mathcal{O}(-2)) \cong H^0(\mathcal{O}(-2))^\vee = 0$). Next let $L = \{p\} \times \mathbf{P}^1$ be a rational line of type $(1, 0)$; twisting $0 \to \mathcal{O}_Q(-1, 0) \to \mathcal{O}_Q \to \mathcal{O}_L \to 0$ by $\mathcal{O}_Q(m, n)$ and using $\mathcal{O}_Q(m, n)|_L = \mathcal{O}_{\mathbf{P}^1}(n)$, Ex. 5.1 and Thm. 5.1 ($\chi(\mathbf{P}^1, \mathcal{O}(n)) = n + 1$ for all $n \in \mathbf{Z}$) give $\chi(m, n) - \chi(m - 1, n) = n + 1$ for all m, n; symmetrically $\chi(m, n) - \chi(m, n - 1) = m + 1$. Hence $\chi(0, n) = \chi(0, 0) + n = n + 1$ and $\chi(m, n) = \chi(0, n) + m(n + 1) = (m + 1)(n + 1)$ for all $m, n \in \mathbf{Z}$. From $0 \to \mathcal{O}_Q(-a, -b) \to \mathcal{O}_Q \to \mathcal{O}_Y \to 0$, $\chi(\mathcal{O}_Y) = 1 - (1 - a)(1 - b) = a + b - ab$, so $p_a(Y) = (-1)^1(\chi(\mathcal{O}_Y) - 1) = ab - a - b + 1$. (E.g. type $(1, 3)$ gives 0, type $(2, 2)$ gives 1.) □

Exercise 5.7

Problem. Let X (respectively, Y) be proper schemes over a noetherian ring A. We denote by $\mathcal{L}$ an invertible sheaf.

(a) If $\mathcal{L}$ is ample on X, and Y is any closed subscheme of X, then $i^*\mathcal{L}$ is ample on Y, where $i\colon Y \to X$ is the inclusion.

(b) $\mathcal{L}$ is ample on X if and only if $\mathcal{L}_{\mathrm{red}} = \mathcal{L} \otimes \mathcal{O}_{X_{\mathrm{red}}}$ is ample on X_{red}.

(c) Suppose X is reduced. Then $\mathcal{L}$ is ample on X if and only if $\mathcal{L} \otimes \mathcal{O}_{X_i}$ is ample on X_i, for each irreducible component X_i of X.

(d) Let $f\colon X \to Y$ be a finite surjective morphism, and let $\mathcal{L}$ be an invertible sheaf on Y. Then $\mathcal{L}$ is ample on Y if and only if $f^*\mathcal{L}$ is ample on X. [*Hints*: Use (5.3) and compare (Ex. 3.1, Ex. 3.2, Ex. 4.1, Ex. 4.2). See also Hartshorne [5, Ch. I Section 4] for more details.]

Solution. We use Prop. 5.3 throughout; note that closed subschemes of proper A-schemes are proper.

(a) Y is proper over A. Let $\mathcal{G}$ be coherent on Y; then $i_*\mathcal{G}$ is coherent on X and, by the projection formula (II, Ex. 5.1(d)) and Lemma 2.10, $H^j(Y, \mathcal{G} \otimes i^*\mathcal{L}^n) = H^j(X, i_*(\mathcal{G} \otimes i^*\mathcal{L}^n)) = H^j(X, i_*\mathcal{G} \otimes \mathcal{L}^n)$, which vanishes for $j > 0$, $n \geq n_0$ because $\mathcal{L}$ is ample on X (5.3(i)⇒(ii)). So $i^*\mathcal{L}$ is ample by 5.3(ii)⇒(i).

(b) (⇒) is (a) with $Y = X_{\mathrm{red}}$. (⇐) Let $\mathcal{N}$ be the nilradical, $\mathcal{N}^m = 0$. Let $\mathcal{F}$ be coherent on X and consider $\mathcal{F} \supseteq \mathcal{N}\mathcal{F} \supseteq \cdots \supseteq \mathcal{N}^m\mathcal{F} = 0$. Each quotient $\mathcal{G}_j = \mathcal{N}^j\mathcal{F}/\mathcal{N}^{j+1}\mathcal{F}$ is coherent and killed by $\mathcal{N}$, so $\mathcal{G}_j = i_*\mathcal{G}'_j$ for a coherent $\mathcal{G}'_j$ on X_{red} ($i\colon X_{\mathrm{red}} \to X$). As above, $H^l(X, \mathcal{G}_j \otimes \mathcal{L}^n) = H^l(X_{\mathrm{red}}, \mathcal{G}'_j \otimes \mathcal{L}_{\mathrm{red}}^n) = 0$ for $l > 0$, $n \geq n_j$, since $\mathcal{L}_{\mathrm{red}}$ is ample on the proper scheme X_{red}. Descending induction on j, using $0 \to \mathcal{N}^{j+1}\mathcal{F} \to \mathcal{N}^j\mathcal{F} \to \mathcal{G}_j \to 0$ tensored with $\mathcal{L}^n$, shows $H^l(X, \mathcal{N}^j\mathcal{F} \otimes \mathcal{L}^n) = 0$ for $l > 0$ and $n \geq \max_{j' \geq j} n_{j'}$; for $j = 0$ this is the claim for $\mathcal{F}$. By 5.3, $\mathcal{L}$ is ample.

(c) (⇒) is (a) (with X_i reduced closed subschemes). (⇐) By induction on the number m of components; $m = 1$ is trivial. Let $Z = X_m$ and $W = X_1 \cup \cdots \cup X_{m-1}$ with reduced structures, with ideal sheaves $\mathcal{I}_Z, \mathcal{I}_W$. Since X is reduced, $\mathcal{I}_Z \cap \mathcal{I}_W = 0$ (a section in both has zero value at all points of $Z \cup W = X$, hence is nilpotent, hence 0), so $\mathcal{I}_Z\mathcal{I}_W = 0$. For $\mathcal{F}$ coherent, $\mathcal{I}_W\mathcal{F}$ is killed by $\mathcal{I}_Z$, hence is a coherent $\mathcal{O}_Z$-module, and $\mathcal{F}/\mathcal{I}_W\mathcal{F}$ is a coherent $\mathcal{O}_W$-module. By induction $\mathcal{L}|_W$ is ample on W (its components are $X_1, \ldots, X_{m-1}$) and $\mathcal{L}|_Z$ is ample, so as in (b) $H^l(\mathcal{I}_W\mathcal{F} \otimes \mathcal{L}^n) = 0 = H^l((\mathcal{F}/\mathcal{I}_W\mathcal{F}) \otimes \mathcal{L}^n)$ for $l > 0$, $n \gg 0$, and the sequence $0 \to \mathcal{I}_W\mathcal{F} \to \mathcal{F} \to \mathcal{F}/\mathcal{I}_W\mathcal{F} \to 0$ gives the same for $\mathcal{F}$. Conclude by 5.3.

(d) (⇒) Let $\mathcal{L}$ be ample on Y. f is finite, hence affine, and X is proper over A. For $\mathcal{F}$ coherent on X, $f_*\mathcal{F}$ is coherent (finiteness of f) and by Ex. 4.1 and the projection formula

$$H^l(X, \mathcal{F} \otimes f^*\mathcal{L}^n) = H^l(Y, f_*(\mathcal{F} \otimes f^*\mathcal{L}^n)) = H^l(Y, f_*\mathcal{F} \otimes \mathcal{L}^n) = 0$$

for $l > 0$, $n \gg 0$ (5.3). So $f^*\mathcal{L}$ is ample. (Surjectivity is not needed here.)

(⇐) Assume $f^*\mathcal{L}$ ample, f finite and surjective. We show that $\mathcal{L}$ satisfies 5.3(ii). Call a closed subset $W \subseteq Y$ *good* if for every coherent $\mathcal{F}$ on Y with $\operatorname{Supp}\mathcal{F} \subseteq W$ there is n_0 with $H^l(Y, \mathcal{F} \otimes \mathcal{L}^n) = 0$ for $l > 0$, $n \geq n_0$. $\emptyset$ is good. Suppose some closed subset is not good; by noetherian induction pick a minimal non-good W, so every proper closed subset of W is good.

W is irreducible. If $W = W_1 \cup W_2$ with $W_1, W_2 \subsetneq W$ closed, let $\mathcal{I}_1$ be the ideal sheaf of W_1 (reduced structure), and $\mathcal{F}$ coherent supported in W. On the open set $Y \setminus W_2$ the support of

$\mathcal{F}$ lies in W_1, so $\mathcal{I}_1^m\mathcal{F}=0$ there for $m\gg 0$ ($Y\setminus W_2$ is quasi-compact). Thus $\operatorname{Supp}(\mathcal{I}_1^m\mathcal{F})\subseteq W_2$, while $\mathcal{F}/\mathcal{I}_1^m\mathcal{F}$ is supported in W_1. The sequence $0\to\mathcal{I}_1^m\mathcal{F}\to\mathcal{F}\to\mathcal{F}/\mathcal{I}_1^m\mathcal{F}\to 0$ shows $\mathcal{F}$ has the vanishing property, i.e. W is good, contradiction.

Reduction to $\mathcal{O}_W$-modules. Let $j\colon W\to Y$ be the inclusion with reduced structure and $\mathcal{F}$ coherent supported in W and not having the vanishing property. Then $\mathcal{I}_W^m\mathcal{F}=0$ for some m, and $\mathcal{F}\supseteq\mathcal{I}_W\mathcal{F}\supseteq\dots$ has quotients that are coherent $\mathcal{O}_W$-modules; one of them fails the vanishing property (else $\mathcal{F}$ would have it). So there is a coherent $\mathcal{O}_W$-module $\mathcal{E}$ with $j_*\mathcal{E}$ failing it. Since $H^l(Y,j_*\mathcal{E}\otimes\mathcal{L}^n)=H^l(W,\mathcal{E}\otimes(\mathcal{L}|_W)^n)$, we get a contradiction as follows.

W is integral. Since f is surjective, $f^{-1}(W)\to W$ is surjective; $f^{-1}(W)_{\mathrm{red}}$ has finitely many irreducible components, whose images (closed, since f is finite) cover W, so one of them, X', maps onto W (irreducibility of W). With the reduced structure, $g=f|_{X'}\colon X'\to W$ is a finite surjective morphism of integral schemes, X' is proper over A, and $g^*(\mathcal{L}|_W)=(f^*\mathcal{L})|_{X'}$ is ample by (a). Put $\mathcal{B}=g_*\mathcal{O}_{X'}$, a coherent $\mathcal{O}_W$-algebra; at the generic point η of W, $\mathcal{B}_\eta=K(X')$ is a field extension of $K=K(W)$ of finite degree ϱ (a finite K-algebra which is a domain). Put $\mathcal{M}=g^*(\mathcal{L}|_W)$, ample on X'. By (II, 7.6) and properness some $\mathcal{M}^s$ gives a closed immersion $X'\to\mathbf{P}_A^M$; let $s_0,\dots,s_M\in\Gamma(X',\mathcal{M}^s)$ be the pulled-back coordinates, indexed so that $s_0\neq 0$ (as X' is integral, s_0 is then a generator of $\mathcal{M}^s$ at the generic point of X'). Put $u_i=s_i/s_0\in K(X')$. The ring $K[u_1,\dots,u_M]\subseteq K(X')$ is a domain finite-dimensional over K, hence a field; it contains the coordinate ring $A[u_1,\dots,u_M]$ of the nonempty affine open $X'\cap D_+(x_0)$, hence its fraction field $K(X')$. So $K(X')=K[u_1,\dots,u_M]$, and for $t\gg 0$ the monomials u^J, $|J|\le t$, span $K(X')$ over K. Now $s^Js_0^{t-|J|}\in\Gamma(X',\mathcal{M}^{st})=\Gamma(W,\mathcal{B}\otimes\mathcal{L}^{st})$ (projection formula) has generic germ u^J after trivializing by s_0^t. Choose ϱ of these whose germs form a K-basis of $K(X')=\mathcal{B}_\eta$, and put $m=st$. This gives a *global* map $\alpha\colon\mathcal{O}_W^\varrho\to\mathcal{B}\otimes\mathcal{L}^m$, an isomorphism at η. Let $\mathcal{H}=\mathcal{H}om_{\mathcal{O}_W}(\mathcal{B},\mathcal{E})$, a coherent $\mathcal{B}$-module via $(b\varphi)(x)=\varphi(bx)$; since g is affine, $\mathcal{H}=g_*\mathcal{G}$ for a coherent $\mathcal{O}_{X'}$-module $\mathcal{G}$ (II, Ex. 5.17). Put $\mathcal{H}'=\mathcal{H}om(\mathcal{B}\otimes\mathcal{L}^m,\mathcal{E})=\mathcal{H}\otimes\mathcal{L}^{-m}=g_*(\mathcal{G}\otimes\mathcal{M}^{-m})$. The map $\mathcal{H}om(\alpha,\mathcal{E})\colon\mathcal{H}'\to\mathcal{E}^\varrho$ is an isomorphism at η (Hom of coherent sheaves commutes with localization). Let $\mathcal{K}$ and $\mathcal{C}$ be its kernel and cokernel; they are coherent and vanish at η, so their supports are proper closed subsets of W, which are good by minimality; and (viewed on Y via j_*) $\mathcal{K},\mathcal{C}$ have the vanishing property. Moreover by Ex. 4.1 and the projection formula

$$H^l(W,\mathcal{H}'\otimes\mathcal{L}^n)=H^l(X',\mathcal{G}\otimes\mathcal{M}^{n-m})=0\qquad(l>0,\ n\gg 0)$$

because $\mathcal{M}$ is ample on X' (5.3). Let $\mathcal{B}'=\operatorname{im}(\mathcal{H}'\to\mathcal{E}^\varrho)$. From $0\to\mathcal{K}\to\mathcal{H}'\to\mathcal{B}'\to 0$: $H^l(\mathcal{B}'\otimes\mathcal{L}^n)$ sits between $H^l(\mathcal{H}'\otimes\mathcal{L}^n)=0$ and $H^{l+1}(\mathcal{K}\otimes\mathcal{L}^n)=0$, so vanishes for $l>0$, $n\gg 0$; from $0\to\mathcal{B}'\to\mathcal{E}^\varrho\to\mathcal{C}\to 0$, $H^l(\mathcal{E}^\varrho\otimes\mathcal{L}^n)=H^l(\mathcal{E}\otimes\mathcal{L}^n)^\varrho$ lies between $H^l(\mathcal{B}'\otimes\mathcal{L}^n)$ and $H^l(\mathcal{C}\otimes\mathcal{L}^n)$, both 0 for $l>0$, $n\gg 0$. So $j_*\mathcal{E}$ does have the vanishing property: contradiction. Hence every closed subset, in particular Y, is good, and $\mathcal{L}$ is ample by 5.3(ii)⇒(i). (Compare EGA II, 6.6.2, and Hartshorne [5, Ch. I, Section 4].) □

Exercise 5.8

Problem. Prove that every one-dimensional proper scheme X over an algebraically closed field k is projective.

(a) If X is irreducible and nonsingular, then X is projective by (II, 6.7).

(b) If X is integral, let $\tilde{X}$ be its normalization (II, Ex. 3.8). Show that $\tilde{X}$ is complete and nonsingular, hence projective by (a). Let $f\colon \tilde{X} \to X$ be the projection. Let $\mathcal{L}$ be a very ample invertible sheaf on $\tilde{X}$. Show that there is an effective divisor $D = \sum P_i$ on $\tilde{X}$ with $\mathcal{L}(D) \cong \mathcal{L}$, and such that $f(P_i)$ is a nonsingular point of X, for each i. Conclude that there is an invertible sheaf $\mathcal{L}_0$ on X with $f^*\mathcal{L}_0 \cong \mathcal{L}$. Then use (Ex. 5.7d), (II, 7.6) and (II, 5.16.1) to show that X is projective.

(c) If X is reduced, but not necessarily irreducible, let $X_1, \ldots, X_r$ be the irreducible components of X. Use (Ex. 4.5) to show $\operatorname{Pic} X \to \bigoplus \operatorname{Pic} X_i$ is surjective. Then use (Ex. 5.7c) to show X is projective.

(d) Finally, if X is any one-dimensional proper scheme over k, use (2.7) and (Ex. 4.6) to show that $\operatorname{Pic} X \to \operatorname{Pic} X_{\mathrm{red}}$ is surjective. Then use (Ex. 5.7b) to show that X is projective.

Solution. (a) A proper scheme that is irreducible and nonsingular (regular local rings are domains) is an integral nonsingular complete curve; by (II, 6.7) it is projective. (If $\dim X = 0$ it is a point $\operatorname{Spec} k$.)

Preliminary. If $\mathcal{M}$ is an ample invertible sheaf on a proper k-scheme X, then by (II, 7.6) some $\mathcal{M}^m$ $(m > 0)$ is very ample over k, i.e. gives an immersion $X \to \mathbf{P}^N_k$, which is a closed immersion since X is proper (II, 4.9; cf. II, 5.16.1). So *a proper X with an ample invertible sheaf is projective.*

(b) $f\colon \tilde{X} \to X$ is finite (II, Ex. 3.8), so $\tilde{X}$ is proper over k; it is integral, of dimension 1, normal, so its local rings at closed points are normal one-dimensional noetherian local domains, i.e. DVRs: $\tilde{X}$ is nonsingular, hence projective by (a). Let $\mathcal{L}$ be very ample on $\tilde{X}$, giving $\tilde{X} \subseteq \mathbf{P}^N_k$.

Let $\Sigma \subset X$ be the set of singular (closed) points; it is finite, being a proper closed subset of the integral curve X (II, 8.16). Over the open set $U = X \setminus \Sigma$ of nonsingular points (local rings DVRs, integrally closed), $\mathcal{O}_X \to f_*\mathcal{O}_{\tilde{X}}$ is an isomorphism, so $f^{-1}(U) \to U$ is an isomorphism. The set $f^{-1}(\Sigma)$ is finite. Choose a hyperplane $H \subset \mathbf{P}^N$ containing none of the finitely many points of $f^{-1}(\Sigma)$ (the hyperplanes through a given point form a proper closed subset of the dual space, and k is infinite); then $H \not\supseteq \tilde{X}$, and $H \cap \tilde{X}$ is the zero scheme of a nonzero section of $\mathcal{L}$, an effective divisor $D = \sum P_i$ with $\mathcal{L}(D) \cong \mathcal{L}$ and $P_i \notin f^{-1}(\Sigma)$, i.e. $f(P_i)$ nonsingular.

Put $D_0 = f(\operatorname{Supp} D) \subset U$. As f is an isomorphism over U, $f^{-1}(D_0) = \operatorname{Supp} D$, and the ideal sheaf $\mathcal{I}_D$ of D corresponds on $f^{-1}(U) \cong U$ to an invertible ideal sheaf $\mathcal{I}_U \subseteq \mathcal{O}_U$. Define $\mathcal{I} \subseteq \mathcal{O}_X$ by $\mathcal{I}|_U = \mathcal{I}_U$ and $\mathcal{I}|_{X \setminus D_0} = \mathcal{O}$; these agree on $U \setminus D_0$ (both are $\mathcal{O}$), so $\mathcal{I}$ is an invertible ideal sheaf. The map $f^*\mathcal{I} \to \mathcal{O}_{\tilde{X}}$ is an isomorphism onto $\mathcal{I}_D$: on $f^{-1}(U)$ by construction and on $\tilde{X} \setminus \operatorname{Supp} D = f^{-1}(X \setminus D_0)$ because both sides are $\mathcal{O}$. Hence with $\mathcal{L}_0 = \mathcal{I}^{-1}$ we get $f^*\mathcal{L}_0 \cong \mathcal{I}_D^{-1} = \mathcal{L}(D) \cong \mathcal{L}$.

Now f is finite and surjective, and $f^*\mathcal{L}_0 \cong \mathcal{L}$ is ample on $\tilde{X}$; by Ex. 5.7(d), $\mathcal{L}_0$ is ample on X, which is proper, so X is projective by the Preliminary.

(c) Let X be reduced, with irreducible components X_i (reduced structure). Each X_i is an integral proper scheme of dimension ≤ 1, so projective by (a), (b) (a 0-dimensional component is a point). The sets $X_i \cap X_j$ $(i \neq j)$ are proper closed subsets of the irreducible X_i of dimension ≤ 1, hence finite sets of closed points; let T_0 be the finite set of points lying on at least two components. Since X is reduced, $\mathcal{O}_X \to \bigoplus_i (\iota_i)_*\mathcal{O}_{X_i}$ $(\iota_i\colon X_i \to X)$ is injective (an element vanishing in each $\mathcal{O}_{X_i}$ lies in all minimal primes, hence is 0), so we get an exact sequence of

sheaves of abelian groups

$$0 \to \mathcal{O}_X^* \to \bigoplus_i (\iota_i)_* \mathcal{O}_{X_i}^* \to \mathcal{T} \to 0.$$

At a point $x \notin T_0$, lying on exactly one component X_i, X coincides near x with X_i (both reduced), and the other summands are trivial near x, so the first map is an isomorphism on stalks at x: $\mathcal{T}$ is supported on the finite set T_0 of closed points, hence $H^1(X, \mathcal{T}) = 0$ (Thm. 2.7 applied to T_0 of dimension 0, and Lemma 2.10). The long exact sequence gives that $H^1(X, \mathcal{O}_X^*) \to \bigoplus H^1(X, (\iota_i)_* \mathcal{O}_{X_i}^*) = \bigoplus H^1(X_i, \mathcal{O}_{X_i}^*)$ is surjective, and by Ex. 4.5 (compatibly with restriction of line bundles) this is $\operatorname{Pic} X \to \bigoplus \operatorname{Pic} X_i$.

Choose very ample $\mathcal{L}_i$ on each X_i and $\mathcal{L}$ on X with $\mathcal{L}|_{X_i} \cong \mathcal{L}_i$ for all i. By Ex. 5.7(c) $\mathcal{L}$ is ample on X; X is proper, so projective.

(d) Let X be any one-dimensional proper scheme, $\mathcal{N}$ its nilradical, $\mathcal{N}^m = 0$. Let $X^{(j)}$ be the closed subscheme with ideal $\mathcal{N}^j$, $1 \le j \le m$; $X^{(1)} = X_{\text{red}}$, $X^{(m)} = X$, all with the same underlying space. $X^{(j)}$ is a closed subscheme of $X^{(j+1)}$ with ideal sheaf $\mathcal{J}_j = \mathcal{N}^j/\mathcal{N}^{j+1}$, and $\mathcal{J}_j^2 = 0$ (as $2j \ge j+1$). By Ex. 4.6 there is an exact sequence $\operatorname{Pic} X^{(j+1)} \to \operatorname{Pic} X^{(j)} \to H^2(X, \mathcal{J}_j)$, and $H^2(X, \mathcal{J}_j) = 0$ by Grothendieck's theorem (2.7) since $\dim X = 1$. So every $\operatorname{Pic} X^{(j+1)} \to \operatorname{Pic} X^{(j)}$ is surjective, and so is the composite $\operatorname{Pic} X \to \operatorname{Pic} X_{\text{red}}$. By (c) X_{red} is projective; let $\mathcal{L}'$ be an ample (e.g. very ample) invertible sheaf on it and $\mathcal{L}$ on X with $\mathcal{L} \otimes \mathcal{O}_{X_{\text{red}}} \cong \mathcal{L}'$. By Ex. 5.7(b) $\mathcal{L}$ is ample on X, hence X is projective (Preliminary). □

Exercise 5.9

Problem. *A Nonprojective Scheme.* We show the result of (Ex. 5.8) is false in dimension 2. Let k be an algebraically closed field of characteristic 0, and let $X = \mathbf{P}_k^2$. Let ω be the sheaf of differential 2-forms on X (II, Section 8). Define an infinitesimal extension X' of X by ω by giving the element $\xi \in H^1(X, \omega \otimes \mathcal{T})$ defined as follows (Ex. 4.10). Let x_0, x_1, x_2 be the homogeneous coordinates of X, let U_0, U_1, U_2 be the standard open covering, and let $\xi_{ij} = (x_i/x_j)\, d(x_j/x_i)$. This gives a Cech 1-cocycle with values in Ω_X^1, and since $\dim X = 2$, we have $\omega \otimes \mathcal{T} \cong \Omega^1$ (II, Ex. 5.16b). Now use the exact sequence

$$\cdots \to H^1(X, \omega) \to \operatorname{Pic} X' \to \operatorname{Pic} X \xrightarrow{\delta} H^2(X, \omega) \to \dots$$

of (Ex. 4.6) and (4.5). We have $\omega \cong \mathcal{O}_X(-3)$ (II, 8.20.1), so $H^2(X, \omega) \cong k$. Since $\operatorname{char} k = 0$, you need only show $\delta(\mathcal{L}(1)) \neq 0$, which can be done by calculating in Cech cohomology. Since $H^1(X, \omega) = 0$, we see that $\operatorname{Pic} X' = 0$. In particular, X' has no ample invertible sheaves, so is not projective.

Note. In fact, this result can be generalized to show that for any nonsingular projective surface X over an algebraically closed field k of characteristic 0, there is an infinitesimal extension X' of X by ω, such that X' is not projective over k. Indeed, let D be an ample divisor on X. Then D determines an element $c_1(D) \in H^1(X, \Omega^1)$ which we use to define X', as above. Then for any divisor E on X one can show that $\delta(\mathcal{L}(E)) = (D.E)$, where $(D.E)$ is the intersection number (Chapter V), considered as an element of k. Hence if E is ample, $\delta(\mathcal{L}(E)) \neq 0$. Therefore X' has no ample divisors. On the other hand, over a field of characteristic $p > 0$, a proper scheme X is projective if and only if X_{red} is!

Solution. Construction. On U_i ($\cong \mathbf{A}^2$, affine) let $\mathcal{A}_i = \mathcal{O}_{U_i} \oplus \omega|_{U_i}$ with multiplication $(f, a)(g, b) = (fg, fb + ga)$ $(\omega^2 = 0)$; $(U_i, \mathcal{A}_i)$ is the affine scheme $\operatorname{Spec}(\mathcal{O}(U_i) \oplus \omega(U_i))$ (same

underlying space, localizations $B_h \oplus M_h$). We identify $\Omega^1 \cong \mathcal{H}om(\Omega^1, \omega) = \omega \otimes \mathcal{T}$ by $\Phi(\eta)(\nu) = \eta \wedge \nu$; this is an isomorphism (in a local basis e_1, e_2 of Ω^1, $\eta = ae_1 + be_2$ goes to $e_1 \mapsto -be_{12}$, $e_2 \mapsto ae_{12}$). (Any other identification differs by a unit of $\mathcal{O}_X$, i.e. by a nonzero constant, which does not affect the argument.) Put $\eta_{ij} = \xi_{ij} = \frac{x_i}{x_j} d\Big(\frac{x_j}{x_i}\Big) = d \log \frac{x_j}{x_i} \in \Omega^1(U_{ij})$; then $\eta_{ij} + \eta_{jk} = \eta_{ik}$, so (η_{ij}) is a cocycle, and $D_{ij}(f) := \eta_{ij} \wedge df$ is an $\mathcal{O}$-derivation $\mathcal{O}_{U_{ij}} \to \omega|_{U_{ij}}$. The maps

$$\varphi_{ij} \colon \mathcal{A}_j|_{U_{ij}} \to \mathcal{A}_i|_{U_{ij}}, \qquad (f, a) \mapsto (f,\, a + D_{ij}(f))$$

are ring isomorphisms ($(fg, fb+ga+D(fg)) = (f, a+Df)(g, b+Dg)$ because $D(fg) = fDg+gDf$ and $\omega^2 = 0$), preserve the ideal $0 \oplus \omega$ (acting as the identity on it) and the quotient $\mathcal{O}$, and satisfy $\varphi_{ij}\varphi_{jk} = \varphi_{ik}$, $\varphi_{ii} = \mathrm{id}$ because $D_{ij} + D_{jk} = D_{ik}$. Gluing gives a scheme X' with $\mathcal{O}_{X'}|_{U_i} \cong \mathcal{A}_i$ (isomorphisms ψ_i, $\psi_i\psi_j^{-1} = \varphi_{ij}$), the same underlying space as X, and an ideal $\omega \subset \mathcal{O}_{X'}$ with $\omega^2 = 0$ and $\mathcal{O}_{X'}/\omega = \mathcal{O}_X$: this is the infinitesimal extension of X by ω with class ξ (Ex. 4.10, II Ex. 8.7). X' is of finite type over k (the rings $\mathcal{O}(U_i) \oplus \omega(U_i)$ are finitely generated k-algebras) and $X'_{\mathrm{red}} = X$. X' is proper over k by the valuative criteria (II, 4.3, 4.7): any morphism from the spectrum of a reduced ring (K a field or R a DVR) to X' factors through $X'_{\mathrm{red}} = X$, where existence and uniqueness of liftings hold since X is proper.

The exact sequence. By Ex. 4.6, with $\mathcal{I} = \omega$, $X_0 = X$, there is an exact sequence of sheaves $0 \to \omega \to \mathcal{O}^*_{X'} \to \mathcal{O}^*_X \to 0$, $a \mapsto 1 + a$, and so $H^1(X, \omega) \to \operatorname{Pic} X' \to \operatorname{Pic} X \xrightarrow{\delta} H^2(X, \omega)$ exact. Here $\omega \cong \mathcal{O}(-3)$, so by Thm. 5.1 $H^1(X, \omega) = 0$ and $H^2(X, \omega) \cong k$. Hence $\operatorname{Pic} X' \to \operatorname{Pic} X$ is injective with image $\ker \delta$. Now $\operatorname{Pic} X = \mathbf{Z} \cdot [\mathcal{O}(1)]$ (II, 6.4) and δ is a homomorphism, so $\delta(\mathcal{O}(m)) = m\, \delta(\mathcal{O}(1))$; as $\operatorname{char} k = 0$ and $H^2(X, \omega) \cong k$, this is nonzero for all $m \neq 0$ as soon as $\delta(\mathcal{O}(1)) \neq 0$. Then $\ker \delta = 0$ and $\operatorname{Pic} X' = 0$.

Computation of $\delta(\mathcal{O}(1))$ by Cech cochains. Use the open affine cover $\mathfrak{U} = (U_i)$. All finite intersections V are affine, and $\mathcal{O}^*_{X'}(V) \to \mathcal{O}^*_X(V)$ is onto (in a chart $V \subseteq U_i$, $\mathcal{O}_{X'}(V) = \mathcal{O}(V) \oplus \omega(V)$ and (g, a) is a unit iff g is). So the sequence of Cech complexes of $0 \to \omega \to \mathcal{O}^*_{X'} \to \mathcal{O}^*_X \to 0$ is exact, and the connecting homomorphism on Cech cohomology is compatible with that of derived-functor cohomology (cf. the proof of Thm. 4.5); moreover $\check{H}^2(\mathfrak{U}, \omega) \cong H^2(X, \omega)$ by Thm. 4.5 (ω quasi-coherent). So it suffices to show that a Cech 2-cocycle representing $\delta(\mathcal{O}(1))$ is not a coboundary.

$\mathcal{O}(1)$ has the generators $e_i = x_i$ on U_i, $e_j = g_{ij}e_i$ with $g_{ij} = x_j/x_i$; $g_{ik} = g_{ij}g_{jk}$. Lift g_{ij} to $\tilde{g}_{ij} \in \mathcal{O}^*_{X'}(U_{ij})$ by $\psi_i(\tilde{g}_{ij}) = (g_{ij}, 0)$. Then $\delta[g]$ is the class of the 2-cocycle $c_{ijk} = \tilde{g}_{ij}\tilde{g}_{jk}\tilde{g}_{ik}^{-1} - 1 \in \omega(U_{ijk})$. Compute in the chart i: $\psi_i(\tilde{g}_{jk}) = \varphi_{ij}(g_{jk}, 0) = (g_{jk}, D_{ij}(g_{jk}))$, so

$$\psi_i\big(\tilde{g}_{ij}\tilde{g}_{jk}\tilde{g}_{ik}^{-1}\big) = (g_{ij}, 0)(g_{jk}, D_{ij}g_{jk})(g_{ik}^{-1}, 0) = \Big(1,\ g_{ij}g_{ik}^{-1}D_{ij}(g_{jk})\Big) = \Big(1,\ \frac{D_{ij}(g_{jk})}{g_{jk}}\Big),$$

so $c_{ijk} = \eta_{ij} \wedge d \log g_{jk} = \eta_{ij} \wedge \eta_{jk}$. For $(i, j, k) = (0, 1, 2)$ put $u = x_1/x_0$, $v = x_2/x_0$, coordinates on U_0, units on U_{012}. Then $\eta_{01} = d \log u$, $\eta_{12} = d \log(v/u) = d \log v - d \log u$, and

$$c_{012} = d \log u \wedge d \log v = u^{-1}v^{-1}\, du \wedge dv \in \omega(U_{012}) = k[u^{\pm 1}, v^{\pm 1}]\, du \wedge dv.$$

(The other convention for $\mathcal{O}(1)$ only changes the sign.) The Cech group $C^2(\mathfrak{U}, \omega) = \omega(U_{012})$ and the coboundaries are $\{\alpha_{12} - \alpha_{02} + \alpha_{01}\}$ with $\alpha_{ij} \in \omega(U_{ij})$. Describe these in the monomial basis $u^a v^b\, du \wedge dv$ of $\omega(U_{012})$:

- $U_{01} = \{u \neq 0\} \subset U_0$: $\omega(U_{01}) = k[u^{\pm 1}, v]\, du \wedge dv$: monomials with $b \geq 0$.

- $U_{02} = \{v \neq 0\} \subset U_0$: $\omega(U_{02}) = k[u, v^{\pm 1}]\, du \wedge dv$: monomials with $a \geq 0$.
- $U_{12} \subset U_1$ with coordinates $s = x_0/x_1 = u^{-1}$, $t = x_2/x_1 = v/u$, $U_{12} = \{t \neq 0\}$: $\omega(U_{12}) = k[s, t^{\pm 1}]\, ds \wedge dt$. Since $ds \wedge dt = -u^{-3} du \wedge dv$, $s^m t^n\, ds \wedge dt = -u^{-m-n-3} v^n\, du \wedge dv$ ($m \geq 0$): monomials with $a + b = -m - 3 \leq -3$.

Each of the three spaces is spanned by monomials, hence so is their sum, which is spanned by the monomials with $b \geq 0$ or $a \geq 0$ or $a + b \leq -3$. The monomial $u^{-1}v^{-1}$ ($a = b = -1$) has $b < 0$, $a < 0$, $a + b = -2 > -3$, so it does not occur. Hence c_{012} is not a coboundary and $\delta(\mathcal{O}(1)) \neq 0$ in $\check{H}^2(\mathfrak{U}, \omega) = H^2(X, \omega)$.

Conclusion. By the above $\operatorname{Pic} X' = 0$, i.e. $\mathcal{O}_{X'}$ is the only invertible sheaf on X'. If X' were projective over k, it would carry a very ample invertible sheaf $\mathcal{O}_{X'}(1) \cong \mathcal{O}_{X'}$ defining a closed immersion $X' \to \mathbf{P}^N_k$ by global sections $s_0, \ldots, s_N \in \Gamma(X', \mathcal{O}_{X'})$. Restricting to $X'_{\mathrm{red}} = X = \mathbf{P}^2$, these become global sections of $\mathcal{O}_X$, i.e. constants, so $X = \mathbf{P}^2 \to \mathbf{P}^N$ would be constant, while it is a closed immersion (composite of closed immersions $X \to X' \to \mathbf{P}^N$), hence injective on points: contradiction. So X' is a proper, two-dimensional, non-projective k-scheme, showing the result of Ex. 5.8 fails in dimension 2. □

Exercise 5.10

Problem. Let X be a projective scheme over a noetherian ring A, and let $\mathcal{F}^1 \to \mathcal{F}^2 \to \cdots \to \mathcal{F}^r$ be an exact sequence of coherent sheaves on X. Show that there is an integer n_0, such that for all $n \geq n_0$, the sequence of global sections

$$\Gamma(X, \mathcal{F}^1(n)) \to \Gamma(X, \mathcal{F}^2(n)) \to \cdots \to \Gamma(X, \mathcal{F}^r(n))$$

is exact.

Solution. Let $d_j \colon \mathcal{F}^j \to \mathcal{F}^{j+1}$ ($1 \leq j \leq r-1$), $\mathcal{Z}^j = \ker d_j$, $\mathcal{B}^{j+1} = \operatorname{im} d_j \subseteq \mathcal{F}^{j+1}$; all are coherent. Exactness at $\mathcal{F}^j$ for $2 \leq j \leq r-1$ means $\mathcal{B}^j = \mathcal{Z}^j$. We have short exact sequences of coherent sheaves

$$0 \to \mathcal{Z}^j \to \mathcal{F}^j \to \mathcal{B}^{j+1} \to 0 \qquad (1 \leq j \leq r-1).$$

By Thm. 5.2(b) applied to the finitely many sheaves $\mathcal{Z}^1, \ldots, \mathcal{Z}^{r-1}$ there is n_0 with $H^1(X, \mathcal{Z}^j(n)) = 0$ for all j and all $n \geq n_0$. Twisting by n (exact) and taking global sections, for $n \geq n_0$ we get exact sequences

$$0 \to \Gamma(\mathcal{Z}^j(n)) \to \Gamma(\mathcal{F}^j(n)) \to \Gamma(\mathcal{B}^{j+1}(n)) \to 0 \qquad (1 \leq j \leq r-1).$$

Fix $2 \leq j \leq r-1$ and $n \geq n_0$. The map $\Gamma(\mathcal{F}^{j-1}(n)) \to \Gamma(\mathcal{F}^j(n))$ is the composite of the surjection $\Gamma(\mathcal{F}^{j-1}(n)) \to \Gamma(\mathcal{B}^j(n)) = \Gamma(\mathcal{Z}^j(n))$ (the sequence for $j-1$) with the inclusion $\Gamma(\mathcal{Z}^j(n)) \hookrightarrow \Gamma(\mathcal{F}^j(n))$; so its image is $\Gamma(\mathcal{Z}^j(n))$. The map $\Gamma(\mathcal{F}^j(n)) \to \Gamma(\mathcal{F}^{j+1}(n))$ is the composite of the surjection onto $\Gamma(\mathcal{B}^{j+1}(n))$ with the injection $\Gamma(\mathcal{B}^{j+1}(n)) \hookrightarrow \Gamma(\mathcal{F}^{j+1}(n))$ (left exactness of Γ), so its kernel is the kernel of $\Gamma(\mathcal{F}^j(n)) \to \Gamma(\mathcal{B}^{j+1}(n))$, which is $\Gamma(\mathcal{Z}^j(n))$. Image equals kernel at every interior spot, so the sequence of global sections is exact for all $n \geq n_0$. □

6. Ext groups and sheaves

Throughout, $\operatorname{Ext}^i(\mathcal{F},\cdot)$ and $\mathcal{E}xt^i(\mathcal{F},\cdot)$ are the right derived functors of $\operatorname{Hom}(\mathcal{F},\cdot)$ and $\mathcal{H}om(\mathcal{F},\cdot)$ on $\mathfrak{Mod}(X)$, as in the text. We use freely: the long exact sequences in the second variable and (Prop. 6.4) in the first variable, which are natural for morphisms of short exact sequences; $\operatorname{Ext}^i(\mathcal{F},\mathcal{I}) = 0 = \mathcal{E}xt^i(\mathcal{F},\mathcal{I})$ for $i > 0$ and $\mathcal{I}$ injective; additivity of Ext^i and $\mathcal{E}xt^i$ in each variable; and, for an affine scheme $\operatorname{Spec} A$ regarded as a ringed space, the module-theoretic Ext^i_A (a ring A is the special case of a one-point space). For a ring A and A-modules the analogue of Prop. 6.4 (long exact sequence in the first variable) holds, with the same proof.

Exercise 6.1

> **Problem.** Let $(X,\mathcal{O}_X)$ be a ringed space, and let $\mathcal{F}',\mathcal{F}'' \in \mathfrak{Mod}(X)$. An *extension* of $\mathcal{F}''$ by $\mathcal{F}'$ is a short exact sequence
>
> $$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$
>
> in $\mathfrak{Mod}(X)$. Two extensions are *isomorphic* if there is an isomorphism of the short exact sequences, inducing the identity maps on $\mathcal{F}'$ and $\mathcal{F}''$. Given an extension as above consider the long exact sequence arising from $\operatorname{Hom}(\mathcal{F}'',\cdot)$, in particular the map
>
> $$\delta\colon \operatorname{Hom}(\mathcal{F}'',\mathcal{F}'') \to \operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}'),$$
>
> and let $\xi \in \operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}')$ be $\delta(1_{\mathcal{F}''})$. Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of $\mathcal{F}''$ by $\mathcal{F}'$, and elements of the group $\operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}')$. For more details, see, e.g., Hilton and Stammbach [1, Ch. III].

Solution. Write $\iota\colon \mathcal{F}' \to \mathcal{F}$ and $p\colon \mathcal{F} \to \mathcal{F}''$ for the maps of an extension E, and let δ_E be its connecting map. Fix once and for all an embedding of $\mathcal{F}'$ in an injective object (Prop. 2.2 of the text; $\mathfrak{Mod}(X)$ has enough injectives): an exact sequence

$$0 \to \mathcal{F}' \xrightarrow{\jmath} \mathcal{I} \xrightarrow{\pi} \mathcal{Q} \to 0, \qquad \mathcal{I} \text{ injective.}$$

Applying $\operatorname{Hom}(\mathcal{F}'',\cdot)$ and using $\operatorname{Ext}^1(\mathcal{F}'',\mathcal{I}) = 0$ we get an exact sequence

$$\operatorname{Hom}(\mathcal{F}'',\mathcal{I}) \xrightarrow{\pi_*} \operatorname{Hom}(\mathcal{F}'',\mathcal{Q}) \xrightarrow{\partial} \operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}') \to 0. \tag{1}$$

So ∂ identifies $\operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}')$ with $\operatorname{Hom}(\mathcal{F}'',\mathcal{Q})/\pi_*\operatorname{Hom}(\mathcal{F}'',\mathcal{I})$.

Step 1: from an extension to a class. Let E be an extension. Since $\mathcal{I}$ is injective, the monomorphism ι and the map $\jmath\colon \mathcal{F}' \to \mathcal{I}$ give $g\colon \mathcal{F} \to \mathcal{I}$ with $g\iota = \jmath$. Then g induces $h\colon \mathcal{F}'' = \mathcal{F}/\mathcal{F}' \to \mathcal{I}/\mathcal{F}' = \mathcal{Q}$ with $hp = \pi g$, and $(1_{\mathcal{F}'}, g, h)$ is a morphism of short exact sequences from E to $E_{\mathcal{I}}\colon 0 \to \mathcal{F}' \to \mathcal{I} \to \mathcal{Q} \to 0$. The connecting maps are natural for morphisms of short exact sequences (this is part of the statement that the long exact sequence is a morphism of δ-functors); for the morphism $(1, g, h)$, with 1 on the first term, this gives the commutative square

$$\delta_E = \partial \circ h_*\colon \operatorname{Hom}(\mathcal{F}'',\mathcal{F}'') \to \operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}'),$$

where $h_*\colon \operatorname{Hom}(\mathcal{F}'',\mathcal{F}'') \to \operatorname{Hom}(\mathcal{F}'',\mathcal{Q})$ is composition with h and $\partial = \delta_{E_{\mathcal{I}}}$. Evaluating at $1_{\mathcal{F}''}$:

$$\xi_E = \delta_E(1_{\mathcal{F}''}) = \partial(h). \tag{2}$$

Another choice g' of the extension of $\jmath$ differs from g by a map $g'-g\colon \mathcal{F}\to\mathcal{I}$ vanishing on $\mathcal{F}'$, hence of the form up with $u\colon \mathcal{F}''\to\mathcal{I}$; then $h'=h+\pi u$, so $h'\equiv h$ modulo $\pi_*\operatorname{Hom}(\mathcal{F}'',\mathcal{I})$, consistent with (2). If $\theta\colon E\to E'$ is an isomorphism of extensions, then $g'\theta$ is an admissible choice of g for E and gives the same h; so $\xi_E=\xi_{E'}$. Thus $E\mapsto\xi_E$ is well defined on isomorphism classes.

Step 2: from h to an extension. Let $h\in\operatorname{Hom}(\mathcal{F}'',\mathcal{Q})$ be arbitrary. Define $\mathcal{F}_h=\operatorname{Ker}(\mathcal{I}\oplus\mathcal{F}''\to\mathcal{Q},\ (a,b)\mapsto\pi(a)-h(b))$ (the fibre product $\mathcal{I}\times_{\mathcal{Q}}\mathcal{F}''$), $\iota_h(a')=(\jmath a',0)$, $p_h(a,b)=b$. The sequence $E_h\colon 0\to\mathcal{F}'\to\mathcal{F}_h\to\mathcal{F}''\to 0$ is exact: ι_h is injective and lands in $\mathcal{F}_h$ since $\pi\jmath=0$; $p_h\iota_h=0$ and $\operatorname{Ker}p_h=\{(a,0):\pi(a)=0\}=\iota_h(\mathcal{F}')$ because $\operatorname{Ker}\pi=\jmath(\mathcal{F}')$ and $\jmath$ is injective; p_h is surjective on stalks: given $b\in\mathcal{F}''_x$, $h(b)\in\mathcal{Q}_x=\pi(\mathcal{I}_x)$, so $h(b)=\pi(a)$ for some a, and $(a,b)\in(\mathcal{F}_h)_x$. For this extension we may take $g=\operatorname{pr}_1\colon\mathcal{F}_h\to\mathcal{I}$ (it satisfies $g\iota_h=\jmath$), and the induced map on quotients is $(a,b)\mapsto\pi(a)=h(b)$, i.e. h itself. By (2), $\xi_{E_h}=\partial(h)$.

Surjectivity. By (1), every element of $\operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}')$ is $\partial(h)$ for some h, hence is ξ_{E_h}.

Injectivity. Suppose $\xi_E=\xi_{E'}$. Choose g,h for E and g',h' for E' as in Step 1. By (2) and (1), $h'-h=\pi u$ for some $u\colon\mathcal{F}''\to\mathcal{I}$. Replacing g by $g+up$ (which still extends $\jmath$, and changes h to $h+\pi u=h'$) we may assume $h=h'$. Now $(g,p)\colon\mathcal{F}\to\mathcal{I}\oplus\mathcal{F}''$ lands in $\mathcal{F}_h$ (because $\pi g=hp$) and gives a morphism of extensions $E\to E_h$ which is the identity on $\mathcal{F}'$ and $\mathcal{F}''$; by the five lemma it is an isomorphism. Likewise $E'\cong E_h$. Hence $E\cong E'$.

Therefore $E\mapsto\xi_E$ is a bijection from isomorphism classes of extensions onto $\operatorname{Ext}^1(\mathcal{F}'',\mathcal{F}')$. □

Exercise 6.2

Problem. Let $X=\mathbf{P}^1_k$, with k an infinite field.

(a) Show that there does not exist a projective object $\mathcal{P}\in\mathfrak{Mod}(X)$, together with a surjective map $\mathcal{P}\to\mathcal{O}_X\to 0$. [*Hint*: Consider surjections of the form $\mathcal{O}_V\to k(x)\to 0$, where $x\in X$ is a closed point, V is an open neighborhood of x, and $\mathcal{O}_V=j_!(\mathcal{O}_X|_V)$, where $j\colon V\to X$ is the inclusion.]

(b) Show that there does not exist a projective object $\mathcal{P}$ in either $\mathfrak{Qco}(X)$ or $\mathfrak{Coh}(X)$ together with a surjection $\mathcal{P}\to\mathcal{O}_X\to 0$. [*Hint*: Consider surjections of the form $\mathcal{L}\to\mathcal{L}\otimes k(x)\to 0$, where $x\in X$ is a closed point, and $\mathcal{L}$ is an invertible sheaf on X.]

Solution. Here $k(x)$ is the skyscraper sheaf at the closed point x with stalk the residue field $\kappa(x)$; it is quasi-coherent and coherent, and $\varepsilon\colon\mathcal{O}_X\to k(x)$, $s\mapsto s(x)$, is a surjection. "Projective" means that $\operatorname{Hom}(\mathcal{P},\cdot)$ is exact on the category in question; in particular maps from $\mathcal{P}$ lift along every surjection of sheaves in that category (a surjective map of sheaves in $\mathfrak{Qco}$ or $\mathfrak{Coh}$ is an epimorphism there). The hypothesis that k is infinite will not be needed: we only use that X is an integral scheme with infinitely many closed points (true for $\mathbf{P}^1_k$ over any field).

(a) Suppose $\pi\colon\mathcal{P}\to\mathcal{O}_X$ is surjective with $\mathcal{P}$ projective in $\mathfrak{Mod}(X)$. Fix a closed point x. Since $\pi_x\colon\mathcal{P}_x\to\mathcal{O}_{X,x}$ is surjective, there are an open neighborhood U of x and $s\in\mathcal{P}(U)$ with $\pi(s)=1|_U$. The set U is the complement of finitely many closed points, so it contains a closed point $y\neq x$. Put $V=U\setminus\{y\}$, an open neighborhood of x, let $j\colon V\to X$ be the inclusion, $\mathcal{O}_V=j_!(\mathcal{O}_X|_V)\subseteq\mathcal{O}_X$ (extension by zero, Ex. 1.19 of Ch. II), with inclusion ι, and let $u=\varepsilon\iota\colon\mathcal{O}_V\to k(x)$. It is surjective: at x the stalk map is $\mathcal{O}_{X,x}\to\kappa(x)$, and at any other point the stalk of $k(x)$ is 0.

The map $f = \varepsilon\pi\colon \mathcal{P} \to k(x)$ lifts, by projectivity, to $\tilde{f}\colon \mathcal{P} \to \mathcal{O}_V$ with $u\tilde{f} = f$. Now $\tilde{f}(s) \in \mathcal{O}_V(U)$ is zero: indeed, $\iota\tilde{f}(s) \in \mathcal{O}_X(U)$ has germ at y equal to the image of $\tilde{f}(s)_y \in (\mathcal{O}_V)_y = 0$ (as $y \notin V$), so $\iota\tilde{f}(s)$ has zero germ at y; since X is integral, $\mathcal{O}_X(U) \to \mathcal{O}_{X,y}$ is injective, hence $\iota\tilde{f}(s) = 0$, and ι is injective, so $\tilde{f}(s) = 0$. Therefore $f(s) = u(\tilde{f}(s)) = 0$. But $f(s) = \varepsilon(\pi(s)) = \varepsilon(1) = 1 \neq 0$ in $k(x)(U) = \kappa(x)$. This contradiction shows no such $\mathcal{P}$ exists.

(b) Let $\mathcal{A}$ be $\mathfrak{Qco}(X)$ or $\mathfrak{Coh}(X)$, suppose $\mathcal{P} \in \mathcal{A}$ is projective in $\mathcal{A}$ and $\pi\colon \mathcal{P} \to \mathcal{O}_X$ is surjective.

Step 1: a coherent piece. There is a coherent subsheaf $\mathcal{C} \subseteq \mathcal{P}$ with $\pi(\mathcal{C}) = \mathcal{O}_X$. If $\mathcal{A} = \mathfrak{Coh}(X)$ take $\mathcal{C} = \mathcal{P}$. If $\mathcal{A} = \mathfrak{Qco}(X)$, $\mathcal{P}$ is the directed union of its coherent subsheaves (Ex. 5.15(e) of Ch. II). The subsheaves $\pi(\mathcal{C}_\alpha)$ of $\mathcal{O}_X$ are quasi-coherent, form a directed family, and their union is $\pi(\mathcal{P}) = \mathcal{O}_X$ (a union of subsheaves is computed on stalks). Cover X by finitely many open affines $U_i = \operatorname{Spec} A_i$ and write $\pi(\mathcal{C}_\alpha)|_{U_i} = \widetilde{I_\alpha^{(i)}}$ with ideals $I_\alpha^{(i)} \subseteq A_i$. The ideals $I_\alpha^{(i)}$ form a directed family in the noetherian ring A_i, so there is a maximal one, $I_{\alpha_i}^{(i)}$; by directedness it contains all the others (for any α pick $\beta \geq \alpha, \alpha_i$; then $I_\beta^{(i)} \supseteq I_{\alpha_i}^{(i)}$ forces equality, so $I_\alpha^{(i)} \subseteq I_{\alpha_i}^{(i)}$). Hence $\widetilde{I_{\alpha_i}^{(i)}} = \mathcal{O}_{U_i}$. Taking $\alpha \geq \alpha_1, \ldots, \alpha_r$ gives $\pi(\mathcal{C}_\alpha) = \mathcal{O}_X$.

Step 2: a global section. Let $\mathcal{K} = \operatorname{Ker}(\pi|_{\mathcal{C}}\colon \mathcal{C} \to \mathcal{O}_X)$, coherent. Twisting by $\mathcal{O}(n)$ is exact: $0 \to \mathcal{K}(n) \to \mathcal{C}(n) \to \mathcal{O}(n) \to 0$. By Thm. 5.2(b) there is $n \geq 0$ with $H^1(X, \mathcal{K}(n)) = 0$, so $\Gamma(X, \mathcal{C}(n)) \to \Gamma(X, \mathcal{O}(n))$ is surjective. Fix a closed point x. As $\mathcal{O}(n)$ ($n \geq 0$) is generated by global sections, there is $\sigma \in \Gamma(X, \mathcal{O}(n))$ with $\sigma(x) \neq 0$ in $\mathcal{O}(n) \otimes k(x)$. Choose $c \in \Gamma(X, \mathcal{C}(n)) \subseteq \Gamma(X, \mathcal{P}(n))$ mapping to σ.

Step 3: twisting is an autoequivalence. The functor $\otimes\mathcal{O}(n)$ is an autoequivalence of $\mathcal{A}$ (with inverse $\otimes\mathcal{O}(-n)$), so $\mathcal{P}(n) = \mathcal{P} \otimes \mathcal{O}(n)$ is projective in $\mathcal{A}$ and $\pi(n)\colon \mathcal{P}(n) \to \mathcal{O}(n)$ is surjective.

Step 4: contradiction. Let $\mathcal{L} = \mathcal{O}(-1)$; then $\Gamma(X, \mathcal{L}) = 0$ (Thm. 5.1). Choose an isomorphism $\theta\colon \mathcal{O}(n) \otimes k(x) \to \mathcal{L} \otimes k(x)$ (both are skyscraper sheaves at x with one-dimensional $\kappa(x)$-vector space as stalk). The map $f = \theta \circ (\mathcal{O}(n) \to \mathcal{O}(n) \otimes k(x)) \circ \pi(n)\colon \mathcal{P}(n) \to \mathcal{L} \otimes k(x)$ lifts, by projectivity of $\mathcal{P}(n)$ and surjectivity of $q\colon \mathcal{L} \to \mathcal{L} \otimes k(x)$, to $\tilde{f}\colon \mathcal{P}(n) \to \mathcal{L}$ with $q\tilde{f} = f$. Then $\tilde{f}(c) \in \Gamma(X, \mathcal{L}) = 0$, so $f(c) = q(\tilde{f}(c)) = 0$. But $f(c) = \theta(\sigma(x)) \neq 0$. Contradiction. So no such $\mathcal{P}$ exists in $\mathfrak{Qco}(X)$ or in $\mathfrak{Coh}(X)$. □

Exercise 6.3

Problem. Let X be a noetherian scheme, and let $\mathcal{F}, \mathcal{G} \in \mathfrak{Mod}(X)$.

(a) If $\mathcal{F}, \mathcal{G}$ are both coherent, then $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is coherent, for all $i \geq 0$.

(b) If $\mathcal{F}$ is coherent and $\mathcal{G}$ is quasi-coherent, then $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is quasi-coherent, for all $i \geq 0$.

Solution. Being coherent (resp. quasi-coherent) is a local property, and $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})|_U \cong \mathcal{E}xt^i_U(\mathcal{F}|_U, \mathcal{G}|_U)$ for U open (Prop. 6.2). Since X is noetherian it is covered by finitely many open affine noetherian subschemes, so we may assume $X = \operatorname{Spec} A$ with A noetherian. Then $\mathcal{F} = \widetilde{M}$ with M finitely generated; since A is noetherian there is a resolution $\cdots \to L_1 \to L_0 \to M \to 0$ by finitely generated free A-modules, and by exactness of $\widetilde{\ }$ (Ch. II, Prop. 5.2) the sheaves $\widetilde{L_j} \cong \mathcal{O}_X^{r_j}$ form a locally free resolution of $\mathcal{F}$ of finite rank. By Prop. 6.5, $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \cong h^i(\mathcal{H}om(\widetilde{L}_\bullet, \mathcal{G}))$. Now $\mathcal{H}om(\widetilde{L_j}, \mathcal{G}) \cong \mathcal{G}^{r_j}$, which is coherent (resp. quasi-coherent) when $\mathcal{G}$ is. The differentials are

$\mathcal{O}_X$-linear maps between such sheaves, and kernels, images and cokernels of morphisms of coherent (resp. quasi-coherent) sheaves on a noetherian scheme are coherent (resp. quasi-coherent) (Ch. II, Prop. 5.7). Hence $h^i = \operatorname{Ker} d^i / \operatorname{Im} d^{i-1}$ is coherent (resp. quasi-coherent). □

Exercise 6.4

> **Problem.** Let X be a noetherian scheme, and suppose that every coherent sheaf on X is a quotient of a locally free sheaf. In this case we say $\mathfrak{Coh}(X)$ has *enough locally frees*. Then for any $\mathcal{G} \in \mathfrak{Mod}(X)$, show that the δ-functor $(\mathcal{E}xt^i(\cdot, \mathcal{G}))_{i \geq 0}$ from $\mathfrak{Coh}(X)$ to $\mathfrak{Mod}(X)$ is a contravariant universal δ-functor. [*Hint*: Show $\mathcal{E}xt^i(\cdot, \mathcal{G})$ is coeffaceable (Sect. 1) for $i > 0$.]

Solution. (Here "locally free" means locally free of finite rank, as in Prop. 6.5.)

It is a contravariant δ-functor on $\mathfrak{Coh}(X)$. A short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ in $\mathfrak{Coh}(X)$ is one in $\mathfrak{Mod}(X)$, so by Prop. 6.4 it gives a long exact sequence

$$0 \to \mathcal{H}om(\mathcal{F}'', \mathcal{G}) \to \mathcal{H}om(\mathcal{F}, \mathcal{G}) \to \mathcal{H}om(\mathcal{F}', \mathcal{G}) \to \mathcal{E}xt^1(\mathcal{F}'', \mathcal{G}) \to \cdots,$$

obtained from the short exact sequence of complexes $0 \to \mathcal{H}om(\mathcal{F}'', \mathcal{I}^\bullet) \to \mathcal{H}om(\mathcal{F}, \mathcal{I}^\bullet) \to \mathcal{H}om(\mathcal{F}', \mathcal{I}^\bullet) \to 0$ for an injective resolution $\mathcal{G} \to \mathcal{I}^\bullet$ (exact by 6.1, as in the proof of 6.4). This construction is functorial in the short exact sequence, so the connecting maps are natural; thus $(\mathcal{E}xt^i(\cdot, \mathcal{G}))_i$ is a contravariant δ-functor.

Vanishing on locally free sheaves. Let $\mathcal{E}$ be locally free of finite rank and $i > 0$. By Prop. 6.7 (with $\mathcal{F} = \mathcal{O}_X$, $\mathcal{L} = \mathcal{E}$), $\mathcal{E}xt^i(\mathcal{E}, \mathcal{G}) = \mathcal{E}xt^i(\mathcal{O}_X \otimes \mathcal{E}, \mathcal{G}) \cong \mathcal{E}xt^i(\mathcal{O}_X, \mathcal{E}^\vee \otimes \mathcal{G}) = 0$ by Prop. 6.3(b).

Coeffaceability. Let $\mathcal{F} \in \mathfrak{Coh}(X)$ and $i > 0$. By hypothesis there is a surjection $u\colon \mathcal{E} \to \mathcal{F}$ with $\mathcal{E}$ locally free, and $\mathcal{E} \in \mathfrak{Coh}(X)$. Then $\mathcal{E}xt^i(u, \mathcal{G})\colon \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \to \mathcal{E}xt^i(\mathcal{E}, \mathcal{G}) = 0$ is zero. So the contravariant δ-functor is coeffaceable for $i > 0$.

By the dual form of Thm. 1.3A (a contravariant δ-functor that is coeffaceable in all degrees $i > 0$ is universal; the proof is the same as for the covariant, effaceable case, with arrows reversed), $(\mathcal{E}xt^i(\cdot, \mathcal{G}))_{i \geq 0}$ is a universal δ-functor on $\mathfrak{Coh}(X)$: it is the universal contravariant δ-functor extending $\mathcal{F} \mapsto \mathcal{H}om(\mathcal{F}, \mathcal{G})$. □

Exercise 6.5

> **Problem.** Let X be a noetherian scheme, and assume that $\mathfrak{Coh}(X)$ has enough locally frees (Ex. 6.4). Then for any coherent sheaf $\mathcal{F}$ we define the *homological dimension* of $\mathcal{F}$, denoted $\operatorname{hd}(\mathcal{F})$, to be the least length of a locally free resolution of $\mathcal{F}$ (or $+\infty$ if there is no finite one). Show:
>
> (a) $\mathcal{F}$ is locally free $\Leftrightarrow \mathcal{E}xt^1(\mathcal{F}, \mathcal{G}) = 0$ for all $\mathcal{G} \in \mathfrak{Mod}(X)$;
>
> (b) $\operatorname{hd}(\mathcal{F}) \leq n \Leftrightarrow \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$ for all $i > n$ and all $\mathcal{G} \in \mathfrak{Mod}(X)$;
>
> (c) $\operatorname{hd}(\mathcal{F}) = \sup_{x \in X} \operatorname{pd}_{\mathcal{O}_x} \mathcal{F}_x$.

Solution. (Resolutions are by locally free sheaves of finite rank, as in Prop. 6.5.) Since $\mathfrak{Coh}(X)$ has enough locally frees, every coherent $\mathcal{F}$ has a locally free resolution $\mathcal{E}_\bullet \to \mathcal{F} \to 0$: choose

$\mathcal{E}_0 \to \mathcal{F}$ onto; the kernel is coherent, and repeat. Put $\mathcal{Z}_0 = \mathcal{F}$ and $\mathcal{Z}_{j+1} = \operatorname{Ker}(\mathcal{E}_j \to \mathcal{Z}_j)$, coherent sheaves with exact sequences $0 \to \mathcal{Z}_{j+1} \to \mathcal{E}_j \to \mathcal{Z}_j \to 0$.

Preliminary 1. *If $\mathcal{K}$ is coherent and $\mathcal{K}_x$ is a free $\mathcal{O}_x$-module, then $\mathcal{K}$ is free on a neighborhood of x.* Let $r = \operatorname{rank} \mathcal{K}_x$; choose generators of $\mathcal{K}_x$ lifting to sections of $\mathcal{K}$ on an open $W \ni x$, giving $\varphi\colon \mathcal{O}_W^r \to \mathcal{K}|_W$ that is an isomorphism at x. Kernel and cokernel are coherent with zero stalk at x; the support of a coherent sheaf is closed (Ch. II, Ex. 5.6), so after shrinking W both vanish. In particular $\{x : \mathcal{K}_x \text{ free}\}$ is open.

Preliminary 2 (dimension shifting). For $\mathcal{G} \in \mathfrak{Mod}(X)$, $i \geq 1$ and $j \geq 0$, the long exact sequence of $\mathcal{E}xt(\cdot, \mathcal{G})$ for $0 \to \mathcal{Z}_{j+1} \to \mathcal{E}_j \to \mathcal{Z}_j \to 0$ together with $\mathcal{E}xt^q(\mathcal{E}_j, \mathcal{G}) = 0$ for $q > 0$ (Ex. 6.4) gives $\mathcal{E}xt^i(\mathcal{Z}_{j+1}, \mathcal{G}) \cong \mathcal{E}xt^{i+1}(\mathcal{Z}_j, \mathcal{G})$. Hence

$$\mathcal{E}xt^1(\mathcal{Z}_n, \mathcal{G}) \cong \mathcal{E}xt^{n+1}(\mathcal{F}, \mathcal{G}) \qquad (n \geq 0), \tag{1}$$

and more generally $\mathcal{E}xt^i(\mathcal{Z}_n, \mathcal{G}) \cong \mathcal{E}xt^{n+i}(\mathcal{F}, \mathcal{G})$ for $i \geq 1$. The same holds verbatim for modules over a ring (free modules have vanishing higher Ext).

(a) ($\Rightarrow$) This was shown in Ex. 6.4: $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$ for $i > 0$ if $\mathcal{F}$ is locally free.

($\Leftarrow$) Suppose $\mathcal{E}xt^1(\mathcal{F}, \mathcal{G}) = 0$ for all $\mathcal{G}$. Let $x \in X$ and let N be any $\mathcal{O}_x$-module. Let $\mathcal{G} = i_*N$ be the skyscraper sheaf: $\mathcal{G}(U) = N$ if $x \in U$ and 0 otherwise; it is an $\mathcal{O}_X$-module via $\mathcal{O}_X(U) \to \mathcal{O}_x$, and $\mathcal{G}_x = N$. By Prop. 6.8 (X noetherian, $\mathcal{F}$ coherent), $\operatorname{Ext}^1_{\mathcal{O}_x}(\mathcal{F}_x, N) = \mathcal{E}xt^1(\mathcal{F}, \mathcal{G})_x = 0$. As N was arbitrary, $\mathcal{F}_x$ is projective (Prop. 6.10A(a)); it is finitely generated over a local ring, hence free. By Preliminary 1, $\mathcal{F}$ is locally free near each x.

(b) ($\Rightarrow$) Let $0 \to \mathcal{E}_n \to \cdots \to \mathcal{E}_0 \to \mathcal{F} \to 0$ be a locally free resolution of length $\leq n$ (pad with zeros). By Prop. 6.5, $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = h^i(\mathcal{H}om(\mathcal{E}_\bullet, \mathcal{G})) = 0$ for $i > n$, since $\mathcal{H}om(\mathcal{E}_j, \mathcal{G}) = 0$ for $j > n$.

($\Leftarrow$) Assume $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$ for all $i > n$ and all $\mathcal{G}$. Take any locally free resolution $\mathcal{E}_\bullet$ and the sheaves $\mathcal{Z}_j$ above. By (1), $\mathcal{E}xt^1(\mathcal{Z}_n, \mathcal{G}) \cong \mathcal{E}xt^{n+1}(\mathcal{F}, \mathcal{G}) = 0$ for all $\mathcal{G}$, and $\mathcal{Z}_n$ is coherent, so $\mathcal{Z}_n$ is locally free by (a). Hence $0 \to \mathcal{Z}_n \to \mathcal{E}_{n-1} \to \cdots \to \mathcal{E}_0 \to \mathcal{F} \to 0$ (for $n = 0$: $\mathcal{Z}_0 = \mathcal{F}$ is locally free) is a locally free resolution of length n, so $\operatorname{hd}(\mathcal{F}) \leq n$.

(c) Put $m = \sup_x \operatorname{pd}_{\mathcal{O}_x} \mathcal{F}_x$ and $h = \operatorname{hd}(\mathcal{F})$.

$m \leq h$: if $h < \infty$ take a locally free resolution of length h. Taking stalks (exact) gives a resolution of $\mathcal{F}_x$ by free $\mathcal{O}_x$-modules of finite rank, of length h; so $\operatorname{pd} \mathcal{F}_x \leq h$ for all x.

$h \leq m$: we may assume $m < \infty$. For each x and $i > m$, $\operatorname{Ext}^i_{\mathcal{O}_x}(\mathcal{F}_x, N) = 0$ for every $\mathcal{O}_x$-module N (Prop. 6.10A(b)). By Prop. 6.8, $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})_x = \operatorname{Ext}^i_{\mathcal{O}_x}(\mathcal{F}_x, \mathcal{G}_x) = 0$ for every $\mathcal{G} \in \mathfrak{Mod}(X)$, every x and every $i > m$. So $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$ for $i > m$, whence $h \leq m$ by (b). □

Exercise 6.6

Problem. Let A be a regular local ring, and let M be a finitely generated A-module. In this case, strengthen the result (6.10A) as follows.

(a) M is projective if and only if $\operatorname{Ext}^i(M, A) = 0$ for all $i > 0$. [*Hint*: Use (6.11A) and descending induction on i to show that $\operatorname{Ext}^i(M, N) = 0$ for all $i > 0$ and all finitely generated A-modules N. Then show M is a direct summand of a free A-module (Matsumura [2, p. 129]).]

(b) Use (a) to show that for any n, $\operatorname{pd} M \le n$ if and only if $\operatorname{Ext}^i(M, A) = 0$ for all $i > n$.

Solution. Let $d = \dim A$. All modules are over A, which is noetherian, so kernels of maps between finitely generated modules are finitely generated.

(a) ($\Rightarrow$) If M is projective then $\operatorname{pd} M = 0$, so $\operatorname{Ext}^i(M, N) = 0$ for $i > 0$ and all N by Prop. 6.10A(b).

($\Leftarrow$) Assume $\operatorname{Ext}^i(M, A) = 0$ for all $i > 0$; hence $\operatorname{Ext}^i(M, A^r) = \operatorname{Ext}^i(M, A)^r = 0$ for all r. By Prop. 6.11A(a), $\operatorname{pd} M \le d$, so $\operatorname{Ext}^i(M, N) = 0$ for all $i > d$ and all N (Prop. 6.10A(b)).

Claim: $\operatorname{Ext}^i(M, N) = 0$ for all $i \ge 1$ and all finitely generated N. We prove it by descending induction on i; it holds for $i > d$. Suppose it holds for $i + 1$ (where $i \ge 1$) and let N be finitely generated. Choose an exact sequence $0 \to N' \to A^r \to N \to 0$ with N' finitely generated. The exact sequence

$$\operatorname{Ext}^i(M, A^r) \to \operatorname{Ext}^i(M, N) \to \operatorname{Ext}^{i+1}(M, N')$$

has both outer terms 0 (the left one by hypothesis, the right one by induction), so $\operatorname{Ext}^i(M, N) = 0$. This proves the claim.

Now choose a surjection $q\colon A^s \to M$ with kernel K, finitely generated. In the exact sequence $\operatorname{Hom}(M, A^s) \to \operatorname{Hom}(M, M) \to \operatorname{Ext}^1(M, K) = 0$ the identity of M lifts to $\sigma\colon M \to A^s$ with $q\sigma = 1_M$. So M is a direct summand of the free module A^s, hence projective.

(b) ($\Rightarrow$) If $\operatorname{pd} M \le n$, then $\operatorname{Ext}^i(M, A) = 0$ for $i > n$ by Prop. 6.10A(b).

($\Leftarrow$) Suppose $\operatorname{Ext}^i(M, A) = 0$ for $i > n$. Take a resolution $\cdots \to F_1 \to F_0 \to M \to 0$ by finitely generated free modules, $K_0 = M$, $K_{j+1} = \operatorname{Ker}(F_j \to K_j)$ (finitely generated). Since $\operatorname{Ext}^q(F_j, N) = 0$ for $q > 0$, the sequences $0 \to K_{j+1} \to F_j \to K_j \to 0$ give $\operatorname{Ext}^i(K_{j+1}, N) \cong \operatorname{Ext}^{i+1}(K_j, N)$ for $i \ge 1$ (Prop. 6.4 for modules). Hence $\operatorname{Ext}^i(K_n, A) \cong \operatorname{Ext}^{i+n}(M, A) = 0$ for $i \ge 1$. By (a), K_n is projective. Then $0 \to K_n \to F_{n-1} \to \cdots \to F_0 \to M \to 0$ is a projective resolution of length n (for $n = 0$ this says $M = K_0$ projective), so $\operatorname{pd} M \le n$. □

Exercise 6.7

Problem. Let $X = \operatorname{Spec} A$ be an affine noetherian scheme. Let M, N be A-modules, with M finitely generated. Then

$$\operatorname{Ext}^i_X(\widetilde{M}, \widetilde{N}) \cong \operatorname{Ext}^i_A(M, N)$$

and

$$\mathcal{E}xt^i_X(\widetilde{M}, \widetilde{N}) \cong \operatorname{Ext}^i_A(M, N)^{\sim}.$$

Solution. Since A is noetherian and M finitely generated, choose a resolution $L_\bullet \to M \to 0$ by finitely generated free A-modules. As $\sim$ is exact (Ch. II, Prop. 5.2), $\widetilde{L}_\bullet \to \widetilde{M} \to 0$ is a locally free resolution with $\widetilde{L_j} \cong \mathcal{O}_X^{r_j}$. Also, by Ch. II, Cor. 5.5, $\sim$ is fully faithful on modules: $\operatorname{Hom}_X(\widetilde{P}, \widetilde{N}) = \operatorname{Hom}_A(P, N)$, compatibly with composition. For the module Ext, Prop. 6.5 applied to the one-point space with ring A gives $\operatorname{Ext}^i_A(M, N) = h^i(\operatorname{Hom}_A(L_\bullet, N))$.

Sheaf Ext. By Prop. 6.5, $\mathcal{E}xt^i_X(\widetilde{M}, \widetilde{N}) = h^i(\mathcal{H}om(\widetilde{L}_\bullet, \widetilde{N}))$. For L finitely generated free, the natural map $\operatorname{Hom}_A(L, N)^{\sim} \to \mathcal{H}om(\widetilde{L}, \widetilde{N})$ (on $D(g)$ it is $\operatorname{Hom}_A(L, N)_g = \operatorname{Hom}_{A_g}(L_g, N_g) =$

$\mathrm{Hom}(\widetilde{L}|_{D(g)}, \widetilde{N}|_{D(g)}))$ is an isomorphism (both sides are $\widetilde{N}^{\,r}$) and is functorial in L for maps between finite free modules. Hence $\mathcal{H}om(\widetilde{L}_\bullet, \widetilde{N}) \cong \mathrm{Hom}_A(L_\bullet, N)^\sim$ as complexes, and since $\sim$ is exact it commutes with taking cohomology: $\mathcal{E}xt^i_X(\widetilde{M}, \widetilde{N}) \cong h^i(\mathrm{Hom}_A(L_\bullet, N))^\sim = \mathrm{Ext}^i_A(M, N)^\sim$.

Global Ext. Put $\mathcal{G} = \widetilde{N}$. First, for $j \geq 0$ and $i \geq 1$,

$$\mathrm{Ext}^i_X(\widetilde{L_j}, \mathcal{G}) \cong \mathrm{Ext}^i_X(\mathcal{O}_X, \mathcal{G})^{r_j} \cong H^i(X, \mathcal{G})^{r_j} = 0, \tag{1}$$

by additivity, Prop. 6.3(c), and Thm. 3.5 (X affine noetherian, $\mathcal{G}$ quasi-coherent).

Let $K_0 = \widetilde{M}$ and $K_{j+1} = \mathrm{Ker}(\widetilde{L_j} \to K_j)$, so $0 \to K_{j+1} \to \widetilde{L_j} \to K_j \to 0$ is exact. By (1) and the long exact sequence in the first variable (Prop. 6.4), for $i \geq 1$ we get $\mathrm{Ext}^i(K_{j+1}, \mathcal{G}) \cong \mathrm{Ext}^{i+1}(K_j, \mathcal{G})$, and, from the beginning of the sequence,

$$\mathrm{Ext}^1(K_j, \mathcal{G}) \cong \mathrm{Coker}\left(\mathrm{Hom}(\widetilde{L_j}, \mathcal{G}) \to \mathrm{Hom}(K_{j+1}, \mathcal{G})\right).$$

Hence for $i \geq 1$: $\mathrm{Ext}^i(\widetilde{M}, \mathcal{G}) \cong \mathrm{Ext}^1(K_{i-1}, \mathcal{G}) \cong \mathrm{Coker}\left(\mathrm{Hom}(\widetilde{L_{i-1}}, \mathcal{G}) \to \mathrm{Hom}(K_i, \mathcal{G})\right)$. Now $\widetilde{L}_i \to K_i$ is onto, and K_{i+1} is the image of $\widetilde{L_{i+1}}$, so $\mathrm{Hom}(K_i, \mathcal{G})$ is the subgroup of $\mathrm{Hom}(\widetilde{L}_i, \mathcal{G})$ of maps killing K_{i+1}, i.e. $\mathrm{Ker}(d^i\colon \mathrm{Hom}(\widetilde{L}_i, \mathcal{G}) \to \mathrm{Hom}(\widetilde{L_{i+1}}, \mathcal{G}))$; and the image of $\mathrm{Hom}(\widetilde{L_{i-1}}, \mathcal{G}) \to \mathrm{Hom}(K_i, \mathcal{G})$ is the image of d^{i-1}. Therefore

$$\mathrm{Ext}^i_X(\widetilde{M}, \widetilde{N}) \cong h^i(\mathrm{Hom}_X(\widetilde{L}_\bullet, \widetilde{N})) = h^i(\mathrm{Hom}_A(L_\bullet, N)) = \mathrm{Ext}^i_A(M, N) \qquad (i \geq 1).$$

For $i = 0$ this is the full faithfulness $\mathrm{Hom}_X(\widetilde{M}, \widetilde{N}) = \mathrm{Hom}_A(M, N)$. □

Exercise 6.8

Problem. Prove the following theorem of Kleiman (see Borelli [1]): if X is a noetherian, integral, separated, locally factorial scheme, then every coherent sheaf on X is a quotient of a locally free sheaf (of finite rank).

(a) First show that open sets of the form X_s, for various $s \in \Gamma(X, \mathcal{L})$, and various invertible sheaves $\mathcal{L}$ on X, form a base for the topology of X. [*Hint*: Given a closed point $x \in X$ and an open neighborhood U of x, to show there is an $\mathcal{L}, s$ such that $x \in X_s \subseteq U$, first reduce to the case that $Z = X - U$ is irreducible. Then let ζ be the generic point of Z. Let $f \in K(X)$ be a rational function with $f \in \mathcal{O}_x$, $f \notin \mathcal{O}_\zeta$. Let $D = (f)_\infty$, and let $\mathcal{L} = \mathcal{L}(D)$, $s \in \Gamma(X, \mathcal{L}(D))$ correspond to D (II, Sect. 6).]

(b) Now use (II, 5.14) to show that any coherent sheaf is a quotient of a direct sum $\bigoplus \mathcal{L}_i^{\otimes n_i}$ for various invertible sheaves $\mathcal{L}_i$ and various integers n_i.

Solution. Recall $X_s = \{z \in X : s_z \notin \mathfrak{m}_z \mathcal{L}_z\}$, the open set where s does not vanish. We have $X_{s \otimes t} = X_s \cap X_t$ for $s \in \Gamma(\mathcal{L})$, $t \in \Gamma(\mathcal{L}')$. Let $K = K(X)$. *Reading of the hint*: the printed hint says "closed point x"; I will prove the statement for an arbitrary point y in place of x, which is what a base of the topology requires (a non-closed point of U need not be handled by the closed-point case).

Facts used. (i) X is locally factorial, so each local ring $\mathcal{O}_z$ is a UFD, hence a noetherian normal domain, and it is the intersection of its localizations at height one primes (Matsumura, Thm. 11.5). The local ring of a prime divisor Y (closure of a point η_Y of codimension one) is

a one-dimensional UFD, hence a DVR, with valuation ν_Y. (ii) Every Weil divisor D on X is locally principal, and $\mathcal{L}(D) \subseteq \mathcal{K}$, $\mathcal{L}(D)(W) = \{g \in K : (g)|_W + D|_W \geq 0\}$ is invertible (Ch. II, Prop. 6.11 and 6.13). If $D = \sum n_Y Y$ is effective, then $1 \in \Gamma(X, \mathcal{L}(D))$ is a section s_D with $X_{s_D} = X \setminus \operatorname{Supp} D$. Indeed, near a point z choose an open $W \ni z$ and $g \in K$ with $D|_W = (g)|_W$; since $D \geq 0$ and X is normal, $g \in \mathcal{O}(W)$, and multiplication by g gives $\mathcal{L}(D)|_W \cong \mathcal{O}_W$ taking s_D to g. So $z \in X_{s_D}$ iff g is a unit in $\mathcal{O}_z$. As $\mathcal{O}_z$ is a UFD, g is a non-unit iff it is divisible by a prime element, i.e. iff $\nu_Y(g) = n_Y > 0$ for some prime divisor $Y \ni z$ (the prime divisors through z correspond to the height one primes of $\mathcal{O}_z$). That is, iff $z \in \operatorname{Supp} D$.

(a) Let $y \in X$ and $U \ni y$ open. We find $\mathcal{L}$ and s with $y \in X_s \subseteq U$. Let $Z = X \setminus U = Z_1 \cup \cdots \cup Z_m$ be the decomposition into irreducible components. If for each ℓ we find $(\mathcal{L}_\ell, s_\ell)$ with $y \in X_{s_\ell} \subseteq X \setminus Z_\ell$, then $\mathcal{L} = \bigotimes \mathcal{L}_\ell$, $s = \bigotimes s_\ell$ satisfy $y \in X_s = \bigcap_\ell X_{s_\ell} \subseteq \bigcap_\ell (X \setminus Z_\ell) = U$. (If $Z = \emptyset$ take $\mathcal{L} = \mathcal{O}_X$, $s = 1$.) So we may assume Z irreducible, with generic point ζ; $y \notin Z$.

Claim 1: $\mathcal{O}_y \not\subseteq \mathcal{O}_\zeta$ in K. Suppose $\mathcal{O}_y \subseteq \mathcal{O}_\zeta$. Consider the two morphisms $\operatorname{Spec} \mathcal{O}_\zeta \to X$: the canonical one ϕ, and ψ, the composite of $\operatorname{Spec} \mathcal{O}_\zeta \to \operatorname{Spec} \mathcal{O}_y$ (from the inclusion) with the canonical $\operatorname{Spec} \mathcal{O}_y \to X$. Both restrict on the dense open $\operatorname{Spec} K \subseteq \operatorname{Spec} \mathcal{O}_\zeta$ to the canonical morphism to the generic point of X. As $\operatorname{Spec} \mathcal{O}_\zeta$ is reduced and X is separated, $\phi = \psi$ (Ch. II, Ex. 4.2). The closed point of $\operatorname{Spec} \mathcal{O}_\zeta$ goes to ζ under ϕ, and under ψ to a point of the image of $\operatorname{Spec} \mathcal{O}_y$, i.e. to a generization of y. Thus ζ is a generization of y, i.e. $y \in \overline{\{\zeta\}} = Z$, a contradiction.

Choose $f \in \mathcal{O}_y \setminus \mathcal{O}_\zeta$, and let $D = (f)_\infty = \sum_{\nu_Y(f)<0} (-\nu_Y(f))Y$ (the polar part of the Weil divisor (f)), an effective divisor.

$y \notin \operatorname{Supp} D$: if $y \in Y$ with $\nu_Y(f) < 0$, then $\mathcal{O}_{\eta_Y}$ is a localization of $\mathcal{O}_y$ (as η_Y is a generization of y), so $f \in \mathcal{O}_y \subseteq \mathcal{O}_{\eta_Y}$ and $\nu_Y(f) \geq 0$, a contradiction.

$Z \subseteq \operatorname{Supp} D$: as $f \notin \mathcal{O}_\zeta = \bigcap_{\mathfrak{p}} (\mathcal{O}_\zeta)_{\mathfrak{p}}$ (over height one primes $\mathfrak{p}$ of $\mathcal{O}_\zeta$), there is such $\mathfrak{p}$ with $f \notin (\mathcal{O}_\zeta)_{\mathfrak{p}}$. This ring is the local ring $\mathcal{O}_{\eta_Y}$ of a prime divisor Y through ζ, a DVR in which $f \notin$, so $\nu_Y(f) < 0$ and $\zeta \in Y \subseteq \operatorname{Supp} D$. Since $\operatorname{Supp} D$ is closed, $Z = \overline{\{\zeta\}} \subseteq \operatorname{Supp} D$.

With $\mathcal{L} = \mathcal{L}(D)$ and $s = s_D$ we get from (ii): $y \in X_s = X \setminus \operatorname{Supp} D \subseteq X \setminus Z = U$. This proves (a).

(b) *The sets X_s used are affine.* By (a), each $y \in X$ has an open affine neighborhood U_y and $(\mathcal{L}_y, s_y)$ with $y \in X_{s_y} \subseteq U_y$. Then X_{s_y} is affine: cover $U_y = \operatorname{Spec} B$ by finitely many $D(g_\ell)$ on which $\mathcal{L}_y$ is trivial; there s_y is a function h_ℓ and $X_{s_y} \cap D(g_\ell) = D(g_\ell h_\ell)$ is affine. So the open immersion $X_{s_y} \to U_y$ is an affine morphism, hence X_{s_y} is affine. As X is noetherian, hence quasi-compact, finitely many X_{s_i}, $i = 1, \ldots, r$ (with $s_i \in \Gamma(X, \mathcal{L}_i)$) cover X.

Let $\mathcal{F}$ be coherent on X. Since X_{s_i} is affine and $\mathcal{F}|_{X_{s_i}}$ coherent, it is generated by finitely many sections $t_{ij} \in \Gamma(X_{s_i}, \mathcal{F})$, $j = 1, \ldots, r_i$. The hypotheses of Lemma 5.14(b) of Ch. II hold (X noetherian: take a finite affine cover on whose members $\mathcal{L}_i$ is free; intersections are quasi-compact). Hence there are integers $n_{ij} > 0$ and global sections $u_{ij} \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}_i^{\otimes n_{ij}})$ whose restriction to X_{s_i} is $t_{ij} \otimes s_i^{n_{ij}}$. Let $\varphi_{ij} \colon \mathcal{L}_i^{\otimes -n_{ij}} \to \mathcal{F}$ be the morphism corresponding to u_{ij} under $\mathcal{F} \otimes \mathcal{L}_i^{\otimes n} = \mathcal{H}om(\mathcal{L}_i^{\otimes -n}, \mathcal{F})$. On X_{s_i} the section s_i trivializes $\mathcal{L}_i$, so $\mathcal{L}_i^{\otimes -n_{ij}}|_{X_{s_i}}$ is generated by the dual generator $s_i^{-n_{ij}}$, and $\varphi_{ij}(s_i^{-n_{ij}}) = t_{ij}$. Thus on X_{s_i} the image of $\bigoplus_j \varphi_{ij}$ contains all t_{ij},

hence is all of $\mathcal{F}|_{X_{s_i}}$. Consequently

$$\Phi = \bigoplus_{i,j} \varphi_{ij} \colon \bigoplus_{i=1}^{r} \bigoplus_{j=1}^{r_i} \mathcal{L}_i^{\otimes -n_{ij}} \to \mathcal{F}$$

is surjective on each X_{s_i}, hence surjective. The source is a finite direct sum of invertible sheaves, so is locally free of finite rank. This proves (b), and with it Kleiman's theorem. □

Exercise 6.9

Problem. Let X be a noetherian, integral, separated, regular scheme. (We say a scheme is *regular* if all of its local rings are regular local rings.) Recall the definition of the Grothendieck group $K(X)$ from (II, Ex. 6.10). We define similarly another group $K_1(X)$ using locally free sheaves: it is the quotient of the free abelian group generated by all locally free (coherent) sheaves, by the subgroup generated by all expressions of the form $\mathcal{E} - \mathcal{E}' - \mathcal{E}''$, whenever $0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$ is a short exact sequence of locally free sheaves. Clearly there is a natural group homomorphism $\varepsilon \colon K_1(X) \to K(X)$. Show that ε is an isomorphism (Borel and Serre [1, Sect. 4]) as follows.

(a) Given a coherent sheaf $\mathcal{F}$, use (Ex. 6.8) to show that it has a locally free resolution $\mathcal{E}_\bullet \to \mathcal{F} \to 0$. Then use (6.11A) and (Ex. 6.5) to show that it has a finite locally free resolution

$$0 \to \mathcal{E}_n \to \cdots \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathcal{F} \to 0.$$

(b) For each $\mathcal{F}$, choose a finite locally free resolution $\mathcal{E}_\bullet \to \mathcal{F} \to 0$, and let $\delta(\mathcal{F}) = \sum(-1)^i \gamma_1(\mathcal{E}_i)$ in $K_1(X)$. Show that $\delta(\mathcal{F})$ is independent of the resolution chosen, that it defines a homomorphism of $K(X)$ to $K_1(X)$, and finally that it is an inverse to ε.

Solution. *Conventions.* Locally free means locally free of finite rank. $\gamma(\mathcal{F}) \in K(X)$ and $\gamma_1(\mathcal{E}) \in K_1(X)$ denote classes; $K(X)$ is generated by the classes of coherent sheaves, with relations $\gamma(\mathcal{F}) = \gamma(\mathcal{F}') + \gamma(\mathcal{F}'')$ for short exact sequences, and $\varepsilon\gamma_1(\mathcal{E}) = \gamma(\mathcal{E})$. A regular local ring is a UFD (Auslander–Buchsbaum; a fact beyond the text), so X is locally factorial and Ex. 6.8 applies. We use the following three elementary facts.

(P1) A coherent sheaf $\mathcal{K}$ with $\mathcal{K}_x$ free is free near x (Preliminary 1 of Ex. 6.5); so $\mathcal{K}$ is locally free iff all its stalks are free, and $\{x : \mathcal{K}_x \text{ free}\}$ is open.

(P2) If $0 \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to 0$ is exact with $\mathcal{B}, \mathcal{C}$ locally free, then $\mathcal{A}$ is locally free (coherent, and stalks are kernels of surjections of free modules over a local ring, which split, so are free; use (P1)).

(P3) *Exact finite complexes.* If $0 \to \mathcal{C}_n \to \cdots \to \mathcal{C}_0 \to 0$ is an exact complex of locally free sheaves then $\sum(-1)^j \gamma_1(\mathcal{C}_j) = 0$ in $K_1(X)$. By induction on n: for $n = 0$, $\mathcal{C}_0 = 0$. For $n \geq 1$, $\mathcal{C}_1 \to \mathcal{C}_0$ is onto with kernel $\mathcal{Z}$ locally free (P2), so $\gamma_1(\mathcal{C}_1) = \gamma_1(\mathcal{Z}) + \gamma_1(\mathcal{C}_0)$ and the alternating sum equals $-[\gamma_1(\mathcal{Z}) - \gamma_1(\mathcal{C}_2) + \gamma_1(\mathcal{C}_3) - \cdots]$, where the bracket is the alternating sum of the exact complex $0 \to \mathcal{C}_n \to \cdots \to \mathcal{C}_2 \to \mathcal{Z} \to 0$ of length $n-1$, zero by induction.

(a) By Ex. 6.8, every coherent sheaf is a quotient of a locally free one. So, as in Ex. 6.5, for coherent $\mathcal{F}$ there is a locally free resolution $\mathcal{E}_\bullet \to \mathcal{F} \to 0$ (choose $\mathcal{E}_0 \to \mathcal{F}$ onto, then $\mathcal{E}_1$ onto the (coherent) kernel, and so on). Let $\mathcal{Z}_0 = \mathcal{F}$ and $\mathcal{Z}_{j+1} = \operatorname{Ker}(\mathcal{E}_j \to \mathcal{Z}_j)$.

Finiteness. Let $V_m = \{x \in X : (\mathcal{Z}_m)_x \text{ is free}\}$, open by (P1). If $x \in V_m$ then $x \in V_{m+1}$, because $(\mathcal{Z}_{m+1})_x$ is the kernel of the surjection $(\mathcal{E}_m)_x \to (\mathcal{Z}_m)_x$ of free modules, which splits. For every x, the local ring $\mathcal{O}_x$ is regular, so $\operatorname{pd}\mathcal{F}_x \le \dim \mathcal{O}_x < \infty$ by Prop. 6.11A(a); and, as in Ex. 6.5, if $\operatorname{pd}\mathcal{F}_x \le m$ then by dimension shifting (Ex. 6.5, Preliminary 2, for modules) $\operatorname{Ext}^1_{\mathcal{O}_x}((\mathcal{Z}_m)_x, N) \cong \operatorname{Ext}^{m+1}_{\mathcal{O}_x}(\mathcal{F}_x, N) = 0$ for all N, so $(\mathcal{Z}_m)_x$ is projective, hence free (Prop. 6.10A(a)). Thus $\bigcup_m V_m = X$. As X is noetherian, hence quasi-compact, and $V_0 \subseteq V_1 \subseteq \cdots$ are open, $X = V_m$ for some m (in fact for all large m). (This is the content of Ex. 6.5(c): $\operatorname{hd}\mathcal{F} = \sup_x \operatorname{pd}\mathcal{F}_x$, and the supremum is finite here by quasi-compactness even if $\dim X = \infty$.) By (P1) $\mathcal{Z}_m$ is locally free, and

$$0 \to \mathcal{Z}_m \to \mathcal{E}_{m-1} \to \cdots \to \mathcal{E}_0 \to \mathcal{F} \to 0$$

is a finite locally free resolution. Moreover this works for any locally free resolution: the syzygy $\mathcal{Z}_m$ is locally free for all large m. $\quad(*)$

(b) *Step 1: comparison for surjective chain maps.* Let $\mathcal{E}_\bullet \to \mathcal{F}$ and $\mathcal{W}_\bullet \to \mathcal{F}$ be finite locally free resolutions and $p\colon \mathcal{W}_\bullet \to \mathcal{E}_\bullet$ a chain map over $1_{\mathcal{F}}$ which is surjective in each degree. Then $\sum(-1)^j\gamma_1(\mathcal{W}_j) = \sum(-1)^j\gamma_1(\mathcal{E}_j)$.

Indeed, let $\mathcal{C}_j = \operatorname{Ker} p_j$, locally free by (P2). Then $0 \to \mathcal{C}_\bullet \to \mathcal{W}_\bullet \to \mathcal{E}_\bullet \to 0$ is a short exact sequence of complexes. The complexes $\mathcal{W}_\bullet, \mathcal{E}_\bullet$ have homology 0 in degrees > 0 and $\mathcal{F}$ in degree 0, and p induces the identity on $H_0 = \mathcal{F}$. The long exact homology sequence shows $\mathcal{C}_\bullet$ is exact in all degrees (in degree 0: $0 = H_1(\mathcal{E}) \to H_0(\mathcal{C}) \to H_0(\mathcal{W}) \xrightarrow{\sim} H_0(\mathcal{E})$). By (P3), $\sum(-1)^j\gamma_1(\mathcal{C}_j) = 0$, and $\gamma_1(\mathcal{W}_j) = \gamma_1(\mathcal{C}_j) + \gamma_1(\mathcal{E}_j)$.

Step 2: any two finite resolutions are dominated by a third. Let $\mathcal{E}_\bullet$, $\mathcal{E}'_\bullet$ be locally free resolutions of $\mathcal{F}$, of lengths $\le n$ ($\mathcal{E}_j = 0 = \mathcal{E}'_j$ for $j > n$). Let $\mathcal{B}_j = \operatorname{Ker}(\mathcal{E}_{j-1} \to \mathcal{E}_{j-2})$ for $j \ge 1$ (with $\mathcal{E}_{-1} = \mathcal{F}$), so $\mathcal{B}_1 = \operatorname{Ker}(\mathcal{E}_0 \to \mathcal{F})$, and by exactness the differential $\mathcal{E}_j \to \mathcal{B}_j$ is onto for $j \ge 1$ (with $\mathcal{B}_{j+1} = \operatorname{Ker}(\mathcal{E}_j \to \mathcal{E}_{j-1})$); define $\mathcal{B}'_j$ likewise. We construct locally free $\mathcal{W}_j$ and maps $d\colon \mathcal{W}_j \to \mathcal{W}_{j-1}$, $p_j\colon \mathcal{W}_j \to \mathcal{E}_j$, $p'_j\colon \mathcal{W}_j \to \mathcal{E}'_j$, all surjective, compatible with the differentials and augmentations, such that $\mathcal{W}_\bullet \to \mathcal{F}$ is a resolution. Put $\mathcal{Y}_j = \operatorname{Ker}(\mathcal{W}_{j-1} \to \mathcal{W}_{j-2})$ ($\mathcal{W}_{-1} = \mathcal{F}$).

$j = 0$: the sheaf $\mathcal{Q}_0 = \mathcal{E}_0 \times_{\mathcal{F}} \mathcal{E}'_0$ is coherent and maps onto $\mathcal{E}_0$ and $\mathcal{E}'_0$ (as both augmentations are onto). Choose a locally free $\mathcal{W}_0$ onto $\mathcal{Q}_0$ (Ex. 6.8) and let p_0, p'_0 be the composites. The kernel of $\mathcal{Q}_0 \to \mathcal{F}$ is $\mathcal{B}_1 \oplus \mathcal{B}'_1$, and its preimage in $\mathcal{W}_0$ is $\mathcal{Y}_1$; so $(p_0, p'_0)\colon \mathcal{Y}_1 \to \mathcal{B}_1 \oplus \mathcal{B}'_1$ is onto.

Inductive step: suppose $\mathcal{W}_0, \dots, \mathcal{W}_{j-1}$ are constructed ($j \ge 1$) and $\mathcal{Y}_j \to \mathcal{B}_j \oplus \mathcal{B}'_j$ is onto. Let $\mathcal{Q}_j = \mathcal{Y}_j \times_{\mathcal{B}_j \oplus \mathcal{B}'_j} (\mathcal{E}_j \oplus \mathcal{E}'_j)$, coherent, mapping onto $\mathcal{Y}_j$ (since $\mathcal{E}_j \oplus \mathcal{E}'_j \to \mathcal{B}_j \oplus \mathcal{B}'_j$ is onto) and onto $\mathcal{E}_j \oplus \mathcal{E}'_j$ (since $\mathcal{Y}_j \to \mathcal{B}_j \oplus \mathcal{B}'_j$ is onto). Choose $\mathcal{W}_j$ locally free onto $\mathcal{Q}_j$, and define $\mathcal{W}_j \to \mathcal{Y}_j \subseteq \mathcal{W}_{j-1}$ and p_j, p'_j via $\mathcal{Q}_j$. All the squares commute by construction of the fibre product, and the maps are onto. The kernel of $\mathcal{Q}_j \to \mathcal{Y}_j$ is $\operatorname{Ker}(\mathcal{E}_j \oplus \mathcal{E}'_j \to \mathcal{B}_j \oplus \mathcal{B}'_j) = \mathcal{B}_{j+1} \oplus \mathcal{B}'_{j+1}$, and its preimage in $\mathcal{W}_j$ is $\mathcal{Y}_{j+1}$; hence $\mathcal{Y}_{j+1} \to \mathcal{B}_{j+1} \oplus \mathcal{B}'_{j+1}$ is onto and the induction continues.

Since $\mathcal{W}_j \to \mathcal{Y}_j$ is onto for every $j \ge 1$ and $\mathcal{W}_0 \to \mathcal{F}$ is onto, $\mathcal{W}_\bullet \to \mathcal{F}$ is a locally free resolution. By $(*)$ choose $m \ge n+2$ with $\mathcal{Y}_m$ locally free, and set $\mathcal{W}^{\mathrm{tr}}_j = \mathcal{W}_j$ for $j < m$, $\mathcal{W}^{\mathrm{tr}}_m = \mathcal{Y}_m$, $\mathcal{W}^{\mathrm{tr}}_j = 0$ for $j > m$. This is a finite locally free resolution of $\mathcal{F}$, and p, p' restrict to chain maps $\mathcal{W}^{\mathrm{tr}}_\bullet \to \mathcal{E}_\bullet$, $\to \mathcal{E}'_\bullet$ (in degrees $\ge m - 1 \ge n + 1$ the target is 0), surjective in every degree.

Step 3: independence. By Steps 1 and 2, $\sum(-1)^j\gamma_1(\mathcal{E}_j) = \sum(-1)^j\gamma_1(\mathcal{W}^{\mathrm{tr}}_j) = \sum(-1)^j\gamma_1(\mathcal{E}'_j)$. So $\delta(\mathcal{F}) \in K_1(X)$ is well defined (it clearly depends only on the isomorphism class of $\mathcal{F}$).

Step 4: additivity. Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be an exact sequence of coherent sheaves. Choose

a finite locally free resolution $\mathcal{E}''_\bullet$ of $\mathcal{F}''$ of length n, with syzygies $\mathcal{Z}''_0 = \mathcal{F}''$, $\mathcal{Z}''_{j+1} = \operatorname{Ker}(\mathcal{E}''_j \to \mathcal{Z}''_j)$; so $\mathcal{Z}''_j = 0$ for $j > n$. Put $\mathcal{A}_0 = \mathcal{F}$, $\mathcal{A}'_0 = \mathcal{F}'$, so that $0 \to \mathcal{A}'_0 \to \mathcal{A}_0 \to \mathcal{Z}''_0 \to 0$ is exact. Suppose we have an exact sequence $0 \to \mathcal{A}'_j \to \mathcal{A}_j \to \mathcal{Z}''_j \to 0$ ($j \geq 0$). Let $\mathcal{Q}_j = \mathcal{E}''_j \times_{\mathcal{Z}''_j} \mathcal{A}_j$; it maps onto $\mathcal{E}''_j$ and onto $\mathcal{A}_j$ (as $\mathcal{A}_j \to \mathcal{Z}''_j$ and $\mathcal{E}''_j \to \mathcal{Z}''_j$ are onto). Choose a locally free $\mathcal{G}_j$ onto $\mathcal{Q}_j$; then $\mathcal{G}_j \to \mathcal{E}''_j$ and $\mathcal{G}_j \to \mathcal{A}_j$ are onto. Let $\mathcal{C}_j = \operatorname{Ker}(\mathcal{G}_j \to \mathcal{E}''_j)$, locally free by (P2); it maps onto $\mathcal{A}'_j$, because $\operatorname{Ker}(\mathcal{Q}_j \to \mathcal{E}''_j) = \mathcal{A}'_j$. We have the commutative diagram with exact rows $0 \to \mathcal{C}_j \to \mathcal{G}_j \to \mathcal{E}''_j \to 0$ and $0 \to \mathcal{A}'_j \to \mathcal{A}_j \to \mathcal{Z}''_j \to 0$ and surjective vertical maps; by the snake lemma the kernels form an exact sequence $0 \to \mathcal{A}'_{j+1} \to \mathcal{A}_{j+1} \to \mathcal{Z}''_{j+1} \to 0$, where $\mathcal{A}_{j+1} = \operatorname{Ker}(\mathcal{G}_j \to \mathcal{A}_j)$, $\mathcal{A}'_{j+1} = \operatorname{Ker}(\mathcal{C}_j \to \mathcal{A}'_j)$. Continue. We obtain resolutions $\mathcal{G}_\bullet \to \mathcal{F}$, $\mathcal{C}_\bullet \to \mathcal{F}'$ by locally free sheaves with $\mathcal{A}_j = \operatorname{Ker}(\mathcal{G}_{j-1} \to \mathcal{G}_{j-2})$, $\mathcal{A}'_j = \operatorname{Ker}(\mathcal{C}_{j-1} \to \mathcal{C}_{j-2})$ and a short exact sequence of complexes $0 \to \mathcal{C}_\bullet \to \mathcal{G}_\bullet \to \mathcal{E}''_\bullet \to 0$ (degreewise exact, with $\mathcal{C}_\bullet \subseteq \mathcal{G}_\bullet$ a subcomplex). For $j > n$ we have $\mathcal{E}''_j = 0 = \mathcal{Z}''_j$, so $\mathcal{C}_j = \mathcal{G}_j$ and $\mathcal{A}'_j = \mathcal{A}_j$. Using $(*)$ choose $m \geq n+2$ with $\mathcal{A}_m = \mathcal{A}'_m$ locally free and truncate both $\mathcal{G}_\bullet$ and $\mathcal{C}_\bullet$ by replacing the degree m term by $\mathcal{A}_m$ and omitting the higher terms. We get finite locally free resolutions $\mathcal{G}^{\mathrm{tr}}_\bullet$ of $\mathcal{F}$ and $\mathcal{C}^{\mathrm{tr}}_\bullet$ of $\mathcal{F}'$ with $0 \to \mathcal{C}^{\mathrm{tr}}_j \to \mathcal{G}^{\mathrm{tr}}_j \to \mathcal{E}''_j \to 0$ exact for every j (for $j \geq n+1$ it reads $0 \to \mathcal{G}_j \to \mathcal{G}_j \to 0 \to 0$, resp. $0 \to \mathcal{A}_m \to \mathcal{A}_m \to 0 \to 0$). Hence $\gamma_1(\mathcal{G}^{\mathrm{tr}}_j) = \gamma_1(\mathcal{C}^{\mathrm{tr}}_j) + \gamma_1(\mathcal{E}''_j)$ for all j, and taking alternating sums and using Step 3 gives $\delta(\mathcal{F}) = \delta(\mathcal{F}') + \delta(\mathcal{F}'')$.

Consequently δ (defined on generators $\gamma(\mathcal{F}) \mapsto \delta(\mathcal{F})$) kills the defining relations of $K(X)$ and gives a group homomorphism $\delta\colon K(X) \to K_1(X)$.

Step 5: δ is inverse to ε. For coherent $\mathcal{F}$ with finite locally free resolution $0 \to \mathcal{E}_n \to \cdots \to \mathcal{E}_0 \to \mathcal{F} \to 0$, set $\mathcal{Z}_0 = \mathcal{F}$, $\mathcal{Z}_{j+1} = \operatorname{Ker}(\mathcal{E}_j \to \mathcal{Z}_j)$, so $\mathcal{Z}_n = \mathcal{E}_n$. The sequences $0 \to \mathcal{Z}_{j+1} \to \mathcal{E}_j \to \mathcal{Z}_j \to 0$ of coherent sheaves give $\gamma(\mathcal{E}_j) = \gamma(\mathcal{Z}_j) + \gamma(\mathcal{Z}_{j+1})$ in $K(X)$ ($j < n$), whence

$$\varepsilon\delta(\mathcal{F}) = \sum_{j=0}^{n} (-1)^j \gamma(\mathcal{E}_j) = \sum_{j=0}^{n-1} (-1)^j \big(\gamma(\mathcal{Z}_j) + \gamma(\mathcal{Z}_{j+1})\big) + (-1)^n \gamma(\mathcal{Z}_n) = \gamma(\mathcal{Z}_0) = \gamma(\mathcal{F}).$$

Thus $\varepsilon \circ \delta = \mathrm{id}_{K(X)}$ (on generators, hence everywhere). Conversely, for a locally free $\mathcal{E}$ the one-term complex $\mathcal{E} \to \mathcal{E}$ is a finite locally free resolution, so $\delta\varepsilon(\gamma_1(\mathcal{E})) = \delta(\mathcal{E}) = \gamma_1(\mathcal{E})$; as $K_1(X)$ is generated by the $\gamma_1(\mathcal{E})$, $\delta \circ \varepsilon = \mathrm{id}_{K_1(X)}$. Hence ε is an isomorphism with inverse δ. $\square$

Exercise 6.10

Problem. *Duality for a Finite Flat Morphism.*

(a) Let $f\colon X \to Y$ be a finite morphism of noetherian schemes. For any quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G}$, $\mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{G})$ is a quasi-coherent $f_*\mathcal{O}_X$-module, hence corresponds to a quasi-coherent $\mathcal{O}_X$-module, which we call $f^!\mathcal{G}$ (II, Ex. 5.17e).

(b) Show that for any coherent $\mathcal{F}$ on X and any quasi-coherent $\mathcal{G}$ on Y, there is a natural isomorphism

$$f_*\mathcal{H}om_X(\mathcal{F}, f^!\mathcal{G}) \xrightarrow{\sim} \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G}).$$

(c) For each $i \geq 0$, there is a natural map

$$\varphi_i\colon \operatorname{Ext}^i_X(\mathcal{F}, f^!\mathcal{G}) \to \operatorname{Ext}^i_Y(f_*\mathcal{F}, \mathcal{G}).$$

[*Hint*: First construct a map $\operatorname{Ext}^i_X(\mathcal{F}, f^!\mathcal{G}) \to \operatorname{Ext}^i_Y(f_*\mathcal{F}, f_*f^!\mathcal{G})$. Then compose with a suitable map from $f_*f^!\mathcal{G}$ to $\mathcal{G}$.]

(d) Now assume that X and Y are separated, $\mathfrak{Coh}(X)$ has enough locally frees, and that $f_*\mathcal{O}_X$ is locally free on Y (this is equivalent to saying that f is flat—see Sect. 9). Show that φ_i is an isomorphism for all i, all $\mathcal{F}$ coherent on X, and all $\mathcal{G}$ quasi-coherent on Y. [*Hints*: First do $i = 0$. Then do $\mathcal{F} = \mathcal{O}_X$, using (Ex. 4.1). Then do $\mathcal{F}$ locally free. Do the general case by induction on i, writing $\mathcal{F}$ as a quotient of a locally free sheaf.]

Solution. Let $\mathcal{A} = f_*\mathcal{O}_X$. As f is finite, it is affine and $\mathcal{A}$ is a coherent $\mathcal{O}_Y$-module: over an open affine $V = \operatorname{Spec} B \subseteq Y$, $f^{-1}V = \operatorname{Spec} A'$ with A' a finite B-algebra. We use Ch. II, Ex. 5.17(e): f_* is an equivalence from quasi-coherent $\mathcal{O}_X$-modules to quasi-coherent $\mathcal{A}$-modules, with inverse $\widetilde{\ }$; and f_* is exact on quasi-coherent sheaves (compute on affine opens V, where $f_*\widetilde{M} = M$ viewed as a B-module).

(a) Over $V = \operatorname{Spec} B$, $\mathcal{G}|_V = \widetilde{N}$, and since A' is a finitely presented B-module (B noetherian), $\mathcal{H}om_Y(\mathcal{A}, \mathcal{G})|_V = \operatorname{Hom}_B(A', N)^{\sim}$ is quasi-coherent. It is an $\mathcal{A}$-module via $(a \cdot \varphi)(b) = \varphi(ab)$ $(a, b \in \mathcal{A})$, compatibly with restrictions; so it is a quasi-coherent $\mathcal{A}$-module and $f^!\mathcal{G}$ is the quasi-coherent $\mathcal{O}_X$-module with $f_*f^!\mathcal{G} \cong \mathcal{H}om_Y(\mathcal{A}, \mathcal{G})$ as $\mathcal{A}$-modules. Let $e\colon f_*f^!\mathcal{G} = \mathcal{H}om_Y(\mathcal{A}, \mathcal{G}) \to \mathcal{G}$ be the evaluation $\varphi \mapsto \varphi(1)$; it is $\mathcal{O}_Y$-linear and will be the "suitable map" of (c).

(b) Define $\Theta\colon f_*\mathcal{H}om_X(\mathcal{F}, f^!\mathcal{G}) \to \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G})$: for $V \subseteq Y$ open and $\psi \in \operatorname{Hom}(\mathcal{F}|_{f^{-1}V}, f^!\mathcal{G}|_{f^{-1}V})$, put $\Theta(\psi) = e \circ f_*\psi\colon f_*\mathcal{F}|_V \to f_*f^!\mathcal{G}|_V \to \mathcal{G}|_V$. It is a natural morphism of sheaves. To see that it is an isomorphism, work over an open affine $V = \operatorname{Spec} B$, with $f^{-1}V = \operatorname{Spec} A'$, $\mathcal{F}| = \widetilde{P}$ for a finitely generated A'-module P (A' is noetherian), $\mathcal{G}|_V = \widetilde{N}$. All sheaves involved are quasi-coherent, and a morphism of quasi-coherent sheaves on an affine scheme is determined by, and an isomorphism iff it is on, global sections. As P is finitely presented over A' and over B, the global sections of the two sides are $\operatorname{Hom}_{A'}(P, \operatorname{Hom}_B(A', N))$ and $\operatorname{Hom}_B(P, N)$, and Θ becomes $\psi \mapsto (m \mapsto \psi(m)(1))$. This is the usual adjunction isomorphism, with inverse $\chi \mapsto (m \mapsto (a \mapsto \chi(am)))$. So Θ is an isomorphism.

(c) Put $\mathcal{H} = f^!\mathcal{G}$, quasi-coherent on X. Choose injective resolutions $\mathcal{H} \to \mathcal{I}^\bullet$ in $\mathfrak{Mod}(X)$ and $\mathcal{G} \to \mathcal{J}^\bullet$ in $\mathfrak{Mod}(Y)$.

Claim: $0 \to f_*\mathcal{H} \to f_*\mathcal{I}^0 \to f_*\mathcal{I}^1 \to \cdots$ *is exact.* Left exactness of f_* gives exactness at $f_*\mathcal{H}$ and $f_*\mathcal{I}^0$. For exactness elsewhere, check on sections over open affines $V \subseteq Y$, which form a base: $\Gamma(V, f_*\mathcal{I}^\bullet) = \Gamma(U, \mathcal{I}^\bullet)$ with $U = f^{-1}V$ affine noetherian, and $\mathcal{I}^\bullet|_U$ is an injective resolution of $\mathcal{H}|_U$ (Lemma 6.1), so its cohomology is $H^i(U, \mathcal{H}|_U) = 0$ for $i > 0$ by Thm. 3.5 ($\mathcal{H}$ quasi-coherent). A complex of sheaves all of whose sections complexes over a base of opens are exact in degree i is exact there (pass to stalks, a direct limit).

Hence $f_*\mathcal{I}^\bullet$ is a resolution of $f_*\mathcal{H}$, and by the comparison theorem (the target $\mathcal{J}^\bullet$ is an injective complex, the source an exact resolution) the map $e\colon f_*\mathcal{H} \to \mathcal{G} \to \mathcal{J}^0$ extends to a chain map $c\colon f_*\mathcal{I}^\bullet \to \mathcal{J}^\bullet$, unique up to homotopy. For any coherent $\mathcal{F}$ on X define

$$\nu_{\mathcal{F}}\colon \operatorname{Hom}_X(\mathcal{F}, \mathcal{I}^\bullet) \to \operatorname{Hom}_Y(f_*\mathcal{F}, \mathcal{J}^\bullet), \qquad \psi \mapsto c \circ f_*\psi,$$

a map of complexes, natural in $\mathcal{F}$ (for $\mathcal{F}_1 \to \mathcal{F}_2$ the evident squares commute). Let $\varphi_i(\mathcal{F}) = h^i(\nu_{\mathcal{F}})\colon \operatorname{Ext}^i_X(\mathcal{F}, f^!\mathcal{G}) \to \operatorname{Ext}^i_Y(f_*\mathcal{F}, \mathcal{G})$. Up to canonical isomorphism, φ_i is independent of the choices (different choices change c by a homotopy), and it is natural in $\mathcal{F}$. This is the composite of "apply f_*", $\operatorname{Ext}^i_X(\mathcal{F}, \mathcal{H}) \to \operatorname{Ext}^i_Y(f_*\mathcal{F}, f_*\mathcal{H})$, with the map induced by $e\colon f_*f^!\mathcal{G} \to \mathcal{G}$, as in the hint.

Compatibility with the long exact sequences in $\mathcal{F}$. Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be exact in $\mathfrak{Coh}(X)$. Then $0 \to f_*\mathcal{F}' \to f_*\mathcal{F} \to f_*\mathcal{F}'' \to 0$ is exact in $\mathfrak{Coh}(Y)$, and the sequences of

complexes $0 \to \mathrm{Hom}_X(\mathcal{F}'', \mathcal{I}^\bullet) \to \mathrm{Hom}_X(\mathcal{F}, \mathcal{I}^\bullet) \to \mathrm{Hom}_X(\mathcal{F}', \mathcal{I}^\bullet) \to 0$ and the analogous one with $\mathrm{Hom}_Y(f_*(\cdot), \mathcal{J}^\bullet)$, which define the connecting maps (Prop. 6.4), are related by the maps $\nu_{\mathcal{F}''}, \nu_{\mathcal{F}}, \nu_{\mathcal{F}'}$, forming a morphism of short exact sequences of complexes. Hence $\varphi_\bullet$ commutes with the connecting homomorphisms: it is a morphism of δ-functors on $\mathfrak{Coh}(X)$ (contravariant in $\mathcal{F}$).

(d) Now assume the hypotheses of (d) (separatedness will not actually be needed in the argument below). Write $T^i(\mathcal{F}) = \mathrm{Ext}^i_X(\mathcal{F}, f^!\mathcal{G})$, $S^i(\mathcal{F}) = \mathrm{Ext}^i_Y(f_*\mathcal{F}, \mathcal{G})$.

Step 0: $i = 0$. Identify $T^0(\mathcal{F}) = \mathrm{Hom}_X(\mathcal{F}, \mathcal{H}) = \Gamma(Y, f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{H}))$ and $S^0(\mathcal{F}) = \mathrm{Hom}_Y(f_*\mathcal{F}, \mathcal{G}) = \Gamma(Y, \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G}))$. For $\psi\colon \mathcal{F} \to \mathcal{H} \subseteq \mathcal{I}^0$, $\nu(\psi) = c^0 f_*(j\psi)$ and $c^0|_{f_*\mathcal{H}} = (\mathcal{G} \to \mathcal{J}^0) \circ e$, so $\varphi_0(\psi) = e \circ f_*\psi = \Gamma(\Theta)(\psi)$. By (b), φ_0 is an isomorphism for every coherent $\mathcal{F}$.

Step 1: $\mathcal{F}$ locally free. First, $f_*\mathcal{F}$ is locally free. Indeed, let $V = \mathrm{Spec}\, B \subseteq Y$ be an open affine so small that $\mathcal{A}|_V$ is free; then $A' = \Gamma(f^{-1}V, \mathcal{O}) \cong B^m$ as B-modules. With $\mathcal{F}|_{f^{-1}V} = \widetilde{P}$, P is a finitely generated module over A' all of whose localizations are free, hence a projective A'-module, a direct summand of some $A'^s \cong B^{ms}$; so P is a finitely generated projective B-module and $f_*\mathcal{F}|_V = \widetilde{P}$ is locally free.

By Lemma 6.6, $\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^j) = \mathcal{F}^\vee \otimes \mathcal{I}^j$ is injective, and $\mathcal{H}om_X(\mathcal{F}, \cdot)$ is exact, so $\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet)$ is an injective resolution of $\mathcal{H}om_X(\mathcal{F}, \mathcal{H})$, and $\mathrm{Hom}_X(\mathcal{F}, \mathcal{I}^\bullet) = \Gamma(X, \mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet)) = \Gamma(Y, f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet))$. Likewise $\Lambda^\bullet := \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{J}^\bullet)$ is an injective resolution of $\mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G})$ and $\mathrm{Hom}_Y(f_*\mathcal{F}, \mathcal{J}^\bullet) = \Gamma(Y, \Lambda^\bullet)$. The complex $f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet)$ is a resolution of $f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{H})$ by flasque sheaves (an injective sheaf is flasque, Lemma 2.4, and f_* preserves flasqueness); exactness is proved as in the Claim of (c), because $\mathcal{H}om_X(\mathcal{F}, \mathcal{H}) \cong \mathcal{F}^\vee \otimes \mathcal{H}$ is quasi-coherent. Define the sheaf chain map

$$\tilde{\nu}\colon f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet) \to \Lambda^\bullet, \qquad \psi \mapsto c \circ f_*\psi.$$

Then $\Gamma(Y, \tilde{\nu}) = \nu_{\mathcal{F}}$, and in degree 0, restricted to $f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{H})$, it equals Θ followed by $\mathcal{H}om(f_*\mathcal{F}, \mathcal{G} \to \mathcal{J}^0)$ (computation of Step 0). Thus $\tilde{\nu}$ is a chain map between two resolutions of the same sheaf $\mathcal{M} = \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G})$ (via the isomorphism Θ of (b)), both consisting of Γ-acyclic (flasque) sheaves, compatible with the identification of $f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{H})$ with $\mathcal{M}$. By Prop. 1.2A (a resolution by acyclic objects computes the derived functors, via the comparison map to an injective resolution), $\Gamma(\tilde{\nu})$ induces isomorphisms $h^i(\Gamma(Y, f_*\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet))) \to h^i(\Gamma(Y, \Lambda^\bullet)) \cong H^i(Y, \mathcal{M})$. These are exactly the maps $\varphi_i(\mathcal{F}) = h^i(\nu_{\mathcal{F}})$. Hence $\varphi_i(\mathcal{F})$ is an isomorphism for all $i \geq 0$ and all locally free $\mathcal{F}$; in particular for $\mathcal{F} = \mathcal{O}_X$, where the identification is $H^i(X, f^!\mathcal{G}) \cong H^i(Y, f_*f^!\mathcal{G})$ (this is Ex. 4.1, which our argument reproves) followed by $H^i(Y, \mathcal{H}om(f_*\mathcal{O}_X, \mathcal{G})) \cong \mathrm{Ext}^i_Y(f_*\mathcal{O}_X, \mathcal{G})$ (Prop. 6.7 and 6.3).

Step 2: induction on i. We show by induction on i that $\varphi_i(\mathcal{F})$ is an isomorphism for all coherent $\mathcal{F}$. The case $i = 0$ is Step 0. Let $i \geq 1$ and assume $\varphi_{i-1}(\mathcal{F})$ is an isomorphism for all coherent $\mathcal{F}$. Let $\mathcal{F}$ be coherent; since $\mathfrak{Coh}(X)$ has enough locally frees, choose $0 \to \mathcal{K} \to \mathcal{E} \to \mathcal{F} \to 0$ with $\mathcal{E}$ locally free and $\mathcal{K}$ coherent. Then φ gives a commutative ladder between the exact rows

$$\begin{array}{ccccccccc} T^{i-1}(\mathcal{E}) & \to & T^{i-1}(\mathcal{K}) & \to & T^i(\mathcal{F}) & \to & T^i(\mathcal{E}) & \to & T^i(\mathcal{K}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ S^{i-1}(\mathcal{E}) & \to & S^{i-1}(\mathcal{K}) & \to & S^i(\mathcal{F}) & \to & S^i(\mathcal{E}) & \to & S^i(\mathcal{K}) \end{array}$$

whose vertical maps are the φ's. The first two verticals are isomorphisms by the induction hypothesis, and the fourth is an isomorphism by Step 1.

Injectivity of $\varphi_i(\mathcal{F})$: by the four lemma applied to the first four columns (first vertical onto, second and fourth injective) $\varphi_i(\mathcal{F})$ is injective. This holds for every coherent $\mathcal{F}$, in particular for $\mathcal{K}$.

Surjectivity: by the four lemma applied to the last four columns (second and fourth vertical onto, fifth injective by the preceding paragraph applied to $\mathcal{K}$), $\varphi_i(\mathcal{F})$ is surjective.

Hence φ_i is an isomorphism for all coherent $\mathcal{F}$ and all quasi-coherent $\mathcal{G}$, completing the induction. □

7. The Serre duality theorem

Exercise 7.1

> **Problem.** Let X be an integral projective scheme of dimension ≥ 1 over a field k, and let $\mathcal{L}$ be an ample invertible sheaf on X. Then $H^0(X, \mathcal{L}^{-1}) = 0$. (This is an easy special case of Kodaira's vanishing theorem.)

Solution. Write $\mathcal{L}^{-1} = \mathcal{L}^\vee$. Since X is projective over k, $A := H^0(X, \mathcal{O}_X)$ is a finite-dimensional k-vector space (Thm. 5.2(a)), and it is an integral domain because X is integral. A finite-dimensional domain over a field is a field, so A is a field; in particular every nonzero element of A is a unit of $\Gamma(X, \mathcal{O}_X)$.

Suppose, for a contradiction, that $0 \neq s \in H^0(X, \mathcal{L}^{-1})$. Let η be the generic point and $K = \mathcal{O}_{X,\eta}$ the function field. For an invertible sheaf $\mathcal{M}$ on the integral scheme X, $\mathcal{M}_\eta$ is a one-dimensional K-vector space and $\Gamma(X, \mathcal{M}) \to \mathcal{M}_\eta$ is injective (a section vanishing at the generic point vanishes on a dense open set, hence is zero, since $\mathcal{M}$ is locally free and X is reduced). Hence a product of nonzero sections of invertible sheaves is nonzero: in $\mathcal{M}_\eta \otimes \mathcal{N}_\eta \cong K$ the product of two nonzero elements of one-dimensional K-spaces is nonzero.

Since $\mathcal{L}$ is ample, $\mathcal{L}^n$ is generated by global sections for some $n \geq 1$ (definition of ample applied to $\mathcal{F} = \mathcal{O}_X$; II, Sect. 7). As $X \neq \emptyset$, $\mathcal{L}^n \neq 0$, so there is $0 \neq t \in H^0(X, \mathcal{L}^n)$. Then $s^n \in H^0(X, \mathcal{L}^{-n})$ is nonzero by the remark above, and so

$$u := t \cdot s^n \in H^0(X, \mathcal{L}^n \otimes \mathcal{L}^{-n}) = H^0(X, \mathcal{O}_X) = A$$

is nonzero, hence a unit of A.

Claim: $s\colon \mathcal{O}_X \to \mathcal{L}^{-1}$ is an isomorphism. Let $x \in X$ and let e generate $\mathcal{L}_x$. Write $s_x = a\, e^{-1}$, $t_x = b\, e^n$ with $a, b \in \mathcal{O}_{X,x}$ (where e^{-1} is the dual generator of $\mathcal{L}_x^{-1}$). Then $u_x = a^n b$ is a unit of $\mathcal{O}_{X,x}$, so a is a unit, i.e. s_x generates $\mathcal{L}_x^{-1}$. Thus s is an isomorphism and $\mathcal{L} \cong \mathcal{O}_X$.

Now ampleness of $\mathcal{L} \cong \mathcal{O}_X$ says that every coherent sheaf on X is generated by global sections. Take a closed point $x \in X$ (X is quasi-compact and nonempty) and let $\mathcal{I}_x \subset \mathcal{O}_X$ be the ideal sheaf of the reduced closed subscheme $\{x\}$. A global section of $\mathcal{I}_x$ is an element of A vanishing at x; a nonzero element of A is a unit and so does not vanish at x. Hence $H^0(X, \mathcal{I}_x) = 0$, and since $\mathcal{I}_x$ is generated by global sections, $\mathcal{I}_x = 0$. Then $\mathcal{O}_X = \mathcal{O}_X/\mathcal{I}_x$ is supported at $\{x\}$; but the stalk of $\mathcal{O}_X$ at the generic point is $K \neq 0$, so $\eta = x$ and $X = \{x\}$ is a point, contradicting $\dim X \geq 1$. Hence $H^0(X, \mathcal{L}^{-1}) = 0$. □

Exercise 7.2

Problem. Let $f\colon X \to Y$ be a finite morphism of projective schemes of the same dimension over a field k, and let ω_Y° be a dualizing sheaf for Y.

(a) Show that $f^!\omega_Y^\circ$ is a dualizing sheaf for X, where $f^!$ is defined as in (Ex. 6.10).

(b) If X and Y are both nonsingular, and k algebraically closed, conclude that there is a natural trace map $t\colon f_*\omega_X \to \omega_Y$.

Solution. Let $n = \dim X = \dim Y$. Recall (Ex. 6.10(a)): f is finite, hence affine, so quasi-coherent $\mathcal{O}_X$-modules correspond to quasi-coherent $f_*\mathcal{O}_X$-modules (II, Ex. 5.17); for a quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G}$, $f^!\mathcal{G}$ is the quasi-coherent $\mathcal{O}_X$-module with $f_*(f^!\mathcal{G}) = \mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, \mathcal{G})$.

(a) *Coherence.* $f_*\mathcal{O}_X$ is coherent (f finite) and ω_Y° is coherent, so $\mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, \omega_Y^\circ)$ is coherent (Ex. 6.3(a)). For a finite morphism, an $\mathcal{O}_X$-module is coherent iff its direct image is (locally B is a finite A-algebra, and a B-module is finitely generated iff it is so over A). So $f^!\omega_Y^\circ$ is coherent. Also X is projective over k of dimension n by hypothesis.

The natural isomorphism of Ex. 6.10(b). For $\mathcal{F}$ coherent on X and $\mathcal{G}$ quasi-coherent on Y there is the map

$$f_*\mathcal{H}om_X(\mathcal{F}, f^!\mathcal{G}) \longrightarrow \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G}), \qquad \varphi \longmapsto \mathrm{ev}_1 \circ f_*\varphi,$$

where $\mathrm{ev}_1\colon f_*f^!\mathcal{G} = \mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{G}) \to \mathcal{G}$ is evaluation at 1. It is an isomorphism: over an affine open $\operatorname{Spec} A \subseteq Y$, with $f^{-1} = \operatorname{Spec} B$, $\mathcal{F} = \tilde{M}$, $\mathcal{G} = \tilde{N}$ (M finitely generated over B), it is the sheaf associated to the usual adjunction isomorphism $\operatorname{Hom}_B(M, \operatorname{Hom}_A(B, N)) \xrightarrow{\sim} \operatorname{Hom}_A(M, N)$, $\varphi \mapsto (m \mapsto \varphi(m)(1))$, with inverse $\psi \mapsto (m \mapsto (b \mapsto \psi(bm)))$. Taking global sections,

$$\operatorname{Hom}_X(\mathcal{F}, f^!\mathcal{G}) \xrightarrow{\sim} \operatorname{Hom}_Y(f_*\mathcal{F}, \mathcal{G}), \qquad \varphi \mapsto \varphi^\flat := \mathrm{ev}_1 \circ f_*\varphi. \qquad \text{(A)}$$

The trace. Since f is finite and X, Y are noetherian and separated, $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$ naturally in quasi-coherent $\mathcal{F}$ (Ex. 4.1). Define $t_X\colon H^n(X, f^!\omega_Y^\circ) = H^n(Y, f_*f^!\omega_Y^\circ) \to k$ as the composite

$$H^n(Y, f_*f^!\omega_Y^\circ) \xrightarrow{H^n(\mathrm{ev}_1)} H^n(Y, \omega_Y^\circ) \xrightarrow{t_Y} k.$$

Duality. Let $\mathcal{F}$ be coherent on X. For $\varphi \in \operatorname{Hom}_X(\mathcal{F}, f^!\omega_Y^\circ)$ and $\xi \in H^n(X, \mathcal{F}) = H^n(Y, f_*\mathcal{F})$ we have, by functoriality of the identification $H^n(X, -) = H^n(Y, f_*-)$,

$$t_X\big(H^n(\varphi)(\xi)\big) = t_Y\big(H^n(\mathrm{ev}_1)H^n(f_*\varphi)(\xi)\big) = t_Y\big(H^n(\varphi^\flat)(\xi)\big).$$

So the pairing $\operatorname{Hom}_X(\mathcal{F}, f^!\omega_Y^\circ) \times H^n(X, \mathcal{F}) \to k$ defining the map $\operatorname{Hom}_X(\mathcal{F}, f^!\omega_Y^\circ) \to H^n(X, \mathcal{F})'$ is the composite of the isomorphism (A) with the corresponding pairing for $f_*\mathcal{F}$ on Y. Since $f_*\mathcal{F}$ is coherent and ω_Y° is dualizing, $\operatorname{Hom}_Y(f_*\mathcal{F}, \omega_Y^\circ) \to H^n(Y, f_*\mathcal{F})'$ is an isomorphism, and therefore so is $\operatorname{Hom}_X(\mathcal{F}, f^!\omega_Y^\circ) \to H^n(X, \mathcal{F})'$. Thus $(f^!\omega_Y^\circ, t_X)$ is a dualizing sheaf for X. (The hypothesis $\dim X = \dim Y$ is what makes both pairings use the same n.)

(b) Now let X, Y be nonsingular projective varieties of the same dimension n, $k = \bar{k}$. By Cor. 7.12, ω_X and ω_Y are dualizing sheaves, with traces t_X, t_Y (which are determined up to scalars; fix them). By (a), $f^!\omega_Y$ is also a dualizing sheaf for X, with trace $t_Y \circ H^n(\mathrm{ev}_1)$. By Prop. 7.2 there is a unique isomorphism $\varphi\colon f^!\omega_Y \xrightarrow{\sim} \omega_X$ compatible with the traces. Define

$$t\colon f_*\omega_X \xrightarrow{f_*\varphi^{-1}} f_*f^!\omega_Y = \mathcal{H}om_Y(f_*\mathcal{O}_X, \omega_Y) \xrightarrow{\mathrm{ev}_1} \omega_Y.$$

This is the natural trace map; it is $t = (\varphi^{-1})^\flat$ in the notation of (A), and by construction $t_Y \circ H^n(t) = t_X$ under $H^n(Y, f_*\omega_X) = H^n(X, \omega_X)$. (One expects that on generic stalks it is the usual trace $K(X) \to K(Y)$ of the field extension, but this is not needed and is not proved here.) $\square$

Exercise 7.3

Problem. Let $X = \mathbf{P}^n_k$. Show that $H^q(X, \Omega^p_X) = 0$ for $p \neq q$, and k for $p = q$, $0 \leq p, q \leq n$.

Solution. Here $\Omega^p = \wedge^p \Omega_{X/k}$. For $p > n$, $\Omega^p = 0$.

Step 1: an exact sequence. The Euler sequence (II, 8.13) is $0 \to \Omega^1 \to \mathcal{O}(-1)^{n+1} \xrightarrow{\pi} \mathcal{O} \to 0$. Let $\mathcal{F} = \mathcal{O}(-1)^{n+1}$ and $\mathcal{F}' = \Omega^1 = \ker \pi$. Define $\iota\colon \wedge^p \mathcal{F} \to \wedge^{p-1}\mathcal{F}$ (contraction with π) by

$$\iota(v_1 \wedge \cdots \wedge v_p) = \sum_{j=1}^{p} (-1)^{j-1} \pi(v_j)\, v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_p.$$

We claim that for $p \geq 1$,

$$0 \to \wedge^p \mathcal{F}' \to \wedge^p \mathcal{F} \xrightarrow{\iota} \wedge^{p-1} \mathcal{F}' \to 0 \qquad \text{(W)}$$

is exact. This is local: on an open set where π is surjective with a section e, $\mathcal{F} = \mathcal{F}' \oplus \mathcal{O}e$ with $\pi(e) = 1$, $\pi|_{\mathcal{F}'} = 0$, and $\wedge^p \mathcal{F} = \wedge^p \mathcal{F}' \oplus e \wedge \wedge^{p-1}\mathcal{F}'$. Then $\iota(a) = 0$ for $a \in \wedge^p \mathcal{F}'$ and $\iota(e \wedge b) = b - e \wedge \iota(b) = b$ for $b \in \wedge^{p-1}\mathcal{F}'$ (as $\iota(b) = 0$), so $\ker \iota = \wedge^p \mathcal{F}'$ and ι maps onto $\wedge^{p-1}\mathcal{F}'$. (Cf. II, Ex. 5.16.) Since $\wedge^p \mathcal{F} \cong \mathcal{O}(-p)^{\binom{n+1}{p}}$ and $\wedge^{p-1}\mathcal{F}' = \Omega^{p-1}$, (W) reads

$$0 \to \Omega^p \to \mathcal{O}(-p)^{\binom{n+1}{p}} \to \Omega^{p-1} \to 0 \qquad (p \geq 1).$$

Step 2: cohomology of $\mathcal{O}(-p)$. By Thm. 5.1, $H^i(\mathbf{P}^n, \mathcal{O}(m)) \neq 0$ only for $i = 0, m \geq 0$ or $i = n$, $m \leq -n - 1$. Hence for $1 \leq p \leq n$ we have $H^i(X, \mathcal{O}(-p)) = 0$ for *all* $i \geq 0$ (for $i = 0$ because $-p < 0$; for $i = n$ because $-p \geq -n > -n - 1$; for $0 < i < n$ always).

Step 3: induction. We show, for $0 \leq p \leq n$: $H^q(X, \Omega^p) = k$ if $q = p$ and 0 otherwise, for every $q \geq 0$. For $p = 0$, $\Omega^0 = \mathcal{O}$ and this is Thm. 5.1(a),(b) (with $m = 0$). For $1 \leq p \leq n$, the long exact sequence of the sequence of Step 1 and the vanishing of Step 2 give $H^0(\Omega^p) = 0$ and isomorphisms (connecting maps)

$$H^{q-1}(X, \Omega^{p-1}) \xrightarrow{\sim} H^q(X, \Omega^p), \qquad q \geq 1,$$

since the neighbors $H^{q-1}(\mathcal{O}(-p)^N)$ and $H^q(\mathcal{O}(-p)^N)$ vanish. By induction, $H^q(\Omega^p) \cong H^{q-1}(\Omega^{p-1})$ is k iff $q - 1 = p - 1$, and 0 otherwise; and $H^0(\Omega^p) = 0$ agrees with $q \neq p$ since $p \geq 1$. $\square$

Exercise 7.4

Problem. *The Cohomology Class of a Subvariety.* Let X be a nonsingular projective variety of dimension n over an algebraically closed field k. Let Y be a nonsingular subvariety of codimension p (hence dimension $n - p$). From the natural map $\Omega_X \otimes \mathcal{O}_Y \to \Omega_Y$ of (II, 8.12)

we deduce a map $\Omega_X^{n-p} \to \Omega_Y^{n-p}$. This induces a map on cohomology $H^{n-p}(X, \Omega_X^{n-p}) \to H^{n-p}(Y, \Omega_Y^{n-p})$. Now $\Omega_Y^{n-p} = \omega_Y$ is a dualizing sheaf for Y, so we have the trace map $t_Y \colon H^{n-p}(Y, \Omega_Y^{n-p}) \to k$. Composing, we obtain a linear map $H^{n-p}(X, \Omega_X^{n-p}) \to k$. By (7.13) this corresponds to an element $\eta(Y) \in H^p(X, \Omega_X^p)$, which we call the *cohomology class* of Y.

(a) If $P \in X$ is a closed point, show that $t_X(\eta(P)) = 1$, where $\eta(P) \in H^n(X, \Omega^n)$ and t_X is the trace map.

(b) If $X = \mathbf{P}^n$, identify $H^p(X, \Omega_X^p)$ with k by (Ex. 7.3), and show that $\eta(Y) = (\deg Y) \cdot 1$, where $\deg Y$ is its degree as a projective variety (I, Sect. 7). [*Hint*: Cut with a hyperplane $H \subseteq X$, and use Bertini's theorem (II, 8.18) to reduce to the case Y is a finite set of points.]

(c) For any scheme X of finite type over k, we define a homomorphism of sheaves of abelian groups $d\log \colon \mathcal{O}_X^* \to \Omega_X$ by $d\log(f) = f^{-1}df$. Here $\mathcal{O}^*$ is a group under multiplication, and Ω_X is a group under addition. This induces a map on cohomology $\operatorname{Pic} X = H^1(X, \mathcal{O}_X^*) \to H^1(X, \Omega_X)$ which we denote by c—see (Ex. 4.5).

(d) Returning to the hypotheses above, suppose $p = 1$. Show that $\eta(Y) = c(\mathcal{L}(Y))$, where $\mathcal{L}(Y)$ is the invertible sheaf corresponding to the divisor Y.

(Starred in the book.)

Solution. Setup, conventions, and what is being assumed. Write $\Omega^i = \Omega_X^i$. Cohomology classes of $\Omega^\bullet$ have a cup product $H^a(X, \Omega^i) \times H^b(X, \Omega^j) \to H^{a+b}(X, \Omega^{i+j})$ (induced by $\wedge$), which is graded-commutative and compatible with restriction $i^* \colon H^a(X, \Omega_X^j) \to H^a(Y, \Omega_Y^j)$ to a closed subvariety $i \colon Y \hookrightarrow X$ (because $\Omega_X^\bullet \to i_*\Omega_Y^\bullet$ is a map of sheaves of graded algebras). We use that the duality isomorphism of Cor. 7.13 is given by the cup product pairing followed by the trace,

$$H^q(X, \Omega^p) \times H^{n-q}(X, \Omega^{n-p}) \xrightarrow{\cup} H^n(X, \Omega^n) \xrightarrow{t_X} k,$$

which is perfect. (This is how the isomorphism $\Omega^{n-p} \cong (\Omega^p)^\vee \otimes \omega$ and (7.7) are combined in the proof of 7.13; that the iso of (7.7) is the cup product pairing is the standard compatibility of the maps θ^i of Thm. 7.6 with cup/Yoneda products, which we take for granted.) Thus $\eta(Y) \in H^p(X, \Omega^p)$ is characterized by

$$t_X\big(\eta(Y) \cup \alpha\big) = t_Y\big(\alpha|_Y\big) \qquad \text{for all } \alpha \in H^{n-p}(X, \Omega^{n-p}). \qquad \text{(E)}$$

For a disjoint union $Y = \bigsqcup Y_j$ of nonsingular varieties of the same dimension we put $t_Y = \sum t_{Y_j}$ and $\eta(Y) = \sum \eta(Y_j)$; (E) still characterizes $\eta(Y)$.

Normalization of the traces. By Prop. 7.2 each trace map t_Y is determined by the dualizing sheaf $\Omega_Y^{\dim Y}$ only up to a nonzero scalar, and Hartshorne does not fix a normalization (cf. Rem. 7.14: only for curves is the trace made explicit, via residues). A statement such as "$t_X(\eta(P)) = 1$" or "$\eta(Y) = c(\mathcal{L}(Y))$" can only hold for compatible normalizations. So the exercise implicitly uses the *canonical* traces (residues, Grothendieck duality), and I make explicit the only two properties of them that are used:

- **(T1)** for $Y = \operatorname{Spec} k$ a rational point, $\omega_Y = \mathcal{O}_Y = k$ and $t_Y = \mathrm{id}_k$;
- **(T2)** (compatibility with Poincare residue) if $D \subset Z$ is a nonsingular subvariety of codimension one in a nonsingular projective variety Z of dimension $m \geq 1$, then $t_D = -\,t_Z \circ \partial$, where $\partial \colon H^{m-1}(D, \Omega_D^{m-1}) \to H^m(Z, \Omega_Z^m)$ is the connecting homomorphism of the residue sequence (R) below (computed with the Cech conventions fixed in (d)).

Since the traces t_Z, t_D are chosen independently by Prop. 7.2, nothing in the axioms of a dualizing sheaf links them; (T2) is precisely the normalization that does, and it is an *assumption* (in dimension 1 it is the residue theorem, Rem. 7.14). Thus what is proved below from first principles in (d) is the Cech computation (D); the equality $\eta(Y) = c(\mathcal{L}(Y))$ then follows only via (T2). The sign in (T2) only reflects the sign conventions for Cech cochains and for $\mathrm{Pic} = H^1(\mathcal{O}^*)$ made in (d); with the opposite convention one would get $\eta(D) = -c(\mathcal{L}(D))$ in (d). For a curve Z and a point $D = P$, (T2) is the statement $t_Z(c(\mathcal{O}(P))) = 1$, i.e. $\mathrm{res}_P(d\log t) = 1$ for a uniformizer t, Rem. 7.14. (Also the identification $H^p(X, \Omega^p) \cong k$ in (b) is determined by Ex. 7.3 only up to a scalar; the one meant is the one in which linear subspaces have class 1, equivalently $\xi \mapsto t_X(\xi \cup c(\mathcal{O}(1))^{n-p})$, see (b).)

(a) Here $p = n$, $Y = \{P\} = \operatorname{Spec} k$ ($k = \bar{k}$), $\Omega^0_Y = \mathcal{O}_Y = k$ and $t_Y = \mathrm{id}$ by (T1). The restriction $H^0(X, \Omega^0_X) = H^0(X, \mathcal{O}_X) = k \to H^0(Y, \mathcal{O}_Y) = k$ is the identity on constants. Apply (E) with $\alpha = 1 \in H^0(X, \mathcal{O}_X)$:

$$t_X(\eta(P)) = t_X(\eta(P) \cup 1) = t_Y(1|_Y) = 1.$$

(c) Let f, g be units. Then $d\log(fg) = (fg)^{-1}(f\,dg + g\,df) = g^{-1}dg + f^{-1}df$, so $d\log\colon \mathcal{O}^*_X \to \Omega_X$ is a morphism of sheaves of abelian groups (multiplicative to additive), and it is compatible with pullback of functions and restriction to closed subschemes $i\colon Y \to X$: $i^*(d\log f) = d\log(i^*f)$. Hence it induces homomorphisms $H^i(X, \mathcal{O}^*_X) \to H^i(X, \Omega_X)$; for $i = 1$ and $\mathrm{Pic}\, X \cong H^1(X, \mathcal{O}^*_X)$ (Ex. 4.5) this is $c\colon \mathrm{Pic}\, X \to H^1(X, \Omega_X)$. In terms of Cech cocycles for an open cover $\mathfrak{U}$: if $\mathcal{L}$ has trivializations $\phi_i\colon \mathcal{L}|_{U_i} \xrightarrow{\sim} \mathcal{O}_{U_i}$ and transition functions $g_{ij} = \phi_i\phi_j^{-1} \in \mathcal{O}^*(U_{ij})$, then $c(\mathcal{L}) = [\,d\log g_{ij}\,]$. Moreover c is a group homomorphism, $c(\mathcal{L} \otimes \mathcal{M}) = c(\mathcal{L}) + c(\mathcal{M})$, and $i^*c(\mathcal{L}) = c(i^*\mathcal{L})$.

(d) Let $Y \subset X$ be a nonsingular prime divisor ($p = 1$); put $\mathcal{L} = \mathcal{L}(Y) = \mathcal{O}_X(Y) \cong \mathcal{I}_Y^{-1}$. By (E) and the perfectness of the pairing it suffices to show

$$t_X\big(c(\mathcal{L}) \cup \alpha\big) = t_Y(\alpha|_Y) \qquad \text{for all } \alpha \in H^{n-1}(X, \Omega^{n-1}). \qquad \text{(G)}$$

The residue sequence. Since Y is nonsingular of codimension one, $\mathcal{I}_Y/\mathcal{I}_Y^2 \to \Omega_X \otimes \mathcal{O}_Y \to \Omega_Y \to 0$ is exact on the left too (II, 8.17), with all terms locally free. Let $x \in Y$ and f a generator of $\mathcal{I}_Y$ near x; then df generates the image of $\mathcal{I}_Y/\mathcal{I}_Y^2$, which is a direct summand, so near x there is a basis $df, dx_2, \ldots, dx_n$ of Ω^1_X, and $dx_2|_Y, \ldots, dx_n|_Y$ is a basis of Ω^1_Y. Hence $\Omega^n_X(Y) = \Omega^n_X \otimes \mathcal{L}$ is generated near x by $\frac{df}{f} \wedge dx_2 \wedge \cdots \wedge dx_n$, every section of $\Omega^n_X(Y)$ near x is of the form $\frac{df}{f} \wedge \beta$ with $\beta \in \Omega^{n-1}_X$, and the kernel of $\Omega^{n-1}_X \to \Omega^{n-1}_Y$ is $f\Omega^{n-1}_X + df \wedge \Omega^{n-2}_X$. One checks that

$$\mathrm{Res}\colon \Omega^n_X(Y) \to i_*\Omega^{n-1}_Y, \qquad \mathrm{Res}\left(\frac{df}{f} \wedge \beta\right) = \beta|_Y$$

is well defined (if $\frac{df}{f} \wedge \beta = \frac{df}{f} \wedge \beta'$ then $\beta - \beta' \in df \wedge \Omega^{n-2}$ modulo $f\Omega^{n-1}$, so $(\beta - \beta')|_Y = 0$; and replacing f by uf, u a unit, changes $\frac{df}{f}$ by the regular form $\frac{du}{u}$, which changes β only by terms restricting to 0 on Y, using the same description of the kernel), $\mathcal{O}_X$-linear, surjective, with kernel Ω^n_X (a form $\frac{df}{f} \wedge \beta$ has residue 0 iff $\beta \in f\Omega^{n-1} + df \wedge \Omega^{n-2}$ iff it lies in Ω^n_X). This is the Poincare residue sequence

$$0 \to \Omega^n_X \to \Omega^n_X(Y) \xrightarrow{\mathrm{Res}} i_*\Omega^{n-1}_Y \to 0, \qquad \text{(R)}$$

all of whose terms are quasi-coherent; let $\partial\colon H^{n-1}(Y,\Omega_Y^{n-1}) \to H^n(X,\Omega_X^n)$ be its connecting homomorphism (note $H^j(X,i_*\mathcal{G}) = H^j(Y,\mathcal{G})$, Lemma 2.10).

Computation of $\partial(\alpha|_Y)$. X is separated, so we may compute with Cech cochains for an affine open cover $\mathfrak{U} = \{U_i\}_{i\in I}$ (I totally ordered) such that $\mathcal{I}_Y|_{U_i} = f_i\mathcal{O}_{U_i}$ (take $f_i = 1$ if $U_i \cap Y = \emptyset$); Cech cohomology computes sheaf cohomology compatibly with connecting maps (Thm. 4.5), and the sequence of Cech complexes of (R) is exact because the terms are quasi-coherent and the $U_{i_0\dots i_r}$ are affine. The sheaf $\mathcal{L}$ is trivialized on U_i by $\phi_i =$ multiplication by f_i (sending the generator $1/f_i$ of $\mathcal{O}(Y)|_{U_i}$ to 1), so its transition functions are $g_{ij} = \phi_i\phi_j^{-1} = f_i/f_j$, and $c(\mathcal{L}) = [d\log g_{ij}]$. Cup products of Cech cochains are $(a\cup b)_{i_0\dots i_{r+s}} = a_{i_0\dots i_r}\wedge b_{i_r\dots i_{r+s}}$.

Let $\alpha = (\alpha_{i_0\dots i_{n-1}})$ be a Cech $(n-1)$-cocycle of Ω_X^{n-1}. A lift of $\alpha|_Y$ (a cocycle of $i_*\Omega_Y^{n-1}$) to a cochain of $\Omega_X^n(Y)$ is

$$\lambda_{i_0\dots i_{n-1}} = d\log f_{i_0}\wedge\alpha_{i_0\dots i_{n-1}}, \qquad \operatorname{Res}\lambda = \alpha|_Y.$$

Now compute $(\delta\lambda)_{i_0\dots i_n} = \sum_{j=0}^n(-1)^j\lambda_{i_0\dots\widehat{i_j}\dots i_n}$. The term $j=0$ is $d\log f_{i_1}\wedge\alpha_{i_1\dots i_n}$, and the terms $j\ge 1$ have first index i_0, so they give $d\log f_{i_0}\wedge\sum_{j\ge1}(-1)^j\alpha_{\dots\widehat{i_j}\dots} = -d\log f_{i_0}\wedge\alpha_{i_1\dots i_n}$, because $\sum_{j=0}^n(-1)^j\alpha_{\dots\widehat{i_j}\dots} = 0$ (α is a cocycle). Hence

$$(\delta\lambda)_{i_0\dots i_n} = -\,d\log\bigl(f_{i_0}/f_{i_1}\bigr)\wedge\alpha_{i_1\dots i_n} = -\,d\log g_{i_0i_1}\wedge\alpha_{i_1\dots i_n} = -(c(\mathcal{L})\cup\alpha)_{i_0\dots i_n}.$$

This is a cocycle with values in Ω_X^n (as it must be), and by definition of the connecting homomorphism

$$\partial(\alpha|_Y) = -\,c(\mathcal{L})\cup\alpha \qquad \text{in } H^n(X,\Omega_X^n). \qquad \text{(D)}$$

By (T2) with $Z=X$, $D=Y$, $t_Y = -t_X\circ\partial$, hence

$$t_Y(\alpha|_Y) = -t_X(\partial(\alpha|_Y)) = t_X(c(\mathcal{L})\cup\alpha),$$

which is (G). Therefore $\eta(Y) = c(\mathcal{L}(Y))$ by the perfectness of the cup product pairing. For $Y = \bigsqcup Y_j$ a nonsingular closed subscheme all of whose components are divisors, we get the same by additivity: $\eta(Y) = \sum c(\mathcal{L}(Y_j)) = c(\mathcal{L}(Y))$, since $\mathcal{L}(Y) = \bigotimes\mathcal{L}(Y_j)$ and c is a homomorphism. (In particular, for a curve X and a point P, (a) and (d) together say $t_X(c(\mathcal{O}(P))) = 1$.)

(b) Let $X = \mathbf{P}^n$ and $c := c(\mathcal{O}_X(1)) \in H^1(X,\Omega^1)$, an element of even total degree 2, which therefore commutes with all cup products. We prove:

Claim A. Let $Y\subset X$ be a nonsingular closed subscheme of pure codimension $p<n$ (a disjoint union of nonsingular varieties), and H a hyperplane such that the scheme $D = Y\cap H$ is nonsingular of pure codimension $p+1$ in X (i.e. it is reduced, and H contains no component of Y). Then $\eta(Y)\cup c = \eta(D)$.

Proof. Let $d = n-p = \dim Y$ and $\alpha\in H^{d-1}(X,\Omega^{d-1})$. Using (E) for Y with $\beta = c\cup\alpha$, and the compatibility of restriction with cup products and with c,

$$t_X\bigl((\eta(Y)\cup c)\cup\alpha\bigr) = t_X\bigl(\eta(Y)\cup(c\cup\alpha)\bigr) = t_Y\bigl((c\cup\alpha)|_Y\bigr) = t_Y\bigl(c(\mathcal{O}_X(1)|_Y)\cup\alpha|_Y\bigr).$$

Now $\mathcal{O}_X(1)|_Y = \mathcal{O}_X(H)|_Y = \mathcal{O}_Y(D)$: a local equation of H restricts to a local equation of the scheme $D = Y\cap H$ in Y, which is a nonsingular, hence reduced, divisor $\sum D_j$ on each component of Y, so $\mathcal{O}_Y(D) = \mathcal{L}_Y(D)$. By (d) applied to the nonsingular projective variety (each component

of) Y and its nonsingular divisor D (with $\alpha|_Y \in H^{d-1}(Y, \Omega_Y^{d-1})$, and with Y of dimension d in place of X, formula (G)),

$$t_Y\big(c(\mathcal{L}_Y(D)) \cup \alpha|_Y\big) = t_D\big((\alpha|_Y)|_D\big) = t_D(\alpha|_D) = t_X\big(\eta(D) \cup \alpha\big),$$

the last equality by (E) for D. As α is arbitrary and the pairing is perfect, $\eta(Y) \cup c = \eta(D)$. $\square$

Claim B. For $Y \subset \mathbf{P}^n$ nonsingular (disjoint union of varieties) of pure codimension p,

$$t_X\big(\eta(Y) \cup c^{n-p}\big) = \deg Y \cdot 1_k.$$

Proof. Put $Y_0 = Y$. If $d = n - p \geq 1$, by Bertini (II, 8.18; applied to each component of Y_0 with its embedding in $\mathbf{P}^n$; the hyperplanes with the required property form a dense open subset of the dual projective space for each component, so a common one exists) there is a hyperplane H_1 containing no component of Y_0 such that $Y_1 = Y_0 \cap H_1$ is regular at every point, hence nonsingular (as $k = \bar{k}$) of pure dimension $d - 1$ (it is nonempty since $d \geq 1$, I, 7.2). Repeat to get $Y_2 = Y_1 \cap H_2, \ldots, Y_d = Y_{d-1} \cap H_d$, a finite set of reduced points of codimension n in X. By Claim A used d times (and c^d commuting with everything), $\eta(Y) \cup c^d = \eta(Y_d) = \sum_{P \in Y_d} \eta(P)$. By (a), $t_X(\eta(Y) \cup c^d) = \#Y_d \cdot 1_k$.

It remains to see that $\#Y_d = \deg Y$. For a variety Z of dimension $r \geq 1$ in $\mathbf{P}^n$ and a hyperplane $H \not\supseteq Z$ with $Z \cap H = Z_1 \cup \cdots \cup Z_s$ (irreducible components), I, Thm. 7.7 says $\deg Z = \sum_j i(Z, H; Z_j) \deg Z_j$, where $i(Z, H; Z_j)$ is the length of the local ring of the scheme $Z \cap H$ at the generic point of Z_j. Since $Z \cap H$ is reduced here, all $i = 1$. Hence $\deg Y_i = \deg Y_{i+1}$ for all i (sum over components; degree is additive for disjoint unions, being the leading coefficient of the Hilbert polynomial times $r!$), and finally $\deg Y_d = \#Y_d$ (a point has Hilbert polynomial 1). Thus $\#Y_d = \deg Y$. $\square$

Identification of $H^p(X, \Omega^p)$ with k. By Ex. 7.3, $H^p(X, \Omega^p)$ is a one-dimensional k-vector space ($0 \leq p \leq n$), so it is isomorphic to k, but not canonically. Let $\iota\colon H^p(X, \Omega^p) \to k$, $\iota(\xi) = t_X(\xi \cup c^{n-p})$. For a linear subspace $L \subset X$ of codimension p (nonsingular, of degree 1), Claim B gives $\iota(\eta(L)) = 1 \neq 0$. So ι is a nonzero linear functional on a one-dimensional space, hence an isomorphism, and it is the identification in which the class of a linear subspace is 1 (for $p = n$ it is just t_X, by (a)). With this identification, Claim B says

$$\eta(Y) = \deg(Y) \cdot 1,$$

i.e. $\iota(\eta(Y)) = \deg Y \cdot 1_k$; equivalently $\eta(Y) = (\deg Y)\, \eta(L)$. (If $\operatorname{char} k = \ell > 0$, this is the statement modulo ℓ.) For $p = 0$, $Y = X$, one has directly $\eta(X) = 1 \in H^0(X, \mathcal{O}_X) = k$ by (E), $\deg X = 1$. $\square$

8. Higher direct images of sheaves

Conventions and results used. For a continuous map (or morphism of ringed spaces) $f : X \to Y$, $R^i f_*$ denotes the right derived functors of f_*. By Prop. 8.4 they may be computed on $\mathfrak{Mod}(X)$ as well as on $\mathfrak{Ab}(X)$. We use: injective $\mathcal{O}_X$-modules are flasque (Prop. 2.4), flasque sheaves are acyclic for $\Gamma(X, \cdot)$ (Prop. 2.5) and for f_* (Cor. 8.3), and acyclic resolutions compute derived functors (Prop. 1.2A). For an $\mathcal{O}_X$-module $\mathcal{F}$ we write $\mathcal{F}(l) = \mathcal{F} \otimes \mathcal{O}_X(l)$, and $\mathcal{G}^\vee = \mathcal{H}om(\mathcal{G}, \mathcal{O})$ (written $\check{}$ in the book).

Exercise 8.1

Problem. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{F}$ be a sheaf of abelian groups on X, and assume that $R^i f_*(\mathcal{F}) = 0$ for all $i > 0$. Show that there are natural isomorphisms, for each $i \geq 0$,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

(This is a degenerate case of the Leray spectral sequence—see Godement [1, II, 4.17.1].)

Solution. Choose a flasque resolution $0 \to \mathcal{F} \to \mathcal{G}^0 \to \mathcal{G}^1 \to \cdots$ of $\mathcal{F}$ on X (for instance an injective resolution in $\mathfrak{Ab}(X)$, or the Godement resolution). Each $f_*\mathcal{G}^j$ is flasque on Y, because $f_*\mathcal{G}^j(V) = \mathcal{G}^j(f^{-1}V)$ and the restriction maps of $\mathcal{G}^j$ are surjective.

Since flasque sheaves are acyclic for f_* (Cor. 8.3), the resolution $\mathcal{G}^\bullet$ computes the derived functors of f_* (Prop. 1.2A): $R^i f_*(\mathcal{F}) = h^i(f_*\mathcal{G}^\bullet)$ for all $i \geq 0$. For $i = 0$ this is the left exactness of f_*: $\ker(f_*\mathcal{G}^0 \to f_*\mathcal{G}^1) = f_* \ker(\mathcal{G}^0 \to \mathcal{G}^1) = f_*\mathcal{F}$. For $i > 0$ it is $R^i f_*\mathcal{F} = 0$ by hypothesis. Hence

$$0 \to f_*\mathcal{F} \to f_*\mathcal{G}^0 \to f_*\mathcal{G}^1 \to \cdots$$

is exact, i.e. it is a flasque resolution of $f_*\mathcal{F}$ on Y. Flasque sheaves are Γ-acyclic (Prop. 2.5), so again by Prop. 1.2A

$$H^i(Y, f_*\mathcal{F}) = h^i\big(\Gamma(Y, f_*\mathcal{G}^\bullet)\big) = h^i\big(\Gamma(X, \mathcal{G}^\bullet)\big) = H^i(X, \mathcal{F}),$$

where the middle equality is $\Gamma(Y, f_*\mathcal{G}^j) = \Gamma(X, \mathcal{G}^j)$, compatibly with the differentials, and the last is again Prop. 1.2A, now on X. The isomorphism is independent of the chosen resolution (any two flasque resolutions are homotopy equivalent over the identity of $\mathcal{F}$) and natural in $\mathcal{F}$ among sheaves satisfying the hypothesis; it is the edge map $H^i(Y, f_*\mathcal{F}) \to H^i(X, \mathcal{F})$ of the Leray spectral sequence (whose only nonzero row is the bottom one). □

Exercise 8.2

Problem. Let $f : X \to Y$ be an affine morphism of schemes (II, Ex. 5.17) with X noetherian, and let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that the hypotheses of (Ex. 8.1) are satisfied, and hence that $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$ for each $i \geq 0$. (This gives another proof of (Ex. 4.1).)

Solution. Recall that f affine means that Y has an open cover by affine open sets V_α with $f^{-1}(V_\alpha)$ affine. Fix $V = V_\alpha = \operatorname{Spec} A$, so $X_V := f^{-1}(V)$ is an open subscheme of the noetherian scheme X, hence noetherian, and it is affine. Let $f_V : X_V \to V$ be the restriction. By Cor. 8.2,

$$(R^i f_*\mathcal{F})|_V = R^i(f_V)_*(\mathcal{F}|_{X_V}).$$

Now V is an affine scheme, X_V is a noetherian scheme and $\mathcal{F}|_{X_V}$ is quasi-coherent, so Prop. 8.5 gives $R^i(f_V)_*(\mathcal{F}|_{X_V}) \cong H^i(X_V, \mathcal{F}|_{X_V})^\sim$. As X_V is a noetherian affine scheme, $H^i(X_V, \mathcal{F}|_{X_V}) = 0$ for $i > 0$ by Thm. 3.5. Therefore $(R^i f_*\mathcal{F})|_{V_\alpha} = 0$ for all α and all $i > 0$, and since the V_α cover Y, $R^i f_*\mathcal{F} = 0$ for $i > 0$. This is the hypothesis of Ex. 8.1, which then yields $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$ for all $i \geq 0$. □

Exercise 8.3

Problem. Let $f : X \to Y$ be a morphism of ringed spaces, let $\mathcal{F}$ be an $\mathcal{O}_X$-module, and let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$-module of finite rank. Prove the projection formula (cf. (II, Ex. 5.1))

$$R^i f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong R^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

Solution. We compare two δ-functors $\mathfrak{Mod}(X) \to \mathfrak{Mod}(Y)$:

$$T^i(\mathcal{F}) = R^i f_*(\mathcal{F} \otimes f^*\mathcal{E}), \qquad U^i(\mathcal{F}) = R^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

Here $R^i f_*$ is the derived functor on $\mathfrak{Mod}(X)$ (Prop. 8.4), and it forms a δ-functor there.

Both are δ-functors. The sheaf $f^*\mathcal{E}$ is locally free of finite rank, hence flat, so $\mathcal{F} \mapsto \mathcal{F} \otimes f^*\mathcal{E}$ is exact on $\mathfrak{Mod}(X)$; composing with the δ-functor $R^i f_*$ gives that T is a δ-functor. Likewise $\mathcal{G} \mapsto \mathcal{G} \otimes \mathcal{E}$ is exact on $\mathfrak{Mod}(Y)$, so U is a δ-functor.

The case $i = 0$ (II, Ex. 5.1(c)). There is a natural map $f_*\mathcal{F} \otimes \mathcal{E} \to f_*(\mathcal{F} \otimes f^*\mathcal{E})$, given over $V \subseteq Y$ by $s \otimes e \mapsto s \otimes f^*e$ (for $s \in \mathcal{F}(f^{-1}V)$ and $e \in \mathcal{E}(V)$, followed by sheafification). It is an isomorphism whenever $\mathcal{E} = \mathcal{O}_V^n$ (both sides are $(f_*\mathcal{F})^n$), hence, the question being local on Y, it is an isomorphism for every locally free $\mathcal{E}$ of finite rank. So $\eta^0 : U^0 \to T^0$ is an isomorphism of functors.

Both are effaceable for $i > 0$. Let $\mathcal{F} \in \mathfrak{Mod}(X)$ and embed $\mathcal{F} \hookrightarrow \mathcal{I}$ with $\mathcal{I}$ injective in $\mathfrak{Mod}(X)$. Then $\mathcal{I}$ is flasque (Prop. 2.4), so $R^i f_*\mathcal{I} = 0$ for $i > 0$ (Cor. 8.3), i.e. $U^i(\mathcal{I}) = 0$. For T^i: let $V \subseteq Y$ be open with $\mathcal{E}|_V \cong \mathcal{O}_V^n$. Then $f^{-1}(V)$ carries

$$(\mathcal{I} \otimes f^*\mathcal{E})|_{f^{-1}V} \cong (\mathcal{I}|_{f^{-1}V})^{\oplus n},$$

which is flasque (restrictions of flasque sheaves to open sets are flasque, finite sums of flasque sheaves are flasque). By Cor. 8.2 and Cor. 8.3, $T^i(\mathcal{I})|_V = R^i(f|_{f^{-1}V})_*\big((\mathcal{I} \otimes f^*\mathcal{E})|_{f^{-1}V}\big) = 0$ for $i > 0$. Such V cover Y, so $T^i(\mathcal{I}) = 0$ for $i > 0$.

Conclusion. A δ-functor all of whose i-th terms ($i > 0$) are effaceable is universal (Thm. 1.3A). Hence U and T are universal δ-functors. The isomorphism $\eta^0 : U^0 \to T^0$ therefore extends uniquely to a morphism of δ-functors $\eta : U \to T$; its inverse $(\eta^0)^{-1}$ extends uniquely to $\theta : T \to U$, and $\theta\eta$, $\eta\theta$ are morphisms of δ-functors restricting to the identity in degree 0, so equal the identity by uniqueness. Thus $\eta^i : R^i f_*(\mathcal{F}) \otimes \mathcal{E} \xrightarrow{\sim} R^i f_*(\mathcal{F} \otimes f^*\mathcal{E})$ for all i, naturally in $\mathcal{F}$. □

Exercise 8.4

Problem. Let Y be a noetherian scheme, and let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$-module of rank $n+1$, $n \geq 1$. Let $X = \mathbf{P}(\mathcal{E})$ (II, Section 7), with the invertible sheaf $\mathcal{O}_X(1)$ and the projection morphism $\pi : X \to Y$.

(a) Then $\pi_*(\mathcal{O}(l)) \cong S^l(\mathcal{E})$ for $l \geq 0$, $\pi_*(\mathcal{O}(l)) = 0$ for $l < 0$ (II, 7.11); $R^i\pi_*(\mathcal{O}(l)) = 0$ for $0 < i < n$ and all $l \in \mathbf{Z}$; and $R^n\pi_*(\mathcal{O}(l)) = 0$ for $l > -n - 1$.

(b) Show there is a natural exact sequence

$$0 \to \Omega_{X/Y} \to (\pi^*\mathcal{E})(-1) \to \mathcal{O} \to 0,$$

cf. (II, 8.13), and conclude that the *relative canonical sheaf* $\omega_{X/Y} = \wedge^n\Omega_{X/Y}$ is isomorphic to $(\pi^* \wedge^{n+1} \mathcal{E})(-n-1)$. Show furthermore that there is a natural isomorphism $R^n\pi_*(\omega_{X/Y}) \cong \mathcal{O}_Y$ (cf. (7.1.1)).

(c) Now show, for any $l \in \mathbf{Z}$, that

$$R^n\pi_*(\mathcal{O}(l)) \cong \pi_*(\mathcal{O}(-l-n-1))^\vee \otimes (\wedge^{n+1}\mathcal{E})^\vee.$$

(d) Show that $p_a(X) = (-1)^n p_a(Y)$ (use (Ex. 8.1)) and $p_g(X) = 0$ (use (II, 8.11)).

(e) In particular, if Y is a nonsingular projective curve of genus g, and $\mathcal{E}$ a locally free sheaf of rank 2, then X is a projective surface with $p_a = -g$, $p_g = 0$, and irregularity g (7.12.3). This kind of surface is called a *geometrically ruled surface* (V, Section 2).

Solution. Preliminaries. Let $V = \operatorname{Spec} A \subseteq Y$ be an affine open set on which $\mathcal{E}|_V \cong \mathcal{O}_V^{n+1}$. Then $\pi^{-1}(V) \cong \mathbf{P}_A^n$ and $\mathcal{O}_X(1)$ corresponds to $\mathcal{O}(1)$ (II, Section 7); A is noetherian. Such V cover Y. By Cor. 8.2 and Prop. 8.5, for every l and i,

$$(R^i\pi_*\mathcal{O}_X(l))|_V \cong H^i(\mathbf{P}_A^n, \mathcal{O}(l))^\sim \qquad (*)$$

naturally in the sheaf $\mathcal{O}(l)$ (i.e. compatibly with morphisms between such sheaves, such as multiplication by a section of some $\mathcal{O}(m)$).

(a) The statements about π_* are given by (II, 7.11). The other statements are local on Y (Cor. 8.2), so by $(*)$ they concern $H^i(\mathbf{P}_A^n, \mathcal{O}(l))$, A a noetherian ring. By Thm. 5.1(b) this vanishes for $0 < i < n$ and all l. For $i = n$, Thm. 5.1(d) says that

$$H^0(\mathbf{P}_A^n, \mathcal{O}(-l-n-1)) \times H^n(\mathbf{P}_A^n, \mathcal{O}(l)) \to H^n(\mathbf{P}_A^n, \mathcal{O}(-n-1)) \cong A$$

is a perfect pairing of finitely generated free A-modules (apply (d) with $m = -l-n-1$ in place of l). If $l > -n-1$ then $-l-n-1 < 0$ and $H^0(\mathbf{P}_A^n, \mathcal{O}(-l-n-1)) = 0$ by Thm. 5.1(a), so $H^n(\mathbf{P}_A^n, \mathcal{O}(l)) = 0$. We shall also need: $R^i\pi_*\mathcal{O}(l) = 0$ for $i > n$ and all l; indeed $\mathbf{P}_A^n$ is covered by the $n+1$ affine open sets $D_+(x_j)$, so the Cech complex of this cover (which computes cohomology of quasi-coherent sheaves, Thm. 4.5) vanishes in degrees $> n$.

(b), the exact sequence. Write $\mathcal{S} = \operatorname{Sym}\mathcal{E}$, so $X = \operatorname{Proj}\mathcal{S}$. The canonical surjection $\varepsilon : \pi^*\mathcal{E} \to \mathcal{O}_X(1)$ (II, 7.11, 7.12) twists to a surjection $\varepsilon(-1) : (\pi^*\mathcal{E})(-1) \to \mathcal{O}_X$; let $\mathcal{K}$ be its kernel, locally free of rank n (a kernel of a surjection of locally free sheaves). We construct an isomorphism $\Omega_{X/Y} \xrightarrow{\sim} \mathcal{K}$.

Work over an affine open $V = \operatorname{Spec} A \subseteq Y$, $M = \mathcal{E}(V)$, $S = \operatorname{Sym}_A M$, $\pi^{-1}V = \operatorname{Proj} S$. For any A-module M one has $\Omega_{S/A} \cong S \otimes_A M$ via $d(m) = 1 \otimes m$ $(m \in M)$; write $d : S \to S \otimes_A M$ for the universal derivation, and extend it to localizations. Let $m \in M$ and $U = D_+(m)$, with $\mathcal{O}(U) = S_{(m)}$. For $h = g/m^e \in S_{(m)}$ $(g \in S_e)$ we have $dh = (dg)/m^e - e\,g\,(1 \otimes m)/m^{e+1}$, an element of $S_m \otimes M$ all of whose S-coordinates have degree -1. That is, $dh \in \Gamma(U, \mathcal{O}_X(-1) \otimes \pi^*\mathcal{E})$. This defines an A-linear derivation $S_{(m)} \to \Gamma(U, (\pi^*\mathcal{E})(-1))$, hence an $\mathcal{O}_X$-linear map $\delta : \Omega_{X/Y}|_U \to (\pi^*\mathcal{E})(-1)|_U$. The maps δ agree on overlaps because they are all restrictions of the single canonical derivation d, and the construction is compatible with change of affine open V; so we get a global map $\delta : \Omega_{X/Y} \to (\pi^*\mathcal{E})(-1)$.

δ lands in $\mathcal{K}$. The composite of d with the multiplication map $S \otimes M \to S$ $(s \otimes m \mapsto sm)$ is a derivation $S \to S$ which equals $m \mapsto m$ on M and vanishes on A; so it is the Euler derivation $s \mapsto (\deg s)\, s$. Hence for h of degree 0 in S_m, $\varepsilon(-1)(dh) = 0$.

$\delta : \Omega_{X/Y} \to \mathcal{K}$ is an isomorphism. This is local on Y, so take M free with basis $e_0, \ldots, e_n$, $S = A[x_0, \ldots, x_n]$ $(x_j = e_j)$. On $U_i = D_+(x_i)$, $\Omega_{X/Y}$ is free with basis $d(x_j/x_i)$, $j \neq i$ (II, 8.13

/ 8.1), and $\delta(d(x_j/x_i)) = x_i^{-1}e_j - x_j x_i^{-2}e_i$. On the other hand $(\pi^*\mathcal{E})(-1)|_{U_i}$ is free with basis $x_i^{-1}e_j$ $(j = 0, \dots, n)$ and $\varepsilon(-1)(x_i^{-1}e_j) = x_j/x_i$; so $\mathcal{K}|_{U_i}$ is free with basis $x_i^{-1}e_j - (x_j/x_i)\, x_i^{-1}e_i$, $j \neq i$, and these are exactly the images of the basis of $\Omega_{X/Y}$. This proves that

$$0 \to \Omega_{X/Y} \xrightarrow{\delta} (\pi^*\mathcal{E})(-1) \xrightarrow{\varepsilon(-1)} \mathcal{O}_X \to 0$$

is exact (it is the sequence of (II, 8.13) made canonical).

(b), $\omega_{X/Y}$**.** For an exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ of locally free sheaves of ranks r', r, r'' we have $\wedge^r\mathcal{F} \cong \wedge^{r'}\mathcal{F}' \otimes \wedge^{r''}\mathcal{F}''$ (II, Ex. 5.16(d)). Applied to the sequence above ($r' = n$, $r'' = 1$):

$$\omega_{X/Y} = \wedge^n\Omega_{X/Y} \cong \wedge^{n+1}\big((\pi^*\mathcal{E})(-1)\big) = \pi^*(\wedge^{n+1}\mathcal{E}) \otimes \mathcal{O}_X(-n-1),$$

using $\wedge^r(\mathcal{G} \otimes \mathcal{L}) = \wedge^r\mathcal{G} \otimes \mathcal{L}^r$ for $\mathcal{G}$ locally free of rank r and $\mathcal{L}$ invertible.

(b), $R^n\pi_*\omega_{X/Y}$**.** Put $G = (\pi^*\mathcal{E})(-1)$ and $\epsilon = \varepsilon(-1) : G \to \mathcal{O}_X$, a surjection. Locally $G = \mathcal{O}_X \oplus \mathcal{K}$ and ϵ is the projection, so the Koszul complex

$$0 \to K_{n+1} \to K_n \to \cdots \to K_1 \to K_0 \to 0, \qquad K_j = \wedge^j G = \pi^*(\wedge^j\mathcal{E})(-j),$$

with differential contraction with ϵ, is exact (locally it is the direct sum of the identity maps $\wedge^{j-1}\mathcal{K} \to \wedge^{j-1}\mathcal{K}$ in the right degrees). Here $K_0 = \mathcal{O}_X$ and $K_{n+1} = \omega_{X/Y}$ by the above. For $1 \le j \le n$, by Ex. 8.3 and (a),

$$R^i\pi_*K_j \cong \wedge^j\mathcal{E} \otimes R^i\pi_*\mathcal{O}(-j) = 0 \quad \text{for all } i \ge 0 :$$

for $i = 0$ because $-j < 0$; for $0 < i < n$ by (a); for $i = n$ because $-j > -n-1$; for $i > n$ by the remark in (a). Let $Z_j = \operatorname{im}(K_{j+1} \to K_j)$, $0 \le j \le n$; then $Z_0 = K_0$, $Z_n \cong K_{n+1}$, and for $1 \le j \le n$ there are exact sequences $0 \to Z_j \to K_j \to Z_{j-1} \to 0$. The long exact sequences, together with $R^i\pi_*K_j = 0$, show that the connecting homomorphisms are isomorphisms $R^i\pi_*Z_{j-1} \xrightarrow{\sim} R^{i+1}\pi_*Z_j$ for all $i \ge 0$, $1 \le j \le n$. Composing for $j = 1, \dots, n$ (with $i = j - 1$):

$$\mathcal{O}_Y = \pi_*\mathcal{O}_X = \pi_*Z_0 \xrightarrow{\sim} R^1\pi_*Z_1 \xrightarrow{\sim} \cdots \xrightarrow{\sim} R^n\pi_*Z_n \cong R^n\pi_*(\omega_{X/Y}).$$

Here $\pi_*\mathcal{O}_X = \mathcal{O}_Y$ by (a) ($l = 0$). This is a natural (indeed canonical) isomorphism $\mathcal{O}_Y \cong R^n\pi_*(\omega_{X/Y})$.

(c). Put $L = \wedge^{n+1}\mathcal{E}$, an invertible sheaf. By (b) and Ex. 8.3,

$$\mathcal{O}_Y \cong R^n\pi_*(\pi^*L \otimes \mathcal{O}(-n-1)) \cong R^n\pi_*(\mathcal{O}(-n-1)) \otimes L,$$

so tensoring with $L^\vee$ gives a canonical isomorphism $\tau : R^n\pi_*(\mathcal{O}(-n-1)) \xrightarrow{\sim} L^\vee$. Fix $l \in \mathbf{Z}$ and put $m = -l-n-1$. Define a pairing

$$\beta : \pi_*(\mathcal{O}(m)) \otimes R^n\pi_*(\mathcal{O}(l)) \longrightarrow R^n\pi_*(\mathcal{O}(l+m)) = R^n\pi_*(\mathcal{O}(-n-1))$$

as follows: for $V \subseteq Y$ open, a section $s \in \Gamma(V, \pi_*\mathcal{O}(m)) = \Gamma(\pi^{-1}V, \mathcal{O}(m))$ gives a morphism $\cdot s : \mathcal{O}(l)|_{\pi^{-1}V} \to \mathcal{O}(l+m)|_{\pi^{-1}V}$, hence $R^n\pi_*(\cdot s)$ maps $H^n(\pi^{-1}V, \mathcal{O}(l)) \to H^n(\pi^{-1}V, \mathcal{O}(-n-1))$; this is bilinear and compatible with restriction, and after sheafification gives β. Adjoining and using τ we get

$$\Theta : R^n\pi_*(\mathcal{O}(l)) \to \mathcal{H}om(\pi_*\mathcal{O}(m), L^\vee) = (\pi_*\mathcal{O}(m))^\vee \otimes L^\vee,$$

the last equality because $\pi_*\mathcal{O}(m)$ is $S^m\mathcal{E}$ or 0, locally free of finite rank. To see that Θ is an isomorphism we may work over $V = \operatorname{Spec} A$ with $\mathcal{E}|_V$ free. By $(*)$ (naturality!), over V the map Θ corresponds to the A-linear map

$$H^n(\mathbf{P}^n_A, \mathcal{O}(l)) \to \operatorname{Hom}_A\big(H^0(\mathbf{P}^n_A, \mathcal{O}(m)), A\big)$$

adjoint to the pairing of Thm. 5.1(d), the identification of $H^n(\mathbf{P}^n_A, \mathcal{O}(-n-1))$ with A being the one induced by the isomorphism τ (any isomorphism with the free rank-one module will do). By Thm. 5.1(d) this adjoint map is an isomorphism of A-modules. Hence Θ is an isomorphism of sheaves, which is (c). (For $m < 0$ both sides are 0, consistent with (a).)

(d). The invariants p_a and p_g are defined for projective varieties over an algebraically closed field k (p_g for nonsingular ones, as the dimension of $\Gamma(\omega)$), so we assume—as the exercise implicitly does—that Y is a projective variety over an algebraically closed field k, nonsingular when we speak of p_g, and let $r = \dim Y$. Then $X = \mathbf{P}(\mathcal{E})$ is an integral scheme of dimension $r+n$, nonsingular if Y is (it is locally $V \times \mathbf{P}^n$), and it is projective over k: $\mathcal{E}(m)$ is generated by global sections for $m \gg 0$, so $X \cong \mathbf{P}(\mathcal{E}(m))$ embeds in $\mathbf{P}^N_Y = \mathbf{P}^N \times Y$ (II, 7.9, 7.12), which is projective by the Segre embedding (I, Ex. 3.16).

p_a. By (a) with $l = 0$: $\pi_*\mathcal{O}_X = \mathcal{O}_Y$ and $R^i\pi_*\mathcal{O}_X = 0$ for $i > 0$ (for $0 < i < n$; for $i = n$ since $0 > -n-1$; for $i > n$ by the remark in (a)). By Ex. 8.1, $H^i(X, \mathcal{O}_X) \cong H^i(Y, \mathcal{O}_Y)$ for all i, so $\chi(\mathcal{O}_X) = \chi(\mathcal{O}_Y)$. With $p_a(Z) = (-1)^{\dim Z}(\chi(\mathcal{O}_Z) - 1)$ (I, Ex. 7.2; Ex. 5.3),

$$p_a(X) = (-1)^{r+n}(\chi(\mathcal{O}_Y) - 1) = (-1)^n p_a(Y).$$

p_g. Let Y be nonsingular. Taking $Z = \operatorname{Spec} k$ in (II, 8.11) we have an exact sequence $\pi^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} \to 0$. The sheaves $\Omega_{Y/k}, \Omega_{X/k}$ are locally free of ranks r and $r+n$ (II, 8.15), and $\Omega_{X/Y}$ is locally free of rank n by (b). Hence the image of $\pi^*\Omega_{Y/k}$ is the kernel of a surjection of locally free sheaves, hence locally free of rank r; a surjection $\pi^*\Omega_{Y/k} \to \text{im}$ between locally free sheaves of the same rank r is an isomorphism. So the sequence $0 \to \pi^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} \to 0$ is exact and, by (II, Ex. 5.16(d)),

$$\omega_X = \wedge^{r+n}\Omega_{X/k} \cong \pi^*\omega_Y \otimes \omega_{X/Y} \cong \pi^*(\omega_Y \otimes L) \otimes \mathcal{O}_X(-n-1).$$

By the projection formula (Ex. 5.1 of Ch. II, or Ex. 8.3 with $i = 0$) and (II, 7.11),

$$\pi_*\omega_X \cong \omega_Y \otimes L \otimes \pi_*\mathcal{O}_X(-n-1) = 0, \qquad \text{since } -n-1 < 0.$$

Therefore $p_g(X) = \dim \Gamma(X, \omega_X) = \dim \Gamma(Y, \pi_*\omega_X) = 0$.

(e). Now Y is a nonsingular projective curve of genus $g = p_a(Y) = \dim H^1(Y, \mathcal{O}_Y)$ (as $H^0(\mathcal{O}_Y) = k$, Ex. 5.3(a)), and $\mathcal{E}$ has rank 2, so $n = 1$. By (d), X is a nonsingular projective variety of dimension 2 (a projective surface) with $p_a(X) = (-1)^1 p_a(Y) = -g$ and $p_g(X) = 0$. By Remark 7.12.3, the irregularity is $q = p_g - p_a = 0 - (-g) = g$ (equivalently $q = \dim H^1(X, \mathcal{O}_X) = \dim H^1(Y, \mathcal{O}_Y) = g$ by Ex. 8.1 as in (d)). □

9. Flat morphisms

Conventions and results used. "Matsumura" means *Commutative Ring Theory*. We use the following standard facts besides those of the text. (i) A flat local homomorphism of local rings

is faithfully flat, and a flat ring map $A \to B$ satisfies going-down, i.e. every prime of A contained in $\mathfrak{q} \cap A$ is of the form $\mathfrak{q}' \cap A$ with $\mathfrak{q}' \subseteq \mathfrak{q}$ (Matsumura, Thm. 9.5). (ii) Local flatness criterion (Matsumura, Thm. 22.3): let A be a noetherian ring, $I \subseteq A$ an ideal and M an A-module which is I-adically ideal-separated (this holds if $A \to B$ is a local homomorphism of noetherian local rings, $I \subseteq \mathfrak{m}_A$ and M is a finite B-module); then M is flat over A if and only if M/IM is flat over A/I and $\mathrm{Tor}_1^A(A/I, M) = 0$. In particular, for $A \to B$ a local homomorphism of noetherian local rings and M a finite B-module, M is A-flat iff $\mathrm{Tor}_1^A(k, M) = 0$, $k = A/\mathfrak{m}_A$. (iii) Generic flatness (Matsumura, Thm. 24.1): if A is a noetherian domain, B a finitely generated A-algebra and M a finite B-module, there is $0 \neq a \in A$ with M_a free over A_a. (iv) A finitely generated flat module over a noetherian local ring is free (9.1A(f)); a finitely generated module over a noetherian ring is flat iff it is locally free (9.2(e)). A localization of a B-module flat over A is flat over A (Ex. 9.1.1). A module over $A_{\mathfrak{p}}$ which is flat over A is flat over $A_{\mathfrak{p}}$.

Exercise 9.1

> **Problem.** A flat morphism $f : X \to Y$ of finite type of noetherian schemes is open, i.e. for every open subset $U \subseteq X$, $f(U)$ is open in Y. [Hint: Show that $f(U)$ is constructible and stable under generization (II, Ex. 3.18) and (II, Ex. 3.19).]

Solution. Let $U \subseteq X$ be open.

$f(U)$ is constructible. U is a constructible subset of X (it is open), and f is of finite type between noetherian schemes, so $f(U)$ is constructible by Chevalley's theorem (II, Ex. 3.19).

$f(U)$ is stable under generization. Let $x \in U$, $y = f(x)$, and let $y' \in Y$ be a generization of y, i.e. $y \in \overline{\{y'\}}$. Put $A = \mathcal{O}_{Y,y}$ and $B = \mathcal{O}_{X,x}$; f flat gives a flat local homomorphism $A \to B$. The point y' corresponds to a prime $\mathfrak{p}' \subseteq A$, and by going-down for flat extensions (Matsumura, Thm. 9.5) there is a prime $\mathfrak{q}' \subseteq B$ with $\mathfrak{q}' \cap A = \mathfrak{p}'$. The prime $\mathfrak{q}'$ is a point $x' \in X$ with $x \in \overline{\{x'\}}$ and $f(x') = y'$. Since U is open and contains x, it contains every generization of x; thus $x' \in U$ and $y' \in f(U)$.

By (II, Ex. 3.18(c)) (the underlying space of a noetherian scheme is a Zariski space), a subset which is constructible and stable under generization is open. Hence $f(U)$ is open. □

Exercise 9.2

> **Problem.** Do the calculation of (9.8.4) for the curve of (I, Ex. 3.14). Show that you get an embedded point at the cusp of the plane cubic curve.

Solution. Recall (I, Ex. 3.14(b)): $Y = X_1 \subseteq \mathbf{P}^3$ is the twisted cubic with parametrization $(x, y, z, w) = (t^3, t^2u, tu^2, u^3)$, $P = (0, 0, 1, 0)$, and the projection φ from P to the plane $z = 0$ is $(x : y : z : w) \mapsto (x : y : w) = (t^3 : t^2u : u^3)$, whose image is the cuspidal cubic $x^2w = y^3$, with cusp at $(0 : 0 : 1)$ (in the chart $w = 1$ it is the curve $(x, y) = (s^3, s^2)$, $s = t/u$). As in (9.8.3) let $\sigma_a(x : y : z : w) = (x : y : az : w)$, $X_a = \sigma_a(X_1)$ $(a \neq 0)$, and let $\overline{X} \subseteq \mathbf{P}^3 \times \mathbf{A}^1$ be the closure of $\bigcup_{a \neq 0} X_a \times \{a\}$; by (9.8) (or by the computation below) it is flat over $\mathbf{A}^1$ and $X_a \cong X_1$ for $a \neq 0$. We compute near the cusp. In the chart $w = 1$ with coordinates (x, y, z) the curve X_1 is $\{(t^3, t^2, t)\}$ (put $u = 1$), and $X_a = \{(t^3, t^2, at)\}$.

The closure of $\{(a, t^3, t^2, at) : a \neq 0\}$ in $\mathbf{A}^4$ is the integral closed subscheme $\operatorname{Spec} R$, where $R = k[a,x,y,z]/I$ and $I = \ker\big(k[a,x,y,z] \to k[a,t]\big)$, $x \mapsto t^3$, $y \mapsto t^2$, $z \mapsto at$ (the map is injective on the open set $a \neq 0$ of the source, which is dense in $\mathbf{A}^2_{a,t}$ since $k[a,t] \hookrightarrow k[a,a^{-1},t]$). Thus $R \cong k[a,t^3,t^2,at] \subseteq k[a,t]$ is a domain, hence torsion-free over the PID $k[a]$, hence flat over $k[a]$ (Ex. 9.1.3).

Claim: $I = J := (z^2 - a^2y,\ yz - ax,\ xz - ay^2,\ x^2 - y^3)$.

The four generators of J visibly map to 0 in $k[a,t]$ (e.g. $z^2 - a^2y \mapsto a^2t^2 - a^2t^2$, $xz - ay^2 \mapsto at^4 - at^4$), so $J \subseteq I$. For the converse use the lexicographic order with $z > x > y > a$. The leading monomials of the four generators are z^2, yz, xz, x^2 respectively. By the division algorithm every $F \in k[a,x,y,z]$ can be written $F = G + N$ with $G \in J$ and N a linear combination of monomials divisible by none of z^2, yz, xz, x^2, i.e. of the monomials a^iy^j, a^ixy^j, a^iz $(i,j \geq 0)$. These have images a^it^{2j}, a^it^{2j+3}, $a^{i+1}t$ in $k[a,t]$, which are pairwise distinct monomials (the exponent of t is even in the first family, odd and ≥ 3 in the second, equal to 1 with a-exponent ≥ 1 in the third). Hence if $F \in I$ then $N \in I$ forces $N = 0$, so $F \in J$. This proves the claim.

Consequently $\overline{X} \cap \{w \neq 0\} = \operatorname{Spec} k[a,x,y,z]/J$ is flat over $\mathbf{A}^1$, and the fibre over $a = 0$ is

$$X_0 = \operatorname{Spec} k[x,y,z]/I_0, \qquad I_0 = (z^2,\ yz,\ xz,\ x^2 - y^3).$$

Structure of X_0. The radical of I_0 is $\mathfrak{p} = (z, x^2 - y^3)$ (as $z^2 \in I_0$ and $x^2 - y^3 \in I_0$), which is prime, so the support of X_0 is the cuspidal cubic $x^2 = y^3$ in the plane $z = 0$, with cusp at the origin 0. At a point of the support other than the origin we have $x \neq 0$ and $y \neq 0$ (as $x = t^3, y = t^2$, $t \neq 0$), so $xz = 0$ gives $z = 0$ in the local ring: X_0 is the reduced cuspidal curve away from the cusp. At the origin: let $\mathfrak{m} = (x,y,z)$ and $B = k[x,y,z]/I_0$. In B, $z \neq 0$ and $\mathfrak{m}z = 0$ (because $z^2, yz, xz \in I_0$). Moreover $z \neq 0$ even in the local ring $B_{\mathfrak{m}}$: otherwise $z \in I_0 \cdot k[x,y,z]_{\mathfrak{m}} \subseteq \mathfrak{m}^2 k[x,y,z]_{\mathfrak{m}}$ (all generators of I_0 lie in $\mathfrak{m}^2$), which is absurd since $z \in \mathfrak{m} \setminus \mathfrak{m}^2$ in the regular local ring. So $\mathfrak{m}$ is an associated prime of B, but it is not minimal (the only minimal prime is $\mathfrak{p} \subsetneq \mathfrak{m}$): the origin is an *embedded point* of X_0. Note also that $z \notin I_0$, so X_0 is not contained in the plane $z = 0$: the nilpotent element z $(z^2 = 0)$ points out of the plane. So the flat limit of the curves $X_a \cong$ twisted cubic is the plane cuspidal cubic together with an embedded point at the cusp, exactly as for the nodal cubic in (9.8.4).

(Consistency check: the cuspidal cubic has $p_a = 1$, the twisted cubic $p_a = 0$, and the embedded point has length 1, so $P_{X_0}(0) = P_{\text{cusp}}(0) + 1 = 0 + 1$, matching $P_{X_1}(0) = 1 - 0$ as required by (9.9).) □

Exercise 9.3

Problem. Some examples of flatness and nonflatness.

(a) If $f : X \to Y$ is a finite surjective morphism of nonsingular varieties over an algebraically closed field k, then f is flat.

(b) Let X be a union of two planes meeting at a point, each of which maps isomorphically to a plane Y. Show that f is not flat. For example, let $Y = \operatorname{Spec} k[x,y]$ and $X = \operatorname{Spec} k[x,y,z,w]/(z,w) \cap (x+z, y+w)$.

(c) Again let $Y = \operatorname{Spec} k[x,y]$, but take $X = \operatorname{Spec} k[x,y,z,w]/(z^2, zw, w^2, xz - yw)$. Show that $X_{\text{red}} \cong Y$, X has no embedded points, but that f is not flat.

Solution. (a) First let $x \in X$ be a closed point, $y = f(x)$; y is a closed point since f is of finite type over k. Put $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$. Both are regular local rings (nonsingular varieties), so A is regular and B is Cohen–Macaulay. Since f is finite and surjective, X and Y have the same dimension (integral dependence), and for closed points of varieties $\dim A = \dim Y$, $\dim B = \dim X$; so $\dim B = \dim A$. The fibre ring $B/\mathfrak{m}_A B$ is the local ring of the fibre X_y at x, and X_y = Spec of a finite k-algebra (finiteness of f), so $\dim B/\mathfrak{m}_A B = 0$. Hence $\dim B = \dim A + \dim B/\mathfrak{m}_A B$, and the "miracle flatness" theorem (Matsumura, Thm. 23.1: for a local homomorphism $A \to B$ of noetherian local rings with A regular and B Cohen–Macaulay, B is A-flat iff $\dim B = \dim A + \dim B/\mathfrak{m}_A B$) shows that B is flat over A.

Now let $x' \in X$ be any point; it specializes to a closed point x (every nonempty closed subset of a scheme of finite type over k contains a closed point), and $\mathcal{O}_{X,x'}$ is a localization of $B = \mathcal{O}_{X,x}$, hence flat over $A = \mathcal{O}_{Y,f(x)}$ (Ex. 9.1.1), and it is a module over $\mathcal{O}_{Y,f(x')} = A_{\mathfrak{p}}$, so it is flat over $\mathcal{O}_{Y,f(x')}$. Thus f is flat.

(b) Let $A_0 = k[x,y]$, $R = k[x,y,z,w]/I$ with $I = (z,w) \cap (x+z, y+w)$, an A_0-algebra via x, y. The two planes are $\{z = w = 0\}$ (coordinates x, y) and $\{x + z = y + w = 0\}$ (coordinates z, w, with $x = -z, y = -w$); each maps isomorphically onto $Y = \mathbf{A}^2$, and they meet only at the origin of $\mathbf{A}^4$. The map $R \to k[x,y,z,w]/(z,w) \times k[x,y,z,w]/(x+z, y+w) \cong A_0 \times A_0$ is injective and A_0-linear (the second factor $\cong A_0$ via $x \mapsto -z$, $y \mapsto -w$), so R is a finitely generated A_0-module (A_0 noetherian) and f is finite.

Suppose f were flat. Then R is a finitely generated flat A_0-module, hence locally free of constant rank on the connected scheme Y (9.2(e)); so $\dim_{\kappa(\mathfrak{p})} R \otimes \kappa(\mathfrak{p})$ is the same for all primes $\mathfrak{p}$ of A_0. We compute it for $\mathfrak{p} = (0)$ and $\mathfrak{p} = (x,y)$.

At the generic point: $K = k(x,y)$. After inverting $A_0 \setminus 0$, the ideals (z,w) and $(x+z, y+w)$ become comaximal (their sum contains x, a unit), so by the Chinese remainder theorem $R \otimes K \cong K[z,w]/(z,w) \times K[z,w]/(x+z, y+w) \cong K \times K$, of dimension 2.

At the origin: put $u_1 = x + z$, $u_2 = y + w$. Then z, w, u_1, u_2 are linear coordinates on $\mathbf{A}^4$, and $(z,w) \cap (u_1, u_2) = (z,w)(u_1, u_2) = (zu_1, zu_2, wu_1, wu_2)$ (a monomial computation in disjoint sets of variables). Reducing modulo $(x,y) = (u_1 - z, u_2 - w)$ identifies $u_1 = z$, $u_2 = w$, so

$$R \otimes k(0) = k[z,w]/(z^2, zw, wz, w^2) = k[z,w]/(z,w)^2,$$

of dimension 3. Since $2 \neq 3$, f is not flat.

(c) Let $A_0 = k[x,y]$, $R = k[x,y,z,w]/J$, $J = (z^2, zw, w^2, xz - yw)$. As $J = (z,w)^2 + (xz - yw)$, we have $R = A_0 \oplus N$ as an A_0-module, where $N = (A_0 z \oplus A_0 w)/A_0(xz - yw)$ is the part of degree 1 in z, w, and $N^2 = 0$ in R. The A_0-linear map $N \to A_0$, $az + bw \mapsto ay + bx$, is well defined (it kills $xz - yw \mapsto xy - yx = 0$), has image the ideal $\mathfrak{m}_0 = (x,y)$, and is injective: the kernel of $A_0^2 \to A_0$, $(a,b) \mapsto ay + bx$, is $\{(a,b) : ay = -bx\} = A_0(x, -y)$ because x, y are coprime in the UFD A_0. So $N \cong \mathfrak{m}_0$, a torsion-free A_0-module.

$X_{\mathrm{red}} \cong Y$. N is a nilpotent ideal of R and $R/N = A_0$ is reduced, so N is the nilradical and $X_{\mathrm{red}} = \operatorname{Spec} A_0 = Y$, the map $X_{\mathrm{red}} \to Y$ being the identity.

No embedded points. Let $\mathfrak{p} = \operatorname{Ann}_R(r)$ be an associated prime, $r = a + n \neq 0$ ($a \in A_0$, $n \in N$). Since $\mathfrak{p}$ is prime it contains the nilradical N, so for all $\nu \in N$: $0 = \nu r = a\nu$. As $N \cong \mathfrak{m}_0$ is a nonzero torsion-free module over the domain A_0, $a = 0$. So $r = n \in N \setminus 0$ and $\operatorname{Ann}_R(n) = \{a' + n' : a'n = 0\} = N$ by torsion-freeness. Hence $\operatorname{Ass}(R) = \{N\}$, the unique

minimal prime. So X has no embedded points.

f is not flat. R is a finitely generated A_0-module (generated by $1, z, w$). If it were flat it would be locally free of constant rank (as in (b)). But $R \otimes_{A_0} K = K \oplus \mathfrak{m}_0 \otimes K \cong K^2$, whereas $R \otimes_{A_0} A_0/\mathfrak{m}_0 = k \oplus \mathfrak{m}_0/\mathfrak{m}_0^2 \cong k^3$. So f is not flat (it is not even flat over 0: $\mathfrak{m}_0$ is not free). $\square$

Exercise 9.4

> **Problem.** *Open Nature of Flatness.* Let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Then $\{x \in X \mid f \text{ is flat at } x\}$ is an open subset of X (possibly empty)—see Grothendieck [EGA IV$_3$, 11.1.1].

Solution. Flatness at x means that $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. The assertion is local on X, so we may cover X by affine opens $\operatorname{Spec} B$ mapping into affine opens $\operatorname{Spec} A$ of Y. Thus let A be a noetherian ring, B a finitely generated A-algebra (so B is noetherian), and

$$U = \{\mathfrak{q} \in \operatorname{Spec} B : \ B_{\mathfrak{q}} \text{ is flat over } A_{\mathfrak{q}\cap A}\}.$$

We prove that U is open. (The argument works verbatim for a finite B-module M in place of B.)

Step 1: U is stable under generization. Let $\mathfrak{q}' \subseteq \mathfrak{q}$ with $\mathfrak{q} \in U$, $\mathfrak{p} = \mathfrak{q} \cap A$, $\mathfrak{p}' = \mathfrak{q}' \cap A \subseteq \mathfrak{p}$. $B_{\mathfrak{q}'}$ is a localization of $B_{\mathfrak{q}}$, which is flat over $A_{\mathfrak{p}}$, so $B_{\mathfrak{q}'}$ is flat over $A_{\mathfrak{p}}$ (Ex. 9.1.1) and, being an $A_{\mathfrak{p}'}$-module, flat over $A_{\mathfrak{p}'}$. So $\mathfrak{q}' \in U$.

Step 2: local statement. **Claim.** If $\mathfrak{q}_0 \in U$ and $Z = V(\mathfrak{q}_0) \subseteq \operatorname{Spec} B$, then there is a nonempty open subset $W \subseteq Z$ with $W \subseteq U$.

Put $\mathfrak{p}_0 = \mathfrak{q}_0 \cap A$, $A_0 = A/\mathfrak{p}_0$ (a noetherian domain), $\bar{B} = B/\mathfrak{p}_0 B$, a finitely generated A_0-algebra. Let $T = \operatorname{Tor}_1^A(A/\mathfrak{p}_0, B)$, a finite B-module (compute it with a resolution of $A/\mathfrak{p}_0$ by finite free A-modules). Since Tor commutes with localization, $T_{\mathfrak{q}_0} = \operatorname{Tor}_1^{A_{\mathfrak{p}_0}}(\kappa(\mathfrak{p}_0), B_{\mathfrak{q}_0}) = 0$ because $B_{\mathfrak{q}_0}$ is flat over $A_{\mathfrak{p}_0}$. Hence $\mathfrak{q}_0 \notin \operatorname{Supp} T$, a closed set. By generic flatness (iii) applied to $A_0 \to \bar{B}$ there is $a \in A$, $a \notin \mathfrak{p}_0$, with $\bar{B}_a$ free over $(A_0)_a$. Let

$$W = \{\mathfrak{q} \in Z : \ \mathfrak{q} \notin \operatorname{Supp} T, \ a \notin \mathfrak{q}\},$$

open in Z and containing $\mathfrak{q}_0$ (as $a \notin \mathfrak{p}_0$ and $\mathfrak{q}_0 \cap A = \mathfrak{p}_0$); so W is nonempty. Let $\mathfrak{q} \in W$ and $\mathfrak{p} = \mathfrak{q} \cap A \supseteq \mathfrak{p}_0$. We apply the local flatness criterion (ii) to $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ and the ideal $I = \mathfrak{p}_0 A_{\mathfrak{p}} \subseteq \mathfrak{p} A_{\mathfrak{p}}$: $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$ iff (1) $B_{\mathfrak{q}}/IB_{\mathfrak{q}} = \bar{B}_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}/I = (A_0)_{\mathfrak{p}}$, and (2) $\operatorname{Tor}_1^{A_{\mathfrak{p}}}(A_{\mathfrak{p}}/\mathfrak{p}_0 A_{\mathfrak{p}}, B_{\mathfrak{q}}) = 0$. Now (2) is $T_{\mathfrak{q}} = 0$, true since $\mathfrak{q} \notin \operatorname{Supp} T$. For (1): $a \notin \mathfrak{p}$, and $\bar{B}_{\mathfrak{q}}$ is a localization of $\bar{B}_a$, which is free over $(A_0)_a$; hence $\bar{B}_{\mathfrak{q}}$ is flat over $(A_0)_a$ (Ex. 9.1.1), and being a module over $(A_0)_{\mathfrak{p}}$ it is flat over $(A_0)_{\mathfrak{p}}$. So $\mathfrak{q} \in U$. This proves the claim.

Step 3: noetherian induction. Suppose U is not open. Among the closed subsets $Z \subseteq \operatorname{Spec} B$ such that $U \cap Z$ is not open in Z (the whole space is one) choose a minimal one, Z. If $Z = Z_1 \cup Z_2$ with $Z_i \subsetneq Z$ closed, then $Z_i \setminus U$ is closed in Z_i, hence in Z, so $Z \setminus U = (Z_1 \setminus U) \cup (Z_2 \setminus U)$ is closed in Z and $U \cap Z$ is open in Z, a contradiction. So Z is irreducible, with generic point $\mathfrak{q}_0$. If $\mathfrak{q}_0 \notin U$, then $U \cap Z = \emptyset$ by Step 1 (every point of Z specializes $\mathfrak{q}_0$), which is open in Z: contradiction. So $\mathfrak{q}_0 \in U$ and Step 2 provides $W \subseteq U \cap Z$ nonempty open in Z. Then $Z \setminus W \subsetneq Z$ is closed, so by minimality $U \cap (Z \setminus W)$ is open in $Z \setminus W$, i.e. $(Z \setminus W) \setminus U$ is closed in $Z \setminus W$,

hence in Z. But $Z \setminus U = (Z \setminus W) \setminus U$ because $W \subseteq U$; so $U \cap Z$ is open in Z, contradicting the choice of Z. Hence U is open. □

Exercise 9.5

Problem. *Very Flat Families.* For any closed subscheme $X \subseteq \mathbf{P}^n$, we denote by $C(X) \subseteq \mathbf{P}^{n+1}$ the projective cone over X (I, Ex. 2.10). If $I \subseteq k[x_0, \ldots, x_n]$ is the (largest) homogeneous ideal of X, then $C(X)$ is defined by the ideal generated by I in $k[x_0, \ldots, x_{n+1}]$.

(a) Give an example to show that if $\{X_t\}$ is a flat family of closed subschemes of $\mathbf{P}^n$, then $\{C(X_t)\}$ need not be a flat family in $\mathbf{P}^{n+1}$.

(b) To remedy this situation, we make the following definition. Let $X \subseteq \mathbf{P}^n_T$ be a closed subscheme, where T is a noetherian integral scheme. For each $t \in T$, let $I_t \subseteq S_t = k(t)[x_0, \ldots, x_n]$ be the homogeneous ideal of X_t in $\mathbf{P}^n_{k(t)}$. We say that the family $\{X_t\}$ is *very flat* if for all $d \geq 0$, $\dim_{k(t)}(S_t/I_t)_d$ is independent of t. Here $(\ )_d$ means the homogeneous part of degree d.

(c) If $\{X_t\}$ is a very flat family in $\mathbf{P}^n$, show that it is flat. Show also that $\{C(X_t)\}$ is a very flat family in $\mathbf{P}^{n+1}$, and hence flat.

(d) If $\{X_{(t)}\}$ is an algebraic family of projectively normal varieties in $\mathbf{P}^n_k$, parametrized by a nonsingular curve T over an algebraically closed field k, then $\{X_{(t)}\}$ is a very flat family of schemes.

Solution. Conventions: "the homogeneous ideal of a closed subscheme $X' \subseteq \mathbf{P}^n_K$ (K a field)" is the largest homogeneous ideal defining it, namely $I(X') = \bigoplus_{d \geq 0} \Gamma(\mathbf{P}^n_K, \mathcal{I}_{X'}(d))$ (II, Ex. 5.10(c),(d); it is the saturated ideal). So $(S/I)_d$ is the image of $\Gamma(\mathbf{P}^n, \mathcal{O}(d)) \to \Gamma(X', \mathcal{O}_{X'}(d))$ (injective on $(S/I)_d$ by left exactness of Γ applied to $0 \to \mathcal{I}(d) \to \mathcal{O}(d) \to \mathcal{O}_{X'}(d) \to 0$). For $d \gg 0$ this map is surjective ($H^1(\mathcal{I}_{X'}(d)) = 0$, Thm. 5.2(b)) and $\dim \Gamma(X', \mathcal{O}_{X'}(d)) = P_{X'}(d)$, the Hilbert polynomial of X' (Thm. 5.2, Ex. 5.2). Hence

$$\dim_K (S/I)_d = P_{X'}(d) \quad (d \gg 0) \qquad\qquad (\dagger)$$

(a) Take k algebraically closed, $T = \mathbf{A}^1_k = \operatorname{Spec} k[t]$, $n = 2$, and the three sections $P_1 = (1:0:0)$, $P_2 = (0:1:0)$, $P_3(t) = (1:1:t)$ of $\mathbf{P}^2_T \to T$. They are pairwise disjoint (at every $t \in k$ the three points are distinct), so $X = s_1(T) \sqcup s_2(T) \sqcup s_3(T) \subseteq \mathbf{P}^2_T$ is a closed subscheme isomorphic to three disjoint copies of T: it is flat over T, with $X_t = \{P_1, P_2, P_3(t)\}$ three reduced points. For $t \neq 0$ these points are not collinear (the determinant of the matrix with rows $(1,0,0)$, $(0,1,0)$, $(1,1,t)$ is t), while for $t = 0$ they lie on the line $z = 0$.

The Hilbert function $h_t(d) = \dim(S_t/I_t)_d$ of three distinct points is the rank of the evaluation map $S_{t,d} \to k^3$ at the three points. $h_t(0) = 1$; $h_t(1)$ is the rank of the 3×3 matrix of the coordinates of the points, i.e. 3 for $t \neq 0$ and 2 for $t = 0$; and $h_t(d) = 3$ for $d \geq 2$ (for each i take the product $\ell_i \lambda^{d-2}$ with ℓ_i a product of two linear forms each vanishing at one of the two other points but not at P_i, and λ a linear form not vanishing at P_i: it takes a nonzero value at P_i and vanishes at the other two points). So

$$h_t = (1,3,3,3,\ldots)\ (t \neq 0), \qquad h_0 = (1,2,3,3,\ldots).$$

The cone $C(X_t) \subseteq \mathbf{P}^3$ is defined by $I_t S'_t$, $S'_t = S_t[x_3]$, and $(S'_t/I_t S'_t)_d = \bigoplus_{j=0}^{d} (S_t/I_t)_j x_3^{d-j}$ has dimension $\sum_{j \leq d} h_t(j)$, which is $3d+1$ for $t \neq 0$ and $3d$ for $t = 0$ ($d \geq 1$). By (†) the Hilbert

polynomials of the cones are $3d+1$ ($t \neq 0$) and $3d$ ($t = 0$). By Thm. 9.9, no closed subscheme of $\mathbf{P}^3_T$ with fibres $C(X_t)$ is flat over T. (Geometrically: the cone over three non-collinear points is three non-coplanar concurrent lines, $p_a = 0$; over three collinear points it is three coplanar concurrent lines, $p_a = 1$.)

(b) This is a definition (with I_t as in the conventions above).

(c) *Very flat ⇒ flat.* If the $\dim(S_t/I_t)_d$ are independent of t, then by (†) the Hilbert polynomials P_t of the fibres are independent of t, so X is flat over T by Thm. 9.9 (T is integral noetherian).

The cone. Let $X \subseteq \mathbf{P}^n_T$. The cone $C(X) \subseteq \mathbf{P}^{n+1}_T$ is defined over an affine open $\operatorname{Spec} A \subseteq T$ as $\operatorname{Proj} A[x_0,\dots,x_{n+1}]/\mathfrak{a}A[x_0,\dots,x_{n+1}]$ for a homogeneous ideal $\mathfrak{a} \subseteq A[x_0,\dots,x_n]$ with $\operatorname{Proj} A[x_0,\dots,x_n]/\mathfrak{a} = X \cap \mathbf{P}^n_A$; it is independent of the choice of $\mathfrak{a}$ (two such ideals agree in all large degrees, $\mathfrak{a}_d = \Gamma(\tilde{\mathfrak{a}}(d))$ for $d \gg 0$, hence so do their extensions), so these glue, and the fibre over t is $C(X_t)$ (Proj commutes with base change, and the image of $\mathfrak{a}$ in S_t agrees with I_t in large degrees).

Let now $\{X_t\}$ be very flat. Fix t and write $S = S_t$, $I = I_t$, $R = S/I$, $S' = S[x_{n+1}]$, $R' = S'/IS' = R[x_{n+1}]$. Since I is saturated, R has no nonzero element killed by a power of $\mathfrak{m} = (x_0,\dots,x_n)$. If $g = \sum g_j x_{n+1}^j \in R'$ is killed by a power of $(x_0,\dots,x_{n+1})$, then $x_i^N g = 0$ for all $i \leq n$, so $x_i^N g_j = 0$ for all i, j and hence $g_j = 0$. So IS' is saturated, i.e. it is the homogeneous ideal of $C(X_t)$ (II, Ex. 5.10(c),(d)). Then

$$\dim_{k(t)}(S'/IS')_d = \sum_{j=0}^{d} \dim_{k(t)}(S_t/I_t)_j$$

is independent of t, so $\{C(X_t)\}$ is very flat, hence flat by the first part (applied in $\mathbf{P}^{n+1}_T$).

(d) Let $X \subseteq \mathbf{P}^n_T$ be the closed subscheme giving the family, $f : X \to T$. Since projective normality implies normality (I, Ex. 3.18(a)), the $X_{(t)}$ are normal varieties, so by Thm. 9.11 f is flat and $X_t = X_{(t)}$ for every closed point t (the scheme-theoretic fibre is the variety). Let η be the generic point of T, $K = k(\eta)$, and $s_t(d) = \dim_{k(t)}(S_t/I_t)_d$ for $t \in T$. We show $s_t(d) = s_\eta(d)$ for all closed t and all $d \geq 0$.

Let $\mathcal{J} \subseteq \mathcal{O}_{\mathbf{P}^n_T}$ be the ideal sheaf of X. Fix a nonempty affine open $\operatorname{Spec} A \subseteq T$; A is a regular noetherian domain of dimension 1, hence a Dedekind domain. Let $S_A = A[x_0,\dots,x_n]$, $\mathfrak{a}_d = \Gamma(\mathbf{P}^n_A, \mathcal{J}(d)) \subseteq (S_A)_d$ and $R_d = (S_A)_d/\mathfrak{a}_d$.

(1) *Fibres of $\mathcal{J}$.* $\mathcal{O}_X$ is flat over A, so $\operatorname{Tor}_1^A(\mathcal{O}_X, k(t)) = 0$ and $0 \to \mathcal{J}\otimes k(t) \to \mathcal{O}_{\mathbf{P}^n_{k(t)}} \to \mathcal{O}_{X_t} \to 0$ is exact for every t (also for $t = \eta$): $\mathcal{J} \otimes k(t) = \mathcal{I}_{X_t}$. Hence the image of $\mathfrak{a}_d$ in $(S_t)_d$ lies in $I_{t,d} = \Gamma(\mathbf{P}^n, \mathcal{I}_{X_t}(d))$, and $R_d \otimes k(t) = (S_t)_d/\overline{\mathfrak{a}_d} \twoheadrightarrow (S_t/I_t)_d$. Thus $s_t(d) \leq \dim_{k(t)} R_d \otimes k(t)$.

(2) *R_d is locally free.* $R_d \hookrightarrow \Gamma(X, \mathcal{O}_X(d))$ (left exactness), and $\Gamma(X, \mathcal{O}_X(d))$ is torsion-free over A since $\mathcal{O}_X(d)$ is A-flat. So R_d is a finitely generated torsion-free module over the Dedekind domain A, hence projective, of some constant rank r_d. Therefore $\dim_{k(t)} R_d \otimes k(t) = r_d$ for all $t \in \operatorname{Spec} A$.

(3) *Generic point.* By flat base change (Prop. 9.3) $\Gamma(\mathbf{P}^n_K, \mathcal{I}_{X_\eta}(d)) = \mathfrak{a}_d \otimes_A K$, so $(S_\eta/I_\eta)_d = R_d \otimes K$ and $s_\eta(d) = r_d$. With (1) and (2): $s_t(d) \leq s_\eta(d)$ for all closed $t \in \operatorname{Spec} A$.

(4) *The reverse inequality.* Let t be closed. Since X_t is projectively normal, Ex. II.5.14(d) says that $\Gamma(\mathbf{P}^n, \mathcal{O}(d)) \to \Gamma(X_t, \mathcal{O}_{X_t}(d))$ is surjective for all $d \geq 0$; with the injectivity noted in the

conventions, $s_t(d) = h^0(X_t, \mathcal{O}_{X_t}(d))$. Now $\mathcal{O}_X(d)$ is flat over T and f is projective, so $s \mapsto h^0(X_s, \mathcal{O}_{X_s}(d))$ is upper semicontinuous on T (Thm. 12.8). The closed set $\{s : h^0(X_s, \mathcal{O}(d)) \geq h^0(X_\eta, \mathcal{O}(d))\}$ contains η, hence is all of T. Also $(S_\eta/I_\eta)_d \subseteq \Gamma(X_\eta, \mathcal{O}(d))$. Therefore $s_t(d) = h^0(X_t, \mathcal{O}(d)) \geq h^0(X_\eta, \mathcal{O}(d)) \geq s_\eta(d)$.

Together, $s_t(d) = s_\eta(d)$ for every closed point t and every $d \geq 0$, i.e. the family is very flat. $\square$

Exercise 9.6

> **Problem.** Let $Y \subseteq \mathbf{P}^n$ be a nonsingular variety of dimension ≥ 2 over an algebraically closed field k. Suppose $\mathbf{P}^{n-1}$ is a hyperplane in $\mathbf{P}^n$ which does not contain Y, and such that the scheme $Y' = Y \cap \mathbf{P}^{n-1}$ is also nonsingular. Prove that Y is a complete intersection in $\mathbf{P}^n$ if and only if Y' is a complete intersection in $\mathbf{P}^{n-1}$. [Hint: See (II, Ex. 8.4) and use (9.12) applied to the affine cones over Y and Y'.]

Solution. Recall (II, Ex. 8.4) that a closed subscheme $Z \subseteq \mathbf{P}^N$ of codimension r is a (strict, global) complete intersection if its homogeneous ideal $I(Z) \subseteq k[x_0, \ldots, x_N]$ (the largest homogeneous ideal defining Z) can be generated by r elements.

Setup. Choose coordinates with $H = \mathbf{P}^{n-1} = \{x_n = 0\}$; $S = k[x_0, \ldots, x_n]$, $S' = S/(x_n) = k[x_0, \ldots, x_{n-1}]$, $\delta = \dim Y \geq 2$, $r = n - \delta = \operatorname{codim}(Y, \mathbf{P}^n)$. Let $I = I(Y)$, a homogeneous prime ideal, and $R = S/I$, a domain of dimension $\delta + 1$ (the coordinate ring of the affine cone). As $Y \not\subseteq H$, $x_n \notin I$, so x_n is a nonzerodivisor on R. We have $R/x_nR = S'/J_0$, where $J_0 \subseteq S'$ is the image of I, and $\operatorname{Proj}(S'/J_0) = Y \cap H = Y'$ as schemes. By hypothesis Y' is nonsingular; it is connected by Cor. 7.9 and Remark 7.9.1 (Y' is a hyperplane section of Y, $\dim Y \geq 2$), hence integral, of dimension $\delta - 1 \geq 1$, and $\operatorname{codim}(Y', \mathbf{P}^{n-1}) = n - \delta = r$. Let $I' = I(Y') \subseteq S'$ (a homogeneous prime). Then $J_0 \subseteq I'$ and J_0, I' agree in large degrees (they define the same subscheme).

($\Rightarrow$) Let $I = (f_1, \ldots, f_r)$. Since S is Cohen–Macaulay and $\operatorname{ht} I = r$, the f_i form a regular sequence (Matsumura, Thm. 17.4), so R is Cohen–Macaulay. As x_n is a nonzerodivisor on R, $R/x_nR = S'/J_0$ is Cohen–Macaulay of dimension $\delta \geq 2$, in particular of depth ≥ 1. Hence the irrelevant ideal $\mathfrak{m}'$ of S' is not an associated prime of S'/J_0, i.e. J_0 is saturated, and a saturated ideal defining Y' is the largest homogeneous ideal: $J_0 = I'$ (II, Ex. 5.10). But $J_0 = (\bar{f}_1, \ldots, \bar{f}_r)$ is generated by $r = \operatorname{codim}(Y', \mathbf{P}^{n-1})$ elements. So Y' is a complete intersection.

($\Leftarrow$) Assume $I' = (g_1, \ldots, g_r)$. Let $\mathfrak{m} = R_+$ be the irrelevant maximal ideal of R and apply Lemma 9.12 to

$$A = R_{\mathfrak{m}}, \qquad t = x_n \in \mathfrak{m}.$$

A is a local noetherian domain which is a localization of a finitely generated k-algebra. Let $\mathfrak{p} \subseteq R$ be the kernel of $R \to R/x_nR = S'/J_0 \to S'/I'$ (a prime of R with $R/\mathfrak{p} = S'/I'$, the coordinate ring of the cone over Y').

(1) *tA has a unique minimal prime $\mathfrak{p}A$.* The minimal primes of x_nR correspond to the minimal primes of J_0, i.e. to the irreducible components of the cone $V(J_0) \subseteq \mathbf{A}^n$. This cone is a union of lines through 0; $V(J_0) \setminus \{0\}$ is the cone over Y' minus the vertex, which is irreducible since Y' is; and every component contains the origin and has positive dimension since $Y' \neq \emptyset$, so every component meets $V(J_0) \setminus \{0\}$. Hence $V(J_0)$ is irreducible, with prime ideal I', so $\mathfrak{p}$ is the only minimal prime of x_nR.

(2) *t generates the maximal ideal of $A_{\mathfrak{p}} = R_{\mathfrak{p}}$.* Choose $i < n$ with $x_i \notin I'$. The ring $(S'/J_0)_{x_i}$ is the Laurent extension $\mathcal{O}(Y' \cap D_+(x_i))[x_i, x_i^{-1}]$, which is reduced because Y' is nonsingular (reduced). $R_{\mathfrak{p}}/x_nR_{\mathfrak{p}}$ is a localization of this ring, hence reduced; it is a zero-dimensional local ring ($\mathfrak{p}$ is minimal over x_nR), hence a field. So $x_nR_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}$.

(3) *$A/\mathfrak{p}A$ is normal.* $A/\mathfrak{p}A = (S'/I')_{\mathfrak{m}'}$. Now Y' is a nonsingular (hence normal) complete intersection of dimension ≥ 1 in $\mathbf{P}^{n-1}$, hence projectively normal (II, Ex. 8.4(b)), i.e. S'/I' is an integrally closed domain; so is its localization.

Lemma 9.12 gives $\mathfrak{p}A = x_nA$ (and A normal). Both $x_nR \subseteq \mathfrak{p}$ are homogeneous ideals of R, and $N = \mathfrak{p}/x_nR$ is a finitely generated graded R-module with $N_{\mathfrak{m}} = 0$. Since a nonzero homogeneous element has a proper homogeneous annihilator, contained in $\mathfrak{m} = R_+$, $N_{\mathfrak{m}} = 0$ forces $N = 0$. Hence $x_nR = \mathfrak{p}$, i.e. $R/x_nR = S'/I'$, i.e. $J_0 = I' = (g_1, \ldots, g_r)$.

Now choose homogeneous $G_i \in I$ with image g_i in S'. For $f \in I$ write $\bar{f} = \sum c_i g_i$ in S', lift c_i to $C_i \in S$; then $f - \sum C_iG_i \in x_nS \cap I = x_nI$ (since I is prime and $x_n \notin I$). So $I = (G_1, \ldots, G_r) + x_nI$. The graded S-module $I/(G_1, \ldots, G_r)$ is finitely generated and equal to x_n times itself, so it is 0 by (graded) Nakayama's lemma. Thus $I = (G_1, \ldots, G_r)$, with $r = \operatorname{codim}(Y, \mathbf{P}^n)$: Y is a complete intersection. □

Exercise 9.7

> **Problem.** Let $Y \subseteq X$ be a closed subscheme, where X is a scheme of finite type over a field k. Let $D = k[t]/t^2$ be the ring of dual numbers, and define an *infinitesimal deformation of Y as a closed subscheme of X*, to be a closed subscheme $Y' \subseteq X \times_k D$, which is flat over D, and whose closed fibre is Y. Show that these Y' are classified by $H^0(Y, \mathcal{N}_{Y/X})$, where $\mathcal{N}_{Y/X} = \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y)$.

Solution. Write $\mathcal{I}$ for the ideal sheaf of Y in X and ε for t. The scheme $X \times_k D$ has the same underlying space as X and structure sheaf $\mathcal{O}_X[\varepsilon] = \mathcal{O}_X \oplus \varepsilon\mathcal{O}_X$. A closed subscheme Y' is given by an ideal sheaf $\mathcal{I}' \subseteq \mathcal{O}_X[\varepsilon]$; "closed fibre is Y" means $Y' \cap (X \times \operatorname{Spec} k) = Y$, i.e. $\mathcal{I}' + (\varepsilon) = \mathcal{I} + (\varepsilon)$, i.e. the image of $\mathcal{I}'$ under $\varepsilon \mapsto 0$ is $\mathcal{I}$.

Flatness over D. For a D-module Q: Q is flat over D iff for each ideal $\mathfrak{b}$ of D, $\mathfrak{b} \otimes_D Q \to Q$ is injective (9.1A(a)). The ideals of D are $0, (\varepsilon), D$, and $(\varepsilon) \cong D/(\varepsilon) = k$, so $(\varepsilon) \otimes Q = Q/\varepsilon Q \to Q$ is multiplication by ε. Hence Q is D-flat iff $\ker(\varepsilon : Q \to Q) = \varepsilon Q$.

Local computation. Let $W = \operatorname{Spec} P \subseteq X$ be an affine open set, $I = \mathcal{I}(W) \subseteq P$, and $I' = \mathcal{I}'(W) \subseteq P[\varepsilon]$ with image I in P (closed fibre Y). Put $J' = \{a \in P : \varepsilon a \in I'\}$, an ideal of P. For $f \in I$ pick g with $f + \varepsilon g \in I'$; then $\varepsilon f = \varepsilon(f + \varepsilon g) \in I'$, so $f \in J'$: $I \subseteq J'$. Let $Q = P[\varepsilon]/I'$. Then $\ker(\varepsilon|Q) = \{a + \varepsilon b : \varepsilon a \in I'\} = \{a + \varepsilon b : a \in J'\} + I'/I'$, and $\varepsilon Q = \{\varepsilon c\} + I'/I'$. As $\varepsilon \cdot \varepsilon Q = 0$, always $\varepsilon Q \subseteq \ker \varepsilon$. The reverse inclusion says: for each $a \in J'$ and $b \in P$, $a + \varepsilon b \in \varepsilon P + I'$, i.e. $a \in$ image of I' in $P = I$. So

$$Q \text{ is flat over } D \iff J' \subseteq I \iff J' = I.$$

Assume $J' = I$. For $f \in I$ the element g with $f + \varepsilon g \in I'$ is unique modulo $J' = I$, so we get a well defined map $\varphi : I \to P/I$, $\varphi(f) = g \bmod I$. It is additive and P-linear ($p(f + \varepsilon g) = pf + \varepsilon pg \in I'$), and it kills I^2 since P/I is killed by I ($\varphi(ff') = f\varphi(f') = 0$). So $\varphi \in \operatorname{Hom}_P(I/I^2, P/I) =$

$\mathrm{Hom}_{P/I}(I/I^2, P/I)$. Moreover $I' = \{f + \varepsilon g : f \in I,\ g \bmod I = \varphi(f)\}$, since $I' \supseteq \varepsilon I$ (as $J' = I$) and I' maps onto I.

Conversely, given $\varphi \in \mathrm{Hom}_P(I/I^2, P/I)$, the set $I'_\varphi = \{f + \varepsilon g : f \in I,\ \bar{g} = \varphi(f)\}$ is an ideal of $P[\varepsilon]$: $(p+\varepsilon q)(f+\varepsilon g) = pf + \varepsilon(pg+qf)$ with $qf \in I$, and $\overline{pg+qf} = p\varphi(f) = \varphi(pf)$. It maps onto I, and $J' = I$ (if $\varepsilon a \in I'_\varphi$ then $0 \in I$, $\bar{a} = \varphi(0) = 0$). So $P[\varepsilon]/I'_\varphi$ is D-flat with closed fibre P/I. We have shown: on the affine open W, deformations $\leftrightarrow \mathrm{Hom}_{P/I}(I/I^2, P/I) = \Gamma(W \cap Y, \mathcal{N}_{Y/X})$ (the Hom sheaf of the coherent sheaves $\mathcal{I}/\mathcal{I}^2$, $\mathcal{O}_Y$ over an affine is the Hom of modules).

Globalization. The construction $\mathcal{I}' \mapsto \varphi$ commutes with restriction to smaller affine opens (localize I' at $h \in P$: $(I')_h$ and J'_h, φ_h), so a deformation $\mathcal{I}'$ yields compatible local sections of $\mathcal{N}_{Y/X}$, i.e. a global section; conversely a global section φ yields compatible local ideals $I'_{\varphi|W}$ which glue to a sheaf of ideals, whose flatness and closed fibre are checked locally. The two constructions are inverse. Hence the set of infinitesimal deformations of Y in X is in canonical bijection with $H^0(Y, \mathcal{N}_{Y/X})$, the trivial deformation $Y \times_k D$ corresponding to $\varphi = 0$. □

Exercise 9.8

> **Problem.** Let A be a finitely generated k-algebra. Write A as a quotient of a polynomial ring P over k, and let J be the kernel:
>
> $$0 \to J \to P \to A \to 0.$$
>
> Consider the exact sequence of (II, 8.4A)
>
> $$J/J^2 \to \Omega_{P/k} \otimes_P A \to \Omega_{A/k} \to 0.$$
>
> Apply the functor $\mathrm{Hom}_A(\cdot, A)$, and let $T^1(A)$ be the cokernel:
>
> $$\mathrm{Hom}_A(\Omega_{P/k} \otimes A, A) \to \mathrm{Hom}_A(J/J^2, A) \to T^1(A) \to 0.$$
>
> Now use the construction of (II, Ex. 8.6) to show that $T^1(A)$ classifies infinitesimal deformations of A, i.e., algebras A' flat over $D = k[t]/t^2$, with $A' \otimes_D k \cong A$. It follows that $T^1(A)$ is independent of the given representation of A as a quotient of a polynomial ring P. (For more details, see Lichtenbaum and Schlessinger [1].) (Starred in the book.)

Solution. Let $P = k[x_1, \ldots, x_m]$, $P_D = P \otimes_k D = P[t]/t^2$, and let $\mathrm{Def}(A)$ be the set of isomorphism classes of pairs (A', α), A' a D-algebra flat over D and $\alpha : A'/tA' \xrightarrow{\sim} A$ an isomorphism of k-algebras; an isomorphism of pairs is a D-algebra isomorphism compatible with the α's. We construct a bijection $\mathrm{Def}(A) \to T^1(A)$. Note that $\mathrm{Hom}_A(\Omega_{P/k} \otimes A, A) = \mathrm{Der}_k(P, A)$ and that the first map above sends a derivation δ to $\delta|_J : J \to A$ (this is P-linear and kills J^2 because $\delta(fg) = \bar{f}\delta g + \bar{g}\delta f = 0$ for $f \in J$, $g \in J$, and $\delta(pf) = \bar{p}\delta f$ for $f \in J$).

Step 1: lifting. Let (A', α) be a deformation and identify $A'/tA' = A$. Since P is a polynomial ring and $A' \to A$ is surjective, there is a k-algebra homomorphism $\psi : P \to A'$ lifting $P \to A$ (lift the images of the x_i). It extends to a D-algebra map $\Psi : P_D \to A'$. It is surjective: $A' = \psi(P) + tA'$ and $tA' = t\psi(P)$ (as $t^2 = 0$), so $A' = \psi(P) + t\psi(P) = \Psi(P_D)$. Put $I'_\psi = \ker \Psi$, so $A' \cong P_D/I'_\psi$. This is flat over D with closed fibre $P_D/(I'_\psi + tP_D) = A = P/J$; so I'_ψ is exactly an infinitesimal deformation of the closed subscheme $\mathrm{Spec}\, A \subseteq \mathbf{A}^m_k$ in the sense of Ex. 9.7 (with $X = \mathbf{A}^m_k$, $Y = \mathrm{Spec}\, A$). By the computation in Ex. 9.7 it corresponds to $\varphi_\psi \in \mathrm{Hom}_A(J/J^2, A)$

defined by: for $f \in J$ and $f + tg \in I'_\psi$, $\varphi_\psi(f) = \bar{g}$ (the class of g in A); every φ arises from a unique ideal.

Step 2: change of lifting. Because A' is D-flat, multiplication by t gives an isomorphism $A = A'/tA' \xrightarrow{\sim} tA'$ (tensor $0 \to k \xrightarrow{t} D \to k \to 0$ with A'). If ψ, ψ' are two lifts, $\psi' - \psi$ takes values in tA', so $\psi' = \psi + t\delta$ for a unique $\delta : P \to A$, and expanding $\psi'(pq) = \psi'(p)\psi'(q)$ (with $t^2 = 0$) shows $\delta(pq) = \bar{p}\delta(q) + \bar{q}\delta(p)$: $\delta \in \mathrm{Der}_k(P, A)$. Conversely every $\delta \in \mathrm{Der}_k(P, A)$ gives a lift $\psi + t\delta$ by the same computation. Now let $f \in J$ and $f + tg \in I'_\psi$, i.e. $\psi(f) = -t\bar{g}$. Then

$$\Psi'(f + tg') = \psi(f) + t\delta(f) + t\bar{g}' = t(\delta(f) + \bar{g}' - \bar{g})$$

vanishes iff $\bar{g}' = \bar{g} - \delta(f)$. Hence

$$\varphi_{\psi'} = \varphi_\psi - \delta|_J.$$

Therefore the class $[\varphi_\psi] \in T^1(A) = \mathrm{Hom}_A(J/J^2, A)/\{\delta|_J\}$ does not depend on ψ; call it $\Phi(A', \alpha)$.

Step 3: Φ is well defined on isomorphism classes. If $\sigma : A'_1 \to A'_2$ is an isomorphism of deformations and ψ_1 is a lift for A'_1, then $\psi_2 = \sigma\psi_1$ is a lift for A'_2 and $\ker \Psi_2 = \ker \Psi_1$; so $\varphi_{\psi_2} = \varphi_{\psi_1}$.

Step 4: surjectivity. Given $\varphi \in \mathrm{Hom}_A(J/J^2, A)$, Ex. 9.7 gives an ideal $I'_\varphi \subseteq P_D$ with P_D/I'_φ flat over D and $P_D/(I'_\varphi + tP_D) = P/J = A$; this deformation has $\Phi = [\varphi]$ (use $\psi =$ the canonical map $P \to P_D/I'_\varphi$).

Step 5: injectivity. Let $\Phi(A'_1) = \Phi(A'_2)$. Choose lifts ψ_1, ψ_2; then $\varphi_{\psi_1} - \varphi_{\psi_2} = \delta|_J$ for some $\delta \in \mathrm{Der}_k(P, A)$; replacing ψ_2 by $\psi'_2 = \psi_2 - t\delta$ we get $\varphi_{\psi'_2} = \varphi_{\psi_2} + \delta|_J = \varphi_{\psi_1}$ (apply the formula of Step 2). By Ex. 9.7 the ideal is determined by φ, so $I'_{\psi_1} = I'_{\psi'_2}$, and $\Psi'_2 \circ \Psi_1^{-1} : A'_1 = P_D/I'_{\psi_1} \to A'_2$ is an isomorphism of D-algebras compatible with the identifications of closed fibres with A.

Hence $\Phi : \mathrm{Def}(A) \to T^1(A)$ is a bijection. In particular $T^1(A)$, which was defined using the representation $A = P/J$, is in bijection with $\mathrm{Def}(A)$, a set defined only in terms of A; this identification is natural, and $T^1(A)$ is independent (up to canonical bijection, indeed up to canonical k-isomorphism for its natural A-module structure) of the presentation. □

Exercise 9.9

> **Problem.** A k-algebra A is said to be *rigid* if it has no infinitesimal deformations, or equivalently, by (Ex. 9.8) if $T^1(A) = 0$. Let $A = k[x, y, z, w]/(x, y) \cap (z, w)$, and show that A is rigid. This corresponds to two planes in $\mathbf{A}^4$ which meet at a point.

Solution. Let $P = k[x, y, z, w]$ and $J = (x, y) \cap (z, w) = (xz, xw, yz, yw)$ (the intersection of monomial ideals is generated by the common multiples of the generators, which are the products). Put $g_1 = xz$, $g_2 = xw$, $g_3 = yz$, $g_4 = yw$. We must show that every $\varphi \in \mathrm{Hom}_A(J/J^2, A) = \mathrm{Hom}_P(J, A)$ is induced by a derivation $\delta \in \mathrm{Der}_k(P, A)$, i.e. $\varphi(f) = \delta(f)$ for $f \in J$; then the cokernel $T^1(A)$ is 0.

Description of A. The map $f \mapsto (f(0, 0, z, w), f(x, y, 0, 0))$ identifies A with the subring of $k[z, w] \times k[x, y]$ of pairs (u, v) with $u(0, 0) = v(0, 0)$ (injective since $J = (x, y) \cap (z, w)$; the image is as stated, since (u, v) is the image of $u(z, w) + v(x, y) - u(0, 0)$). We write $a = (a^{(1)}, a^{(2)})$ with $a^{(1)} \in k[z, w]$ (restriction to the plane $x = y = 0$) and $a^{(2)} \in k[x, y]$ (restriction to $z = w = 0$).

Homomorphisms. Let $\varphi \in \mathrm{Hom}_P(J, A)$ and $a_i = \varphi(g_i) \in A$. The syzygies $wg_1 = zg_2$, $wg_3 = zg_4$, $yg_1 = xg_3$, $yg_2 = xg_4$ give in A:

$$wa_1 = za_2, \quad wa_3 = za_4, \quad ya_1 = xa_3, \quad ya_2 = xa_4.$$

Applying the two restriction maps: the first two relations are trivial on $z = w = 0$ (both sides vanish), and the last two are trivial on $x = y = 0$. So we get

$$wa_1^{(1)} = za_2^{(1)}, \quad wa_3^{(1)} = za_4^{(1)} \text{ in } k[z, w]; \qquad ya_1^{(2)} = xa_3^{(2)}, \quad ya_2^{(2)} = xa_4^{(2)} \text{ in } k[x, y].$$

Since z, w are coprime in the UFD $k[z, w]$, there are $u, u' \in k[z, w]$ with $(a_1^{(1)}, a_2^{(1)}) = (zu, wu)$ and $(a_3^{(1)}, a_4^{(1)}) = (zu', wu')$; similarly there are $v, v' \in k[x, y]$ with $(a_1^{(2)}, a_3^{(2)}) = (xv, yv)$ and $(a_2^{(2)}, a_4^{(2)}) = (xv', yv')$. Thus

$$a_1 = (zu, xv), \quad a_2 = (wu, xv'), \quad a_3 = (zu', yv), \quad a_4 = (wu', yv')$$

(all with zero constant terms, so these are indeed elements of A).

Derivations. A derivation $\delta \in \mathrm{Der}_k(P, A)$ is given by arbitrary elements $\alpha_x = \delta(x)$, α_y, α_z, $\alpha_w \in A$. Write $\alpha_x = (p_x, q_x)$, etc. Then

$$\delta(g_1) = z\alpha_x + x\alpha_z = (zp_x,\ xq_z), \quad \delta(g_2) = w\alpha_x + x\alpha_w = (wp_x,\ xq_w),$$

$$\delta(g_3) = z\alpha_y + y\alpha_z = (zp_y,\ yq_z), \quad \delta(g_4) = w\alpha_y + y\alpha_w = (wp_y,\ yq_w),$$

because x, y vanish on the plane $x = y = 0$ and z, w on the plane $z = w = 0$. Choose $\alpha_x = (u, u(0))$, $\alpha_y = (u', u'(0))$, $\alpha_z = (v(0), v)$, $\alpha_w = (v'(0), v')$ (pairs with equal constant terms, so elements of A). Then $\delta(g_1) = (zu, xv) = a_1$, $\delta(g_2) = (wu, xv') = a_2$, $\delta(g_3) = (zu', yv) = a_3$, $\delta(g_4) = (wu', yv') = a_4$. So φ and $\delta|_J$ agree on the generators g_i of J, hence $\varphi = \delta|_J$.

Therefore $\mathrm{Der}_k(P, A) \to \mathrm{Hom}_A(J/J^2, A)$ is surjective, $T^1(A) = 0$, and A is rigid. □

Exercise 9.10

Problem. A scheme X_0 over a field k is *rigid* if it has no infinitesimal deformations.

(a) Show that $\mathbf{P}^1_k$ is rigid, using (9.13.2).

(b) One might think that if X_0 is rigid over k, then every global deformation of X_0 is locally trivial. Show that this is not so, by constructing a proper, flat morphism $f : X \to \mathbf{A}^2$ over k algebraically closed, such that $X_0 \cong \mathbf{P}^1_k$, but there is no open neighborhood U of 0 in $\mathbf{A}^2$ for which $f^{-1}(U) \cong U \times \mathbf{P}^1$.

(c) Show, however, that one can trivialize a global deformation of $\mathbf{P}^1$ after a flat base extension, in the following sense: let $f : X \to T$ be a flat projective morphism, where T is a nonsingular curve over k algebraically closed. Assume there is a closed point $t \in T$ such that $X_t \cong \mathbf{P}^1_k$. Then there exists a nonsingular curve T', and a flat morphism $g : T' \to T$ whose image contains t, such that if $X' = X \times_T T'$ is the base extension, then the new family $f' : X' \to T'$ is isomorphic to $\mathbf{P}^1_{T'} \to T'$.

(Part (c) is starred in the book.)

Solution. (a) Let k be algebraically closed (as in 9.13.2). By (9.13.2) the set of infinitesimal deformations of the nonsingular variety $X = \mathbf{P}^1$, up to isomorphism, is in bijection with $H^1(X, \mathcal{T}_X)$, where $\mathcal{T}_X = \mathcal{H}om(\Omega_X, \mathcal{O}_X)$. For $\mathbf{P}^1$, $\Omega \cong \mathcal{O}(-2)$ (the canonical sheaf of $\mathbf{P}^n$ is $\mathcal{O}(-n-1)$, II, 8.20.1), so $\mathcal{T} \cong \mathcal{O}(2)$. By Thm. 5.1(d), $H^1(\mathbf{P}^1, \mathcal{O}(2))$ is dual to $H^0(\mathbf{P}^1, \mathcal{O}(-4)) = 0$. So $H^1(\mathcal{T}) = 0$: there is only the trivial infinitesimal deformation, and $\mathbf{P}^1$ is rigid.

(b) Here k is algebraically closed (also in (c)). Let $T = \mathbf{A}^2 = \operatorname{Spec} k[a,b]$ and let $X \subseteq \mathbf{P}^2_T = \operatorname{Proj} k[a,b][x,y,z]$ be the conic bundle

$$X: \ F = x^2 + xy + a\,y^2 - (1+b)\,z^2 = 0, \qquad f : X \to T.$$

Then f is projective, hence proper. For every point $s \in T$ (closed or not), $F_s \in k(s)[x,y,z]$ is a nonzero form of degree 2 (the coefficient of x^2 is 1), so X_s is a plane conic and its Hilbert polynomial is $2m+1$; since T is integral noetherian, f is flat by Thm. 9.9.

The fibre over 0. $X_0 = \{x^2 + xy - z^2 = 0\} \subseteq \mathbf{P}^2_k$ is a smooth conic: the partial derivatives $2x + y,\ x,\ -2z$ vanish simultaneously only if $x = 0, y = 0$ and then $z = 0$ when $\operatorname{char} k \neq 2$; in characteristic 2 the partials are $y, x, 0$, which vanish only at $x = y = 0$, where $F = z^2$ forces $z = 0$. A smooth plane conic over an algebraically closed field is isomorphic to $\mathbf{P}^1$ (e.g. project from one of its points), so $X_0 \cong \mathbf{P}^1_k$.

No trivialization. Suppose $U \ni 0$ is open in $\mathbf{A}^2$ and $f^{-1}(U) \cong U \times \mathbf{P}^1$ as U-schemes. Then $f^{-1}(U) \to U$ has a section, hence the generic fibre X_η (over $K = k(a,b)$) has a K-rational point: there is $(x : y : z) \in \mathbf{P}^2(K)$ with $F = 0$. We show this is impossible. Put $q = 1 + b$, so $k[a,b] = k[a,q]$ is a polynomial ring (a UFD). Clearing denominators and dividing by common factors, we may assume $x, y, z \in k[a,q]$ have no common prime factor and $x^2 + xy + ay^2 = qz^2$. Reduce modulo q: with $\bar{x}, \bar{y} \in k[a]$ we get $\bar{x}^2 + \bar{x}\bar{y} + a\bar{y}^2 = 0$. If $\bar{y} \neq 0$ let $g = \gcd(\bar{x}, \bar{y})$, $\bar{x} = gu$, $\bar{y} = gv$, $\gcd(u,v) = 1$; then $u^2 + uv + av^2 = 0$, so $v \mid u^2$, hence $v = c \in k^*$ and $u^2 + cu + ac^2 = 0$ in $k[a]$. If u is constant, the left side has degree 1 in a; if $\deg u \geq 1$, $\deg(u^2 + cu) = 2 \deg u \geq 2 > 1$. Contradiction. So $\bar{y} = 0$, then $\bar{x}^2 = 0$, $\bar{x} = 0$: $q \mid x$ and $q \mid y$. Then $q^2 \mid qz^2$, so $q \mid z$, contradicting the choice of x, y, z. Hence X_η has no K-point, and no neighborhood U of 0 has $f^{-1}(U) \cong U \times \mathbf{P}^1$ over U. (Of course f is a smooth conic bundle over a neighborhood of 0, since X_0 is smooth and f is flat; the obstruction to Zariski local triviality is arithmetic: the generic conic has no rational point.)

(c) *Step 1: shrinking T.* Since X_t is smooth over k and f is flat, f is smooth at every point of X_t (EGA IV_4 17.5.1; this is the local form of Thm. 10.2). The smooth locus $V \subseteq X$ is open and contains X_t, and f is closed, so $T_1 = T \setminus f(X \setminus V)$ is an open neighborhood of t with $f^{-1}(T_1) \subseteq V$. By Thm. 12.8 the functions $s \mapsto h^i(X_s, \mathcal{O}_{X_s})$ are upper semicontinuous, and at t we have $h^0 = 1$, $h^1 = 0$; shrinking T_1 we get $h^0(\mathcal{O}_{X_s}) \leq 1$ and $h^1(\mathcal{O}_{X_s}) = 0$ for all $s \in T_1$. Since f is flat and T_1 is connected, $\dim X_s = \dim X_t = 1$ for all $s \in T_1$ (Cor. 9.10), in particular $X_s \neq \emptyset$ and $h^0(\mathcal{O}_{X_s}) = 1$. Thus for each closed point $s \in T_1$, X_s is a smooth (it lies in the smooth locus V) projective connected curve of genus 0, i.e. $X_s \cong \mathbf{P}^1_k$. Replace T by T_1; then f is smooth, projective, with all closed fibres $\mathbf{P}^1$.

Step 2: a finite flat cover over which f has a section. Let $X \subseteq \mathbf{P}^N_T$ (projective morphism). $X_t \subseteq \mathbf{P}^N_k$ is an irreducible curve; choose a hyperplane $H \subseteq \mathbf{P}^N_k$ not containing it. Then $X_t \cap H$ is nonempty (a projective variety of dimension ≥ 1 meets every hypersurface, Ex. I.3.7) and finite. Let $C = X \cap (H \times T)$. By upper semicontinuity of fibre dimension (II, Ex. 3.22) and properness of f, shrinking T around t we may assume $C \to T$ has finite fibres, hence is finite

(C is proper over T). Let $c \in C_t$. Since f is flat with fibre dimension 1, $\dim \mathcal{O}_{X,c} = 1 + 1 = 2$ (Prop. 9.5), and C is defined near c by one equation in X, so $\dim \mathcal{O}_{C,c} \geq 1$. Choose a prime $\mathfrak{q}$ of $\mathcal{O}_{C,c}$ with $\dim \mathcal{O}_{C,c}/\mathfrak{q} \geq 1$ and let $C_1 \subseteq C$ be the closure of the corresponding point with the reduced structure: an integral closed subscheme through c of dimension ≥ 1. $f(C_1)$ is an irreducible closed subset of T containing t; if it were $\{t\}$ then $C_1 \subseteq C_t$ would be finite, contradicting $\dim C_1 \geq 1$. So $f(C_1) = T$ and C_1 is an integral curve, finite and surjective over T. Let $T' \to C_1$ be its normalization: a nonsingular curve, finite over C_1, and $g : T' \to T$ is finite and surjective. T' is integral and dominates the nonsingular curve T, so $\mathcal{O}_{T'}$ is torsion-free, hence flat, over the local rings of T (Dedekind): g is flat, and its image is all of $T \ni t$.

Let $X' = X \times_T T'$, $f' : X' \to T'$. It is smooth and projective, and for each closed $s' \in T'$ the fibre is $X_{g(s')} \cong \mathbf{P}^1_k$. The morphism $T' \to C_1 \hookrightarrow X$ together with $\mathrm{id}_{T'}$ defines a section $\sigma : T' \to X'$ of f'.

Step 3: X' is a $\mathbf{P}^1$-bundle. X' is regular (smooth over the regular scheme T'), so the closed integral curve $D = \sigma(T')$ is an effective Cartier divisor (II, 6.11); put $\mathcal{M} = \mathcal{O}_{X'}(D)$. D meets each closed fibre $X'_{s'}$ transversally in the single reduced point $\sigma(s')$ ($D \to T'$ is an isomorphism), so $\mathcal{M}|_{X'_{s'}} \cong \mathcal{O}_{\mathbf{P}^1}(\sigma(s')) \cong \mathcal{O}_{\mathbf{P}^1}(1)$. Hence $h^0(X'_{s'}, \mathcal{M}_{s'}) = 2$ and $h^1 = 0$ for all closed s', hence (semicontinuity; a nonempty closed subset of T' contains a closed point) for all points $s' \in T'$. By Cor. 12.9, $\mathcal{E}' = f'_*\mathcal{M}$ is locally free of rank 2 and $\mathcal{E}' \otimes k(s') \cong H^0(X'_{s'}, \mathcal{M}_{s'})$. The evaluation map $f'^*\mathcal{E}' \to \mathcal{M}$ is surjective on each closed fibre ($H^0(\mathcal{O}(1)) \otimes \mathcal{O} \to \mathcal{O}(1)$ is surjective), so by Nakayama it is surjective on a neighborhood of every closed fibre, hence everywhere (the support of the cokernel is closed and its image in T' is closed and contains no closed point). By (II, 7.12) it defines a T'-morphism $\psi : X' \to \mathbf{P}(\mathcal{E}')$ with $\psi^*\mathcal{O}(1) = \mathcal{M}$, which on each closed fibre is the morphism $\mathbf{P}^1 \to \mathbf{P}(H^0(\mathbf{P}^1, \mathcal{O}(1))) = \mathbf{P}^1$ given by the complete linear system $|\mathcal{O}(1)|$, an isomorphism.

ψ is an isomorphism. Indeed ψ is proper, and all its fibres over closed points of $\mathbf{P}(\mathcal{E}')$ are single points; by semicontinuity of fibre dimension the locus of points of X' where ψ has positive-dimensional fibre is closed, and its image is a closed subset of $\mathbf{P}(\mathcal{E}')$ containing no closed point, hence empty. So ψ is proper and quasi-finite, hence finite. Put $\mathcal{A} = \psi_*\mathcal{O}_{X'}$, a coherent $\mathcal{O}_{\mathbf{P}(\mathcal{E}')}$-algebra, flat over T' (as ψ is finite and X' is T'-flat). The map $\mathcal{O}_{\mathbf{P}(\mathcal{E}')} \to \mathcal{A}$ becomes an isomorphism after $\otimes k(s')$ for each closed s' (the fibre of ψ over s'). By Nakayama its cokernel vanishes near the fibres over closed points, hence everywhere; for its kernel $\mathcal{K}$, flatness of $\mathcal{A}$ over T' gives that $\mathcal{K} \otimes k(s') \to \mathcal{O} \otimes k(s') \to \mathcal{A} \otimes k(s')$ is exact on the left, so $\mathcal{K} \otimes k(s') = 0$ and again $\mathcal{K} = 0$. So ψ is an isomorphism: $X' \cong \mathbf{P}(\mathcal{E}')$ over T'.

Step 4: trivialization. Pick a closed point $t' \in g^{-1}(t)$. The locally free sheaf $\mathcal{E}'$ is free on a neighborhood $T'' \subseteq T'$ of t'. Then $X'|_{T''} \cong \mathbf{P}(\mathcal{O}^2_{T''}) = \mathbf{P}^1_{T''}$. The morphism $T'' \hookrightarrow T' \xrightarrow{g} T$ is flat (open immersion followed by a flat map), T'' is a nonsingular curve, its image contains t, and the base extension $X \times_T T'' = X'|_{T''}$ is isomorphic to $\mathbf{P}^1_{T''} \to T''$. This proves (c).

(Remark: the base extension g constructed is finite and surjective onto a neighborhood of t before the final shrinking; the family becomes trivial near a point over t. In contrast, (b) shows that over a two-dimensional base even this kind of shrinking cannot trivialize the family on a Zariski neighborhood of the point itself.) □

Exercise 9.11

Problem. Let Y be a nonsingular curve of degree d in $\mathbf{P}^n_k$, over an algebraically closed field k. Show that

$$0 \le p_a(Y) \le \tfrac{1}{2}(d-1)(d-2).$$

[Hint: Compare Y to a suitable projection of Y into $\mathbf{P}^2$, as in (9.8.3) and (9.8.4).]

Solution. Here a curve is an integral (equivalently, connected nonsingular) projective scheme of dimension 1, $p_a(Z) = 1 - P_Z(0)$ for a 1-dimensional Z (I, Ex. 7.2; Ex. III 5.3), where $P_Z(m) = dm + (1 - p_a)$ is the Hilbert polynomial. Let $g = p_a(Y)$.

Lower bound. $H^0(Y, \mathcal{O}_Y) = k$ since Y is integral projective over the algebraically closed k, so $p_a(Y) = 1 - \chi(\mathcal{O}_Y) = h^1(\mathcal{O}_Y) \ge 0$.

Upper bound. If $n = 1$, then $Y = \mathbf{P}^1$, $d = 1$, $g = 0$. If $n = 2$, Y is a plane curve of degree d and the sequence $0 \to \mathcal{O}(-d) \to \mathcal{O} \to \mathcal{O}_Y \to 0$ gives $\chi(\mathcal{O}_Y(m)) = \chi(\mathcal{O}(m)) - \chi(\mathcal{O}(m-d)) = dm + 1 - \frac{1}{2}(d-1)(d-2)$, so $g = \frac{1}{2}(d-1)(d-2)$. Let $n \ge 3$.

Step A: a good projection. *Claim:* there is a linear subspace $L \subseteq \mathbf{P}^n$ of dimension $n-3$ with $L \cap Y = \emptyset$ such that the projection $\pi_L : Y \to \mathbf{P}^2$ from L is birational onto its image Γ, which is an integral plane curve. It suffices to project successively from general points: let $\Gamma_0 \subseteq \mathbf{P}^m$ ($m \ge 3$) be an integral curve. For a point $P \in \mathbf{P}^m \setminus \Gamma_0$ the image of π_P is a curve (not a point: otherwise Γ_0 lies in a line through P, hence is that line, contradicting $P \notin \Gamma_0$). Let N be the (dense open) nonsingular locus of Γ_0. The union S of the secant lines $\overline{xx'}$, $x \ne x' \in N$, is a constructible set of dimension ≤ 3, and the union of the tangent lines to Γ_0 at points of N has dimension ≤ 2. If $m \ge 4$, a general P lies on no secant and no tangent line. If $m = 3$, a general P lies on no tangent line and on only finitely many secant lines (the incidence variety $\{(x, x', P)\}$ has dimension 3 so its fibres over a general $P \in \mathbf{P}^3$ are finite, or empty). Fix such a general P (also $P \notin \Gamma_0$) and put $\Gamma_1 = \pi_P(\Gamma_0)$. Then: (i) there is a finite set $F \subseteq N$ such that π_P is injective on $N \setminus F$ (take F to be the points lying on the finitely many secant lines through P if $m = 3$, and $F = \emptyset$ if $m \ge 4$); hence the finite surjective map $\pi_P : \Gamma_0 \to \Gamma_1$ has exactly one point in the fibre over every point of the dense open set $\Gamma_1 \setminus \pi_P(F \cup (\Gamma_0 \setminus N))$, so the field extension $k(\Gamma_0)/k(\Gamma_1)$ has trivial separable degree; (ii) since P lies on no tangent line at a point of N, the differential of $N \to \mathbf{P}^{m-1}$ is injective at each $x \in N$, i.e. $\Omega_{N/\mathbf{P}^{m-1}} = 0$, and since Ω_{N/Γ_1} is a quotient of $\Omega_{N/\mathbf{P}^{m-1}}$ (II, 8.11) we get $\Omega_{k(\Gamma_0)/k(\Gamma_1)} = 0$, so the extension is separable. Hence $k(\Gamma_0) = k(\Gamma_1)$ and $\pi_P : \Gamma_0 \to \Gamma_1$ is birational. Iterating from $\mathbf{P}^n$ down to $\mathbf{P}^2$ ($n-2$ projections) and composing gives the claim, the composite being the projection from a linear space L of dimension $n-3$ disjoint from Y.

Choose coordinates such that $L = \{x_0 = x_1 = x_2 = 0\}$, so $\pi_L(x_0 : \cdots : x_n) = (x_0 : x_1 : x_2)$, defined on all of Y.

Step B: the family (cf. 9.8.3). For $a \in k^*$ let $\sigma_a(x_0 : \cdots : x_n) = (x_0 : x_1 : x_2 : ax_3 : \cdots : ax_n)$, an automorphism of $\mathbf{P}^n$. The map

$$\Phi : Y \times \mathbf{A}^1 \to \mathbf{P}^n \times \mathbf{A}^1, \qquad (y, a) \mapsto (\sigma_a(y), a)$$

(the formula $(x_0 : x_1 : x_2 : ax_3 : \cdots : ax_n)$ makes sense for $a = 0$ too, because x_0, x_1, x_2 have no common zero on Y) is a morphism over $\mathbf{A}^1$, hence proper. Its image is closed and irreducible, and it is the closure of $\Phi(Y \times \mathbf{G}_m)$ ($Y \times \mathbf{G}_m$ is dense in $Y \times \mathbf{A}^1$). Let $\overline{X} \subseteq \mathbf{P}^n \times \mathbf{A}^1$ be this image with its reduced structure; $\overline{X}$ is an integral closed subscheme, hence flat over $\mathbf{A}^1$ (Ex. 9.1.3, $k[a]$ a PID), and:

- for $a \neq 0$, $\overline{X}_a = \sigma_a(Y) \cong Y$ (over $a \neq 0$, Φ is an isomorphism of $Y \times \mathbf{G}_m$ onto $\overline{X}|_{a\neq 0}$, since $(y, a) \mapsto (\sigma_a y, a)$ is the restriction of an automorphism of $\mathbf{P}^n \times \mathbf{G}_m$);
- the underlying set of $\overline{X}_0$ is $\Phi(Y \times \{0\}) = \pi_L(Y) = \Gamma \subseteq \mathbf{P}^2 \subseteq \mathbf{P}^n$.

By Thm. 9.9 all fibres have Hilbert polynomial $P_Y(m) = dm+1-g$; in particular $\overline{X}_0$ has Hilbert polynomial $dm + 1 - g$.

Step C: the multiplicity of $\overline{X}_0$ along Γ is 1. The map $\Phi : Y \times \mathbf{A}^1 \to \overline{X}$ is proper, quasi-finite (the fibre over $(z, 0)$ is $\{y : \pi_L(y) = z\}$, finite since $\pi_L : Y \to \Gamma$ is a non-constant morphism of curves) and birational (an isomorphism over $a \neq 0$), hence finite and birational. As $Y \times \mathbf{A}^1$ is normal, it is the normalization of $\overline{X}$. Let ξ be the generic point of Γ in $\overline{X}_0$ and $\mathcal{O} = \mathcal{O}_{\overline{X},\xi}$, a one-dimensional local domain, in which $a \neq 0$ is a nonzerodivisor with $\mathcal{O}/a\mathcal{O} = \mathcal{O}_{\overline{X}_0,\xi}$. The normalization B of $\mathcal{O}$ is the semi-local ring of $Y \times \mathbf{A}^1$ at the points lying over ξ. A point of $Y \times \{0\}$ over ξ must (by finiteness, dimensions) be the generic point η_Y of $Y \times \{0\}$, so $B = \mathcal{O}_{Y\times\mathbf{A}^1,(\eta_Y,0)} = k(Y)[a]_{(a)}$, a DVR with uniformizer a and residue field $k(Y)$. Since π_L is birational onto Γ, $k(Y) = k(\Gamma) = \kappa(\xi)$: the residue fields of B and $\mathcal{O}$ coincide. $B/\mathcal{O}$ is a finitely generated torsion $\mathcal{O}$-module over a one-dimensional local domain, hence of finite length; multiplication by a on $0 \to \mathcal{O} \to B \to B/\mathcal{O} \to 0$ and the snake lemma (for a finite-length module Q, $\ell(\ker a|_Q) = \ell(Q/aQ)$) give $\ell(\mathcal{O}/a\mathcal{O}) = \ell(B/aB) = \ell(\kappa(B)) = 1$. So $\mathcal{O}_{\overline{X}_0,\xi} = \mathcal{O}/a\mathcal{O}$ is the field $\kappa(\xi)$: $\overline{X}_0$ is generically reduced along Γ.

Step D: comparison of genera. Let $\mathcal{H} \subseteq \mathcal{O}_{\overline{X}_0}$ be the subsheaf of sections supported on a finite set of points (a coherent ideal with finite support, of length $\ell \geq 0$) and $C \subseteq \overline{X}_0$ the closed subscheme with $\mathcal{O}_C = \mathcal{O}_{\overline{X}_0}/\mathcal{H}$. Then C has no embedded or isolated points, its support is Γ (irreducible), and by Step C it is reduced at the generic point of Γ; a scheme with no embedded points that is reduced at its generic points is reduced, so C is the reduced integral curve Γ, a plane curve contained in $\mathbf{P}^2 \subseteq \mathbf{P}^n$. From $0 \to \mathcal{H} \to \mathcal{O}_{\overline{X}_0} \to \mathcal{O}_\Gamma \to 0$: $\chi(\mathcal{O}_{\overline{X}_0}(m)) = \chi(\mathcal{O}_\Gamma(m)) + \ell$, i.e.

$$dm + 1 - g = P_\Gamma(m) + \ell.$$

So P_Γ has leading coefficient d: Γ is a plane curve of degree d (a plane curve of degree e has leading coefficient e, by the computation for $n = 2$ above), and $P_\Gamma(0) = 1 - g - \ell$. But $P_\Gamma(0) = 1 - \frac{1}{2}(d-1)(d-2)$ by the plane case. Therefore

$$g = \tfrac{1}{2}(d-1)(d-2) - \ell \ \leq \ \tfrac{1}{2}(d-1)(d-2).$$

(The number ℓ measures the embedded points of the flat limit, as in (9.8.4) and Ex. 9.2.) Together with the lower bound, $0 \leq p_a(Y) \leq \frac{1}{2}(d-1)(d-2)$. □

10. Smooth morphisms

Conventions. As in the text, all schemes are of finite type over a field k. The definition of "smooth of relative dimension n" makes sense verbatim for a scheme of finite type over any field K (flatness is automatic; condition (2) says every irreducible component has dimension n; condition (3) says $\dim_{k(x)} \Omega_{X/K} \otimes k(x) = n$ for all points x), and it is used in this form for fibres X_y over $K = k(y)$. For a morphism $f : X \to Y$ and $x \in X$, $y = f(x)$, base change (II, 8.10)

gives $\Omega_{X_y/k(y)} = \Omega_{X/Y}|_{X_y}$, hence

$$\Omega_{X/Y} \otimes k(x) = \Omega_{X_y/k(y)} \otimes k(x).$$

The sheaf $\Omega_{X/Y}$ is coherent, so by Nakayama $\Omega_{X/Y} = 0$ if and only if $\Omega_{X/Y} \otimes k(x) = 0$ for all x, and for each r the set $\{x : \dim_{k(x)} \Omega_{X/Y} \otimes k(x) \geq r\}$ is closed (II, Ex. 5.8).

Exercise 10.1

> **Problem.** Over a nonperfect field, smooth and regular are not equivalent. For example, let k_0 be a field of characteristic $p > 0$, $k = k_0(t)$, and let $X \subseteq \mathbf{A}^2_k$ be the curve defined by $y^2 = x^p - t$. Show that every local ring of X is a regular local ring, but X is not smooth over k.

Solution. *Regularity.* Put $f = y^2 - x^p + t$ and $S = k_0[t] \setminus \{0\}$. Since $k[x,y] = S^{-1}k_0[t][x,y]$ and localization commutes with quotients,

$$\mathcal{O}(X) = k[x,y]/(f) = S^{-1}\big(k_0[x,y,t]/(t - (x^p - y^2))\big).$$

The ring $k_0[x,y,t]/(t - x^p + y^2)$ is isomorphic to $k_0[x,y]$ by $t \mapsto x^p - y^2$. Under this isomorphism S goes to the multiplicative set $S' = \{g(x^p - y^2) : 0 \neq g \in k_0[T]\}$, which does not contain 0 because $x^p - y^2$ is transcendental over k_0. Hence

$$\mathcal{O}(X) \cong S'^{-1}k_0[x,y]$$

is a localization of the polynomial ring $k_0[x,y]$. So X is integral and each local ring $\mathcal{O}_{X,x}$ is a localization of $k_0[x,y]$ at a prime ideal, hence a regular local ring (a polynomial ring over a field is regular at its maximal ideals, and a localization of a regular local ring at a prime ideal is regular by Serre's theorem, Matsumura, Thm. 19.3). The function field of X is $k_0(x,y)$, of transcendence degree $2 - 1 = 1$ over $k = k_0(t)$ (as $t = x^p - y^2$ there), so $\dim X = 1$.

Non-smoothness. Let $A = \mathcal{O}(X)$. By (II, 8.4A) we have the exact sequence $(f)/(f^2) \to \Omega_{k[x,y]/k} \otimes A \to \Omega_{A/k} \to 0$ with $\Omega_{k[x,y]/k} = k[x,y]\,dx \oplus k[x,y]\,dy$. Since k-derivations kill $t \in k$ and $p = 0$,

$$df = 2y\,dy - px^{p-1}dx + dt = 2y\,dy, \qquad \text{so} \qquad \Omega_{X/k} \cong A\,dx \oplus (A/2yA)\,dy.$$

The polynomial $x^p - t$ is irreducible over k (t is not a p-th power in $k_0(t)$: if $th^p = g^p$ then comparing t-adic valuations gives $1 \equiv 0 \pmod p$). Hence $\mathfrak{m} = (y, x^p - t)$ is a maximal ideal of $k[x,y]$, and $f \in \mathfrak{m}$, so $\mathfrak{m}$ gives a closed point $x_0 \in X$ with $y(x_0) = 0$. Then $2y\,dy \otimes k(x_0) = 0$, whence

$$\dim_{k(x_0)} \Omega_{X/k} \otimes k(x_0) = 2.$$

If X were smooth of relative dimension n over k, condition (2) (with $Y = \operatorname{Spec} k$, X integral) would force $n = \dim X = 1$, and condition (3) would force this dimension to be 1 at every point. It is 2 at x_0, so X is not smooth over k. (For $p = 2$ the dimension is 2 at every point.) $\square$

Exercise 10.2

Problem. Let $f : X \to Y$ be a proper, flat morphism of varieties over k. Suppose for some point $y \in Y$ that the fibre X_y is smooth over $k(y)$. Then show that there is an open neighborhood U of y in Y such that $f : f^{-1}U \to U$ is smooth.

Solution. Here varieties are integral, separated, of finite type over an algebraically closed field k, and "X_y smooth over $k(y)$" means smooth of relative dimension n over $k(y)$ for some n (see the conventions).

Lemma A. Let Z be a scheme of finite type over a field K and $z \in Z$. Then $\dim_{k(z)} \Omega_{Z/K} \otimes k(z) \geq \dim_z Z$, where $\dim_z Z$ is the maximum of the dimensions of the irreducible components of Z through z.

Proof. Let Z' be such a component with its reduced structure; it is integral of finite type over K and $\dim Z' = \operatorname{tr.deg}_K K(Z')$. By (II, 8.6A), the coherent sheaf $\Omega_{Z'/K}$ has rank $\dim_{K(Z')} \Omega_{K(Z')/K} \geq \dim Z'$, so it needs at least $\dim Z'$ generators at every point (its stalk at z surjects onto its generic stalk after tensoring). By (II, 8.12) $\Omega_{Z/K} \otimes k(z) \to \Omega_{Z'/K} \otimes k(z) \to 0$ is surjective. □

If $X_y = \emptyset$, then $U = Y \setminus f(X)$ works, because f is proper, so $f(X)$ is closed; $f^{-1}U = \emptyset$. So assume $X_y \neq \emptyset$.

Step 1: all nonempty fibres are equidimensional of dimension $e := \dim X - \dim Y$, and $n = e$. Let $y' \in Y$ and let x be a closed point of the fibre $X_{y'}$; thus $k(x)$ is finite over $k(y')$. By (9.5), $\dim \mathcal{O}_{X_{y'},x} = \dim \mathcal{O}_{X,x} - \dim \mathcal{O}_{Y,y'}$. By (II, Ex. 3.20), $\dim \mathcal{O}_{X,x} = \dim X - \dim \overline{\{x\}}$ and $\dim \mathcal{O}_{Y,y'} = \dim Y - \dim \overline{\{y'\}}$. Since $k(x)/k(y')$ is finite, $\dim \overline{\{x\}} = \operatorname{tr.deg}_k k(x) = \operatorname{tr.deg}_k k(y') = \dim \overline{\{y'\}}$. Hence $\dim \mathcal{O}_{X_{y'},x} = e$ for every closed point of every nonempty fibre. If Z is an irreducible component of $X_{y'}$, choose a closed point $x \in Z$ lying on no other component; then $\dim Z = \dim \mathcal{O}_{Z,x} = \dim \mathcal{O}_{X_{y'},x} = e$. So all irreducible components of all fibres have dimension e. Applied to the fibre X_y, which is smooth of relative dimension n, this gives $n = e$.

Step 2: shrinking. Let $W = \{x \in X : \dim_{k(x)} \Omega_{X/Y} \otimes k(x) \geq n + 1\}$, a closed subset of X. For $x \in X_y$ we have $\Omega_{X/Y} \otimes k(x) = \Omega_{X_y/k(y)} \otimes k(x)$ of dimension n (condition (3) for X_y), so $W \cap X_y = \emptyset$, i.e. $y \notin f(W)$. As f is proper, $f(W)$ is closed. Put $U = Y \setminus f(W)$, an open neighborhood of y, and $X_U = f^{-1}U$, a nonempty open subset of X.

Step 3: $X_U \to U$ is smooth of relative dimension n. (1) f is flat, so is its restriction. (2) X_U and U are integral, hence are their own unique irreducible components, and $\dim X_U = \dim X = \dim Y + n = \dim U + n$ (nonempty opens of varieties have the full dimension). (3) Let $x \in X_U$, $y' = f(x)$. Since $x \notin W$, $\dim \Omega_{X/Y} \otimes k(x) \leq n$. On the other hand, with $Z = X_{y'}$, Lemma A and Step 1 give $\dim_{k(x)} \Omega_{X/Y} \otimes k(x) = \dim \Omega_{Z/k(y')} \otimes k(x) \geq \dim_x Z = e = n$. So the dimension equals n at every point. □

Exercise 10.3

Problem. A morphism $f : X \to Y$ of schemes of finite type over k is *etale* if it is smooth of relative dimension 0. It is *unramified* if for every $x \in X$, letting $y = f(x)$, we have $\mathfrak{m}_y \cdot \mathcal{O}_x = \mathfrak{m}_x$, and $k(x)$ is a separable algebraic extension of $k(y)$. Show that the following conditions are equivalent:

(i) f is etale;

(ii) f is flat, and $\Omega_{X/Y} = 0$;

(iii) f is flat and unramified.

Solution. Fix $x \in X$, $y = f(x)$, $K = k(y)$, $L = k(x)$, and let $B_0 = \mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x = \mathcal{O}_{X_y,x}$, a local K-algebra with residue field L and maximal ideal $\mathfrak{m}_0$. Note that L is a finitely generated field extension of k, hence of K.

Claim. $(\Omega_{X/Y})_x = 0$ if and only if f is unramified at x (i.e. $\mathfrak{m}_y\mathcal{O}_x = \mathfrak{m}_x$ and L/K is separable algebraic).

($\Leftarrow$) If $\mathfrak{m}_y\mathcal{O}_x = \mathfrak{m}_x$ then $B_0 = L$. As L/K is finite separable, $\Omega_{L/K} = 0$ by (II, 8.6A) (it is separably generated of transcendence degree 0). Using the base change (II, 8.10), $(\Omega_{X/Y})_x \otimes k(x) = \Omega_{B_0/K} \otimes L = \Omega_{L/K} = 0$, so $(\Omega_{X/Y})_x = 0$ by Nakayama.

($\Rightarrow$) If $(\Omega_{X/Y})_x = 0$ then $\Omega_{B_0/K} = 0$ (base change). By (II, 8.12) applied to $B_0 \to L$ the sequence $\mathfrak{m}_0/\mathfrak{m}_0^2 \to \Omega_{B_0/K} \otimes L \to \Omega_{L/K} \to 0$ is exact, so $\Omega_{L/K} = 0$. By (II, 8.6A), L is separably generated over K of transcendence degree $\leq \dim \Omega_{L/K} = 0$; so L/K is finite and separable. Now let $B' = B_0/\mathfrak{m}_0^2$, an artinian local K-algebra with residue field L. By (II, 8.25A) (or Hensel's lemma, as B' is complete and L/K is finite separable), B' contains a field of representatives $L' \supseteq K$. We have $\Omega_{B'/K} \otimes L$ a quotient of $\Omega_{B_0/K} \otimes L = 0$ (II, 8.12), and $\Omega_{B'/K} \to \Omega_{B'/L'}$ is surjective (II, 8.3A for $K \to L' \to B'$), so $\Omega_{B'/L'} = 0$. By (II, 8.7), $\mathfrak{m}'/\mathfrak{m}'^2 \cong \Omega_{B'/L'} \otimes L = 0$ where $\mathfrak{m}' = \mathfrak{m}_0/\mathfrak{m}_0^2$; since $\mathfrak{m}'^2 = 0$ this gives $\mathfrak{m}' = 0$, i.e. $\mathfrak{m}_0 = \mathfrak{m}_0^2$, so $\mathfrak{m}_0 = 0$ by Nakayama. Thus $B_0 = L$, i.e. $\mathfrak{m}_y\mathcal{O}_x = \mathfrak{m}_x$. □

(ii)⇔(iii): immediate from the Claim, applied at every point x (and $\Omega_{X/Y} = 0$ iff all stalks vanish).

(i)⇒(ii): f is flat by definition, and condition (3) with $n = 0$ says $\Omega_{X/Y} \otimes k(x) = 0$ for all x; since $\Omega_{X/Y}$ is coherent, Nakayama gives $\Omega_{X/Y} = 0$.

(ii)⇒(i): f is flat and, by (ii), condition (3) holds with $n = 0$. It remains to check condition (2): let $X' \subseteq X$ and $Y' \subseteq Y$ be irreducible components with $f(X') \subseteq Y'$; we must show $\dim X' = \dim Y'$. Let ξ', η' be their generic points. Since $f(\xi') \in Y'$, η' is a generization of $f(\xi')$; by flatness generizations lift (going down, Matsumura, Thm. 9.5), so there is a generization x_1 of ξ' with $f(x_1) = \eta'$. But ξ' is a generic point of an irreducible component, so $x_1 = \xi'$ and $f(\xi') = \eta'$. By the Claim, f is unramified at ξ', so $k(\xi')$ is finite over $k(\eta')$. Therefore $\operatorname{tr.deg}_k k(\xi') = \operatorname{tr.deg}_k k(\eta')$, i.e. $\dim X' = \dim Y'$ (I, 1.8A, for the integral schemes X'_{red}, Y'_{red}). Hence f is smooth of relative dimension 0. □

Exercise 10.4

Problem. Show that a morphism $f : X \to Y$ of schemes of finite type over k is etale if and only if the following condition is satisfied: for each $x \in X$, let $y = f(x)$. Let $\hat{\mathcal{O}}_x$ and $\hat{\mathcal{O}}_y$ be the completions of the local rings at x and y. Choose fields of representatives (II, 8.25A) $k(x) \subseteq \hat{\mathcal{O}}_x$ and $k(y) \subseteq \hat{\mathcal{O}}_y$, so that $k(y) \subseteq k(x)$ via the natural map $\hat{\mathcal{O}}_y \to \hat{\mathcal{O}}_x$. Then our condition is that for every $x \in X$, $k(x)$ is a separable algebraic extension of $k(y)$, and the natural map

$$\hat{\mathcal{O}}_y \otimes_{k(y)} k(x) \to \hat{\mathcal{O}}_x$$

is an isomorphism.

Solution. Remarks on the statement. (a) If $k(x)/k(y)$ is algebraic then it is finite, since $k(x)$ is finitely generated over k, hence over $k(y)$. (b) Existence of compatible fields of representatives: $\hat{\mathcal{O}}_y$ contains a field, so by (II, 8.25A) applied with the prime field (over which every extension is separably generated) it has a field of representatives $K \cong k(y)$. Its image in $\hat{\mathcal{O}}_x$ is a subfield K' mapping onto the copy of $k(y)$ in $k(x)$. If $k(x)/k(y)$ is finite separable, (II, 8.25A) applied to $\hat{\mathcal{O}}_x$ and the subfield K' gives a field of representatives $L \supseteq K'$ of $\hat{\mathcal{O}}_x$. So the choices described exist whenever $k(x)/k(y)$ is separable algebraic, and below any such compatible choice may be used. Then $\hat{\mathcal{O}}_y \otimes_{k(y)} k(x)$ is a finite free $\hat{\mathcal{O}}_y$-module (call it T), hence a complete local ring with maximal ideal $\mathfrak{m}_y T$ (since $T/\mathfrak{m}_y T = k(x)$) and residue field $k(x)$.

Lemma B. Let $A \to B$ be a local homomorphism of noetherian local rings. Then B is flat over A if and only if $\hat{B}$ is flat over $\hat{A}$.

Proof. Let $\kappa = A/\mathfrak{m}_A$. By the local criterion of flatness (Matsumura, Thm. 22.3; B is ideal-separated for $\mathfrak{m}_A$ since $\mathfrak{m}_A B \subseteq \mathfrak{m}_B$ and Krull's theorem applies to the finitely generated B-modules $B \otimes_A J$), B is flat over A iff $\operatorname{Tor}_1^A(\kappa, B) = 0$; likewise $\hat{B}$ is flat over $\hat{A}$ iff $\operatorname{Tor}_1^{\hat{A}}(\kappa, \hat{B}) = 0$. Since $\hat{A}$ is flat over A with $\hat{A} \otimes_A \kappa = \kappa$, $\operatorname{Tor}_1^{\hat{A}}(\kappa, \hat{B}) = \operatorname{Tor}_1^A(\kappa, \hat{B})$, and since $\hat{B}$ is flat over B, this equals $\operatorname{Tor}_1^A(\kappa, B) \otimes_B \hat{B}$. As $\hat{B}$ is faithfully flat over B, the latter vanishes iff $\operatorname{Tor}_1^A(\kappa, B) = 0$. □

Now fix x, $y = f(x)$, $A = \mathcal{O}_y$, $B = \mathcal{O}_x$, $K = k(y)$, $L = k(x)$. By Ex. 10.3, f is etale iff f is flat and unramified.

($\Rightarrow$) Let f be etale. Then f is unramified, so L/K is separable algebraic (finite) and $\mathfrak{m}_A B = \mathfrak{m}_B$. Choose compatible fields of representatives $K \subseteq \hat{A}$, $L \subseteq \hat{B}$, and let $T = \hat{A} \otimes_K L$, with the natural map $\varphi : T \to \hat{B}$ (an $\hat{A}$-algebra map and L-linear). We have $\mathfrak{m}_{\hat{A}} \hat{B} = \mathfrak{m}_A \hat{B} = \mathfrak{m}_B \hat{B} = \mathfrak{m}_{\hat{B}}$.

φ is surjective. T is complete for the topology defined by its maximal ideal $\mathfrak{n} = \mathfrak{m}_A T$, and $\varphi(\mathfrak{n})\hat{B} = \mathfrak{m}_{\hat{B}}$. Consider $\hat{B}$ as a T-module via φ: it is separated for the $\mathfrak{n}$-adic topology (as that is the $\mathfrak{m}_{\hat{B}}$-adic topology), and $\hat{B}/\mathfrak{n}\hat{B} = \hat{B}/\mathfrak{m}_{\hat{B}} = L$ is generated by the image of 1 (indeed $L \subseteq T$ maps isomorphically onto $\hat{B}/\mathfrak{m}_{\hat{B}}$). By the complete Nakayama lemma (Matsumura, Thm. 8.4), $\hat{B} = T \cdot 1 = \varphi(T)$.

φ is injective. $\hat{B}$ is flat over $\hat{A}$ (Lemma B, f being flat), and T is free over $\hat{A}$; $\hat{B}$ is a finitely generated $\hat{A}$-module since it is a quotient of T. Let $N = \operatorname{Ker} \varphi$, finitely generated over $\hat{A}$. Tensoring $0 \to N \to T \to \hat{B} \to 0$ with $\kappa = \hat{A}/\mathfrak{m}_{\hat{A}}$, and using $\operatorname{Tor}_1^{\hat{A}}(\kappa, \hat{B}) = 0$, gives the exact sequence $0 \to N/\mathfrak{m}_{\hat{A}} N \to T/\mathfrak{m}_{\hat{A}} T = L \to \hat{B}/\mathfrak{m}_{\hat{A}}\hat{B} = L \to 0$, in which the last map is an isomorphism. Hence $N/\mathfrak{m}_{\hat{A}} N = 0$ and $N = 0$ by Nakayama.

So φ is an isomorphism.

($\Leftarrow$) Suppose L/K is separable algebraic and $\varphi : T = \hat{A} \otimes_K L \to \hat{B}$ is an isomorphism, for all x. Then $\hat{B} \cong T$ is free over $\hat{A}$, so B is flat over A by Lemma B. Moreover $\mathfrak{m}_A \hat{B}$ corresponds to $\mathfrak{m}_A T = \mathfrak{m}_T$, the maximal ideal of T, so $\mathfrak{m}_A \hat{B} = \mathfrak{m}_{\hat{B}} = \mathfrak{m}_B \hat{B}$. By faithful flatness of $B \to \hat{B}$ we have $I\hat{B} \cap B = I$ for ideals I of B; applying this to $I = \mathfrak{m}_A B$ and $I = \mathfrak{m}_B$ gives $\mathfrak{m}_A B = \mathfrak{m}_B$. Thus f is flat at every x and unramified, i.e. etale by Ex. 10.3. □

Exercise 10.5

Problem. If x is a point of a scheme X, we define an *etale neighborhood* of x to be an etale morphism $f : U \to X$, together with a point $x' \in U$ such that $f(x') = x$. As an example

of the use of etale neighborhoods, prove the following: if $\mathcal{F}$ is a coherent sheaf on X, and if every point of X has an etale neighborhood $f : U \to X$ for which $f^*\mathcal{F}$ is a free $\mathcal{O}_U$-module, then $\mathcal{F}$ is locally free on X.

Solution. Let $x \in X$ and let $f : U \to X$, $x' \mapsto x$, be an etale neighborhood with $f^*\mathcal{F}$ free. Put $A = \mathcal{O}_{X,x}$, $B = \mathcal{O}_{U,x'}$, $M = \mathcal{F}_x$, a finitely generated A-module ($\mathcal{F}$ is coherent). Since f is etale it is flat, so $A \to B$ is a flat local homomorphism of noetherian local rings, hence faithfully flat (Matsumura, Thm. 7.2). The stalk $(f^*\mathcal{F})_{x'} = M \otimes_A B$ is free over B, of finite rank r since it is finitely generated.

M is flat over A. Let $N' \hookrightarrow N$ be an injection of A-modules. Then $(N' \otimes_A M) \otimes_A B = (N' \otimes_A B) \otimes_B (M \otimes_A B) \to (N \otimes_A B) \otimes_B (M \otimes_A B)$ is injective, since $N' \otimes_A B \to N \otimes_A B$ is injective (B flat over A) and $M \otimes_A B$ is free over B. By faithful flatness, $N' \otimes_A M \to N \otimes_A M$ is injective.

A finitely generated flat module over a noetherian local ring is free (Matsumura, Thm. 7.10), so $M = \mathcal{F}_x \cong A^s$. Take sections $s_1, \dots, s_s$ of $\mathcal{F}$ over a neighbourhood V of x whose germs generate $\mathcal{F}_x$ and (shrinking V, using coherence) such that $\mathcal{O}_V^s \to \mathcal{F}|_V$ is surjective. Its kernel $\mathcal{K}$ is coherent with $\mathcal{K}_x = \operatorname{Ker}(A^s \to A^s) = 0$ (the map $A^s \to M \cong A^s$ is a surjective endomorphism of a finitely generated module, hence an isomorphism). Since $\operatorname{Supp}\mathcal{K}$ is closed and misses x, after shrinking V we get $\mathcal{F}|_V \cong \mathcal{O}_V^s$. So $\mathcal{F}$ is locally free. □

Exercise 10.6

Problem. Let Y be the plane nodal cubic curve $y^2 = x^2(x+1)$. Show that Y has a finite etale covering X of degree 2, where X is a union of two irreducible components, each one isomorphic to the normalization of Y (Fig. 12).

Solution. We assume $\operatorname{char} k \neq 2$ (as is implicit in the picture: in characteristic 2 the singularity $y^2 = x^2(x+1)$ is a cusp). k need not be algebraically closed.

The curve Y and its normalization. Let $A = k[x,y]/(y^2 - x^2(x+1))$, a domain (since $x^2(x+1)$ is not a square in $k(x)$). Define $A \to k[t]$ by $x \mapsto u := t^2 - 1$, $y \mapsto v := t^3 - t = tu$. This is injective ($A$ is a domain of dimension 1 and the image contains the non-constant u). Its image is

$$k[u,v] = \{g \in k[t] : g(1) = g(-1)\} = k + (t^2-1)k[t].$$

Indeed "$\subseteq$" is clear; conversely, if $g(1) = g(-1)$ then $g - g(1)$ vanishes at ± 1 (which are distinct as $\operatorname{char} k \neq 2$) so lies in $(t^2-1)k[t]$, and writing $h = h_e(t^2) + t\,h_o(t^2)$ we get $(t^2-1)h = (u)h_e(u+1) + v\,h_o(u+1) \in k[u,v]$. We henceforth identify A with this subring of $k[t]$. As $k[t]$ is integrally closed, with the same fraction field $k(t) = k(x,y)$ ($t = y/x$) and integral over A ($t^2 = x+1$), $\mathbf{A}^1 = \operatorname{Spec} k[t] \to Y$ is the normalization.

The covering. Let

$$R = \{(g_1, g_2) \in k[t] \times k[t] :\ g_1(1) = g_2(-1),\ g_1(-1) = g_2(1)\}, \qquad X = \operatorname{Spec} R,$$

i.e. two copies of $\mathbf{A}^1$ glued crosswise at $t = \pm 1$ (two ordinary double points). Map $A \to R$ by $g \mapsto (g, g)$; this lands in R since $g(1) = g(-1)$ for $g \in A$. Let $\pi : X \to Y$ be the corresponding morphism.

Components. The projections $p_i : R \to k[t]$ are surjective (for given g_1 choose g_2 with the prescribed values at ± 1). Put $I_1 = \operatorname{Ker} p_1 = \{(0, g_2)\}$ and $I_2 = \operatorname{Ker} p_2$. Then $I_1 \cap I_2 = 0$ and $I_1 I_2 = 0$, so $X = V(I_1) \cup V(I_2)$ with $V(I_i) \cong \operatorname{Spec} k[t] = \mathbf{A}^1$, and R is reduced; hence X has exactly two irreducible components, each isomorphic to the normalization of Y, and π restricts to the normalization map on each.

Finiteness. $k[t]^2$ is a finite module over $k[x]$ ($t^2 = x + 1$), so R (a $k[x]$-submodule) is finite over $k[x]$ and hence over A.

Etaleness. The elements $x = u$ and $x + 1 = t^2$ of A generate the unit ideal ($(x + 1) - x = 1$), so $Y = D(x) \cup D(x+1)$; since by Ex. 10.3 "etale" means "flat and $\Omega_{X/Y} = 0$", which are conditions local on Y, it suffices to show that $\pi^{-1}D(x) \to D(x)$ and $\pi^{-1}D(x + 1) \to D(x + 1)$ are etale.

Over $D(x)$. Inverting $u = t^2 - 1$: the conditions at $t = \pm 1$ disappear. Indeed $R_x \subseteq k[t]_u \times k[t]_u$, and any (h_1, h_2) in the right side satisfies $(u^N h_1, u^N h_2) \in k[t]^2$ with all values at ± 1 equal to 0 (for $N \geq 1$ large), so is in R, giving $(h_1, h_2) = (u^N h_1, u^N h_2)/(u, u)^N \in R_x$. In the same way $A_x = k[t]_u$. So $R_x = A_x \times A_x$ and $\pi^{-1}D(x) = D(x) \sqcup D(x) \to D(x)$ is the trivial double cover, which is etale (an open immersion is smooth of relative dimension 0 by Prop. 10.1(a)).

Over $D(x + 1)$. Put $A' = A_{x+1}$, $R' = R_{x+1}$. As above (inverting t^2, which equals 1 at $t = \pm 1$) we get

$$A' = \{h \in k[t, t^{-1}] : h(1) = h(-1)\}, \qquad R' = \{(g_1, g_2) \in k[t, t^{-1}]^2 : \, g_1(1) = g_2(-1), \, g_1(-1) = g_2(1)\}.$$

Let $w = (t, -t) \in R'$ (it lies in R: $g_1(1) = 1 = g_2(-1)$, $g_1(-1) = -1 = g_2(1)$); then $w^2 = (t^2, t^2) = x + 1$. Claim: $R' = A' \oplus A'w$ (free with basis $1, w$). Given $(g_1, g_2) \in R'$ put $a = \frac{1}{2}(g_1 + g_2)$, $b = \frac{1}{2t}(g_1 - g_2)$ in $k[t, t^{-1}]$; then $(g_1, g_2) = a \cdot 1 + b \cdot w$. We check $a, b \in A'$:

$$a(1) = \tfrac{1}{2}(g_1(1) + g_2(1)) = \tfrac{1}{2}(g_2(-1) + g_1(-1)) = a(-1),$$

$$b(-1) = \tfrac{1}{-2}(g_1(-1) - g_2(-1)) = \tfrac{1}{-2}(g_2(1) - g_1(1)) = \tfrac{1}{2}(g_1(1) - g_2(1)) = b(1).$$

The sum is direct: $a + bw = (a + bt, a - bt) = 0$ forces $a = b = 0$. Therefore $R' \cong A'[W]/(W^2 - (x + 1))$ (a surjection between free A'-modules of rank 2), which is free, hence flat, over A', and $\Omega_{R'/A'}$ is generated by dW with $2W\,dW = 0$; since $W^2 = x + 1$ is a unit in A', W and 2 are units, so $\Omega_{R'/A'} = 0$. By Ex. 10.3 ((ii)⇒(i)), $\operatorname{Spec} R' \to \operatorname{Spec} A'$ is etale.

Thus π is finite and etale; it is locally free of rank 2 (trivial over $D(x)$, free over $D(x + 1)$), so it is a finite etale covering of degree 2 with the required properties. □

Exercise 10.7

Problem. (Serre). *A linear system with moving singularities.* Let k be an algebraically closed field of characteristic 2. Let $P_1, \ldots, P_7 \in \mathbf{P}^2_k$ be the seven points of the projective plane over the prime field $\mathbb{F}_2 \subseteq k$. Let $\mathfrak{d}$ be the linear system of all cubic curves in X passing through $P_1, \ldots, P_7$.

(a) $\mathfrak{d}$ is a linear system of dimension 2 with base points $P_1, \ldots, P_7$, which determines an inseparable morphism of degree 2 from $X - \{P_i\}$ to $\mathbf{P}^2$.

(b) Every curve $C \in \mathfrak{d}$ is singular. More precisely, either C consists of 3 lines all passing through one of the P_i, or C is an irreducible cuspidal cubic with cusp $P \neq$ any P_i. Furthermore, the correspondence $C \mapsto$ the singular point of C is a 1-1 correspondence between $\mathfrak{d}$ and $\mathbf{P}^2$. Thus the singular points of elements of $\mathfrak{d}$ move all over.

Solution. Here $X = \mathbf{P}^2_k$ with homogeneous coordinates x, y, z (an evident misprint: the curves lie in $\mathbf{P}^2$). Put

$$F_1 = x^2y + xy^2, \quad F_2 = y^2z + yz^2, \quad F_3 = z^2x + zx^2.$$

(a) *The system.* A cubic form $G = \sum c_m m$ over k (sum over the 10 monomials of degree 3) vanishes at the seven points of $\mathbf{P}^2(\mathbb{F}_2)$ iff the function $\mathbb{F}_2^3 \setminus 0 \to k$ it defines is zero. As functions on $\mathbb{F}_2^3$, x^3, y^3, z^3, xyz are x, y, z, xyz, the two monomials x^2y, xy^2 both give xy, the monomials y^2z, yz^2 both give yz, and z^2x, zx^2 both give zx. The functions x, y, z, xy, yz, zx, xyz are linearly independent over k on $\mathbb{F}_2^3 \setminus 0$: the eight multilinear monomials $1, x, \ldots, xyz$ form a basis of all k-valued functions on $\mathbb{F}_2^3$, and the only linear combinations vanishing off the origin are multiples of $(1+x)(1+y)(1+z) = 1 + x + y + z + xy + yz + zx + xyz$, which has a nonzero constant term. Consequently the evaluation map from cubic forms to k^7 is onto, with kernel of dimension $10 - 7 = 3$, namely the span of $x^2y + xy^2,\ y^2z + yz^2,\ z^2x + zx^2$. So $\mathfrak{d} = \{aF_1 + bF_2 + cF_3\} \cong \mathbf{P}^2$, a linear system of dimension 2.

Base points. $F_1 = xy(x+y)$, $F_2 = yz(y+z)$, $F_3 = zx(z+x)$. If $x = 0$ then $F_1 = F_3 = 0$ and $F_2 = 0$ gives $y = 0$, $z = 0$ or $y = z$: the points $(0:0:1), (0:1:0), (0:1:1)$. If $y = 0$ (and $x \neq 0$), $F_1 = F_2 = 0$ and $F_3 = 0$ gives $z = 0$ or $z = x$: $(1:0:0), (1:0:1)$. If $xy \neq 0$ and $x = y$, then $F_2 = 0$ gives $z = 0$ or $z = y$: $(1:1:0), (1:1:1)$ (and then $F_3 = 0$ holds). So the base locus is exactly $\{P_1, \ldots, P_7\}$.

The morphism. On $X - \{P_i\}$ the three cubics have no common zero, so $\varphi = (F_1 : F_2 : F_3) : X - \{P_i\} \to \mathbf{P}^2$ is a morphism. Use $z = 1$ on the source and the affine coordinates $U = F_1/F_3$, $V = F_2/F_3$ on the target; with $K = k(x, y)$, $L = k(U, V) \subseteq K$:

$$U = \frac{x^2y + xy^2}{x^2 + x}, \qquad V = \frac{y^2 + y}{x^2 + x}.$$

Compute $Vx + U = \dfrac{(y^2 + y)x + x^2y + xy^2}{x^2 + x} = \dfrac{xy(1+x)}{x(x+1)} = y$. Squaring, $y^2 = V^2x^2 + U^2$; inserting this into $y^2 + y = V(x^2 + x)$ gives $V^2x^2 + U^2 + Vx + U = Vx^2 + Vx$, i.e.

$$x^2 = \frac{U^2 + U}{V^2 + V} \in L, \qquad y = Vx + U \in L(x).$$

Hence $K = L(x)$ with $x^2 \in L$: K/L is purely inseparable of degree 1 or 2. It is not of degree 1: in characteristic 2 one computes $d(x^2y + xy^2) = x^2dy + y^2dx$ and $d(x^2 + x) = dx$, giving

$$dU = x^2\,\frac{(x^2 + x)\,dy + (y^2 + y)\,dx}{(x^2 + x)^2} = x^2\,dV.$$

If $L = K$, then $\Omega_{K/k}$ would be generated by dU, dV, hence by dV, a space of dimension ≤ 1; but $\Omega_{K/k} = K\,dx \oplus K\,dy$ has dimension 2 (II, 8.6A). So $[K : L] = 2$: φ is dominant of degree 2 and the extension of function fields is purely inseparable, i.e. φ is inseparable of degree 2.

(b) *Singular points.* Let $G = aF_1 + bF_2 + cF_3$, $(a, b, c) \neq 0$, $C = V(G)$. Since $\operatorname{char} k = 2$,

$$G_x = ay^2 + cz^2, \qquad G_y = ax^2 + bz^2, \qquad G_z = by^2 + cx^2,$$

and Euler's relation gives $xG_x + yG_y + zG_z = 3G = G$. So G lies in the ideal of its partials and, by the Jacobian criterion, the singular points of C are the common zeros of G_x, G_y, G_z. These

are linear equations in (x^2, y^2, z^2) with matrix $\begin{pmatrix} 0 & a & c \\ a & 0 & b \\ c & b & 0 \end{pmatrix}$; its 2×2 minors include a^2, b^2, c^2 (up to sign), so the rank is ≥ 2, while $(x^2, y^2, z^2) = (b, c, a)$ is a solution. So the solution space is 1-dimensional and

$$C \text{ has exactly one singular point,} \qquad S(C) = (\sqrt{b} : \sqrt{c} : \sqrt{a}).$$

In particular G is squarefree (a multiple component would give infinitely many singular points), so C is a reduced cubic, singular. As k is perfect, $(a : b : c) \mapsto (\sqrt{b} : \sqrt{c} : \sqrt{a})$ is a bijection $\mathfrak{d} \cong \mathbf{P}^2 \to \mathbf{P}^2$. This proves the last assertion: $C \mapsto S(C)$ is a bijection between $\mathfrak{d}$ and $\mathbf{P}^2$.

The shape of C. Write $S = (s_1 : s_2 : s_3) = S(C)$, so $(a, b, c) = (s_3^2, s_1^2, s_2^2)$. Note: $g \in GL_3(\mathbb{F}_2)$ permutes the P_i, so $G \mapsto G \circ g$ preserves $\mathfrak{d}$ and carries the singular point S to $g^{-1}S$; the group acts transitively on the seven points and on the seven $\mathbb{F}_2$-lines of $\mathbf{P}^2$.

(i) S is one of the P_i. By transitivity it suffices to take $S = (0 : 0 : 1)$, i.e. $(a, b, c) = (1, 0, 0)$, $G = F_1 = xy(x + y)$: three lines through $(0 : 0 : 1)$. So C is three lines through a single P_i (this is the first alternative of the printed statement).

(ii) $S \notin \mathbf{P}^2(\mathbb{F}_2)$, but S lies on an $\mathbb{F}_2$-rational line. By transitivity take the line $z = 0$: $s_3 = 0$, $s_1 s_2 (s_1 + s_2) \neq 0$, $(a, b, c) = (0, s_1^2, s_2^2) =: (0, \beta, \alpha)$ with $\alpha\beta \neq 0$, $\alpha \neq \beta$. Then $G = \beta F_2 + \alpha F_3 = z\,Q$ with $Q = \alpha x^2 + \beta y^2 + z(\alpha x + \beta y)$. So $C = \{z = 0\} \cup V(Q)$ is reducible. The partials $Q_x = \alpha z$, $Q_y = \beta z$, $Q_z = \alpha x + \beta y$ vanish only at $(\beta : \alpha : 0)$, where $Q = \alpha\beta(\alpha + \beta) \neq 0$; so Q is a smooth, irreducible conic. On $z = 0$, $Q = (\sqrt{\alpha}x + \sqrt{\beta}y)^2$, so the line is tangent to Q at S. Thus these curves are *a line plus a smooth conic tangent to it*.

(iii) S lies on no $\mathbb{F}_2$-rational line. Then C is irreducible. Otherwise $G = \ell \cdot Q'$ with ℓ a line. Then $\emptyset \neq \ell \cap V(Q') \subseteq \operatorname{Sing} C = \{S\}$, so $S \in \ell$ and ℓ is not $\mathbb{F}_2$-rational. A non-rational line contains at most one point of $\mathbf{P}^2(\mathbb{F}_2)$ (two such points span a rational line), so the conic Q' passes through at least six of the seven P_i. Omitting at most one point of the Fano plane leaves a full $\mathbb{F}_2$-line L_0 of three points on $V(Q')$, whence $L_0 \mid Q'$ and L_0 is a component of C. Applying the same argument to this component gives $S \in L_0$, a contradiction.

Now we show C is a cusp at S. Expanding G at S monomial by monomial (e.g. $(s_1 + v_1)^2(s_2 + v_2) = s_1^2 s_2 + s_1^2 v_2 + s_2 v_1^2 + v_1^2 v_2$ in characteristic 2), one finds

$$G(S + v) = G(S) + \sum G_i(S) v_i + Q_S(v) + G(v),$$

$$Q_S(v) = s_2 s_3 (s_2 + s_3) v_1^2 + s_3 s_1 (s_3 + s_1) v_2^2 + s_1 s_2 (s_1 + s_2) v_3^2.$$

Here $G(S) = 0$ and $G_i(S) = 0$. Q_S has no cross terms, so it is the square of a linear form m. The coefficients vanish simultaneously iff all ratios $s_i/s_j \in \{0, 1, \infty\}$, i.e. iff S is rational; so $m \neq 0$ for S non-rational. Also $Q_S(S) = s_1 s_2 s_3 \cdot 2(\cdots) = 0$ so $m(S) = 0$. Choose a complement W of kS in k^3. Then $m|_W \neq 0$ (otherwise $W \subseteq \ker m$ and $S \in \ker m$ would give $m = 0$), so we may take coordinates (u, v) on W with $u = m|_W$. The points $[S + w]$, $w \in W$, form an affine plane $\mathbf{A}^2 \subseteq \mathbf{P}^2$ with coordinates (u, v) centred at S, and by the expansion above $C \cap \mathbf{A}^2$ has the exact equation

$$u^2 + g(u, v) = 0, \qquad g = G|_W = \alpha u^3 + \beta u^2 v + \gamma u v^2 + \delta v^3.$$

If $\delta = 0$ this would factor as $u\,(u + \alpha u^2 + \beta uv + \gamma v^2)$, and C (the closure of this dense open piece) would be reducible, contrary to (iii). So $\delta \neq 0$. Thus S is a double point with tangent cone the double line $u^2 = 0$, and the tangent line $u = 0$ meets C at S with multiplicity 3. To see that this is a cusp, project from S: the line $u = sv$ meets C where $s^2v^2 + v^3 g(s,1) = 0$, i.e. at S (doubly) and at $v = s^2/g(s,1)$, where $g(s,1) = \alpha s^3 + \beta s^2 + \gamma s + \delta$ and $g(0,1) = \delta \neq 0$. So near $s = 0$ the curve is parametrized by $(u,v) = (sv, v)$ with $v = s^2 \cdot (\text{unit})$, i.e. v has order 2 and u order 3 in the local parameter s. Hence C is unibranch at S and $\widehat{\mathcal{O}}_{C,S} = k[[s^2, s^3]]$ (the normalization is $\mathbf{P}^1$, via s): S is a cusp. So C is an irreducible cuspidal cubic with cusp $S \neq P_i$ (this is the second alternative of the printed statement).

Remark (a correction). Case (ii) really occurs: for example take $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$ and $G = z(x^2 + \omega y^2 + zx + \omega zy)$, which vanishes at all seven $\mathbb{F}_2$-points (checked directly) and lies in $\mathfrak{d}$ ($a = 0$). So the dichotomy printed in the book omits the case of a line plus a tangent smooth conic. Every other assertion in (b) is unaffected: every member is singular, has a unique singular point, and $C \mapsto S(C)$ is a bijection $\mathfrak{d} \to \mathbf{P}^2$; the cuspidal members correspond exactly to points of $\mathbf{P}^2$ not on the seven $\mathbb{F}_2$-lines. □

Exercise 10.8

> **Problem.** *A linear system with moving singularities contained in the base locus (any characteristic).* In affine 3-space with coordinates x, y, z, let C be the conic $(x-1)^2 + y^2 = 1$ in the xy-plane, and let P be the point $(0,0,t)$ on the z-axis. Let Y_t be the closure in $\mathbf{P}^3$ of the cone over C with vertex P. Show that as t varies, the surfaces $\{Y_t\}$ form a linear system of dimension 1, with a moving singularity at P. The base locus of this linear system is the conic C plus the z-axis.

Solution. *Hypothesis.* I assume $\operatorname{char} k \neq 2$, with k algebraically closed. This is necessary: in characteristic 2 the equation $(x-1)^2 + y^2 = 1$ reads $(x+y)^2 = 0$, a double line, and the computation below degenerates (the second generator $B = 2xz$ vanishes, all the Y_t coincide). The heading "any characteristic" is to be understood as "any characteristic $\neq 2$" (including 0).

Use homogeneous coordinates $(x : y : z : w)$ on $\mathbf{P}^3$ (affine part $w = 1$) and set

$$A = x^2 + y^2 - 2xw, \qquad B = 2xz, \qquad F_t = tA + B = t(x^2 + y^2) - 2txw + 2xz.$$

$Y_t = V(F_t)$ for $t \neq 0$. The points of the line through $P = (0,0,t)$ and $(q_1, q_2, 0) \in C$ are $(sq_1, sq_2, t(1-s))$, and

$$F_t(sq_1, sq_2, t(1-s), 1) = ts^2(q_1^2 + q_2^2) - 2tsq_1 + 2sq_1 t(1-s) = ts^2(q_1^2 + q_2^2 - 2q_1) = 0,$$

since $q \in C$. So the cone, hence Y_t, lies in $V(F_t)$. With $z' = z - tw$ we get $F_t = t(x^2+y^2) + 2xz'$, a quadratic form in x, y, z' only, with Gram matrix $\begin{pmatrix} t & 0 & 1 \\ 0 & t & 0 \\ 1 & 0 & 0 \end{pmatrix}$ of determinant $-t \neq 0$. So $V(F_t)$ is an irreducible quadric cone of rank 3, with unique singular point $x = y = z' = 0$, i.e. $P_t = (0:0:t:1) = P$, the vertex. Y_t is an irreducible closed surface (closure of the image of $C \times \mathbf{A}^1$) contained in the irreducible surface $V(F_t)$, so $Y_t = V(F_t)$.

Linear system. Thus $\{Y_t\}_{t \neq 0}$ are members of the pencil $\Lambda = \{\mu A + \nu B\}$ spanned by the linearly independent quadrics A and B (char $\neq 2$): a linear system of dimension 1. (The remaining

members are $t = \infty$, the cylinder $A = 0$ over C with vertex $(0 : 0 : 1 : 0)$, and $t = 0$, the reducible quadric $xz = 0$ containing the plane $z = 0$ of C.) Any member has the form $\mu A + \nu B$, $(\mu : \nu) \in \mathbf{P}^1$.

Moving singularity. By the above, for each $t \neq 0$ the surface Y_t is singular exactly at $P_t = (0 : 0 : t : 1)$, and P_t moves along the z-axis as t varies.

Base locus. A point lies on all Y_t ($t \neq 0$) iff $t_1 A + B = t_2 A + B = 0$ for two distinct values, i.e. iff $A = B = 0$. Now $B = 2xz = 0$ means $x = 0$ or $z = 0$. If $z = 0$ then $A = 0$ is the conic C (closure of C in the plane $z = 0$ of $\mathbf{P}^3$). If $x = 0$ then $A = y^2 = 0$, so $x = y = 0$: the z-axis (the line $x = y = 0$ in $\mathbf{P}^3$, including its point at infinity). Hence the base locus is $C \cup \{z\text{-axis}\}$, as claimed; note that every P_t lies in it. (Scheme-theoretically the z-axis occurs with multiplicity 2, because $x = 0$ forces $y^2 = 0$.) This is consistent with Bertini (Cor. 10.9, characteristic 0): the singularities of the general member can only occur in the base locus. □

Exercise 10.9

> **Problem.** Let $f : X \to Y$ be a morphism of varieties over k. Assume that Y is regular, X is Cohen–Macaulay, and that every fibre of f has dimension equal to $\dim X - \dim Y$. Then f is flat. [*Hint*: Imitate the proof of (10.4), using (II, 8.21A).]

Solution. Put $n = \dim X - \dim Y$. Flatness means $\mathcal{O}_x$ is flat over $\mathcal{O}_{f(x)}$ for all $x \in X$. As in the proof of (10.4), it suffices to treat closed points $x \in X$: a general point x' specializes to a closed point x with $\mathcal{O}_{x'}$ a localization of $\mathcal{O}_x$ and $\mathcal{O}_{f(x')}$ a localization of $\mathcal{O}_{f(x)}$, and flatness localizes.

Let $x \in X$ be closed and $y = f(x)$, also closed ($k(x)$ is finite over k, so $k(y) \subseteq k(x)$ is finite over k). Let $A = \mathcal{O}_{Y,y}$ and $B = \mathcal{O}_{X,x}$ with $r = \dim A = \dim Y$ and $\dim B = \dim X = r + n$ (closed points of varieties, I, 1.8A / II, Ex. 3.20). A is a regular local ring, and B is Cohen–Macaulay by hypothesis. Let $t_1, \ldots, t_r$ be a regular system of parameters of A, so $\mathfrak{m}_y = (t_1, \ldots, t_r)$ and $A/(t_1, \ldots, t_r) = k(y) = k$; they form an A-regular sequence by (II, 8.21A(a),(c)), since $\dim A/(t_1, \ldots, t_r) = 0 = \dim A - r$. Denote by the same letters their images in $\mathfrak{m}_x \subseteq B$.

The t_i form a B-regular sequence. $B/(t_1, \ldots, t_r)B = B/\mathfrak{m}_y B = \mathcal{O}_{X_y,x}$, the local ring of the fibre X_y at x, whose dimension is $\leq \dim X_y = n$ by hypothesis. By Krull's height theorem $\dim B/(t_1, \ldots, t_r)B \geq \dim B - r = n$. So $\dim B/(t_1, \ldots, t_r)B = \dim B - r$, and since B is Cohen–Macaulay, (II, 8.21A(c)) says that $t_1, \ldots, t_r$ is a B-regular sequence.

Flatness by descending induction. For $1 \leq i \leq r + 1$ let $A_i = A/(t_1, \ldots, t_{i-1})$ and $B_i = B/(t_1, \ldots, t_{i-1})B$; so $A_1 = A$, $B_1 = B$, and $A_i \to B_i$ is a local homomorphism of noetherian local rings. We show by descending induction that B_i is flat over A_i. For $i = r+1$: $A_{r+1} = k(y)$ is a field, so every module over it is flat. Suppose $B_{i+1} = B_i/t_i B_i$ is flat over $A_{i+1} = A_i/t_i A_i$. The element $t_i \in A_i$ is a nonunit and a non-zerodivisor ($t_1, \ldots, t_r$ is A-regular), and it is a non-zerodivisor on B_i ($t_1, \ldots, t_r$ is B-regular). Apply Lemma 10.3.A to $A_i \to B_i$, $M = B_i$ (a finitely generated B_i-module), $t = t_i$: conditions (1) and (2) hold, so B_i is flat over A_i. At $i = 1$ we get that $B = \mathcal{O}_x$ is flat over $A = \mathcal{O}_y$.

Hence f is flat at all closed points, hence everywhere. □

11. The theorem on formal functions

Throughout this section, "Thm. 11.1", "Cor. 11.2–11.5" refer to the text, "12.8–12.11" to the next section (Semicontinuity, Grauert, cohomology and base change), and "II, Ex. ..." etc. to exercises of other chapters. A closed point of a scheme of finite type over an algebraically closed field k has residue field k. The following standard fact is used repeatedly (and its proof is repeated each time it is needed in a slightly different form): a scheme of finite type over a field is a Jacobson scheme, so a nonempty closed subset contains a closed point, and the closed points are dense in every closed subset.

Exercise 11.1

> **Problem.** Show that the result of (11.2) is false without the projective hypothesis. For example, let $X = \mathbf{A}^n_k$, let $P = (0, \dots, 0)$, let $U = X - P$, and let $f\colon U \to X$ be the inclusion. Then the fibres of f all have dimension 0, but $R^{n-1}f_*\mathcal{O}_U \neq 0$.

Solution. We assume $n \geq 2$ (needed: then $n-1 > 0 = r$; for $n = 1$ the statement $R^0 f_*\mathcal{O}_U \neq 0$ is no contradiction to (11.2), which concerns R^i with $i > 0$).

Fibres. f is an open immersion, hence injective on points, and the fibre over $x \in U$ is $\operatorname{Spec} k(x)$ (an open immersion is a monomorphism, so $U \times_X \operatorname{Spec} k(x) = \operatorname{Spec} k(x)$); the fibre over P is empty. So every fibre has dimension 0 (or is empty), i.e. $r = 0$, and (11.2) would give $R^i f_*\mathcal{O}_U = 0$ for all $i > 0$.

The higher direct image. Write $X = \operatorname{Spec} S$, $S = k[x_1, \dots, x_n]$. Then $U = \bigcup_{i=1}^n U_i$ with $U_i = D(x_i)$. Since X is affine, $R^i f_*\mathcal{O}_U = \widetilde{H^i(U, \mathcal{O}_U)}$ (Prop. 8.5). The scheme U is separated and the open sets $U_{i_0 \cdots i_p} = D(x_{i_0} \cdots x_{i_p})$ are affine, so $H^i(U, \mathcal{O}_U)$ is computed by the Cech complex of the covering $\mathfrak{U} = \{U_1, \dots, U_n\}$ (Thm. 4.5). This complex has $C^p = \bigoplus_{i_0 < \cdots < i_p} S_{x_{i_0} \cdots x_{i_p}}$ for $0 \leq p \leq n-1$. In the top degree $p = n-1$ there is the single term $S_{x_1 \cdots x_n}$, and $d\colon C^{n-2} \to C^{n-1}$ maps the summand $S_{x_1 \cdots \widehat{x_i} \cdots x_n}$ into it by $\pm$ the inclusion. Hence

$$H^{n-1}(U, \mathcal{O}_U) = S_{x_1 \cdots x_n} \Big/ \sum_{i=1}^n S_{x_1 \cdots \widehat{x_i} \cdots x_n}.$$

The ring $S_{x_1 \cdots x_n}$ has the k-basis of Laurent monomials $x^a = x_1^{a_1} \cdots x_n^{a_n}$, $a \in \mathbf{Z}^n$, and the subring $S_{x_1 \cdots \widehat{x_i} \cdots x_n}$ is spanned by those with $a_i \geq 0$. So the quotient has the basis of monomials with all $a_i < 0$; it is nonzero (it contains the class of $(x_1 \cdots x_n)^{-1}$). Therefore $R^{n-1}f_*\mathcal{O}_U = \widetilde{H^{n-1}(U, \mathcal{O}_U)} \neq 0$ for $n \geq 2$, although all fibres have dimension 0. Of course f is not projective (not even proper) here. □

Exercise 11.2

> **Problem.** Show that a projective morphism with finite fibres (= quasi-finite (II, Ex. 3.5)) is a finite morphism.

Solution. As throughout Section 11 we assume the base Y is noetherian. Let $f\colon X \to Y$ be projective with finite fibres. By Cor. 11.5 (Stein factorization) $f = g \circ f'$ with $f'\colon X \to Y' = \operatorname{Spec} f_*\mathcal{O}_X$ projective, $f'_*\mathcal{O}_X = \mathcal{O}_{Y'}$, with connected fibres, and $g\colon Y' \to Y$ finite. It suffices to show that f' is an isomorphism.

f' is surjective. f' is projective, hence proper, hence closed. If $y' \in Y'$ were not in the closed set $f'(X)$, there would be an open neighborhood $V \ni y'$ with $f'^{-1}(V) = \emptyset$, so $0 = \mathcal{O}_X(f'^{-1}V) = (f'_*\mathcal{O}_X)(V) = \mathcal{O}_{Y'}(V) \neq 0$, a contradiction.

f' is injective. Let $y' \in Y'$ and $y = g(y')$. Then $f'^{-1}(y') \subseteq f^{-1}(y)$ and the latter is a finite set. Its topology is that of the underlying space of the fibre X_y (II, Ex. 3.10), a scheme of finite type over $k(y)$ with finitely many points; such a space is discrete: if some point x_1 were not closed in X_y, then $\overline{\{x_1\}}$ would be an integral scheme of finite type over $k(y)$ of dimension ≥ 1, and it has infinitely many points (by Noether normalization it maps finitely and surjectively onto some $\mathbf{A}^d_{k(y)}$, $d \geq 1$, which has infinitely many points). So all points of X_y are closed, and a finite T_1 space is discrete. So a connected subset of $f^{-1}(y)$ is a single point; by Cor. 11.5, $f'^{-1}(y')$ is connected and nonempty, so it is a point.

Hence f' is a continuous, closed, bijective map, i.e. a homeomorphism, and $f'^{\#}\colon \mathcal{O}_{Y'} \to f'_*\mathcal{O}_X$ is an isomorphism of sheaves. A homeomorphism of ringed spaces with an isomorphism $f'^{\#}$ is an isomorphism of locally ringed spaces, hence of schemes. So $f = g \circ f'$ is finite. □

Exercise 11.3

> **Problem.** Let X be a normal, projective variety over an algebraically closed field k. Let $\mathfrak{d}$ be a linear system (of effective Cartier divisors) without base points, and assume that $\mathfrak{d}$ is *not composite with a pencil*, which means that if $f\colon X \to \mathbf{P}^n_k$ is the morphism determined by $\mathfrak{d}$, then $\dim f(X) \geq 2$. Then show that every divisor in $\mathfrak{d}$ is connected. This improves Bertini's theorem (10.9.1). [*Hints*: Use (11.5), (Ex. 5.7) and (7.9).]

Solution. Let $f\colon X \to \mathbf{P}^n = \mathbf{P}^n_k$ be the morphism determined by $\mathfrak{d}$. It is projective (X is projective over k and $\mathbf{P}^n$ is separated over k; II, Ex. 4.9). Every $D \in \mathfrak{d}$ is $f^*H = \operatorname{div}(f^*h)$ for a hyperplane $H = \{h = 0\}$, so $|D| = f^{-1}(H)$, and $f^*h \neq 0$ since D is an effective Cartier divisor.

Step 1: Stein factorization. By Cor. 11.5, $f = g \circ f'$ with $f'\colon X \to Y' = \operatorname{Spec} f_*\mathcal{O}_X$ projective, $f'_*\mathcal{O}_X = \mathcal{O}_{Y'}$, with connected fibres (in particular f' is surjective, as in Ex. 11.2) and $g\colon Y' \to \mathbf{P}^n$ finite. Let $Z = f(X) = g(Y')$, a closed integral subscheme (reduced structure) of dimension ≥ 2.

Step 2: Y' is an integral normal projective variety of dimension ≥ 2. Let $V \subseteq \mathbf{P}^n$ be an affine open. Then $g^{-1}V = \operatorname{Spec} B_V$ with $B_V = \mathcal{O}_X(f^{-1}V)$. As X is integral, B_V is a domain, and $B_V = \bigcap_{x \in f^{-1}V} \mathcal{O}_{X,x} \subseteq K(X)$. Since X is normal, every $\mathcal{O}_{X,x}$ is integrally closed in $K(X)$, so B_V is integrally closed in $K(X)$, hence in its own fraction field. Thus Y' is integral and normal, and it is of finite type over k (finite over $\mathbf{P}^n$), separated, and proper. Since Y' is reduced, g factors through Z, and $g\colon Y' \to Z$ is finite and surjective; hence $\dim Y' = \dim Z \geq 2$. The sheaf $\mathcal{O}_Z(1)$ is ample on Z (Ex. 5.7(a)), so $g^*\mathcal{O}_Z(1) = g^*\mathcal{O}_{\mathbf{P}^n}(1)$ is ample on Y' (Ex. 5.7(d), Y' and Z being proper over k), and therefore (II, 7.6) some power of it is very ample and Y' is projective over k.

Step 3: $g^{-1}(H)$ is connected. Let $D \in \mathfrak{d}$ correspond to h as above. The section $\sigma = g^*h$ of the ample invertible sheaf $\mathcal{M} = g^*\mathcal{O}(1)$ on the integral scheme Y' is nonzero (otherwise $f^*h = f'^*\sigma = 0$). Its zero scheme is therefore an effective Cartier divisor E with $\mathcal{O}_{Y'}(E) \cong \mathcal{M}$ ample and $|E| = g^{-1}(H)$. It is nonempty (if $E = \emptyset$ then $\mathcal{M} \cong \mathcal{O}_{Y'}$ would be ample on a projective variety of positive dimension, impossible), hence a closed subset of codimension 1. By Cor. 7.9, applied to the integral, normal, projective variety Y' of dimension ≥ 2, the set $S := |E| = g^{-1}(H)$ is connected.

Step 4: $f'^{-1}(S) = |D|$ is connected. Indeed $f'^{-1}(g^{-1}(H)) = f^{-1}(H) = |D|$. Suppose $f'^{-1}(S) = A \sqcup B$ with A, B disjoint closed. Every fibre $f'^{-1}(s)$, $s \in S$, is connected, so lies in A or in B. Hence $S = f'(A) \sqcup f'(B)$, a disjoint union (a point cannot have its fibre in both), and $f'(A)$, $f'(B)$ are closed because f' is proper. As S is connected, one of them is empty, say $A = \emptyset$. So $|D|$ is connected. □

Exercise 11.4

> **Problem.** *Principle of Connectedness.* Let $\{X_t\}$ be a flat family of closed subschemes of $\mathbf{P}^n_k$ parametrized by an irreducible curve T of finite type over k. Suppose there is a nonempty open set $U \subseteq T$, such that for all closed points $t \in U$, X_t is connected. Then prove that X_t is connected for all $t \in T$.

Solution. *Reading of the statement.* k is algebraically closed (as in the surrounding discussion); the family is a closed subscheme $X \subseteq \mathbf{P}^n_T$ flat over T, and "for all $t \in T$" is proved for the closed points t (at the end we remark on the generic point). A connected space is nonempty.

Step 0: reduction to T a nonsingular affine curve. The base change $X \times_T T_{\mathrm{red}} \to T_{\mathrm{red}}$ is still a flat family with the same fibres at closed points, and so is its pullback to the normalization $\nu\colon \widetilde{T} \to T_{\mathrm{red}}$ (ν is finite and surjective, $\widetilde{T}$ is an integral normal curve of finite type over k). For a closed point $\tilde{t}$ of $\widetilde{T}$ lying over t, the residue fields are both k and the fibre is unchanged: $\widetilde{X}_{\tilde{t}} = X_t$. The set $\nu^{-1}(U)$ is nonempty and open. Given a closed point $t_0 \in T$ choose a closed point $\tilde{t}_0 \in \nu^{-1}(t_0)$ and an affine open neighborhood $\operatorname{Spec} R$ of $\tilde{t}_0$ in $\widetilde{T}$. So we may assume: $T = \operatorname{Spec} R$, R a Dedekind domain which is a finitely generated k-algebra, $K = \operatorname{Frac} R$, $U \neq \emptyset$ open (so $T \setminus U$ is a finite set of closed points), and we must show that X_{t_0} is connected for a given closed point t_0 ($\mathfrak{m}_t$ denotes the maximal ideal of a closed point t).

Step 1: connected components of closed fibres. Let $p\colon X \to T$ be the projection (projective), and let $B = \Gamma(X, \mathcal{O}_X) = \Gamma(T, p_*\mathcal{O}_X)$, a finite R-algebra. Cor. 11.5 gives $p = g \circ p'$ with $p'\colon X \to T' = \operatorname{Spec} B$ projective, surjective, with connected fibres, and $g\colon T' \to T$ finite. For a closed point t, $g^{-1}(t)$ is a finite set of closed points and $X_t = \bigsqcup_{t' \in g^{-1}(t)} p'^{-1}(t')$ is a finite disjoint union of nonempty closed connected subsets, which are therefore its connected components. Hence

$(*)$ $\quad X_t$ is connected $\iff$ $g^{-1}(t)$ is a single point $\iff$ $\operatorname{Spec} B$ has exactly one point over t.

Step 2: algebra. X is flat over the Dedekind domain R, so $\mathcal{O}_X$ is torsion-free over R and so is B; thus $B \subset B \otimes_R K$. Let N be the nilradical of B and $C = B/N$. C is R-torsion-free: if $a \in R \setminus 0$, $c \in B$ and $ac \in N$, then $a^m c^m = 0$ for some m, so $c^m = 0$ and $c \in N$. Hence C is a finite torsion-free reduced R-algebra, $\operatorname{Spec} C = \operatorname{Spec} B$ as spaces, and $L := C \otimes_R K \supseteq C$ is a

finite reduced K-algebra, i.e. $L = \prod_{i=1}^{s} L_i$ with L_i finite field extensions of K. Let $\widetilde{C} = \prod \widetilde{C}_i$ be the integral closure of R in L ($\widetilde{C}_i$ = integral closure of R in L_i). It is a finite R-module (I, 3.9A), $C \subseteq \widetilde{C}$, $\operatorname{Spec} \widetilde{C} \to \operatorname{Spec} C$ is surjective (lying over), and $\widetilde{C}/C$ is a finite torsion R-module, annihilated by some $a \neq 0$. So the set S of closed points where a vanishes is finite, and for closed $t \notin S$, C and $\widetilde{C}$ coincide after localizing at $\mathfrak{m}_t$, so they have the same number of primes over $\mathfrak{m}_t$.

By hypothesis and $(*)$, $\operatorname{Spec} C$ has exactly one point over each closed $t \in U$. Choose a closed $t_1 \in U \setminus S$ (possible: $U \setminus S$ is the complement of a finite set in the infinite set T, and nonempty open sets are cofinite). We derive two facts.

(i) $s = 1$. Each $\widetilde{C}_i$ is a domain, finite over R and containing R, so $\operatorname{Spec} \widetilde{C}_i \to T$ is surjective, giving at least one point over t_1 in each of the s components. As $\operatorname{Spec} \widetilde{C}$ has exactly one point over t_1, $s = 1$ and L is a field.

(ii) L/K is purely inseparable. Let L_s be the separable closure of K in L, $n_s = [L_s : K]$. Write $L_s = K(\alpha)$ with α integral over R and let $h \in R[x]$ be its minimal polynomial (monic, in $R[x]$ since R is normal), $\delta = \operatorname{disc}(h) \neq 0$. Replace t_1 by a closed point of $U \setminus S$ with $\delta(t_1) \neq 0$ (we excluded finitely many more points). The fibre of $R[\alpha] = R[x]/(h)$ over t_1 is $k[x]/(\bar{h})$, where $\bar{h}$ has n_s distinct roots in k (since $\operatorname{disc} \bar{h} = \delta(t_1) \neq 0$ and k is algebraically closed), i.e. $R[\alpha]$ has n_s points over t_1. Now $R[\alpha] \subseteq \widetilde{C}$ is an integral extension, so $\operatorname{Spec} \widetilde{C} \to \operatorname{Spec} R[\alpha]$ is surjective and $\widetilde{C}$ has $\geq n_s$ points over t_1. Hence $n_s = 1$.

So L/K is purely inseparable: there is q (a power of the characteristic, or $q = 1$ in characteristic 0) with $x^q \in K$ for all $x \in L$.

Step 3: the special point. Let P_1, P_2 be primes of $\widetilde{C}$ over $\mathfrak{m}_{t_0}$. For $x \in \widetilde{C}$, $x^q \in K$ is integral over R, so $x^q \in R$ (R normal). Then $x \in P_j \iff x^q \in P_j \cap R = \mathfrak{m}_{t_0}$. Thus $P_1 = P_2$: $\operatorname{Spec} \widetilde{C}$ has exactly one point over t_0, and as $\operatorname{Spec} \widetilde{C} \to \operatorname{Spec} C$ is surjective, $\operatorname{Spec} C = \operatorname{Spec} B$ has at most one (and at least one, as $R \subseteq C$ is integral) point over t_0. By $(*)$, X_{t_0} is connected.

Remark (generic point). The argument also shows L is a field, and $H^0(X_\eta, \mathcal{O}_{X_\eta}) = B \otimes K$ has nilradical quotient L, so it is a local ring; hence X_η has no nontrivial idempotent global function, i.e. it is connected. Thus all fibres, closed or not, are connected. □

Exercise 11.5

> **Problem.** Let Y be a hypersurface in $X = \mathbf{P}_k^N$ with $N \geq 4$. Let $\widehat{X}$ be the formal completion of X along Y (II, Section 9). Prove that the natural map $\operatorname{Pic} \widehat{X} \to \operatorname{Pic} Y$ is an isomorphism. [*Hint*: Use (II, Ex. 9.6), and then study the maps $\operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n$ for each n using (Ex. 4.6) and (Ex. 5.5).] (Starred in the book.)

Solution. Let $Y = V(F)$, F a form of degree $d \geq 1$, $\mathcal{I} = F \cdot \mathcal{O}_X(-d) \cong \mathcal{O}_X(-d)$ its ideal sheaf, and $X_n = (Y, \mathcal{O}_X/\mathcal{I}^n)$ the n-th infinitesimal neighborhood, $X_1 = Y$. Each X_n is projective over k, so the Mittag-Leffler condition of (II, Ex. 9.6(d)) holds and $\operatorname{Pic} \widehat{X} = \varprojlim \operatorname{Pic} X_n$. The natural map to $\operatorname{Pic} Y$ is the projection to the term $n = 1$. So it suffices to show that $\operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n$ is an isomorphism for every $n \geq 1$.

Let $\mathcal{J}_n = \mathcal{I}^n/\mathcal{I}^{n+1} \subseteq \mathcal{O}_{X_{n+1}}$; as $2n \geq n+1$, $\mathcal{J}_n^2 = 0$, and $\mathcal{O}_{X_{n+1}}/\mathcal{J}_n = \mathcal{O}_{X_n}$. By (Ex. 4.6) there is an exact sequence

$$H^1(X_{n+1}, \mathcal{J}_n) \to \operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n \to H^2(X_{n+1}, \mathcal{J}_n),$$

where the cohomology of $\mathcal{J}_n$ is that of an $\mathcal{O}_Y$-module on Y. Since $\mathcal{I}^n = \mathcal{O}_X(-nd)$, we have $\mathcal{J}_n = \mathcal{O}_X(-nd)/\mathcal{O}_X(-(n+1)d) \cong \mathcal{O}_Y(-nd)$, so it suffices to prove $H^i(Y, \mathcal{O}_Y(m)) = 0$ for $i = 1, 2$ and all $m \in \mathbf{Z}$. Use $0 \to \mathcal{O}_X(m-d) \xrightarrow{F} \mathcal{O}_X(m) \to \mathcal{O}_Y(m) \to 0$ and Thm. 5.1(b): $H^i(X, \mathcal{O}_X(j)) = 0$ for $0 < i < N$, all j. The long exact sequence gives $H^i(Y, \mathcal{O}_Y(m)) = 0$ whenever $i \geq 1$ and $i + 1 < N$, i.e. $1 \leq i \leq N - 2$. Since $N \geq 4$ this includes $i = 1, 2$. (This is Ex. 5.5(c), Y being a complete intersection of dimension $N - 1 \geq 3$.) Therefore $\operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n$ is an isomorphism for all $n \geq 1$, and $\operatorname{Pic} \widehat{X} \to \operatorname{Pic} X_1 = \operatorname{Pic} Y$ is an isomorphism. □

Exercise 11.6

Problem. Again let Y be a hypersurface in $X = \mathbf{P}_k^N$, this time $N \geq 2$.

(a) If $\mathcal{F}$ is a locally free sheaf on X, show that the natural map $H^0(X, \mathcal{F}) \to H^0(\widehat{X}, \widehat{\mathcal{F}})$ is an isomorphism.

(b) Show that the following conditions are equivalent: (i) for each locally free sheaf $\mathfrak{J}$ on $\widehat{X}$, there exists a coherent sheaf $\mathcal{F}$ on X such that $\mathfrak{J} \cong \widehat{\mathcal{F}}$ (i.e., $\mathfrak{J}$ is *algebraizable*); (ii) for each locally free sheaf $\mathfrak{J}$ on $\widehat{X}$, there is an integer n_0 such that $\mathfrak{J}(n)$ is generated by global sections for all $n \geq n_0$. [*Hint*: For (ii)⇒(i), show that one can find sheaves $\mathcal{E}_0, \mathcal{E}_1$ on X, which are direct sums of sheaves of the form $\mathcal{O}(-q_i)$, and an exact sequence $\widehat{\mathcal{E}_1} \to \widehat{\mathcal{E}_0} \to \mathfrak{J} \to 0$ on $\widehat{X}$. Then apply (a) to the sheaf $\mathcal{H}om(\mathcal{E}_1, \mathcal{E}_0)$.]

(c) Show that the conditions (i) and (ii) of (b) imply that the natural map $\operatorname{Pic} X \to \operatorname{Pic} \widehat{X}$ is an isomorphism.

Note. In fact, (i) and (ii) always hold if $N \geq 3$. This fact, coupled with (Ex. 11.5) leads to Grothendieck's proof [SGA 2] of the Lefschetz theorem which says that if Y is a hypersurface in $\mathbf{P}^N$ with $N \geq 4$, then $\operatorname{Pic} Y \cong \mathbf{Z}$, and it is generated by $\mathcal{O}_Y(1)$. See Hartshorne [5, Ch. IV] for more details.

Solution. Notation: $Y = V(F)$, $\deg F = d \geq 1$, $\mathcal{I} = \mathcal{O}_X(-d) \subseteq \mathcal{O}_X$, $\mathcal{F}_n = \mathcal{F}/\mathcal{I}^n\mathcal{F}$ and $\widehat{\mathcal{F}} = \varprojlim \mathcal{F}_n$, so $\Gamma(\widehat{X}, \widehat{\mathcal{F}}) = \varprojlim H^0(Y, \mathcal{F}_n)$. For a sheaf $\mathfrak{J}$ on $\widehat{X}$, $\mathfrak{J}(n) = \mathfrak{J} \otimes \widehat{\mathcal{O}_X(n)}$, and $\widehat{\mathcal{F}}(n) = \widehat{\mathcal{F}(n)}$. Completion is an exact functor on coherent sheaves (II, Section 9).

(a) Let $\mathcal{F}$ be locally free. Then $\mathcal{I}^n\mathcal{F} = \mathcal{F}(-nd)$ and there is an exact sequence $0 \to \mathcal{F}(-nd) \to \mathcal{F} \to \mathcal{F}_n \to 0$, whence

$$0 \to H^0(X, \mathcal{F}(-nd)) \to H^0(X, \mathcal{F}) \to H^0(\mathcal{F}_n) \to H^1(X, \mathcal{F}(-nd)).$$

(1) $H^1(X, \mathcal{F}(-nd)) = 0$ for $n \gg 0$ by Cor. 7.8 ($X = \mathbf{P}^N$ is normal projective of dimension $N \geq 2$). (2) $H^0(X, \mathcal{F}(-nd)) = 0$ for $n \gg 0$: the sheaf $\mathcal{F}^\vee(q)$ is generated by global sections for some q (II, 5.17), so there is a surjection $\mathcal{O}_X^s \to \mathcal{F}^\vee(q)$; dualizing the surjection of locally free sheaves gives an injection $\mathcal{F}(-q) \hookrightarrow \mathcal{O}_X^s$, so $H^0(\mathcal{F}(-q-j)) \subseteq H^0(\mathcal{O}(-j))^s = 0$ for $j \geq 1$.

Hence there is n_0 such that for $n \geq n_0$ the map $H^0(X, \mathcal{F}) \to H^0(\mathcal{F}_n)$ is bijective. These maps are compatible with the transition maps $H^0(\mathcal{F}_{n+1}) \to H^0(\mathcal{F}_n)$, which are therefore bijective for $n \geq n_0$, and $\varprojlim H^0(\mathcal{F}_n) = H^0(\mathcal{F}_{n_0}) \cong H^0(X, \mathcal{F})$ compatibly with the natural map. So $H^0(X, \mathcal{F}) \to H^0(\widehat{X}, \widehat{\mathcal{F}})$ is an isomorphism.

(b) (i)⇒(ii). Let $\mathfrak{J} \cong \widehat{\mathcal{F}}$ with $\mathcal{F}$ coherent. By II, 5.17 there is n_0 with $\mathcal{F}(n)$ generated by global sections for $n \geq n_0$, i.e. there is a surjection $\mathcal{O}_X^s \to \mathcal{F}(n)$. Completion is exact, so $\mathcal{O}_{\widehat{X}}^s \to \widehat{\mathcal{F}(n)} = \mathfrak{J}(n)$ is surjective: $\mathfrak{J}(n)$ is generated by global sections for $n \geq n_0$.

(ii)⇒(i). Let $\mathfrak{J}$ be locally free on $\widehat{X}$. By (ii), $\mathfrak{J}(n_0)$ is generated by global sections, hence (II, Ex. 9.5) by finitely many of them: there is a surjection $\mathcal{O}_{\widehat{X}}^s \to \mathfrak{J}(n_0)$, i.e. $\widehat{\mathcal{E}}_0 \to \mathfrak{J} \to 0$ with $\mathcal{E}_0 = \mathcal{O}_X(-n_0)^s$. Its kernel $\mathfrak{J}'$ is coherent (II, Cor. 9.9) and locally free (the kernel of a surjection of locally free sheaves is locally a direct summand). Applying (ii) to $\mathfrak{J}'$ we find in the same way a surjection $\widehat{\mathcal{E}}_1 \to \mathfrak{J}'$ with $\mathcal{E}_1 = \bigoplus \mathcal{O}_X(-q_j)$ a finite direct sum. Composing, we get an exact sequence $\widehat{\mathcal{E}}_1 \xrightarrow{\varphi} \widehat{\mathcal{E}}_0 \to \mathfrak{J} \to 0$.

The sheaf $\mathcal{H} = \mathcal{H}om(\mathcal{E}_1, \mathcal{E}_0) = \bigoplus \mathcal{O}_X(q_j - n_0)$ is locally free, and $\widehat{\mathcal{H}} = \bigoplus \mathcal{O}_{\widehat{X}}(q_j - n_0) = \mathcal{H}om(\widehat{\mathcal{E}}_1, \widehat{\mathcal{E}}_0)$. So $\varphi \in H^0(\widehat{X}, \widehat{\mathcal{H}})$ and by (a) there is $u \in H^0(X, \mathcal{H}) = \operatorname{Hom}(\mathcal{E}_1, \mathcal{E}_0)$ with $\widehat{u} = \varphi$. Put $\mathcal{F} = \operatorname{coker}(u)$, a coherent sheaf on X. By exactness of completion, $\widehat{\mathcal{F}} = \operatorname{coker}(\widehat{u}) = \operatorname{coker} \varphi \cong \mathfrak{J}$. This proves (i).

(c) The natural map sends $\mathcal{L} \in \operatorname{Pic} X$ to $\widehat{\mathcal{L}}$, which is invertible on $\widehat{X}$ (II, Section 9). Recall $\operatorname{Pic} X = \mathbf{Z} \cdot [\mathcal{O}(1)]$ (II, 6.4).

Injectivity. If $\widehat{\mathcal{O}(m)} \cong \mathcal{O}_{\widehat{X}}$, then $H^0(\widehat{X}, \widehat{\mathcal{O}(m)})$ contains a nowhere vanishing section, so is nonzero, and by (a) $H^0(X, \mathcal{O}(m)) \neq 0$, i.e. $m \geq 0$. Dualizing, $\widehat{\mathcal{O}(-m)} \cong \mathcal{O}_{\widehat{X}}$ and the same argument gives $-m \geq 0$. So $m = 0$.

Surjectivity. Let $\mathfrak{J}$ be an invertible sheaf on $\widehat{X}$. By (i) there is a coherent $\mathcal{F}$ on X with $\widehat{\mathcal{F}} \cong \mathfrak{J}$. Locally on $X = \bigcup \operatorname{Spec} A$ (we may take X covered by such) $\mathcal{F} = \widetilde{M}$ and $\widehat{\mathcal{F}}$ corresponds to the $\mathfrak{a}$-adic completion $\widehat{M} = M \otimes_A \widehat{A}$ over $\widehat{A}$ ($\mathfrak{a}$ the ideal of Y); that $\widehat{\mathcal{F}}$ is invertible means $\widehat{M}$ is locally free of rank one. For a prime $P \supseteq \mathfrak{a}$ (a point of Y), $\mathfrak{a} \subseteq P$ gives $\widehat{A}/P\widehat{A} = A/P$, so $P\widehat{A}$ is prime, and $A_P \to \widehat{A}_{P\widehat{A}}$ is a flat local, hence faithfully flat, homomorphism. As $M_P \otimes_{A_P} \widehat{A}_{P\widehat{A}} = \widehat{M}_{P\widehat{A}}$ is free of rank one, M_P is free of rank one (faithfully flat descent for a finitely generated module). So $\mathcal{F}$ is invertible at every point of Y, hence on an open set $W \supseteq Y$ (the locally free locus of a coherent sheaf is open).

The closed set $X \setminus W$ is disjoint from Y. It has dimension 0: an irreducible closed subset $C \subseteq X \setminus W$ of positive dimension (with its reduced structure) would be a projective variety contained in the affine scheme $X \setminus Y = D_+(F)$, impossible, since a closed subscheme of an affine scheme which is proper over k is finite. So $X \setminus W$ is a finite set of closed points, of codimension $N \geq 2$. Since X and W are regular, (II, 6.5 and 6.16) give $\operatorname{Pic} X = \operatorname{Cl} X \xrightarrow{\sim} \operatorname{Cl} W = \operatorname{Pic} W$. Hence there is $\mathcal{M} \in \operatorname{Pic} X$ with $\mathcal{M}|_W \cong \mathcal{F}|_W$. Completion along $Y \subseteq W$ only involves W, so $\widehat{\mathcal{M}} \cong \widehat{\mathcal{F}} \cong \mathfrak{J}$. □

Remark used in Ex. 11.7. The surjectivity argument shows, for $N \geq 2$ and without assuming (i) or (ii): *if an invertible sheaf $\mathfrak{J}$ on $\widehat{X}$ is isomorphic to $\widehat{\mathcal{F}}$ for a coherent $\mathcal{F}$, then $\mathfrak{J} \cong \widehat{\mathcal{M}}$ for some $\mathcal{M} \in \operatorname{Pic} X$.*

Exercise 11.7

Problem. Now let Y be a curve in $X = \mathbf{P}_k^2$.

(a) Use the method of (Ex. 11.5) to show that $\operatorname{Pic}\widehat{X} \to \operatorname{Pic} Y$ is surjective, and its kernel is an infinite-dimensional vector space over k.

(b) Conclude that there is an invertible sheaf $\mathfrak{L}$ on $\widehat{X}$ which is not algebraizable.

(c) Conclude also that there is a locally free sheaf $\mathfrak{J}$ on $\widehat{X}$ so that no twist $\mathfrak{J}(n)$ is generated by global sections. Cf. (II, 9.9.1).

Solution. Notation as in Ex. 11.5: $Y = V(F)$, $\deg F = d \geq 1$, $\mathcal{I} = \mathcal{O}_X(-d)$, X_n, $\mathcal{J}_n = \mathcal{I}^n/\mathcal{I}^{n+1} \cong \mathcal{O}_Y(-nd)$ $(n \geq 1)$, and $\operatorname{Pic}\widehat{X} = \varprojlim \operatorname{Pic} X_n$ (II, Ex. 9.6(d)).

Cohomology of $\mathcal{O}_Y(m)$. From $0 \to \mathcal{O}_X(m-d) \to \mathcal{O}_X(m) \to \mathcal{O}_Y(m) \to 0$ and Thm. 5.1 ($H^1(\mathcal{O}_X(j)) = 0$ for all j; $H^0(\mathcal{O}_X(j)) = 0$ for $j < 0$; $H^2(\mathcal{O}_X(-j))$ is dual to $H^0(\mathcal{O}_X(j-3))$, so has dimension $\binom{j-1}{2}$ for $j \geq 1$, where $\binom{0}{2} = \binom{1}{2} = 0$):

- $H^2(Y, \mathcal{O}_Y(m)) = 0$ for all m (indeed $\dim Y = 1$, Grothendieck vanishing, 2.7).
- $H^0(Y, \mathcal{O}_Y) = k$, and $H^0(Y, \mathcal{O}_Y(-m)) = 0$ for $m \geq 1$ (as $H^0(\mathcal{O}_X(-m)) = 0 = H^1(\mathcal{O}_X(-m-d))$).
- For $m \geq 1$: $0 = H^1(\mathcal{O}_X(-m)) \to H^1(\mathcal{O}_Y(-m)) \to H^2(\mathcal{O}_X(-m-d)) \to H^2(\mathcal{O}_X(-m)) \to H^2(\mathcal{O}_Y(-m)) = 0$, so with $c(j) = \binom{j-1}{2} = \dim H^2(\mathcal{O}_X(-j))$ (this is 0 for $j = 1, 2$),

$$h^1(Y, \mathcal{O}_Y(-m)) = c(m+d) - c(m).$$

For $m = nd$: $h^1(Y, \mathcal{O}_Y(-nd)) = \frac{1}{2}\big[((n+1)d-1)((n+1)d-2) - (nd-1)(nd-2)\big] = \frac{1}{2}d\big((2n+1)d-3\big)$, which tends to ∞ with n.

The groups $\operatorname{Pic} X_n$. Put $V_n = H^1(Y, \mathcal{J}_n) = H^1(Y, \mathcal{O}_Y(-nd))$, a k-vector space with $\dim V_n \to \infty$. We claim $H^0(X_n, \mathcal{O}_{X_n}) = k$ for all $n \geq 1$: from $0 \to \mathcal{J}_n \to \mathcal{O}_{X_{n+1}} \to \mathcal{O}_{X_n} \to 0$ and $H^0(\mathcal{J}_n) = 0$, $H^0(\mathcal{O}_{X_{n+1}}) \hookrightarrow H^0(\mathcal{O}_{X_n})$, and induction from $H^0(\mathcal{O}_Y) = k$ (the constants k lie in all of them). Therefore $H^0(X_{n+1}, \mathcal{O}^*) = k^* \to H^0(X_n, \mathcal{O}^*) = k^*$ is bijective. Ex. 4.6 ($\mathcal{J}_n^2 = 0$) gives the exact sequence

$$k^* \xrightarrow{\sim} k^* \to V_n \to \operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n \to H^2(Y, \mathcal{J}_n) = 0,$$

so the connecting map $k^* \to V_n$ is zero and

$$0 \to V_n \to \operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n \to 0 \quad \text{is exact} \quad (n \geq 1).$$

(a) The transition maps $\operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n$ are surjective, so the projection $\operatorname{Pic}\widehat{X} = \varprojlim \operatorname{Pic} X_n \to \operatorname{Pic} X_1 = \operatorname{Pic} Y$ is surjective (lift step by step). Let $K_n = \ker(\operatorname{Pic} X_n \to \operatorname{Pic} Y)$ and $K = \ker(\operatorname{Pic}\widehat{X} \to \operatorname{Pic} Y) = \varprojlim K_n$. Since $\operatorname{Pic} X_{n+1} \to \operatorname{Pic} X_n$ is onto and V_n maps to $0 \in \operatorname{Pic} Y$, we get that $K_{n+1} \to K_n$ is surjective with kernel V_n. Hence $K \to K_n$ is surjective for every n, and $V_{n-1} \subseteq K_n$, so K surjects onto groups containing vector spaces of arbitrarily large dimension: K is infinite-dimensional in the following sense.

K is a k-vector space of infinite dimension if $\operatorname{char} k = 0$. Let $\mathfrak{n}_n = \mathcal{I}/\mathcal{I}^n \subseteq \mathcal{O}_{X_n}$, a nilpotent ideal. Then $\mathcal{O}_{X_n}^* \to \mathcal{O}_Y^*$ is surjective with kernel $1 + \mathfrak{n}_n$, and as $H^0(\mathcal{O}_{X_n}^*) = k^* \to H^0(\mathcal{O}_Y^*) = k^*$ is bijective, $K_n = H^1(Y, 1 + \mathfrak{n}_n)$. In characteristic 0 the truncated logarithm and exponential

series give isomorphisms of sheaves of abelian groups $1+\mathfrak{n}_n \cong \mathfrak{n}_n$, compatible with the ring maps $\mathcal{O}_{X_{n+1}} \to \mathcal{O}_{X_n}$, so $K_n \cong H^1(Y, \mathfrak{n}_n)$ is a k-vector space, the transition maps are k-linear, and $K = \varprojlim K_n$ is a k-vector space, which surjects onto $K_n \supseteq V_{n-1}$ with $\dim V_{n-1} \to \infty$. So $\dim_k K = \infty$.

Honest remark. In characteristic $p > 0$, $1+\mathfrak{n}_n$ need not be killed by p (e.g. $(1+a\varepsilon)^2 = 1+a^2\varepsilon^2$ in characteristic 2), so K is in general an abelian group, an inverse limit of successive extensions of k-vector spaces of unbounded dimension, not obviously a k-vector space; the statement "vector space" is proved above in characteristic 0. In all characteristics we obtain what is needed below: K is uncountable. Indeed, $V_n \neq 0$ for all large n, so in the inverse system of surjections $K_{n+1} \to K_n$ the fibres over every element have ≥ 2 elements for large n, and one builds an injection $\{0,1\}^{\mathbf{N}} \hookrightarrow K$.

(b) $\operatorname{Pic} X = \mathbf{Z}$ (II, 6.4) is countable, while $\operatorname{Pic} \widehat{X} \supseteq K$ is uncountable. So there is $\mathfrak{L} \in \operatorname{Pic} \widehat{X}$ not in the image of $\operatorname{Pic} X \to \operatorname{Pic} \widehat{X}$. If $\mathfrak{L}$ were algebraizable, $\mathfrak{L} \cong \widehat{\mathcal{F}}$ with $\mathcal{F}$ coherent, then by the Remark at the end of Ex. 11.6(c) (valid for $N = 2$) $\mathfrak{L} \cong \widehat{\mathcal{M}}$ with $\mathcal{M} \in \operatorname{Pic} X$, a contradiction. So $\mathfrak{L}$ is not algebraizable.

(c) First, the sheaves $\mathcal{O}_{\widehat{X}}(j) = \widehat{\mathcal{O}_X(j)}$, $j \geq 0$, are generated by global sections ($\mathcal{O}_X^3 \to \mathcal{O}_X(j)$ by x_0^j, x_1^j, x_2^j is surjective, and completion is exact). If $\mathfrak{J}(n)$ is generated by global sections, then so is $\mathfrak{J}(n+j) = \mathfrak{J}(n) \otimes \mathcal{O}_{\widehat{X}}(j)$ for all $j \geq 0$ (a tensor product of sheaves generated by global sections is generated by the products of the sections). Hence "some twist $\mathfrak{J}(n)$, $n \in \mathbf{Z}$, is generated by global sections" implies "$\mathfrak{J}(n)$ is generated by global sections for all $n \geq n_0$".

Suppose every locally free $\mathfrak{J}$ on $\widehat{X}$ had some twist generated by global sections. Then condition (ii) of Ex. 11.6(b) holds, hence so does (i), and the invertible sheaf $\mathfrak{L}$ of (b) would be algebraizable, contradicting (b). Hence there is a locally free sheaf $\mathfrak{J}$ on $\widehat{X}$ such that no twist $\mathfrak{J}(n)$, $n \in \mathbf{Z}$, is generated by global sections. □

Exercise 11.8

> **Problem.** Let $f\colon X \to Y$ be a projective morphism, let $\mathcal{F}$ be a coherent sheaf on X which is flat over Y, and assume that $H^i(X_y, \mathcal{F}_y) = 0$ for some i and some $y \in Y$. Then show that $R^i f_*(\mathcal{F})$ is 0 in a neighborhood of y.

Solution. (Y is noetherian, as throughout.) Let $A = \mathcal{O}_{Y,y}$, $\mathfrak{m}$ its maximal ideal, $k(y) = A/\mathfrak{m}$. By the flat base change $\operatorname{Spec} A \to Y$ (Prop. 9.3) we may replace Y by $\operatorname{Spec} A$, X by $X \times_Y \operatorname{Spec} A$ and $\mathcal{F}$ by its pull-back (still coherent and flat over A, and the fibre over y is unchanged). Let $X_n = X \times \operatorname{Spec} A/\mathfrak{m}^n$, $\mathcal{F}_n = \mathcal{F} \otimes_A A/\mathfrak{m}^n$ (a sheaf on the space X_y), so $X_1 = X_y$, $\mathcal{F}_1 = \mathcal{F}_y$.

Claim: $H^i(X_n, \mathcal{F}_n) = 0$ for all $n \geq 1$. For $n = 1$ this is the hypothesis. Tensoring $0 \to \mathfrak{m}^n/\mathfrak{m}^{n+1} \to A/\mathfrak{m}^{n+1} \to A/\mathfrak{m}^n \to 0$ with the flat A-module $\mathcal{F}$ gives an exact sequence of sheaves on X_y:

$$0 \to \mathcal{F} \otimes_A (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \to \mathcal{F}_{n+1} \to \mathcal{F}_n \to 0.$$

As $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is a finite-dimensional $k(y)$-vector space, say of dimension r, the left term is $\mathcal{F}_y^{\oplus r}$, whose H^i is 0. The exact sequence $H^i(\mathcal{F}_y^{\oplus r}) \to H^i(\mathcal{F}_{n+1}) \to H^i(\mathcal{F}_n)$ and induction on n give the claim.

Now by Thm. 11.1, $(R^i f_*\mathcal{F})^\wedge_y = \varprojlim H^i(X_n, \mathcal{F}_n) = 0$. The stalk $M = (R^i f_*\mathcal{F})_y$ is a finitely generated A-module (Prop. 8.8(b)) and its $\mathfrak{m}$-adic completion $M \otimes_A \widehat{A}$ is 0; since $A \to \widehat{A}$ is faithfully flat, $M = 0$. A coherent sheaf with zero stalk at y vanishes in a neighborhood of y. (Alternatively: $H^i(X_y, \mathcal{F}_y) = 0$ makes $\varphi^i(y)\colon R^i f_*\mathcal{F} \otimes k(y) \to H^i(X_y, \mathcal{F}_y)$ surjective, hence an isomorphism by Cor. 12.11(a); so $R^i f_*\mathcal{F} \otimes k(y) = 0$ and Nakayama gives the same conclusion.) □

12. The semicontinuity theorem

Notation as in the text: for $f\colon X \to Y$ projective and $\mathcal{F}$ coherent and flat over Y, $h^i(y, \mathcal{F}) = \dim_{k(y)} H^i(X_y, \mathcal{F}_y)$, and $\varphi^i(y)\colon R^i f_*\mathcal{F} \otimes k(y) \to H^i(X_y, \mathcal{F}_y)$ is the natural map. Thm. 12.8 says that $h^i(y, \mathcal{F})$ is upper semicontinuous on Y, so by Remark 12.7.1 each set $\{y \mid h^i(y, \mathcal{F}) \geq n\}$ is closed. We use freely that on a scheme of finite type over a field k every nonempty closed subset contains a closed point, so that a closed subset containing all closed points is everything.

Exercise 12.1

> **Problem.** Let Y be a scheme of finite type over an algebraically closed field k. Show that the function $\varphi(y) = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2)$ is upper semicontinuous on the set of closed points of Y.

Solution. Let y be a closed point. Then $A = \mathcal{O}_{Y,y}$ is a local ring containing k with residue field $A/\mathfrak{m}_y = k$ (k algebraically closed, y closed). By (II, Prop. 8.7) the map $d\colon \mathfrak{m}_y/\mathfrak{m}_y^2 \to \Omega_{A/k} \otimes_A k$ is an isomorphism. Moreover $\Omega_{A/k} = (\Omega_{Y/k})_y$ (II, 8.2A), so

$$\varphi(y) = \dim_k \big(\Omega_{Y/k} \otimes k(y)\big).$$

Since Y is of finite type over k, $\Omega_{Y/k}$ is coherent (II, Rem. 8.9.1), and by Example 12.7.2 the function $y \mapsto \dim_{k(y)}(\Omega_{Y/k} \otimes k(y))$ is upper semicontinuous on Y. Its restriction to the subspace of closed points is therefore upper semicontinuous (the induced topology on the closed points consists of the intersections with open sets of Y). □

Exercise 12.2

> **Problem.** Let $\{X_t\}$ be a family of hypersurfaces of the same degree in $\mathbf{P}^n_k$. Show that for each i, the function $h^i(X_t, \mathcal{O}_{X_t})$ is a constant function of t.

Solution. Interpretation: t runs through the points of a parameter scheme T (of finite type over k, say) and $X_t \subseteq \mathbf{P}^n_{k(t)}$ is the hypersurface $V(f_t)$ defined by a nonzero form f_t of the fixed degree $d \geq 1$ (f_t in $k(t)[x_0, \ldots, x_n]$); h^i is computed over $k(t)$. We show that $h^i(X_t, \mathcal{O}_{X_t})$ depends only on n, d, i (and not on f_t, on t, or on whether the family is flat).

Fix a field $K = k(t)$, $P = \mathbf{P}^n_K$, $X = V(f)$, $f \neq 0$ homogeneous of degree $d \geq 1$. Multiplication by f is injective on $\mathcal{O}_P(-d)$, so $0 \to \mathcal{O}_P(-d) \to \mathcal{O}_P \to \mathcal{O}_X \to 0$ is exact. By Thm. 5.1: $H^0(\mathcal{O}_P) = K$ and $H^j(\mathcal{O}_P) = 0$ for $j > 0$; $H^0(\mathcal{O}_P(-d)) = 0$; $H^j(\mathcal{O}_P(-d)) = 0$ for $0 < j < n$; and $H^n(\mathcal{O}_P(-d))$ is dual to $H^0(\mathcal{O}_P(d - n - 1))$, of dimension $\binom{d-1}{n}$.

If $n = 1$: $0 \to K \to H^0(\mathcal{O}_X) \to H^1(\mathcal{O}_P(-d)) \to H^1(\mathcal{O}_P) = 0$, so $h^0 = 1 + (d-1) = d$ and $h^i = 0$ for $i \geq 1$ (as $\dim X = 0$).

If $n \geq 2$: $H^1(\mathcal{O}_P(-d)) = 0$, so $H^0(\mathcal{O}_X) \cong H^0(\mathcal{O}_P) = K$ and $h^0 = 1$. For $1 \leq i \leq n-2$: $H^i(\mathcal{O}_P) = 0 = H^{i+1}(\mathcal{O}_P(-d))$, so $h^i = 0$. For $i = n-1$: $0 = H^{n-1}(\mathcal{O}_P) \to H^{n-1}(\mathcal{O}_X) \to H^n(\mathcal{O}_P(-d)) \to H^n(\mathcal{O}_P) = 0$, so $h^{n-1} = \binom{d-1}{n}$; and $h^i = 0$ for $i \geq n$ ($\dim X = n-1$).

In all cases the values depend only on (n, d, i), so each $h^i(X_t, \mathcal{O}_{X_t})$ is constant in t. □

Exercise 12.3

Problem. Let $X_1 \subseteq \mathbf{P}^4_k$ be the *rational normal quartic curve* (which is the 4-uple embedding of $\mathbf{P}^1$ in $\mathbf{P}^4$). Let $X_0 \subseteq \mathbf{P}^3_k$ be a nonsingular rational quartic curve, such as the one in (I, Ex. 3.18b). Use (9.8.3) to construct a flat family $\{X_t\}$ of curves in $\mathbf{P}^4$, parametrized by $T = \mathbf{A}^1$, with the given fibres X_1 and X_0 for $t = 1$ and $t = 0$. Let $\mathcal{I} \subseteq \mathcal{O}_{\mathbf{P}^4 \times T}$ be the ideal sheaf of the total family $X \subseteq \mathbf{P}^4 \times T$. Show that $\mathcal{I}$ is flat over T. Then show that

$$h^0(t, \mathcal{I}) = \begin{cases} 0 & t \neq 0 \\ 1 & t = 0 \end{cases} \qquad \text{and also} \qquad h^1(t, \mathcal{I}) = \begin{cases} 0 & t \neq 0 \\ 1 & t = 0. \end{cases}$$

This gives another example of cohomology groups jumping at a special point.

Solution. Misprint. As printed (with the untwisted ideal sheaf) the displayed values are false: $H^0(\mathbf{P}^4, \mathcal{I}_t) = 0$ always (it consists of constants vanishing on a nonempty set) and, as computed below, $H^1(\mathbf{P}^4, \mathcal{I}_t) = 0$ for *all* t, because $H^0(\mathcal{O}_{X_t}) = k$ for every t. The statement is true, and is what the example is meant to illustrate, for the twist $\mathcal{I}(1)$: $h^0(t, \mathcal{I}(1)) =$ number of hyperplanes containing X_t, which jumps at $t = 0$ because X_0 lies in a hyperplane. We prove the statement with $\mathcal{I}(1)$ (and we also verify the untwisted vanishing).

Construction of the family. Let $\nu\colon \mathbf{P}^1 \xrightarrow{\sim} X_0 \subseteq \mathbf{P}^3$, $\nu = (g_0 : g_1 : g_2 : g_3)$ with g_i binary quartic forms in v, w (for the quartic of I, Ex. 3.18b: $g = (v^4, v^3w, vw^3, w^4)$). Since $\nu^*\mathcal{O}(1) = \mathcal{O}_{\mathbf{P}^1}(4)$ (the curve has degree 4), $g_0, \ldots, g_3$ are sections of $\mathcal{O}_{\mathbf{P}^1}(4)$; they are linearly independent, since X_0 is not contained in a plane (a plane quartic has arithmetic genus 3, while $X_0 \cong \mathbf{P}^1$). Complete them by a form g_4 to a basis of the 5-dimensional space $H^0(\mathbf{P}^1, \mathcal{O}(4))$ (for the standard example $g_4 = v^2w^2$). Then $(g_0 : \cdots : g_4)$ is the complete linear system $|\mathcal{O}(4)|$, i.e. the 4-uple embedding followed by a linear change of coordinates, and its image is a rational normal quartic $X_1 \subseteq \mathbf{P}^4$ (we choose the coordinates $z_0, \ldots, z_4$ of $\mathbf{P}^4$ so that this is the given X_1). The point $Q = (0 : 0 : 0 : 0 : 1)$ is not on X_1 (the $g_0, \ldots, g_3$ have no common zero), and the projection of X_1 from Q to the hyperplane $H = \{z_4 = 0\} \cong \mathbf{P}^3$ is X_0.

As in (9.8.3), for $t \in k^*$ (or t a point of $\mathbf{G}_m = \mathbf{A}^1 \setminus 0$) let $\sigma_t(z_0 : \cdots : z_3 : z_4) = (z_0 : \cdots : z_3 : tz_4)$ and $X_t = \sigma_t(X_1)$. Over $\mathbf{G}_m$ the automorphism $\sigma\colon \mathbf{P}^4 \times \mathbf{G}_m \to \mathbf{P}^4 \times \mathbf{G}_m$, $(z, t) \mapsto (\sigma_t z, t)$, carries $X_1 \times \mathbf{G}_m$ to a closed subscheme $X' \cong X_1 \times \mathbf{G}_m$, flat over $\mathbf{G}_m$, with fibres $X_t = \sigma_t(X_1) \cong X_1$, and X' is reduced and irreducible. By Prop. 9.8 (with $Y = T = \mathbf{A}^1$, $P = 0$) the scheme-theoretic closure X of X' in $\mathbf{P}^4_T$ is flat over T (and is the unique such extension); X_1 is the fibre at $t = 1$.

The fibre at 0. Let $\Phi\colon \mathbf{P}^1 \times \mathbf{A}^1 \to \mathbf{P}^4 \times \mathbf{A}^1$, $((v : w), t) \mapsto ((g_0 : g_1 : g_2 : g_3 : tg_4), t)$ (a morphism since $g_0, \ldots, g_3$ have no common zero). It is proper, so its image Z is closed and irreducible, and contains $X' = \Phi(\mathbf{P}^1 \times \mathbf{G}_m)$ as a dense subset. Hence $|X| = Z$ and $|X_0| = Z \cap \{t = 0\} = \Phi(\mathbf{P}^1 \times \{0\}) = \{(g_0 : \cdots : g_3 : 0)\} = X_0^{\mathrm{set}}$, the set underlying the curve

$X_0 \subseteq H \subseteq \mathbf{P}^4$. (We write C for X_0 regarded as a reduced subscheme of $\mathbf{P}^4$, and X_0^{fib} for the fibre at 0 of the family just constructed; so far we know X_0^{fib} has underlying set C.) The scheme X_0^{fib} has the same Hilbert polynomial as X_1 (Thm. 9.9): $P(m) = \chi(\mathcal{O}_{\mathbf{P}^1}(4m)) = 4m+1$. Since C is reduced with the same support, $C \subseteq X_0^{\mathrm{fib}}$ is the reduced subscheme, with ideal $\mathcal{N}$ (nilpotent). Also $C \cong \mathbf{P}^1$ and $\mathcal{O}_C(1) = \mathcal{O}_{\mathbf{P}^1}(4)$, so $P_C(m) = 4m+1$. From $0 \to \mathcal{N} \to \mathcal{O}_{X_0^{\mathrm{fib}}} \to \mathcal{O}_C \to 0$ we get $P_{\mathcal{N}} = 0$; but a nonzero coherent sheaf $\mathcal{N}$ has $\chi(\mathcal{N}(m)) = h^0(\mathcal{N}(m)) > 0$ for $m \gg 0$ (Serre). So $\mathcal{N} = 0$ and

$$X_0^{\mathrm{fib}} = C = X_0.$$

Thus $\{X_t\}_{t \in \mathbf{A}^1}$ is a flat family of curves in $\mathbf{P}^4$ with fibres X_1 at $t=1$ and X_0 at $t=0$ (here X_0 is viewed in $\mathbf{P}^3 \subseteq \mathbf{P}^4$), and $X_t \cong$ rational normal quartic for $t \neq 0$.

Flatness of $\mathcal{I}$. In $0 \to \mathcal{I} \to \mathcal{O}_{\mathbf{P}^4_T} \to \mathcal{O}_X \to 0$ both $\mathcal{O}_{\mathbf{P}^4_T}$ and $\mathcal{O}_X$ are flat over T, so $\mathcal{I}$ is flat over T (Prop. 9.1A(e)). Moreover, tensoring with $k(t)$ over $\mathcal{O}_T$ and using $\mathrm{Tor}_1^{\mathcal{O}_T}(\mathcal{O}_X, k(t)) = 0$ gives for every point $t \in T$ the exact sequence $0 \to \mathcal{I}_t \to \mathcal{O}_{\mathbf{P}^4_{k(t)}} \to \mathcal{O}_{X_t} \to 0$, i.e. $\mathcal{I}_t$ is the ideal sheaf of the fibre X_t and $h^i(t, \mathcal{I}(m)) = \dim H^i(\mathbf{P}^4_{k(t)}, \mathcal{I}_t(m))$. (Below $k(t) = k$ for closed t; the generic point is treated identically over $k(t)$.)

Untwisted groups. From $0 \to H^0(\mathcal{I}_t) \to H^0(\mathcal{O}_{\mathbf{P}^4}) = k \to H^0(\mathcal{O}_{X_t}) \to H^1(\mathcal{I}_t) \to H^1(\mathcal{O}_{\mathbf{P}^4}) = 0$, and $H^0(\mathcal{O}_{X_t}) = k$ for every t (each fibre is $\cong \mathbf{P}^1$, including $X_0 = C$), we get $h^0(t, \mathcal{I}) = h^1(t, \mathcal{I}) = 0$ for all t.

Twisted groups. Twisting by $\mathcal{O}(1)$: $0 \to H^0(\mathcal{I}_t(1)) \to H^0(\mathcal{O}_{\mathbf{P}^4}(1)) \xrightarrow{\rho_t} H^0(\mathcal{O}_{X_t}(1)) \to H^1(\mathcal{I}_t(1)) \to H^1(\mathcal{O}_{\mathbf{P}^4}(1)) = 0$. Here $H^0(\mathcal{O}_{\mathbf{P}^4}(1))$ has dimension 5 (basis $z_0, \dots, z_4$), and $X_t \cong \mathbf{P}^1$ with $\mathcal{O}_{X_t}(1) \cong \mathcal{O}_{\mathbf{P}^1}(4)$, so $h^0(\mathcal{O}_{X_t}(1)) = 5$.

- $t \neq 0$: ρ_t sends $z_i \mapsto g_i$ ($i \leq 3$) and $z_4 \mapsto t g_4$. As $g_0, \dots, g_4$ is a basis of $H^0(\mathcal{O}_{\mathbf{P}^1}(4))$ and $t \neq 0$ in $k(t)$, ρ_t is an isomorphism. So $h^0(t, \mathcal{I}(1)) = 0$ and $h^1(t, \mathcal{I}(1)) = 0$.
- $t = 0$: ρ_0 sends $z_i \mapsto g_i$ ($i \leq 3$) and $z_4 \mapsto 0$, so $\ker \rho_0 = k \cdot z_4$ is 1-dimensional and the image is 4-dimensional, so the cokernel is 1-dimensional. So $h^0(0, \mathcal{I}(1)) = 1$ and $h^1(0, \mathcal{I}(1)) = 1$.

This is the stated jump: $h^0(t, \mathcal{I}(1)) = h^1(t, \mathcal{I}(1)) = 0$ for $t \neq 0$ and $= 1$ for $t = 0$. (Consistent with Thm. 12.8, and with χ constant: $h^0 - h^1 = 0$ for all t.) □

Exercise 12.4

> **Problem.** Let Y be an integral scheme of finite type over an algebraically closed field k. Let $f\colon X \to Y$ be a flat projective morphism whose fibres are all integral schemes. Let $\mathcal{L}, \mathcal{M}$ be invertible sheaves on X, and assume for each $y \in Y$ that $\mathcal{L}_y \cong \mathcal{M}_y$ on the fibre X_y. Then show that there is an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}$. [*Hint*: Use the results of this section to show that $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank 1 on Y.]

Solution. (We use the hypothesis $\mathcal{L}_y \cong \mathcal{M}_y$ only for the closed points y.) Put $\mathcal{G} = \mathcal{L} \otimes \mathcal{M}^{-1}$, an invertible sheaf on X, flat over Y (locally $\cong \mathcal{O}_X$, and X is flat over Y). By hypothesis $\mathcal{G}_y \cong \mathcal{O}_{X_y}$ for all closed $y \in Y$. Every fibre is integral, in particular nonempty.

Step 1: $h^0(y, \mathcal{G}) = 1$ for all $y \in Y$. For y closed, X_y is an integral projective scheme over $k(y) = k$, so $H^0(X_y, \mathcal{O}_{X_y})$ is a finite-dimensional k-algebra which is a domain, hence a field finite over

k, hence $= k$. So $h^0(y, \mathcal{G}) = 1$ for every closed point y. By Thm. 12.8 the set $\{y \mid h^0(y, \mathcal{G}) \geq 2\}$ is closed and contains no closed point, hence is empty; the set $\{y \mid h^0(y, \mathcal{G}) \geq 1\}$ is closed and contains all closed points, hence equals Y. So $h^0(y, \mathcal{G}) = 1$ for all $y \in Y$.

Step 2: $\mathcal{N} := f_*\mathcal{G}$ is invertible. Y is integral and $h^0(y, \mathcal{G})$ is constant. By Cor. 12.9, $f_*\mathcal{G}$ is locally free on Y and $\varphi^0(y)\colon f_*\mathcal{G} \otimes k(y) \to H^0(X_y, \mathcal{G}_y)$ is an isomorphism for every y. Hence $\mathcal{N}$ is locally free of rank $h^0(y, \mathcal{G}) = 1$.

Step 3: $f^*\mathcal{N} \cong \mathcal{G}$. Consider the natural map $\alpha\colon f^*f_*\mathcal{G} \to \mathcal{G}$, a map of invertible sheaves on X; it suffices to show it is surjective (a surjective map of invertible sheaves is an isomorphism). Its cokernel has closed support; if nonempty, it contains a closed point $x \in X$ (finite type over k), lying over a closed point $y = f(x)$. Restricted to the fibre X_y, α becomes the map $\mathcal{O}_{X_y} \otimes_k (\mathcal{N} \otimes k(y)) \to \mathcal{G}_y$ adjoint to $\varphi^0(y)$, i.e. it sends a generator e of $\mathcal{N} \otimes k(y)$ to the section $\varphi^0(y)(e) \in H^0(X_y, \mathcal{G}_y)$, which is nonzero because $\varphi^0(y)$ is an isomorphism. Now $\mathcal{G}_y \cong \mathcal{O}_{X_y}$ and $H^0(X_y, \mathcal{O}_{X_y}) = k$, so this section is a nonzero constant times a generator, hence a unit: $\alpha|_{X_y}$ is an isomorphism, in particular surjective at x. By Nakayama α is surjective at x, a contradiction. So α is surjective, hence an isomorphism, and $\mathcal{G} \cong f^*\mathcal{N}$, i.e.

$$\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}.$$

Remark. Every step is local on Y (semicontinuity, Grauert's corollary, the formation of f_*), so the argument applies equally if f is only locally projective over Y; this is used in Ex. 12.5. □

Exercise 12.5

> **Problem.** Let Y be an integral scheme of finite type over an algebraically closed field k. Let $\mathcal{E}$ be a locally free sheaf on Y, and let $X = \mathbf{P}(\mathcal{E})$—see (II, Section 7). Then show that $\operatorname{Pic} X \cong (\operatorname{Pic} Y) \times \mathbf{Z}$. This strengthens (II, Ex. 7.9).

Solution. Hypothesis. We assume $\operatorname{rank} \mathcal{E} = r \geq 2$, as in (II, Ex. 7.9): for $r = 1$, $X = Y$ and the statement is false (it would give $\operatorname{Pic} Y \cong \operatorname{Pic} Y \times \mathbf{Z}$ for, e.g., $Y = \operatorname{Spec} k$). Let $\pi\colon X \to Y$ be the projection and $\mathcal{O}_X(1)$ the tautological invertible sheaf (II, 7.11). Locally on Y (where $\mathcal{E}$ is free), $X \cong \mathbf{P}^{r-1}_V$, so π is flat, locally projective, with all closed fibres $X_y \cong \mathbf{P}^{r-1}_k$ and $\mathcal{O}_X(1)|_{X_y} = \mathcal{O}_{\mathbf{P}^{r-1}}(1)$; also $\pi_*\mathcal{O}_X = \mathcal{O}_Y$.

Define $\psi\colon \operatorname{Pic} Y \times \mathbf{Z} \to \operatorname{Pic} X$, $\psi(\mathcal{N}, m) = \pi^*\mathcal{N} \otimes \mathcal{O}_X(m)$, clearly a homomorphism. We show it is bijective.

Injective. Suppose $\pi^*\mathcal{N} \otimes \mathcal{O}_X(m) \cong \mathcal{O}_X$. Restrict to the fibre over a closed point y: $\pi^*\mathcal{N}$ is trivial on X_y, so $\mathcal{O}_{\mathbf{P}^{r-1}}(m) \cong \mathcal{O}$, forcing $m = 0$ since $r - 1 \geq 1$ (II, 6.4). Then $\pi^*\mathcal{N} \cong \mathcal{O}_X$; applying π_* and the projection formula, $\mathcal{N} \cong \pi_*\pi^*\mathcal{N} \cong \mathcal{N} \otimes \pi_*\mathcal{O}_X$ and $\pi_*\pi^*\mathcal{N} \cong \pi_*\mathcal{O}_X = \mathcal{O}_Y$. So $\mathcal{N} \cong \mathcal{O}_Y$.

Surjective. Let $\mathcal{L} \in \operatorname{Pic} X$. For a closed point $y \in Y$, $\mathcal{L}|_{X_y} \cong \mathcal{O}_{\mathbf{P}^{r-1}}(m_y)$ for a unique $m_y \in \mathbf{Z}$ (II, 6.4).

Claim: m_y is independent of y. For $c \in \mathbf{Z}$ let $A_c = \{y \in Y \mid h^0(y, \mathcal{L}(-c)) \geq 1\}$ and $B_c = \{y \in Y \mid h^0(y, \mathcal{L}^{-1}(c)) \geq 1\}$. These are closed in Y by Thm. 12.8 (semicontinuity is local on Y, and $\mathcal{L}(-c)$, $\mathcal{L}^{-1}(c)$ are invertible and flat over Y). For closed y, $h^0(y, \mathcal{L}(-c)) = h^0(\mathbf{P}^{r-1}, \mathcal{O}(m_y - c)) \geq 1 \iff m_y \geq c$, and $y \in B_c \iff m_y \leq c$. Hence $A_c \cup B_{c-1}$ is closed and contains all closed

points, so equals Y; as Y is irreducible, $Y = A_c$ or $Y = B_{c-1}$. Fix a closed y_0 and take $c = m_{y_0}$: $y_0 \notin B_{c-1}$, so $Y = A_c$, whence $m_y \geq m_{y_0}$ for all closed y. Exchanging the roles of the points, $m_{y_0} \geq m_y$. So $m_y = m$ for all closed y.

Now $\mathcal{L}(-m) = \mathcal{L} \otimes \mathcal{O}_X(-m)$ restricts to $\mathcal{O}$ on every closed fibre $\mathbf{P}_k^{r-1}$, i.e. it agrees on closed fibres with $\mathcal{M} = \mathcal{O}_X$. All fibres are integral, so Ex. 12.4 (applied with $\mathcal{L}(-m)$ and $\mathcal{O}_X$, which only needs the closed fibres and works locally on Y) gives $\mathcal{N} \in \operatorname{Pic} Y$ with $\mathcal{L}(-m) \cong \pi^*\mathcal{N}$, i.e. $\mathcal{L} \cong \psi(\mathcal{N}, m)$.

Hence $\operatorname{Pic} X \cong \operatorname{Pic} Y \times \mathbf{Z}$. □

Exercise 12.6

Problem. Let X be an integral projective scheme over an algebraically closed field k, and assume that $H^1(X, \mathcal{O}_X) = 0$. Let T be a connected scheme of finite type over k.

(a) If $\mathcal{L}$ is an invertible sheaf on $X \times T$, show that the invertible sheaves $\mathcal{L}_t$ on $X_t = X \times \{t\}$ are isomorphic, for all closed points $t \in T$.

(b) Show that $\operatorname{Pic}(X \times T) = \operatorname{Pic} X \times \operatorname{Pic} T$. (Do *not* assume that T is reduced!) Cf. (IV, Ex. 4.10) and (V, Ex. 1.6) for examples where $\operatorname{Pic}(X \times T) \neq \operatorname{Pic} X \times \operatorname{Pic} T$. [*Hint*: Apply (12.11) with $i = 0, 1$ for suitable invertible sheaves on $X \times T$.]

(Starred in the book.)

Solution. Let $p\colon X \times T \to T$ be the second projection: it is projective and flat (base change of $X \to \operatorname{Spec} k$), T is noetherian, and every invertible sheaf on $X \times T$ is flat over T. For a closed point t we have $k(t) = k$ and $X_t = X$. Since X is integral and projective over the algebraically closed field k, $H^0(X, \mathcal{O}_X) = k$ (a finite-dimensional domain over k is k). Also: a nonzero section of an invertible sheaf on an integral scheme is nonzero at the generic point.

Lemma. Let $\mathcal{G}$ be an invertible sheaf on $X \times T$ and $t_1 \in T$ a closed point with $\mathcal{G}_{t_1} \cong \mathcal{O}_X$. Then there is an open neighborhood W of t_1 such that $p_*\mathcal{G}|_W$ is invertible, and $\varphi^0(t)\colon p_*\mathcal{G} \otimes k(t) \to H^0(X_t, \mathcal{G}_t)$ is an isomorphism for all $t \in W$; in particular $h^0(t, \mathcal{G}) = 1$ for all $t \in W$.

Proof. $H^1(X_{t_1}, \mathcal{G}_{t_1}) = H^1(X, \mathcal{O}_X) = 0$, so $\varphi^1(t_1)$ is surjective, hence (Cor. 12.11(a)) an isomorphism near t_1, and $R^1p_*\mathcal{G} \otimes k(t_1) = 0$; by Nakayama ($R^1p_*\mathcal{G}$ is coherent) $R^1p_*\mathcal{G} = 0$ in a neighborhood of t_1, in particular it is locally free there. Now Cor. 12.11(b) with $i = 1$ ($\varphi^1(t_1)$ surjective and $R^1p_*\mathcal{G}$ locally free near t_1) shows that $\varphi^0(t_1)$ is surjective, hence an isomorphism by 12.11(a) (for $i = 0$), and the same holds for all t in a neighborhood W of t_1. Also $p_*\mathcal{G}$ is locally free near t_1: since $\varphi^0(t_1)$ is surjective, T^0 is right exact at t_1 (Prop. 12.10); it is always left exact ($T^0(M) = H^0(X, \mathcal{G} \otimes M)$ and $\mathcal{G}$ is flat), hence exact at t_1, so $T^0(A) = H^0(X, \mathcal{G})\widetilde{\ }$ is projective locally at t_1 (Cor. 12.6, 12.7), and $p_*\mathcal{G} = \widetilde{T^0(A)}$ is locally free near t_1. Its rank is $\dim_k H^0(X_{t_1}, \mathcal{G}_{t_1}) = \dim H^0(X, \mathcal{O}_X) = 1$. Shrinking W gives the claim. □

(a) Fix a closed point t_0 and put $\mathcal{M} = \mathcal{L}_{t_0}$ (an invertible sheaf on X) and $\mathcal{G} = \mathcal{L} \otimes p_1^*\mathcal{M}^{-1}$, so $\mathcal{G}_t = \mathcal{L}_t \otimes \mathcal{M}^{-1}$ for closed t and $\mathcal{G}_{t_0} \cong \mathcal{O}_X$. We must show $\mathcal{G}_t \cong \mathcal{O}_X$ for all closed t.

For an invertible sheaf $\mathcal{H}$ on X: $\mathcal{H} \cong \mathcal{O}_X$ if and only if $H^0(X, \mathcal{H}) \neq 0$ and $H^0(X, \mathcal{H}^{-1}) \neq 0$. (If s, s' are nonzero sections, then $ss' \in H^0(\mathcal{O}_X) = k$ is nonzero at the generic point, hence a nonzero constant, so s is nowhere zero and trivializes $\mathcal{H}$; the converse is clear.) Let

$$Z = \{t \in T \mid h^0(t, \mathcal{G}) \geq 1 \text{ and } h^0(t, \mathcal{G}^{-1}) \geq 1\},$$

a closed subset of T (Thm. 12.8). For closed t: $t \in Z \iff \mathcal{G}_t \cong \mathcal{O}_X$.

Let t_1 be a closed point of Z. The Lemma applied to $\mathcal{G}$ and to $\mathcal{G}^{-1}$ (both trivial at t_1) gives an open $W_{t_1} \ni t_1$ with $h^0(t, \mathcal{G}) = h^0(t, \mathcal{G}^{-1}) = 1$ on W_{t_1}, so $W_{t_1} \subseteq Z$. Let $W = \bigcup W_{t_1}$ over all closed $t_1 \in Z$: it is open, $W \subseteq Z$, and Z minus W is a closed subset of T containing no closed point, hence empty. So $Z = W$ is open and closed, nonempty ($t_0 \in Z$), and T is connected: $Z = T$. Thus $\mathcal{L}_t \cong \mathcal{L}_{t_0}$ for all closed t.

(b) Let $\psi\colon \operatorname{Pic} X \times \operatorname{Pic} T \to \operatorname{Pic}(X \times T)$, $\psi(\mathcal{M}, \mathcal{N}) = p_1^*\mathcal{M} \otimes p_2^*\mathcal{N}$ ($p_2 = p$), a homomorphism. We show it is bijective.

Injective. Let $p_1^*\mathcal{M} \otimes p_2^*\mathcal{N} \cong \mathcal{O}_{X\times T}$. Restricting to $X \times \{t_0\} = X$ (t_0 a closed point; $T \neq \emptyset$) gives $\mathcal{M} \otimes$ (trivial sheaf) $\cong \mathcal{O}_X$, so $\mathcal{M} \cong \mathcal{O}_X$. Then $p_2^*\mathcal{N} \cong \mathcal{O}_{X\times T}$. Applying p_{2*} and the projection formula, $\mathcal{N} \otimes p_{2*}\mathcal{O}_{X\times T} \cong p_{2*}\mathcal{O}_{X\times T}$, and $p_{2*}\mathcal{O}_{X\times T} = H^0(X, \mathcal{O}_X) \otimes_k \mathcal{O}_T = \mathcal{O}_T$ by flat base change (Prop. 9.3: over an affine open $\operatorname{Spec} B \subseteq T$, $H^0(X \times \operatorname{Spec} B, \mathcal{O}) = H^0(X, \mathcal{O}_X) \otimes_k B = B$). So $\mathcal{N} \cong \mathcal{O}_T$. (This part uses neither $H^1(X, \mathcal{O}_X) = 0$ nor any reducedness of T.)

Surjective. Let $\mathcal{L} \in \operatorname{Pic}(X \times T)$. By (a), $\mathcal{L}_t \cong \mathcal{L}_{t_0} =: \mathcal{M}$ for all closed t. Put $\mathcal{G} = \mathcal{L} \otimes p_1^*\mathcal{M}^{-1}$; then $\mathcal{G}_t \cong \mathcal{O}_X$ for every closed $t \in T$. By the Lemma, every closed point t_1 has an open neighborhood W_{t_1} on which $p_*\mathcal{G}$ is invertible and $\varphi^0(t)$ is an isomorphism. The union of the W_{t_1} is an open set containing all closed points, so equals T. Therefore $\mathcal{N} := p_*\mathcal{G}$ is an invertible sheaf on T (no reducedness of T is used: the rank is read off from the fibre at closed points, and the Lemma gives local freeness directly from 12.11), and $\varphi^0(t)$ is an isomorphism for every closed t.

Consider $\alpha\colon p^*\mathcal{N} = p^*p_*\mathcal{G} \to \mathcal{G}$, a map of invertible sheaves on $X \times T$. Its cokernel has closed support in $X \times T$ (finite type over k); if nonempty it contains a closed point x, lying over a closed point t of T. On the fibre $X_t = X$, α is $\mathcal{O}_X \otimes_k (\mathcal{N} \otimes k(t)) \to \mathcal{G}_t \cong \mathcal{O}_X$, sending a generator to $\varphi^0(t)$(generator), a nonzero element of $H^0(X, \mathcal{G}_t) = k \cdot$ (generator of $\mathcal{G}_t$), hence a unit multiple of a generator. So $\alpha|_{X_t}$ is an isomorphism, and by Nakayama α is surjective at x, a contradiction. So α is a surjection of invertible sheaves, hence an isomorphism: $\mathcal{G} \cong p_2^*\mathcal{N}$, i.e.

$$\mathcal{L} \cong p_1^*\mathcal{M} \otimes p_2^*\mathcal{N}.$$

Hence ψ is surjective, and $\operatorname{Pic}(X \times T) = \operatorname{Pic} X \times \operatorname{Pic} T$. □